UN GTR 9 (version 0.3.0)

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I. Statement of technical rationale and justification

A. Phase 1

0 bis. Sections 1. to 10. reflect the development of Phase 1 of UN Global Techincal Regulation (GTR) No. 9 and concern the legform test procedure with the lower legform impactor designed by the European Enhanced Vehicle-Safety Committee (EEVC) and the upper legform impactor for the high bumper vehicles as well as the headform test procedure.

0. Foreword:

UN Global Technical Regulation (UN GTR) No. 9, Pedestrian Safety, was established in the Global Registry on 12 November 2008. The development of the original GTR was discussed as "Phase 1" beginning with paragraph 0 bis. The GTR was amended by Corrigendum 1 on 12 November 2009, and by Corrigendum 2 and Amendment 1, affecting only the scope of the UN GTR, on 10 November 2010. Amendment 2 was established on 14 November 2018 and replaced the European Enhanced Vehicle-safety Committee (EEVC) lower legform impactor used for the bumper test with the flexible pedestrian legform impactor (FlexPLI). Development of Amendment 2 is discussed as "Phase 2" beginning with paragraph 133. Amendment 3 was established on [insert date when established] and added new requirements for vehicles equipped with a Deployable Pedestrian Protection System (DPPS). Amendment 3 is discussed as "Phase 3" beginning with paragraph 228. The informal working group that developed Amendment 3/Phase 3 notes that paragraph 122 in the discussion of Phase 1 has been superseded for vehicles equipped with a DPPS.

1. Safety need

1. Each year, thousands of pedestrians and cyclists are struck by motor vehicles. Most of these accidents take place in urban areas where serious or fatal injuries can be sustained at relatively low speed, particularly in the case of children. This global technical regulation (gtr) will significantly reduce the levels of injury sustained by pedestrians involved in frontal impacts with motor vehicles.

2. This gtr is based on data from a number of sources, including the International Harmonized Research Activities (IHRA) 1/ Pedestrian Safety working group (IHRA/PS). The data was sourced from Australia, Germany, Japan and the United States of America. Data from IHRA 2/, Germany 3/, Italy 4/, the UNECE 5/, Spain 6/, Canada 7/, the Netherlands 8/, Sweden 9/, and Korea 10/ indicate that, annually: in the European Union about 8,000 pedestrians and cyclists are killed and about 300,000 injured; in North America approximately 5,000 pedestrians are killed and 85,000 injured; in Japan approximately 3,300 pedestrians and cyclists are killed and 27,000 seriously injured; and in Korea around 3,600 pedestrians are killed and 90,000 injured.

1/ IHRA is an inter-governmental initiative that aims to facilitate greater harmony of vehicle safety policies through multi-national collaboration in research.

2/ A number of reference documents is listed in the appendix to this global technical regulation. The documents are available on the UNECE WP.29 website at the address:

http://www.unece.org/trans/main/welcwp29.htm.

IHRA data are set forth in working paper No. 3 of the informal group on pedestrian safety (INF GR/PS/3) at http://www.unece.org/trans/doc/2002/wp29grsp/inf-gr-ps-3e.ppt#262,1, 1st meeting of the Informal Group on Pedestrian Safety, and in working paper No. 31 (INF GR/PS/31).

3/ INF GR/PS/12, /13 and /25

4/ INF GR/PS/14

5/ INF GR/PS/15

6/ INF GR/PS/16

7/ INF GR/PS/20

8/ INF GR/PS/21

9/ INF GR/PS/41

10/ INF GR/PS/70

3. The IHRA/PS study indicates the following:

(a) Distribution of the injuries

4. Comparing the ages of those involved, statistics show the highest frequency of accidents is for children of five to nine years old, and for adults over 60 years old. Children (aged 15 and under) account for nearly one-third of all injuries in the dataset, even though they constitute only 18 percent of the population in the four countries included in the IHRA data.

5. The frequency of fatal and serious injuries (Abbreviated Injury Scale: AIS 2-6) is highest for the child and adult head and adult leg body regions (INF GR/PS/3).

(b) Crash speeds

6. Crash speeds between vehicles and pedestrians were collected from pedestrian accident data. The cumulative frequency of the crash speeds shows that a crash speed of up to 40 km/h can cover more than 75 percent of total pedestrian injuries (AIS 1+) in all regions.

(c) Target population for this gtr

7. The IHRA injury data indicate the injury distribution by body regions. Fatal and serious head injuries (AIS2+) of children and adults as well as AIS2+ adult leg injuries were extracted from the IHRA data base for clearly identified injury causing parts on the vehicle and on the road (INF GR/PS/131 and 169). It was found that bonnet/wing contacts caused 41 percent of child head injuries of AIS2+ and 19 percent of the adult AIS2+ head injuries. Bumper contacts lead to 64 percent of adult AIS2+ leg injuries. The cumulative frequency curves versus vehicle impact speed for these injuries and their respective injury causing parts show that 58 percent of the child head AIS2+ injuries are addressed to a vehicle impact speed up to 40 km/h, 40 percent to adult head AIS2+ injuries and 50 percent of the adult leg AIS2+ injuries respectively. Based on these figures of injuries by injury source and vehicle contact area, the target population of the above-mentioned AIS2+ injuries for this proposed gtr is 24 percent of child pedestrian head injuries, 8 percent of adult pedestrian head injuries, and 32 percent of adult leg injuries.

8. Each of these body regions, i.e. head of child/adult and adult leg, covers more than 30 percent of total fatal and severe injuries (INF GR/PS/3). This gtr focuses on protecting these body regions.

9. The major source of child head injuries is the top surface of the bonnet/wing, while adult head injuries result from impacts to the top surface of bonnet/wing and windscreen area. For adult leg injuries, the major source is the front bumper of vehicles.

(d) Applicability to motor vehicle categories

10. The maximum benefit from making vehicles pedestrian friendly would occur if all types of vehicles comply with these technical provisions, but it is recognized that their application to heavier vehicles (large trucks and buses) as well as to very small and light vehicles could be of limited value and may not be technically appropriate in their present form. The tests proposed in this gtr have been developed on the basis of current light vehicles,

taking into account the pedestrian kinematics when impacted by such vehicles. For this reason, the scope of application is limited to passenger cars, sport utility vehicles (SUV), light trucks and other light commercial vehicles. Since these vehicle categories represent the vast majority of vehicles currently in use, the proposed measures will have the widest practicable effect in reducing pedestrian injuries.

II. SUMMARY: DESCRIPTION OF THE PROPOSED REGULATION

(a) Introduction

11. It is generally accepted that in the most representative pedestrian to car accident the pedestrian is in normal walking posture, meaning that the pedestrian is standing sideways to the vehicle, and is struck by the vehicle from the side. This scenario is therefore also the basis for this gtr.

12. When an adult pedestrian is struck by vehicles addressed by this gtr, the first impact is generally between the pedestrian knee region and the vehicle's front bumper. Because this initial contact is usually below the pedestrian's centre of gravity, the upper body in such a case begins to rotate toward the vehicle. The pedestrian's body accelerates linearly relative to the ground because the pedestrian is being carried along by the vehicle. The second contact is usually between the upper part of the grille or front edge of the bonnet and the pedestrian's pelvic area. The pedestrian's legs and pelvis have reached the linear velocity of the vehicle at this point and the upper body (head and thorax) are still rotating toward the vehicle. The final phase of the collision involves the head and thorax striking the vehicle with a linear velocity approaching that of the initial striking velocity of the vehicle. IHRA research has shown that the linear head impact velocity averages about 80 percent of the initial contact velocity.

13. Through the pedestrian accidents analysis, it has been concluded that child and adult heads and adult legs are the body regions most affected by contact with the front end of vehicles. On vehicles, the bumper, the bonnet top and the windscreen area, including the A-pillars, are the vehicle regions mostly identified with a high potential for contact. According to the IHRA/PS study, the abovementioned areas can cover more than 65 percent of the fatal and serious injuries.

14. Based on these study results, the informal group prioritized the development of approaches to simulate a pedestrian impact and encourage countermeasures that will improve pedestrian protection. This gtr would improve pedestrian safety by requiring vehicle bonnets and bumpers to absorb energy more efficiently when impacted in a 40 kilometre per hour (km/h) vehicle-to-pedestrian impact, which accounts for more than 75 percent of the pedestrian injured accidents (AIS 1+) reported by IHRA/PS of the injury frequency.

15. During the activities of the gtr informal group, headform to windscreen testing was proposed to be included in the gtr. IHRA accident studies mentioned above identify the windscreen, windscreen frame and A-pillars as injury causing parts of the vehicle in pedestrian to car collisions.

16. After several discussions on this issue the group decided not to include these kinds of tests into the gtr at this stage for the following reasons:

(i) The group recognized that the A-pillars, windscreen roof and lower frame have to be very stiff vehicle parts due to their functional requirements. As an example in the lower windscreen area, the required deformation space to meet a head impact requirement is restricted by the instrument panel. Some components that are required to meet governmental safety standards, such as defrost/demist etc., make it impossible to lower the dashboard significantly. In addition, the structural components of the dashboard represent important load paths in front or side crashes. On the other hand, the entire windscreen frame would need to be softened extremely to pass any HIC (Head Injury Criterion) requirement. This strongly contradicts roll over requirements and other existing legal and consumer demands (see INF GR/PS/059 and INF GR/PS/035). In addition the group received detailed data showing that extremely high HIC measurements are frequently obtained in these difficult areas (see INF GR/PS/072, 094, 102 and 103) and agreed that there are no technical design solutions or countermeasures available so far to drastically reduce HIC levels.

(ii) The windscreen glass itself does not produce severe injuries and therefore the amount of saved casualties will be very low. The effectiveness of testing the windscreen inner part is seen as very questionable.

(iii) In addition, it was noted that vehicle manufacturers believed that there are problems related to scatter of HIC when performing tests in the windscreen and that the reasons for the scatter are not yet fully understood (see INF GR/PS/134, 163 and 164).

17. However, some delegates expressed interest in having domestic regulations that apply head protection requirements to the windscreen area. The informal group did not believe the gtr would foreclose any jurisdiction from applying head protection requirements to the windscreen area by way of domestic regulations.

(b) Overview

18. This gtr consists of two sets of performance criteria applying to: (a) the bonnet top and wings; and (b) the front bumper. Test procedures have been developed for each region using sub-system impacts for adult and child head protection and adult leg protection. 11/

11/ To develop these test procedures, the group carefully studied the availability of the pedestrian dummy as an alternative method for the test procedures. The group concluded that there is no test dummy presently available that could be considered suitable for regulatory use. Accordingly, the informal group decided to select subsystem test methods which are readily available, and which have the necessary reliability, repeatability and simplicity.

19. The head impact requirements will ensure that bonnet tops and wings will provide head protection when struck by a pedestrian. The bonnet top and wings would be impacted with a child headform and an adult headform at 35 kilometres per hour (km/h). The HIC must not exceed 1,000 over one half of a child headform test area and must not exceed 1,000 over two thirds of a combined child and adult headform test areas. The HIC for the remaining areas must not exceed 1,700 for both headforms.

20. The leg protection requirements for the front bumper would require bumpers to subject pedestrians to lower impact forces. This gtr specifies that the vehicle bumper is struck at 40 km/h with a legform that simulates the impact response of an adult's leg. Vehicles with a lower bumper height of less than 425 millimetre (mm) are tested with a lower legform, while vehicles with a lower bumper height of more than 500 mm are tested with an upper legform test device. Vehicles with a lower bumper height between 425 mm and 500 mm are tested with either legform chosen by the manufacturer. In the lower legform to bumper test, vehicles must meet limits on lateral knee bending angle, knee shearing displacement, and lateral tibia acceleration. In the upper legform to bumper test, limits are placed on the instantaneous sum of the impact forces with respect to time and the bending moment of the test.

21. The performance requirements, test procedures and supplementary information explaining the rationale for this gtr are discussed in detail in later sections of this preamble.

III. PROCEDURAL BACKGROUND

22. During the one-hundred-and-twenty-sixth session of WP.29 in March 2002, AC.3 concluded their considerations of priorities for developing future global technical regulations. WP.29 adopted the 1998 Global Agreement Programme of Work, which included pedestrian safety, and decided to start the work on pedestrian safety at the thirty-first session of GRSP in May 2002, by establishing an informal group to draft the gtr. The formal proposal to develop a gtr (TRANS/WP.29/AC.3/7) was considered and adopted by the AC.3 at its tenth session, in March 2004. It is based on document TRANS/WP.29/2004/26, which had been submitted by the European Community, the technical sponsor of the project.

23. Informal document No. 10 of the thirty-first session of GRSP lays down the terms of reference of the group and the document was adopted by GRSP (INF GR/PS/2).

24. Informal document No. 7 of the thirty-second session of GRSP reported on the result of the first meeting of the informal group (INF GR/PS/9).

25. Informal document No. 2 of the thirty-third session of GRSP (INF GR/PS/47 Rev.1) was the first preliminary report of the informal group and responds to paragraph 5 of documents TRANS/WP.29/2002/24 and TRANS/WP.29/2002/49 as adopted by AC.3 and endorsed during the one-hundred-and-twenty-seventh session of WP.29. The documents were consolidated in the final document TRANS/WP.29/882. The preliminary report was adopted as TRANS/WP.29/2003/99 by AC.3 in November 2003.

26. Informal document No.GRSP-34-2 of the thirty-fourth session of GRSP reported on the action plan of the informal group (INF GR/PS/62).

27. Informal document No. GRSP-35-5 of the thirty-fifth session of GRSP was the second preliminary report of the informal group (INF GR/PS/86 Rev2 and PS/88). This report was considered by AC.3 in June 2004 as informal document No. WP.29-133-7.

28. Informal document No. GRSP-36-1 of the thirty-sixth session of GRSP was the first draft gtr of the informal group (INF GR/PS/116).

29. TRANS/WP.29/GRSP/2005/3 was proposed at the thirty-seventh session of GRSP and was a revised draft gtr including the preamble, of the informal group (INF GR/PS/117).

30. The group had held the following meetings:

(a) 4-5 September, 2002, Paris, France

(b) 10 December, 2002, Geneva, Switzerland

(c) 15-16 January, 2003, Santa Oliva, Spain

(d) 15-16 May, 2003, Tokyo, Japan

(e) 10-12 September, 2003, Ottawa, Canada

(f) 24-26 February, 2004, Paris, France

(g) 28-30 September, 2004, Paris, France

(h) 11-13 July, 2005, Brussels, Belgium

(i) 5-6 December, 2005, Geneva, Switzerland

(j) 16-19 January, 2006, Washington DC, USA

31. The meetings were attended by representatives of:

32. Canada, France, Germany, European Community (EC), Italy, Japan, Korea, the Netherlands, Spain, Turkey, the United States of America (USA), Consumers International (CI), the European Enhanced Vehicle-safety Committee (EEVC) 12/, the European Association of Automotive Suppliers (CLEPA) and the International Organization of Motor Vehicle Manufacturers (OICA).

12/ The steering committee of the EEVC is composed of representatives from European national governments. The EEVC conducts research in motor vehicle safety and develops recommendations for test devices and procedures that governments can decide to adopt into national regulations.

33. The meetings were chaired by Mr. Mizuno (Japan) and Mr. Friedel/Mr. Cesari (EC), while the secretariat was provided by Mr. Van der Plas (OICA).

IV. EXISTING REGULATIONS, DIRECTIVES, AND INTERNATIONAL VOLUNTARY STANDARDS

34. At the present time, there are no regulations concerning the provision of improved protection for pedestrians and other vulnerable road users in the Compendium of Candidates.

35. The following is a summary of national and regional legislation and of work in international forums:

36. The Japanese Government has established a regulation on pedestrian protection. The regulation addresses the issues of providing protection for the child and adult heads. It applies to passenger cars with up to 9 seats and to small trucks of up to 2,500 kg Gross Vehicle Mass (GVM) with application from 2005 for new vehicle types and from 2010 for existing vehicle types (certain other vehicles have a timetable which is postponed by two years). The regulation requires compliance with test requirements using representative head impactors.

37. The European Parliament and Council adopted the Directive 2003/102/EC which provides for the introduction of requirements for leg injuries, and adult and child head injuries. The Directive and its requirements are incorporated into Community legislation under the European Union (EU) whole vehicle type approval system set up by EU Framework Directive 70/156/EEC. It applies to passenger cars of category M_1 and to light commercial vehicles derived from passenger cars of M_1 category, both up to 2,500 kg gross vehicle mass, with application dates in two phases starting in 2005 and 2010. The requirements and the tests are based on the research results that were published by EEVC in the 1990's and that were introduced in a less severe form for the first phase and intended to be introduced in the originally proposed form for the second phase. However, since EEVC results have never been fully accepted by all involved parties, the Directive provided for a feasibility review of the requirements for the planned second phase in 2004. This feasibility review has taken place and may result in amendments to the European requirements in its second phase, starting in 2010.

38. Canada is currently reviewing its bumper regulation. The Canadian bumper regulation is one of the most stringent in the world (all the safety features of the vehicle have to be functional after an 8 km/h impact). In addition, Canada and the USA are conducting a preliminary investigation of the effects of bumper design on different leg test devices (Transport Research Laboratory (TRL) legform impactor; Polar dummy and flexible pedestrian legform impactor (Flex-PLI)).

39. The United States research programmes have addressed how vehicles could be modified to reduce the severity of head and leg impacts. 13/ 14/ The current US pedestrian protection research programme supports the IHRA objectives. Current activities include (1) pedestrian field data analysis to develop test conditions, (2) evaluation of pedestrian head and leg test tools, (3) experimental impact testing of vehicle structures to assess aggressivity, (4) pedestrian case reconstructions using a combination of field data, computer simulation, and testing to better understand injury mechanisms, (5) computer model development using available biomechanical literature, and (6) completion of other IHRA Pedestrian Safety Working Group action items.

13/ Saul, R.A., Edlefson, J.F., Jarrett, K.L., Marcus, J.R.; "Vehicle Interactions with Pedestrians," Accidental Injury: Biomechanics and Prevention, New York: Springer-Verlag, 2002.

14/ "Report to Congress: Pedestrian Injury Reduction Research," NHTSA Report DOT HS 808 026, June 1993.

40. The IHRA Pedestrian Safety working group has conducted in-depth accident studies based on pedestrian accident data collected from the member countries. In addition, this group carefully studied the front shape of passenger vehicles including SUVs, and used best available computer simulation models to study the effective head mass, adult and child head impacting speed during the impact with vehicles and the impact angles.

41. Based on these research results, the IHRA group developed test procedures and test devices for adult and child head protection and for adult leg protection.

42. The International Organization for Standardization (ISO) created the pedestrian protection working group (ISO/TC22/SC10/WG2) in 1987 to develop test methods for the reduction of serious injuries and fatalities for pedestrian to car accidents. The mandate for ISO/WG2 was to produce test methods, covering crash speeds up to 40 km/h, which will contribute to make cars pedestrian friendly. Since then, the WG2 has developed pedestrian test procedures and has described the necessary test tools. The study results were fully used in the IHRA/PS group when IHRA/PS developed the adult and child impactors.

43. The ISO standards and draft standards are:

(a) ISO 11096 2002 Road vehicles - Pedestrian protection - Impact test method for pedestrian thigh, leg and knee,

(b) ISO/DIS 14513 2006 Road vehicles - Pedestrian protection - Head impact test method,

(c) ISO 16850 2007 Road vehicles - Pedestrian protection - Child head impact test method.

V. GENERAL ISSUES

(a) Scope

44. From the review of pedestrian fatality and injury statistics from several countries, it was shown that the head and the legs are the most frequently injured body regions in pedestrian accidents. It was recommended that the gtr would encompass tests for the adult head and leg, and the child head. The studies also showed that the majority of pedestrian injuries are occurring in urban environments. Therefore, the gtr should test those vehicles found in this environment, including passenger vehicles, vans, and light trucks.

45. As suggested by the terms of reference of the informal group, consideration was given to the use of the best available technology and improvements in technology that will provide significant steps in developing methods and in achieving and improving benefits, including both active and passive safety measures (TRANS/WP.29/GRSP/30). There was a discussion on whether the proposed pedestrian gtr should regulate passive and/or active safety systems. Active safety systems, such as brake assist, anti-lock brakes and day-light running lights were suggested as solutions for the reduction of pedestrian injuries, but it was ultimately counselled by GRSP and WP.29 to concentrate on passive systems for this gtr, as this is the main domain of expertise of the GRSP working party, and only to provide advice on the use of active systems.

46. The group understood that active safety and infrastructure measures were not within the remit of the group, but determined that it could be useful and efficient to inform WP.29/AC.3 as well as other authorities of the need to take these issues into account for real world safety improvements. The group also noted the importance of educational measures and the need to enforce existing road traffic legislation. Some experts noted that consideration of other safety measures, if properly balanced with the passive safety requirements, might help in ensuring that the vehicle passive safety requirements are kept at a realistic and feasible level.

47. OICA, in particular, mentioned brake assist systems which can, in emergency situations, substantially improve the braking performance and consequently reduce the impact speed when the impact is unavoidable. A study on the effectiveness of such a system was presented by OICA using the German In-Depth Accident Studies (GIDAS) database (INF GR/PS/25). This showed that if the vehicle speed is 50 km/h at the start of braking, the collision speed (car versus pedestrian) would be reduced to 40 km/h in general, to 35 km/h for an experienced driver and to 25 km/h for cars equipped with brake assist systems. Another study performed by the Technical University of Dresden on behalf of the European Automobile Manufacturers' Association (ACEA) was presented by OICA (INF GR/PS/92). This study confirmed the positive effect of brake assist systems on pedestrian fatalities and injuries.

48. As for infrastructure measures that could improve pedestrian safety, OICA presented the results of a 1998 study conducted on behalf of ACEA by the consultants ORIENTATIONS (France) and TMS Consultancy (United Kingdom) (INF GR/PS/29). This study, which evaluated the effect of infrastructural measures based on real data evaluations, concluded that such measures could dramatically reduce the number of pedestrian victims (fatalities/injuries) at low cost.

(b) Applicability

49. The application of the requirements of this gtr refers, to the extent possible, to the revised vehicle classification and definitions outlined in the 1998 Global Agreement Special Resolution No. 1 (S.R.1) concerning the common definitions of vehicle categories, masses and dimensions.

50. Difficulties, due to differing existing regulations and divergent vehicle fleets, were encountered in determining which vehicles would be included in the scope. The Japanese regulation applies to passenger cars for up to nine occupants and commercial vehicles up to a GVM of 2,500 kg. The IHRA recommends tests and procedures for passenger vehicles of GVM 2,500 kg or less. The European Union (EU) Directive applies to M_1 vehicles up to

2,500 kg and N_1 vehicles up to 2,500 kg, which are derived from M_1. The ISO recommendations are for M_1 and N_1 vehicles that have a GVM of 3,500 kg or less. In addition, some countries, taking into account their current fleet composition, wanted to ensure that larger vehicles, such as light trucks and sport utility vehicles with a GVM of 4,500 kg or less, were not excluded.

51. The group originally reviewed in detail the IHRA recommendation in detail to take into account the shape of the front of the vehicle as an important parameter when discussing the types of pedestrian injuries to be mitigated. IHRA specifies three groups of vehicle shape: sedan, SUV, and 1-box. For the adult and child head impacts, IHRA foresees different impact test speeds and different impact angles. The Japanese legislation is based on the IHRA recommended method. The EU requirements, on the contrary, do not differentiate between the various test speeds and impact angles.

52. The group compared these various considerations and, on the basis of simulations (INF GR/PS/129), concluded that the EU requirements in effect are more severe than the Japanese proposals. For safety reasons, the group therefore uses the EU approach, not taking into account the shape of the vehicle front in defining the requirements. Furthermore, the group also determined that the IHRA recommendations would be difficult to put in place in the context of a regulatory and certification approach.

53. There was considerable discussion over the mass of vehicles to which this gtr should apply. Using the categories described in S.R.1, several options were examined. Some delegates wanted to limit application of the gtr to vehicles in category 1-1 with a vehicle mass of less than 2,500 kg GVM. Other delegates did not agree with a 2,500 kg limit on GVM, believing that since the front-end structure of vehicles with a mass up to 4,500 kg GVM is usually similar to that of lighter vehicles, the application of the gtr should include the heavier vehicles. In addition, some delegates sought to limit application of the gtr to vehicles of a GVM of more than 500 kg, while other delegates expressed concern about having a lower mass limit, believing that a particular jurisdiction might determine there is a need to apply the gtr requirements in that jurisdiction to vehicles with a GVM of less than 500 kg. There was a suggestion that the gtr should also apply to vehicles in category 2 that had the "same" general structure and shape forward of the A-pillars as vehicles in category 1-1. However, some were concerned that it would be unfeasible to define objectively what was meant by "same".

54. After considering these issues, it was recommended that the gtr should be drafted to have a wide application to vehicles, to maximize the ability of jurisdictions to effectively address regional differences in pedestrian accident crash characteristics. The gtr would establish that if a jurisdiction determines that its domestic regulatory scheme is such that full applicability is inappropriate, it may limit domestic regulation to certain vehicle categories or mass limits. The jurisdiction could also decide to phase-in the requirements for certain vehicles. A footnote was added to the gtr text to make it clear that jurisdictions can decide to limit the applicability of the regulation. This approach recognizes that niche vehicles that are unique to a jurisdiction would best be addressed by that jurisdiction, without affecting the ability or need for other jurisdictions to regulate the vehicles. When a Contracting Party proposes to adopt the gtr into its domestic regulations, it is expected that the Contracting Party will provide reasonable justification concerning the application of the standard.

55. While this approach maximizes the discretion of jurisdictions to decide whether vehicles should be excluded from the gtr for feasibility or practical reasons, or because there is no safety need to regulate the vehicles, the group also decided to recommend excluding one unique vehicle type from the regulation. The test procedures in the gtr are based largely on the classic vehicle shape with a long bonnet. Certain vehicles, generally cargo vehicles, have a very short bonnet and a front shape that is very close to the vertical. Because of the short, vertical bonnet, the design leaves very little soft space once the hinges, latches, and other hardware are considered. There are additionally feasibility concerns for the bumpers of these vehicles. The different shape of the front leaves little room to incorporate existing countermeasures, such as those used on passenger vehicles, and new countermeasures have not been identified. The pedestrian kinematics with these vehicles may be very different. The head to bonnet impact is occurring earlier and leg injuries are occurring at a reduced frequency than with

traditional long bonnet vehicles. Inaddition, there are difficulties in applying the head tests to these vehicles, particularly with regard to determination of test zone reference lines.[1] For these reason, the group recommends that those vehicles of category 1-2 and category 2, where the distance, measured longitudinally on a horizontal plane, between the transverse centre line of the front axle and the R-point of the driver's seat is less than 1,100 mm, be exempt from the requirements of the regulation. To prevent in consistencies in the market, Contracting Parties can exempt category 1-1 vehicles if they have components of the front structure that are interchangeable with exempted category 1-2 and category 2 vehicles. The group agreed to recommend allowing Contracting Parties this option even though not all Contracting Parties have these vehicles in their fleet and were therefore not able to fully evaluate the exemption.

[1] Informal document No. GRSP-45-25

56. For these reasons, with the exception of the exemption discussed above, the gtr is recommended to apply to category 1-1 vehicles with a GVM exceeding 500 kg; and to category 1-2 and category 2 vehicles with a GVM exceeding 500 kg but not exceeding 4,500 kg. In addition, the group recommends that a Contracting Party may restrict application of the requirements in its domestic legislation if it decides that such restriction is appropriate.

57. Regarding the applicability of this gtr, it should be noted that the requirements of the draft gtr are substantially more severe than any existing legislation at the time of adoption of the gtr. In addition, many countries do not yet have pedestrian safety requirements. It is therefore recommended that Contracting Parties implementing this gtr allow adequate lead time before full mandatory application, considering the necessary vehicle development time and product lifecycle.

58. Furthermore, during the development phase of this gtr, the main focus was on vehicles of a GVM of 2,500 kg or less, that are also addressed in all existing legislation. The later extension to other vehicles however needs to recognise that some additional lead-time may be necessary, because many current vehicles, exempted from existing national or regional requirements, are now included. In addition, while the test procedures and requirements of this gtr were based on requirements originally developed for "classical" (sedan type) passenger cars, the gtr now also covers vehicles with specific shapes or features (High Front Vehicles, special purpose vehicles, etc.), for which it is recognised that special consideration may be needed.

(c) Implementation generally

59. The informal group considers all tests in this proposed gtr to be technically feasible and able to evaluate objectively the ability of vehicle bonnets and bumpers to absorb energy more efficiently. However, pedestrian accident crash characteristics and vehicle baseline performance may differ regionally. It will be the decision of each jurisdiction to determine whether the benefits achieved by requiring these tests justify the costs. Based on this determination, a jurisdiction can choose to limit the application in its own regulation to specific vehicle categories, specific tests, and/or it may decide to phase in the regulations over time.

(d) Points tested

60. The informal group considered whether to specify both the number of test points and the minimum spacing of such test points. On consideration, the group determined that the specification of such points did not have a place within this proposed gtr for the following reasons:

(i) For governments that use a self-certification regulatory framework, it was not considered necessary to mention the number of tests required for testing or their spacing, as it would be incumbent on vehicle manufacturers to ensure that vehicles comply with all the impact zone requirements defined within this proposed gtr when tested by the regulating authority.

(ii) For type approval, the number of tests that need to be carried out to satisfy the relevant authority that vehicles meet the requirements is an issue for that authority, which may specify the number of tests and the spacing between the test points.

(iii) The mention of a minimum number of tests or a minimum distance apart between tests could result in manufacturers being burdened with unnecessary tests and/or authorities being unnecessarily restricted in test programs, as it would be difficult to set a target that would encompass both the largest and smallest test zones, and the situation could arise where test zones could be smaller than the minimum number of tests required that could be fitted into that zone.

(e) Vehicle design position

61. As vehicles come in many variants and modifications, the ride height may vary greatly. Taking into account the differences between type approval and self certification, it is recommended that Contracting Parties take this into account upon national implementation of the gtr. As guidance to Contracting Parties, the EU addresses this issue by defining the concept of "primary reference marks". This definition (paragraph 2.2 of EU Commission Decision of 23 December 2003) reads: "Primary reference marks" means holes, surfaces, marks and identification signs on the vehicle body. The type of reference mark used and the vertical (Z) position of each mark relative to the ground shall be specified by the vehicle manufacturer according to the running conditions specified in paragraph 2.3. These marks shall be selected such as to be able to easily check the vehicle front and rear ride heights and vehicle attitude.

62. If the primary reference marks are found to be within ± 25 mm of the design position in the vertical (Z) axis, then the design position shall be considered to be the normal ride height. If this condition is met, either the vehicle shall be adjusted to the design position, or all further measurements shall be adjusted, and tests performed, to simulate the vehicle being at the design position.

(f) Future consideration

63. During the discussions, it became clear that some issues could not be fully resolved within the timeframe of the terms of reference for the informal group. The group determined that the following issues should be considered further beyond this gtr.

1. Lower legform impactor

64. The lower legform impactor currently used for testing in Europe was designed by the Transport Research Laboratory (TRL) in the United Kingdom. However, it is known to also have certain limitations regarding the biofidelity and the repeatability of the test results. Therefore, Japan proposed to use a completely new legform, the so-called Flexible Pedestrian Legform Impactor (FlexPLI). As the FlexPLI legform is considered by some to have high biofidelity and an excellent ability to assess potential leg injuries, the FlexPLI should be considered to replace the TRL lower legform impactor in the future. However, because of the lack of experience in using the FlexPLI as a certification tool, a further confirmation process is needed. Therefore, a Technical Evaluation Group (TEG) was established to evaluate the reliability of the FlexPLI as a certification tool (TRANS/WP.29/GRSP/36). The TEG is currently assessing the FlexPLI and will advise GRSP by the end of 2007 as to the suitability of the FlexPLI for testing and compliance verification purposes (TRANS/WP.29/GRSP/37). The TEG is also expected to provide its recommendation as to the effective date of entry into force and the date on which the FlexPLI could replace the rigid lower legform impactor. TEG will also consider a transitional period during which the FlexPLI and the rigid lower legform impactor can be used as alternatives.

2. Upper legform impactor to high bumper test

65. Some delegates had concerns about the biofidelity of the upper legform impactor and the limitations of the test tool in assessing injury. IHRA/PS is working on recommendations for an improved upper legform impactor for possible future use.

3. Upper legform impactor to bonnet leading edge test

66. Several accident studies from some regions comparing modern "streamline" vehicle fronts registered in or after 1990 and old vehicles from the eighties or seventies indicate a decrease in AIS 2+ upper leg and pelvis injuries caused by the bonnet leading edge. The accident studies were performed by the LAB using French data

(INF GR/PS/30), and by the University of Dresden using German GIDAS data (INF GR/PS/92). In addition, EEVC Working Group 17 (WG17) summarized in their 1998 report that no serious (AIS 2+) upper leg or pelvis injuries caused by the bonnet leading edge were found for post-1990 car models impacting a pedestrian at a speed up to 40 km/h (INF GR/PS/187 Rev.1). In contrast, data from the United States of America indicate a high incidence of above-the-knee injuries due to the prevalence of light trucks and vans in the United States fleet, and that consideration should be given to evaluating thigh, hip, and pelvis injuries in future test procedures.

67. Despite the desire to address any potential injuries in the upper leg or pelvic area, the group was also concerned that there was a serious lack of biofidelity for the existing test device and the respective test procedure to assess injury caused by the bonnet leading edge of high profile vehicles. Therefore, the group recommended excluding the upper legform impactor to bonnet leading edge test at this stage. IHRA/PS is carrying out further research into an improved impactor and test procedures for this test.

VI. PEDESTRIAN HEAD PROTECTION

68. IHRA data show that a major source of child and adult pedestrian head injuries is the top surface of the bonnet/wing of the striking vehicle. As explained in this section, this gtr requires the bonnet/wing to perform at levels that decrease the likelihood that head impact with the bonnet/wing in a 40 kilometre per hour (km/h) pedestrian-to-vehicle impact will result in fatal or serious injury.

69. The bonnet/wing would be impacted with a headform at 35 km/h. The bonnet/wing would be divided into a "child headform test area" and an "adult headform test area". The child headform test area is the area of the bonnet/wing that is likely to be impacted by the head of a 6-year-old child in a pedestrian impact. A child headform is used to evaluate the bonnet/wing in that area. Likewise, the adult headform test area corresponds to the area of the bonnet/wing that the head of a mid-size adult male pedestrian is likely to impact. An adult headform is used to test the bonnet/wing in the latter area.

(a) Test areas

70. The bonnet top is an area bounded by reference lines corresponding to the bonnet leading edge, the sides of the vehicle, and the rear of the bonnet (all terms are objectively defined in this gtr). The gtr divides the bonnet top into test areas using a parameter called the "wrap around distance" (WAD). The WAD is the distance from a point on the ground directly below the bumper's leading edge to a designated point on the bonnet, as measured with a flexible device, such as a cloth tape measure. A WAD of a specified distance, measured as described in the gtr, defines points on the vehicle's bonnet from which test areas can be determined.

71. The WAD is a good indicator of where head impacts are likely to occur on the bonnet. Head impact locations on the bonnet are largely explained by the standing height of the pedestrian and the frontal geometry of the striking vehicle. The WAD measurement is based on both pedestrian height and vehicle configuration. By use of the WAD, it can reasonably be estimated where on a vehicle a child or adult pedestrian's head may impact.

72. The WAD boundaries were selected based in large part on accident data from Australia, Europe, Japan and the United States of America of areas commonly struck by the head of a child and adult pedestrian. They were also selected considering the technical feasibility of regulating within a test area. The child headform test area is bounded in the front by a boundary determined by a WAD of 1,000 mm, and at the rear by a WAD of 1,700 mm line. A WAD of 1,000 mm was selected for the front boundary of the child headform test area because accident data show that most child pedestrian head contacts are above a WAD of 1,000 mm. Discussion also took place on possibly using a WAD of 900 mm. However, it was concluded that for many vehicles, a WAD of 900 mm would be located in the headlamp assembly area, where there would be feasibility problems in meeting the head protection requirements of this gtr.

73. Based on accident studies of adult pedestrian head impacts to the bonnet area, the adult headform test area begins in the front at a wraparound distance of 1,700 mm, and ends at the rear with a boundary determined by a WAD of 2,100 mm (or the rear edge of the bonnet for shorter vehicles). The child and adult headform test zones

cover approximately 62 percent of the pedestrian cases (United States of America). Although 35 percent of the cases (United States of America) occurred at WADs exceeding 2,100 mm, many of these occurred in impacts greater than 40 km/h (Three percent occurred below WAD 1,000). An overlap area was also considered with a WAD of 1,400 mm to 1,700 mm, where both adults and children have received head injuries in actual accidents. However, a defined boundary at 1,700 mm was determined to be more suitable, because little difference in the life-saving rate was perceived between the two approaches and because the boundary method provided a clearer approach.

(b) Head Injury criterion

74. The majority of pedestrian fatalities in road accidents are caused by head injuries. The informal group determined that the head protection performance should be based on the Head Injury Criterion (HIC) 16/, given the ability of HIC to estimate the risk of serious to fatal head injury in motor vehicle crashes. An HIC value of 1,000 is equivalent to approximately a 15 percent risk of AIS 4+ head injury.

16/ The resultant acceleration at the location of the accelerometer mounted in the headform will be limited so that, for any two points in time, t_1 and t_2, during the event which are separated by not more than a 15 millisecond time interval and where t_1 is less than t_2, the maximum calculated head injury criterion (HIC) shall not exceed 1,000, determined using the resultant head acceleration at the centre of gravity of the headform, a_r, expressed as a multiple of g (the acceleration of gravity), calculated using the expression:

$$\mathrm{HIC} = \left[\frac{1}{(t_2 - t_1)} \int_{t_1}^{t_2} a\,dt \right]^{2.5} (t_2 - t_1)$$

75. The gtr specifies that HIC must not exceed 1,000 over one half of the child headform test area and must not exceed 1,000 over two thirds of the combined child and adult headform test areas. The HIC for the remaining areas must not exceed 1,700 for both headforms (The need for "relaxation zones," in which the HIC limit is 1,700, is discussed in the next section of this preamble).

76. HIC would be calculated within a 15 ms interval. The main reason that a longer interval was not used was that head impacts to external car structures are very short, occurring within a few milliseconds of contact. As the pulse itself is so short in time, there is no risk to lose part of the pulse during the HIC calculation--and no risk of a lower calculated HIC value--if a 15 ms interval were used rather than a longer interval (INF GR/PS/168). Accordingly, using either a 15 ms or a 36 ms pulse window will provide the same HIC value. Moreover, the test is not intended to record more than one impact. A short time duration avoids the risk that a second impact could be recorded after rebound. A longer duration for the time interval could result in distortions in the data recorded by the headform, which may lead to inaccurate HIC values.

(c) Relaxation zones

77. Within the child and adult headform test areas are "relaxation zones" in which the HIC threshold is 1,700 instead of 1,000. Virtually all bonnets have hard substructures (e.g., shock struts) beneath them that prevent attainment of a 1,000 HIC performance criterion at all areas within the test area. The feasibility study detailed in INF GR/PS/91 and 101 showed the problem areas on the bonnet. Also, the feasibility study conducted on behalf of the European Commission (INF GR/PS/89) acknowledged the need to define an area on the bonnet for which a higher HIC limit is needed. As the problems on the bonnet are not the same for every vehicle model, it was felt necessary to set a maximum area with relaxed requirements that could be defined for every vehicle by the manufacturer.

78. The informal group considered the feasibility of applying the relaxation zone separately for the child and adult headform test areas, i.e., applying an HIC 1700 limit to a maximum of one third of the child test zone. It was determined that, because the location of necessary under-bonnet components, such as locks and suspension towers, cannot be fundamentally changed; they need to be located in the child headform test area. For vehicle

types with small child headform test areas, the under bonnet components which are essential for functionality will be located in this test area. As a result, the relaxation zone for the child headform test area may be greater than one third of the zone (see illustrations 1 and 2 of INF GR/PS/158). 17/

17/ While the example in working paper INF GR/PS/158 shows a needed relaxation zone of around 40 percent in the child headform test area, the example represents only one vehicle. The informal group determined that a maximum 50 percent relaxation area in the child headform test area is a reasonable approach taking into account all envisaged vehicle types.

79. The relaxed value of 1,700 HIC was the subject of much discussion within the informal group. It was noted that IHRA/PS recognized the technical infeasibility of a headform test area without a relaxation zone with a HIC of more than 1,000 (INF GR/PS/5). The group also considered that current regulations in Japan and Europe, as enforced since 2005, limit the HIC in the relaxation zone to 2000. Although the European legislation also defines a Phase 2, beginning in 2010 and containing the original EEVC WG17 requirements, that does not allow any relaxation in the headform test area, the European legislation has provided for a review of the Phase 2 requirements. This review is ongoing, and has led to the conclusion that the Phase 2 requirements (that specify no relaxation zone) are technically infeasible (INF GR/PS/89, 91, 92 and 120). The European legislation also requires any new requirements replacing the existing ones to be more effective than those proposed by the EEVC WG17. The feasibility study performed on behalf of the European Commission (INF GR/PS/120) has determined that, among other requirements, a HIC of 1,700 in the relaxation zone not only represents the maximum achievable criterion, but will also lead to higher effectiveness, taking also into account feasibility aspects, whereas HIC values of less than 1,700 would bring back the feasibility problems. The group therefore decided to adopt the most demanding and maximum achievable criterion for the relaxation zone, a HIC of 1,700.

(d) Headform

80. A child headform is used to test the bonnet in the child headform test area, and an adult headform is used in the adult headform test area. The appropriate headform impactor size and mass, determined based on the characteristics of the human body, are explained below (INF GR/PS/46, 74 and 93).

1. Headform diameter

81. The diameter of the child headform is 165 mm. Due to the fact that the majority of child pedestrian victims are 5 or 6 years old, this value was determined based on the average head diameter of a 6-year-old child (by averaging the diameter obtained from the circumference of the head and the longitudinal and lateral measurements of the head).

82. The diameter of the adult headform is 165 mm, which is the same diameter used in the test procedures of EEVC and ISO. The value was considered to represent the diameter mainly of the forehead portion of the 50th percentile adult male, rather than the maximum outer diameter of the head. The average height and weight of all adult pedestrian victims in the IHRA dataset is about the same as those of the 50th percentile male.

83. Thus the diameter of both the child and adult headforms is 165 mm. Although the diameter is different from the diameter of actual child and adult human heads, the diameter and moment of inertia are appropriately designed so that the child and adult headforms can properly estimate severity of injury to actual child and adult human heads (see IHRA/PS N231).

84. Testing experience with the headforms show them to be highly repeatable and reproducible.

2. Headform mass and moment of inertia

85. Computer simulations conducted in the IHRA study show that the effective mass of the head in an impact with vehicles is identical to the actual mass of the head. Accordingly, the headform mass was therefore determined as follows:

(a) The mass of the child headform is 3.5 kilograms (kg), representing the mass of the head of a 6-year-old child.

(b) The mass of the adult headform is 4.5 kg, representing the mass of the head of a 50th percentile adult male.

86. This proposed gtr specifies the moment of inertia of the child and adult headforms as analyzed by IHRA (INF GR/PS/177) based on the EEVC/WG17 proposal (INF GR/PS/148) and the Japanese proposal (INF GR/PS/149). The IHRA proposal is 0.0084 – 0.0102 kgm^2 for the child headform and 0.0103 – 0.0127 kgm^2 for the adult headform. The informal working group on pedestrian safety adopted the proposal using the following rounded numbers: 0.008 – 0.010 kgm^2 for the child headform and 0.010 – 0.013 kgm^2 for the adult headform at the tenth informal meeting. Some members wanted to check these new limits and, after that meeting, it was revealed the above moment of inertia requirement is difficult to achieve for a child headform impactor which is developed in Europe. Therefore, the informal group slightly adjusted the upper limit for the child head impactor and finally adopted following values for the gtr headform impactors: 0.008 – 0.012 kgm^2 for the child headform and 0.010 – 0.013 kgm^2 for the adult headform.

87. The informal group noted that the mass of the child headform impactor (3.5 kg) differs from that specified in the corresponding EU Directive (2.5 kg) and that the European Commission intends to modify the latter to provide consistency.

3. Headform accelerometer

88. This proposed gtr recommends a damped accelerometer (as specified in INF GR/PS/133) in the adult and child headform impactors. As explained in INF GR/PS/96, in a research program in 2002 using the Japanese New Car Assessment Program (J-NCAP) headform test with undamped accelerometers, abnormal acceleration signals with high HIC values were recorded frequently in windshield impacts, and also in bonnet impacts. It was determined that this was due to the resonance vibration of the undamped accelerometer, which would occur if the spectrum of the impact waveform was near to the resonance frequency of the accelerometer. Once a high resonance, over the Channel Amplitude Class (CAC) setting level, occurs, it has a high chance to deform the acceleration waveform, i.e. one cannot obtain a correct acceleration waveform from the undamped accelerometer.

(e) Headform test speed and angle

89. The head impact conditions (speed and angle) were considered together. The head impact test is representative of a vehicle-to-pedestrian impact at 40 km/h.

90. The gtr specifies that the child headform impacts the bonnet top at 35 km/h at an angle of 50 degrees to the horizontal. The adult headform impacts the bonnet at 35 km/h at a 65 degree angle.

91. In determining test speeds and angles of impact, the informal group considered the findings of IHRA and the EEVC. IHRA had explored whether various vehicle shapes influenced the angle at which a pedestrian's head impacted the bonnet top. Computer simulations were conducted, as part of the IHRA study, by the Japan Automobile Research Institute (JARI), the United States National Highway Traffic Safety Administration (NHTSA), and the Road Accident Research Unit of Adelaide University (RARU). The simulations used a 50th percentile adult male model and a 6-year-old child model. The distribution of headform impact speeds and angles in various impacts was obtained by simulating head impacts using three types of walking positions, three types of vehicle frontal shapes and two types of bonnet stiffness as parameters. The studies showed that the same headform impact speed could be used for any type of vehicle frontal shape. Further, the interpretation of the results indicated an average speed of 32 km/h, which is 0.8 times the vehicle impact speed of 40 km/h. In addition, various angles for adult and child impact conditions and for the three different shapes were defined as well.

92. In contrast, EEVC had concluded that one set of angles (50 degrees for the child headform test and 65 degrees for the adult headform test) for all vehicles is reasonable, simplifying any head test procedure dramatically. EEVC's decisions concerning head impact angles for child and adult tests were based on two reports used as working documents: Glaeser K.P. (1991), "Development of a Head Impact Test Procedure for Pedestrian Protection," BASt Report under contract N° ETD/89/7750/M1/28 to the E.C. (INF GR/PS/150); and Janssen E.G., Nieboer J.J. (1990), "Protection of vulnerable road users in the event of a collision with a passenger car, part 1 – computer simulations," TNO Report N° 75405002/1.

93. The EEVC values were based on post-mortem human subject (PMHS) tests and simulation results. The PMHS tests indicated a peak of the distribution of adult head impact angles to be 60 degrees, with all the results falling between 50° and 80°. Simulations gave a result around 67 degrees for adults, and indicated that vehicle shape had little influence on the angle of impact. EEVC chose a value of 65 degrees, which was close to the 67 degree angle resulting from the simulation and to the average of the PMHS results.

94. For child head impacts, EEVC considered simulations of a small adult female (close in anthropometry to a 12-year-old child) and of a 6-year-old child. Results of the small adult female simulations were very close to the results of the simulations for the 50th percentile male adult, while the simulations involving the 6-year-old child suggested a value around 50 degrees. EEVC picked the value of 50 degrees, believing that the simulations of a 6-year-old child were more relevant than those of a 12-year-old child for child pedestrian protection.

95. The informal group noted that the one set of angles from EEVC involves a different (higher) impact speed than that specified by IHRA. The group considered which of the two approaches of EEVC and IHRA was most stringent and thus offering the most protection to pedestrians. Working paper INF GR/PS/129 showed, by both numerical calculation and by simulation, that the set of requirements defined by EEVC is more severe than the requirements defined by IHRA. The group thus decided to use the EEVC 50 degrees and 65 degrees impact angle for child and adult head testing while maintaining the higher EEVC impact speed to the bonnet of 35 km/h (compared to the IHRA speed of 32 km/h).

96. The informal group noted that the headform velocity, at the time of impact, was lower than specified in the corresponding EU Directive and that the European Commission intends to modify the latter to provide consistency.

VII. PEDESTRIAN LEG PROTECTION

(a) General

1. Purpose

97. This proposed gtr would specify minimum performance requirements for vehicle bumpers to provide leg protection by subjecting pedestrians to lower impact forces. As the majority of victims of leg injuries are adults, this proposed gtr specifies use of a legform impactor that simulates the leg of a mid-size adult male. The performance of the bumper would be evaluated by impacting the bumper with either of the two legforms, a lower legform impactor or an upper legform impactor, depending on the height of the bumper. The impact speed for both legform tests is the same as that of the striking vehicle in a 40 km/h impact, and thus determined to be 40 km/h.

98. The lower legform impactor is used to test vehicles with low bumpers, i.e., bumpers of heights less than 425 mm to a reference line on the lower surface of the bumper. The large majority of current passenger cars of the sedan type, as well as monospace (mini-van) type vehicles, have a lower bumper height around 200 to 250 mm above the ground. Therefore, these vehicles will be tested using the lower legform test procedure. Upper legform to bumper tests shall be carried out if the lower bumper height is more than 500 mm, typically represented by vehicles with off-road capabilities (SUVs). For vehicles that have a lower bumper height between 425 mm and 500 mm, the vehicle manufacturer can elect to perform either a lower legform test or an upper legform test.

2. Rationale for limiting the lower legform test

99. The reason that the lower legform test would not be applied to certain vehicles is due to the height limitations of the impactor, and the feasibility limitations of high-bumper vehicles to meet the test. The contact point between impactor and bumper should be below the knee, due to the impactor's structure and characteristics. 18/ The EEVC WG17 states in its report, paragraph 7.2.1. (INF GR/PS/159):

Some vehicles, like off-road vehicles, have high bumpers for certain functional reasons. These high bumpers will impact the femur part of the legform impactor, where no acceleration is measured to assess the risk of fractures.

Moreover, there is often no structure below the bumper to restrain the tibia part of the legform, for instance because an off-road vehicle needs a certain ramp angle and ground clearance. Therefore WG17 decided to include an optional, alternative horizontal upper legform test with an impact speed of 40 km/h, when the lower bumper height is more than 500 mm above the ground.

18/ There is also a concern that the lower leg test could readily be met by simply allowing the lower legform to slide and/or rotate beneath the high bumper. This could have an unintended consequence of encouraging high bumpers as a way to meet the requirements, and lead to more pedestrian injury due to run-over.

100. The informal group concurs with the determination of WG17 that the lower leg impactor test would be inappropriate for vehicles with high bumpers 19/.

19/ WG17 stated that the alternative legform test should be available for vehicles with a lower bumper height of more than 500 mm. However, WG17 referenced a value that WG10 had associated with the upper (rather than lower) bumper reference line. Also, since pedestrians are usually wearing shoes, the bottom of the legform impactor was determined to be 25 mm above the ground, the same height as the sole of a shoe (INF GR/PS/98). Accordingly, the informal group believes that the value of 425 mm (not 500 mm), measured to the lower bumper reference line is consistent with WG17's provision.

101. At the same time, the informal group believes that high bumpers should be more energy absorbing, and for that reason adopts in this gtr the upper legform test for vehicles with a lower bumper height of more than 500 mm.

102. For vehicles that have a lower bumper height between 425 mm and 500 mm, the gtr provides that the vehicle manufacturer can elect to perform either a lower legform test or an upper legform test. Investigations conducted with vehicles with lower bumper heights between 400 and 500 mm indicate that a large majority of these vehicles have features for off-road capability. For these off-road vehicles, it is technically not feasible to have a countermeasure that will enable the vehicle to support the tibia part of the lower legform. That is, data show (see INF GR/PS/175/Rev.2) that the absence of a lower structure to support the lower part of the leg, due to the necessary off road capacities, make it very difficult for these vehicles to meet the proposed lower leg criteria, especially the bending angle. Therefore, the group recommends to use the upper legform to bumper test as an optional alternative to the lower legform to bumper test for these vehicles.

103. The group recognizes that excluding vehicles from the lower legform test will affect the target population of a lower extremity pedestrian regulation, and will reduce the benefits of the leg protection requirements, particularly with regard to knee injuries.

3. Handling procedures

104. Delegates to the informal group were concerned about the effects of humidity on the foam flesh used in the legforms, recognizing that the material can vary significantly in performance depending on the humidity to which it is exposed. These concerns are addressed by specific controls in the regulatory text of this gtr regarding the humidity and other conditions (such as soaking time and a maximum time between removal from the soaking room and testing) under which the legform tests should be performed.

105. In addition, the group noted that legforms must also be carefully handled, as handling of the legform can affect variability in the bending angle, shearing displacement, and acceleration measured by the impactors due to the sensitivity to humidity. The group believed that handling instructions generally were not necessary to be specified in the gtr text, because the impactors are usually provided with handling instructions, which are the normal working tools for test houses and are therefore believed to be sufficient to cover the normal handling procedures. Nonetheless, the informal group emphasized that careful and controlled handling procedures, such as those developed by the German Federal Highway Research Institute BASt (INF GR PS/154/Rev.1), are highly important to ensure reliable test results.

(b) Lower legform test

1. Impactor

106. It was agreed to recommend using the legform impactor developed by TRL, for the time being, to evaluate the performance of vehicles in protecting the lower leg. However, it was also recommended to consider the possible future use of the Flex-PLI, which is considered by some to be more biofidelic and expected to be highly usable and repeatable, following the evaluation to be conducted by the Technical Evaluation Group (INF GR/PS/106) 20/.

20/ The size and mass of both the present rigid lower legform and the Flex-PLI were determined to be equivalent to those of a 50th percentile adult male (INF GR/PS/79). The results of computer simulation analyses and experimental data indicate that the mass of the upper body need not be taken into consideration for those impacts where the bumper strikes the legs below knee level (INF GR/PS/105).

107. The TRL legform is able to estimate human knee injury risk and has been shown to be a durable and repeatable test tool, provided that handling procedures for the legform are carefully followed 21/.

21/ "Use of the TRL Legform to Assess Lower Leg Injury Risk," Stammen and Mallory, NHTSA VRTC, February 2006. INF GR/PS/XX

2. Injury criteria

108. Knee injuries, which are one of the typical leg injuries in pedestrian to car collisions, most frequently involve the elongation or rupture of knee ligaments, and/or crush of knee articulation surfaces (tibia plateau and/or femur condyle). Lower leg injuries are not typically fatal, but leg injuries generally involve longer periods of recuperation. Knee injuries can be permanently debilitating. The most common mechanism causing pedestrian knee injury is a lateral bending between the thigh and the leg, which can be associated with shearing motion (horizontal displacement between the tibia top and the femur lower extremity in the direction of impact).

109. The 2001 Report of the IHRA/PS Working Group and the September 2002 Report of EEVC WG 17 22/ discuss several experimental research programs that were conducted in Europe, Japan and the United States of America using PMHS components during the last decade. There were also numerical simulations conducted to understand better what happens inside the knee joint during the loading process.

22/ EEVC Working Group 17 Report: "Improved Test Methods to Evaluate Pedestrian Protection Afforded by Passenger Cars (December 1998 with September 2002 updates)," available at www.eevc.org

110. These studies suggest a bending limit in the range of 15° to 21° for knee protection. The informal group determined that a value close to the upper limit (21°) of this range should be considered, and not the average. The absence of muscle tone in the PMHS tests reduced the knee stiffness of the subjects, and the high rigidity of the impactor bones transferred to the knee joint a part of the impact energy normally absorbed by the deformation of human long bones. For these reasons, a bending limit of 19º was selected for this gtr.

111. With regard to knee shearing limits, the informal group selected a limit of 6 mm, based on the analysis of PMHS by EEVC WG17 and WG10 that showed that a 6 mm shear displacement corresponds to a 4 kN shear force. The 4 kN shear force in the TRL device approximates the 3 kN average peak shearing force acting at the knee joint level that was found associated in the PMHS tests with diaphysis/metaphysis failure.

112. With regard to limiting the maximum acceleration on the tibia, results of a series of pedestrian PMHS tests performed with modern cars suggests that the maximum tibia acceleration for the PMHS sustaining a tibia fracture was 170g to 270g, with the average value of 222g. A value of 200g would correspond to a 50 percent injury risk. To protect a higher proportion of the population at risk, the informal group recommends a maximum lateral tibia acceleration limit of 170g.

113. In summary, it was concluded that the acceptance levels for the lower legform test should be set at the following limits:

Maximum lateral knee bending angle ≤ 19.0°;

Maximum lateral knee shearing displacement ≤ 6.0 mm;

Maximum lateral tibia acceleration ≤ 170g.

114. These values are identical to those under consideration by the EC in its review of the Phase 2 requirements of the European directive.

3. Relaxation of acceleration limit

115. In order for the vehicle to provide adequate occupant protection in frontal crashes, portions of the vehicle bumper structure will have to be stiff enough to enable the vehicle to absorb a sufficient amount of the impact energy. In addition, the bumper structure contains towing hooks and other devices. Because of these factors, certain portions of the bumper will not be able to meet the maximum lateral tibia acceleration limit of 170g across the full length of the bumper. For feasibility reasons, this gtr allows manufacturers to nominate bumper test widths up to 264 mm in total where the acceleration measured at the upper end of the tibia shall not exceed 250g. The relaxation zone of 264 mm corresponds to an area that is twice the width of the legform.

(c) Upper legform test for high bumpers

116. As discussed above, the informal group recognized that the lower leg impactor test would be inappropriate for vehicles whose bumpers strike the legs above knee level, but the group believed that vehicles with high bumpers should be subject to a test that would require the bumper to be more energy absorbing. For that reason, the informal working group recommends an upper legform test for vehicles with a lower bumper height of more than 500 mm.

117. Data provided in INF GR/PS/175 Rev.2 indicate that, in order to meet the proposed criteria for the upper legform test, energy absorbing foam will have to be added to the bumper; such modifications are expected to reduce the risk of fractures which also constitutes an important injury risk.

1. Impactor

118. As the majority of victims of upper leg injuries are adults, the informal group generally agreed to recommend a subsystem test using a legform impactor that represents an upper adult leg. The impactor specifications in this proposed gtr are those used in the EU Directive 2003/102/EC for the upper legform impactor.

2. Injury criteria

119. The gtr specifies that the instantaneous sum of the impact forces with respect to time shall not exceed 7.5 kN and that the bending moment on the test impactor shall not exceed 510 Nm. Biomechanical research of WG17 has found an upper leg tolerance in the range of 4 to 7 kN peak force, and 300 to 600 Nm bending moment. These values are based on PMHS test results, for a three-point bending in the middle of the femur. The absence of muscle tone in the PMHS tests and the difference in the impact point between the PMHS tests and the car impact would support a higher tolerance, especially for the peak force value. Accordingly, the informal group decided to recommend a 7.5 kN peak force limit, and a limit on bending moment of 510 Nm. These limits are the same as those under consideration by the EC for Phase 2 of the European directive.

3. Limits of the upper legform test

120. Some delegates to the informal group were concerned that, although a test that encourages high bumpers to be more energy absorbing has merit, the benefits resulting from the existing upper legform test for high bumpers should be more quantified and the suitability of the existing upper legform as a test device should be further examined. Interest was also expressed in developing a test device that could be used to test high bumpers to assess their potential for knee, thigh, hip, and pelvis injuries. Research is continuing in this area.

VIII. OTHER ISSUES

1. Systems or components that change position

121. Any vehicle system or component which could change shape or position, such as pop-up headlights or headlamp cleaners, other than active devices to protect pedestrians, were considered by the informal group to possibly create additional injury risks for pedestrians. It was therefore discussed whether such systems need to

meet the requirements in their "normal position of use" or in any other possible position. During the discussion it became clear that currently, due to the fact of the latest developments in lighting technology, no vehicles with pop-up headlights exist on the market. Other systems such as headlamp cleaners move back to their stowed position under a small preload. Finally, the informal group decided to recommend such active systems to be set to their stowed position when determining the test areas. When performing the tests, the vehicle shall comply with the requirements with the components in each fixed shape or position (e.g. stowed and popped-up).

2. Active devices to protect pedestrians

122. The issue of active devices to protect pedestrians, such as deployable bonnets, was also discussed in detail. The group fully agreed that such devices must not create a higher risk of injuries for the pedestrians. A document entitled "Certification Standard for Type Approval Testing of Active Deployable Systems of the Bonnet/Windscreen Area," proposed by the industry (INF GR/PS/141) was found to be acceptable as a guideline for certification of deployable devices, but the deployable devices clearly also need to satisfy all other requirements of this gtr. Contracting Parties who wish to implement national test procedures for these deployable devices may use the certification standard as its basis.

3. Impact on existing standards

123. During the discussions of the informal group, it was generally recognized that any proposed legislative requirements on pedestrian protection should be assessed against other vehicle parameters.

124. It was pointed out that both existing and future vehicle requirements should be taken into account, internationally as well as nationally, to ensure that potential conflicts are reduced as much as possible. The group also stressed that, in addition to legislative requirements, other vehicle parameters also need verification, in terms of customer satisfaction, repairability, insurance classification, comfort, handling, etc. A list of all necessary parameters was drafted for evaluation (INF GR/PS/35). Impacts of these standards and requirements were taken into account amongst others in the feasibility studies detailed in working papers INF GR/PS/91 and 101.

IX. REGULATORY IMPACT AND ECONOMIC EFFECTIVENESS

125. This global technical regulation is expected to reduce the number of pedestrian fatalities and injuries resulting from head impacts against the bonnet and leg impacts with the bumper. It will also maximize economic effectiveness of pedestrian protection regulations globally.

126. It should not, however, be allowed to impose any restrictions on other measures, either active or passive, which may be utilised by any Contracting Party to provide additional benefits for the safety of vulnerable road users.

(a) Benefits

127. The informal working group recognizes that there are many variables affecting the potential benefits of this gtr, such as region-to-region differences in vehicle fleet composition, in driver behaviour, in the degree to which existing vehicles now meet the pedestrian protection requirements of this gtr or are otherwise equipped with safety features beneficial to pedestrians, and in the prevalence of pedestrian-friendly infrastructure. The group also recognizes that in estimating the potential benefits of this gtr, jurisdictions differed to a degree in their methodology and assumptions so that a direct comparison of benefits between regions was not possible. However, various delegates have made very preliminary estimates of benefits based on limited data currently available to the informal working group.

128. It is estimated that this gtr will prevent between 1 and 5 percent of all pedestrian fatalities, depending on the region. Based on preliminary data, it is estimated that, mainly due to the head protection requirements, the measures would result in the following reduction in pedestrian fatalities annually: in the European Union, about 320 lives saved; in Korea, 175 lives saved; in Japan, 111 lives saved; and in the United States of America, between 61 to 92 lives saved. It is also expected that the head protection requirements would be beneficial for

pedal-cyclists. In the European Union, an estimated 46 pedal-cyclist lives would be saved. Additionally, it is expected that the gtr will provide some level of benefit in impacts with speeds greater than 40 km/h because of a reduction in injury levels (i.e. severe/serious injuries will become moderate/minor injuries).

129. In addition to the fatality estimates, the European Union also provided estimates for the impact of this gtr on serious injuries. In 2003, there were an estimated 68,016 to 160,504 serious pedestrian injuries and 46,286 to 109,226 pedal-cyclist injuries in the 25 European Union countries.

130. The estimated proportional savings in fatalities and injuries are based on data from the GIDAS and IHRA databases and an examination of the European Union vehicle fleet composition. The final resulting analysis estimates an annual reduction of 18,893 serious pedestrian injuries and 5,168 serious pedal-cyclist injuries. 23/

23/ http://ec.europa.eu/enterprise/automotive/pagesbackground/pedestrianprotection/final_trl_2006.pdf

(b) Costs

131. The European Union analysis also provides cost estimates to implement necessary changes to the vehicles to meet the gtr leg and head requirements (Table 1). These costs include the price of parts and the costs to the manufacturer for tooling and assembly line outlays. These costs are dependant on lead time to implement the regulation and advancements in technologies developed to address the gtr requirements. It is expected that some of these costs will decrease with time.

Table 1

Vehicle Style	Cost per vehicle (Euros)
Super Mini	45.98
Small Family Car	27.76
Large Family Car	36.93
Executive Car	37.64
Sports Car	85.77
Small MPV	30.80
Large MPV	34.53
Large Off-Roader	47.41

(c) Other Analysis

132. The group did not have separate assessments of the potential leg/knee injury benefits and costs from each of the other regions. Other countries are currently conducting such studies and will consider the results when the gtr is established in their national legislation. The preamble may be amended to incorporate the completed analyses.

10. Appendix

Number of working paper	*Title of informal document*
INF GR/PS/1 and Rev 1	Agenda 1st meeting
INF GR/PS/2	Terms of Reference of the GRSP Informal Group on Pedestrian Safety adopted by GRSP at its thirty first session
INF GR/PS/3	IHRA accident study presentation

INF GR/PS/4 and Rev 1	JMLIT proposed legislation - Comparison of draft regulations
INF GR/PS/5	IHRA feasibility study (doc. IHRA/PS/224 - Chapter 9)
INF GR/PS/6	Japanese proposal for the scope of Global Technical Regulations on Pedestrian Protection
INF GR/PS/7	Attendance list 1st meeting
INF GR/PS/8 and Rev 1	Draft Meeting Minutes 1st meeting
INF GR/PS/9 and Rev 1	Report of the First Meeting (Informal Document GRSP 32-07)
INF GR/PS/10	Draft GRSP/Pedestrian Safety Ad hoc Action Plan
INF GR/PS/11	Agenda 2nd meeting
INF GR/PS/12	Pedestrian Protection In Europe - The Potential of Car Design and Impact Testing (GIDAS Study)
INF GR/PS/13	Pedestrian Protection In Europe - The Potential of Car Design and Impact Testing (GIDAS Presentation)
INF GR/PS/14	Italy 1999 – 2000 [accident data]
INF GR/PS/15	Pedestrians killed in road traffic accidents [UN Statistics of Road Traffic Accidents in Europe and North America]
INF GR/PS/16	Pedestrians injury profile evaluation in a hospital-based multicenter polytrauma survey [Spanish accident data]
INF GR/PS/17	European Accident Causation Survey (EACS)
INF GR/PS/18	Draft Meeting Minutes 2nd meeting
INF GR/PS/19	Agenda 3rd meeting
INF GR/PS/20	Canadian Pedestrian Fatalities and Injuries 1990 – 2000
INF GR/PS/21	Data from the Netherlands for the years 1990-2001: Pedestrian and Cyclists Casualties
INF GR/PS/22	Vehicle Category (Proposal for the Scope)
INF GR/PS/23	Draft content preliminary report
INF GR/PS/24	Attendance list 3rd meeting
INF GR/PS/25	Extract from the GIDAS study for pedestrian safety
INF GR/PS/26	ITARDA Leg Injury Data
INF GR/PS/27 and Rev 1	Draft Meeting Minutes 3rd meeting
INF GR/PS/28	Technical feasibility – general
INF GR/PS/29	Study of the Efficiency of Infrastructural Measures for Pedestrian Protection
INF GR/PS/30	Frequency of Pelvis/Femur Fractures for Pedestrians more than 11 Years
INF GR/PS/31	IHRA/PS-WG Pedestrian accident data

INF GR/PS/32	ESV summary paper on IHRA/PS-WG report
INF GR/PS/33	Introduction of the regulation of pedestrian head protection in Japan (Nishimoto/Toshiyuki, 18th ESV Conference, Nagoya 2003)
INF GR/PS/34	Proposal for a Directive of the European Parliament and the Council relating to the protection of pedestrians and other vulnerable road users in the event of a collision with a motor vehicle and amending Directive 70/156/EEC; Commission of the European Communities, Brussels, February 2003
INF GR/PS/35	List of conflicts with existing legislation / requirements
INF GR/PS/36	Draft preliminary report
INF GR/PS/37	Agenda 4th meeting
INF GR/PS/38	Technical prescriptions concerning test provisions for pedestrian safety
INF GR/PS/39 and Rev 1	Vehicle safety standards report 1
INF GR/PS/40	US Cumulative 2002 Fleet GVMR
INF GR/PS/41	Swedish pedestrian fatalities 1994-2002
INF GR/PS/42	Proposal for a new draft global technical regulation concerning uniform provisions for common definitions and procedures to be used in Global Technical Regulations (TRANS/WP.29/GRSG/2003/10)
INF GR/PS/43	Vehicles of Category 1-1 in Japan
INF GR/PS/44	Light duty truck
INF GR/PS/45	Analysis of Euro NCAPdata
INF GR/PS/46	Development of JAMA / JARI pedestrian child and adult headform impactors
INF GR/PS/47 and Rev 1	Preliminary report to GRSP 33rd session
INF GR/PS/48 and Rev 1	Draft meeting minutes 4th meeting
INF GR/PS/49	IHRA child head test method
INF GR/PS/50	IHRA adult head test method
INF GR/PS/51	Attendance list 4th meeting
INF GR/PS/52	Provisional agenda for the 5th meeting
INF GR/PS/53	Draft gtr format
INF GR/PS/54	gtr proposal to WP.29
INF GR/PS/55	Proposal for a new draft gtr (Japan)
INF GR/PS/56 and Rev 1	Pedestrian Safety Comparison Table
INF GR/PS/57	Pedestrian Safety gtr Preparation Schedule (Draft)
INF GR/PS/58	Contents of headform test procedure

INF GR/PS/59	Comments on windscreen/A pillars as headform test area
INF GR/PS/60	Document ISO/TC22/SC10/WG2 N613
INF GR/PS/61	IHRA Computer simulation results (document IHRA PS 237)
INF GR/PS/62	Action plan from 5th meeting
INF GR/PS/63	Attendance list 5th meeting
INF GR/PS/64 and Rev 1	Draft meeting minutes 5th meeting
INF GR/PS/65 and Rev 1	Provisional agenda for the 6th meeting
INF GR/PS/66	Australian NCAP pedestrian report
INF GR/PS/67	CLEPA proposal for a test-method - active hood / bonnet systems
INF GR/PS/68	Initial Assessment of Target Population for Potential Reduction of Pedestrian Head Injury in the US (Mallory/Stammen 2004)
INF GR/PS/69 and Rev 1	Proposed draft global technical regulation (gtr) on pedestrian protection - Transmitted by OICA
INF GR/PS/70	Current Status in Korea for Pedestrian Safety Rule-making Researches
INF GR/PS/71	Possibility to define an impact zone in the windscreen/A-pillar area to fulfil HIC criteria
INF GR/PS/72	Head impact to windscreen
INF GR/PS/73	Re-assessment of headform impactor test parameters
INF GR/PS/74	Specification of headform impactor (document IHRA/PS 270)
INF GR/PS/75	"Active hood" systems test method CLEPA proposal
INF GR/PS/76	IHRA Discussions on Legform Test - Reviewing the 14th IHRA Meeting Minutes
INF GR/PS/77	UVA Dynamic Bending Corridors for Mid-Thigh, Knee, and Mid-Leg; Explained by JARI instead of UVA
INF GR/PS/78	Explanation of the Bio-Rating Method of Maltese M. R. (NHTSA) and Application the Method to Flex-PLI 2003R using UVA Dynamic Bending Corridors for Mid-Thigh, Knee, and Mid-Leg
INF GR/PS/79	[IHRA anthropometric leg proposal]
INF GR/PS/80	IHRA developed/IHRA recommendation to gtr [IHRA/PS/278]
INF GR/PS/81	Schedule for legform impactor for gtr
INF GR/PS/82	Discussion on Injury Threshold for Pedestrian Legform Test
INF GR/PS/83	Action plan / decisions resulting from the 6th meeting
INF GR/PS/84	Draft meeting minutes of the 6th meeting
INF GR/PS/85	Attendance list of the 6th meeting
INF GR/PS/86 and Rev 1/2	Proposed draft global technical regulation (gtr) on pedestrian protection

INF GR/PS/87	Development of a biofidelic flexible pedestrian leg form impactor [document IHRA PS 273]
INF GR/PS/88	Second interim report to GRSP 35
INF GR/PS/89	A study on the feasibility of measures relating to the protection of pedestrians and other vulnerable road users. Final Report. [European Commission]
INF GR/PS/90	Provisional agenda for the 7th meeting
INF GR/PS/91	Study on Technical Feasibility of EEVC WG17 (Matra/ACEA)
INF GR/PS/92	Equal Effectiveness Study on Pedestrian Protection (TU Dresden/ACEA)
INF GR/PS/93	Design of JAMA/JARI pedestrian headform impactor
INF GR/PS/94	J-NCAP Pedestrian Headform Test - HIC Values in Windshield Impact
INF GR/PS/95	Proposed Global Technical Regulation (GTR) on Pedestrian Protection - JPN comment on PS-86-Rev 2
INF GR/PS/96	Problem of Undamped Accelerometer in Headform Impact Test - Generation of Abnormal Acceleration in Headform Impact Tests - Causes and Solutions
INF GR/PS/97	Durability and repeatability of headform skin
INF GR/PS/98	IHRA/PS Decisions for the IHRA/PS Legform Test Procedures - IHRA/PS Working Group (IHRA PS 310)
INF GR/PS/99	Aging Effect of PVC Headform Skin on the Drop Certification Testing
INF GR/PS/100	OICA proposed amendments to INF/GR/PS/86/Rev.2 J - 28 September 2004
INF GR/PS/101	JAMA Technical Feasibility Study on EEVC/WG17 - Pedestrian Subsystem Test
INF GR/PS/102	Windscreen Tests according to Euro NCAP Protocol (Example)
INF GR/PS/103	[Windscreen & A-pillar testing on one car model]
INF GR/PS/104	Minimum Standard for Type Approval Testing of Active Deployable Systems of the Bonnet / Windscreen Area (CLEPA/OICA)
INF GR/PS/105	Human Biomechanical Responses to support the Design of a Pedestrian Leg Impactor
INF GR/PS/106	Information on the Flexible Pedestrian Legform Impactor (Flex-PLI) from J-MLIT Research
INF GR/PS/107	Knee ligament figure
INF GR/PS/108	Comment for IHRA or gtr regarding Legform Test
INF GR/PS/109	-[Flex-PLI as a certification tool]
INF GR/PS/110	DRAFT: Definition of the windscreen reference lines
INF GR/PS/111 and Rev 1	Pedestrian Safety Global Technical Regulation Preamble [draft and guideline]
INF GR/PS/112	Action plan resulting from the 7th meeting
INF GR/PS/113	Proposed Draft Global Technical Regulation (gtr) on Pedestrian Protection
INF GR/PS/114	Attendance list 7th meeting

INF GR/PS/115 and Rev 1	Draft meeting minutes of the 7th meeting
INF GR/PS/116	Proposed Draft Global Technical Regulation (gtr) on Pedestrian Protection [working version]
INF GR/PS/117	Proposed Draft Global Technical Regulation (gtr) on Pedestrian Protection [Proposal for 37th GRSP]
INF GR/PS/118 and Rev 1	Provisional agenda for the 8th meeting
INF GR/PS/119	Proposal from the Chairman to include the history of ISO work in the Preamble under item III. Existing Regulations, Directives, and International Voluntary Standards
INF GR/PS/120	A study on the feasibility of measures relating to the protection of pedestrians and other vulnerable road users - Addendum to Final Report (EC)
INF GR/PS/121	TRANS/WP.29/GRSP/2005/3 incl. amendments of the 37th GRSP session
INF GR/PS/122	GRSP-37-18 – USA Comments on Draft GTR on Pedestrian Head and Leg Protection (TRANS/WP.29/GRSP/2005/3)
INF GR/PS/123	GRSP-37-15 - Japan's Comment to TRANS/WP.29/GRSP/2005/3
INF GR/PS/124	GRSP-37-16 - Flex-PLI Technical Evaluation Group (Flex-PLI TEG) Activities
INF GR/PS/125	Short report on comments received during GRSP-37
INF GR/PS/126	Request for submission of the justifications for PS gtr proposal [task list]
INF GR/PS/127	Pedestrian Protection Phase 2 [EU]
INF GR/PS/128	The need for harmonized legislation on pedestrian protection
INF GR/PS/129	Comparison of the severity between the Japanese regulation based on IHRA and Phase 2 proposal based on EEVC
INF GR/PS/130	Derivation of [head] impact direction; extract from Glaeser 1995
INF GR/PS/131	Analysis of pedestrian accident situation and portion addressed by this gtr
INF GR/PS/132	Gtr Head Tests of US Fleet Vehicles
INF GR/PS/133 and Rev 1	Miniature Damped Accelerometer Series Kyowa ASE-A
INF GR/PS/134	Head Impact Tests in the Centre of the Windscreen
INF GR/PS/135	Definition of Windscreen Lower Reference Line
INF GR/PS/136	Action 10 of INF GR/PS/112: Clarification of values upper legform/bumper
INF GR/PS/137	Definition of High Bumper Vehicles
INF GR/PS/138	Economic Appraisal for Technical Regulation on Pedestrian Protection, focused on head protection
INF GR/PS/139	Action items from the 8th meeting
INF GR/PS/140	IHRA Injury Breakdown (All Ages)
INF GR/PS/141 and Rev 1	Certification Standard for Type Approval Testing of Active Deployable Systems of the Bonnet Area

INF GR/PS/142	Relative humidity of Korea
INF GR/PS/143 and Rev 1	Draft gtr based on INF GR/PS/121 as amended during the 8th meeting
INF GR/PS/144 and Rev 1	Draft meeting minutes of the 8th meeting
INF GR/PS/145	Attendance list 8th meeting
INF GR/PS/146	Flex-PLI TEG Activities
INF GR/PS/147	Proposals from Mr Césari for amendments to the preamble as agreed in the action items INF GR/PS/139
INF GR/PS/148	Assessment of the FTSS 4.5 kg aluminium headform as a possible alternative for EEVC WG17
INF GR/PS/149	New Requirement Proposal for the GTR Adult Headform Impactor Specification - Moment of Inertia
INF GR/PS/150	Development of a Head Impact Test Procedure for Pedestrian Protection (Glaeser, 13th ESV Conference, Paris 1991)
INF GR/PS/151	Proposed wording by Japan for the preamble on the headform (damped) accelerometer issue
INF GR/PS/152	Provisional agenda for the 9th meeting
INF GR/PS/153	Explanation of amendments to INF GR/PS/143 resulting in INF GR/PS/143 Rev. 1
INF GR/PS/154 and Rev 1	Handling Guideline for the EEVC WG17 Legform Impactor (Draft) and (Version 1.0)
INF GR/PS/155	Proposal for a Definition of the Lower Windscreen Reference Line and Justification
INF GR/PS/156	Proposal for Impact Angles for Headform to Windscreen Tests and Justification
INF GR/PS/157	Proposal for HIC Limits for Headform to Windscreen Tests and Justification
INF GR/PS/158	Proposal for New Criteria for Headform Impactor to Bonnet Tests and Justification
INF GR/PS/159	Proposal for a Definition of Vehicles with High Bumpers and Justification
INF GR/PS/160	Revised preamble replacing the preamble in doc. INF GR/PS/143 Rev. 1
INF GR/PS/161 and Rev 1	EU proposed amendments to doc. INF GR/PS/143 Rev. 1
INF GR/PS/162	Explanation of EU proposals (in INF GR/PS/161) to amend INF GR/PS/143 Rev. 1
INF GR/PS/163	SUV - Windshield Head Impacts
INF GR/PS/164	Observation of different windscreen glass fracture modes during headform impactor tests
INF GR/PS/165	Leg feasibility testing
INF GR/PS/166	Relaxation zone and GVWR application for US
INF GR/PS/167	EU field data on crossbeam height
INF GR/PS/168	Relationship between HIC15, HIC36, Peak Acceleration and Pulse duration
INF GR/PS/169	Revised Analysis of Pedestrian Accident Situation and Portion Addressed by this gtr
INF GR/PS/170	Draft preamble: Target population for this gtr

INF GR/PS/171	Draft meeting minutes of the 9th meeting
INF GR/PS/172	Attendance list 9th meeting
INF GR/PS/173	Provisional agenda for the 10th meeting
INF GR/PS/174 and Rev 1	Euro NCAP test results, Phase 12 – 17, lower leg tests - OICA presentation for Jan. 2006 meeting
INF GR/PS/175 and Rev 1/2	Lower/Upper Bumper Reference Lines, Data on existing vehicles - OICA presentation for Jan 2006 meeting
INF GR/PS/176 and Rev 1/2	Headform test data OICA presentation for Jan 2006 meeting
INF GR/PS/177	IHRA/PS Proposal for the Moment of Inertia of gtr Adult/Child Headform Impactors
INF GR/PS/178	Expected life-saving of introducing the GTR Head Protection Regulation in Japan
INF GR/PS/179	Ongoing Researches on Pedestrian Leg Injuries Assessment Performed by INRETS in Relation with EEVC WG 17
INF GR/PS/180	OICA position on the change of the definition of the ble reference line
INF GR/PS/181	Lower Extremity Pedestrian Injury in the US: A Summary of PCDS Data (from IHRA/PS 333)
INF GR/PS/182	Factor causing scatter in dynamic certification test results for compliance with EEVC WG17 legform impactor standard (Matsui/Takabayashi, IJCrash 2004 Vol. 9 No. 1 pp. 5–13)
INF GR/PS/183	OICA position on bonnet leading edge 165 mm exemption zone
INF GR/PS/184	Final draft gtr (without preamble)
INF GR/PS/185	Mr Saul letter dated on 3/1/2006
INF GR/PS/186	NHTSA revision of preamble PS/160
INF GR/PS/187 and Rev 1	EEVC WG17 report December 1998 and with September 2002 updates
INF GR/PS/188	Draft meeting minutes of the 10th meeting
INF GR/PS/189	Attendance list 10th meeting
GRSP-47-18/Rev.2	(USA) Proposal for amendments to global technical regulation No. 9 (Pedestrian Safety)

B. Phase 2

133. Sections 1. to 6. reflect the development of Phase 2 of UN GTR No. 9 and concern the legform test procedure with the flexible lower legform impactor (FlexPLI) without changing the requirements for the upper legform impactor and the test procedure for the high bumper vehicles as well as the headform impactors and the respective test procedures.

1. Introduction and general background

134. At the thirty-sixth session of GRSP (7-10 December 2004) the expert from Japan proposed to evaluate the possibility to replace the European Enhanced Vehicle safety Committee (EEVC) lower legform impactor by a flexible lower legform impactor. A technical evaluation group (TEG) was thus established by GRSP.

135. Under the chairmanship of Japan, the TEG prepared a draft proposal submitted by Japan for the May 2011 session of GRSP, to introduce the flexible lower legform impactor in the global technical regulation (UN GTR) No. 9 on pedestrian safety.[1] After the review, GRSP decided that pending issues should addressed by a reconstituted Informal Working Group (IWG).

[1] The final report of the TEG is available as documents ECE/TRANS/WP.29/GRSP/2011/13 and GRSP-49-15: the documents discussed are listed in document GTR9-C-08 of the constitutional meeting of the IWG and are available in the section "FlexPLI subgroup" of the first Informal Group on Pedestrian Safety.

136. The representatives of Germany and Japan proposed the development of Phase 2 (PH2) of the global technical regulation (UN GTR) No. 9 on pedestrian safety. The main objective of PH2 is the development of a draft proposal to amend UN GTR No. 9 by introducing the flexible pedestrian legform impactor (FlexPLI) as a single harmonized test tool aimed at enhancing the level of protection for the lower legs of pedestrians.

137. The work of the IWG shall not be limited to draft proposals to amend the UN GTR No. 9, but shall cover the development of a complementary draft proposal to amend UN Regulation No. 127.

138. The IWG should also review proposals to improve and/or clarify aspects of the legform test procedure.

139. The changes introduced by this amendment do not intend to change the severity of the original requirements significantly. However, with the introduction of the flexible lower legform impactor, Contracting Parties and domestic economic integration organizations are able to adopt, by preference, a particular tool with superior performance into their national or domestic legislation.

2. Procedural background

140. At its forty-ninth session, GRSP considered ECE/TRANS/WP.29/ GRSP/2011/13 and GRSP-49-15 concerning the introduction of the flexible pedestrian legform (FlexPLI) into the UN GTR. The expert from the United States of America (USA) gave a presentation on the outcome of a comparison research study conducted in his country between the FlexPLI and the current lower legform. He concluded that additional research, testing and additional world fleet data would be needed to address the injury criteria concerns and to justify its introduction on the FlexPLI. The expert from Japan gave a presentation (GRSP-49-24), showing that the FlexPLI and the current legform have totally different structures and injury criteria. Therefore, he concluded that direct comparison between the two legforms would give misleading results. GRSP agreed that pending issues should be addressed by an IWG, co-chaired by Germany and Japan. The IWG would finalizing proposals for introducing the FlexPLI into the UN GTR and, simultaneously, into the draft Regulation on pedestrian safety in the same time.

141. GRSP agreed to seek consent from WP.29 and the Executive Committee of the 1998 Agreement (AC.3) at their June 2011 sessions to mandate an IWG on the FlexPLI. GRSP also noted the draft terms of reference of the IWG (GRSP-49-38) and agreed to refer to this group for finalization. Finally, it was agreed to resume consideration on this agenda item on the basis of revised proposals, if any.

142. At the 154th session of the WP.29, the representative of the USA informed AC.3 that, as an outcome of a research study conducted in her country, concerns were expressed by her delegation at the May 2011 session of GRSP on the readiness of FlexPLI as a test tool. She added that GRSP had agreed that pending issues should be addressed by a reconstituted IWG. The representative of Germany clarified that the IWG GTR9-PH2, would be co-chaired and co-sponsored by Germany and Japan with the secretariat tasks assigned to the International Organization of Motor Vehicle Manufacturers (OICA). AC.3 gave its consent to mandate the IWG subject to the submission of appropriate terms of references to AC.3. It was agreed to set up an IWG to solve the pending issues on incorporating the FlexPLI into PH2 of UN GTR No. 9 and in Regulation No. 127 on pedestrian safety.

143. The Chair of GRSP reported on the forty-ninth session where GRSP agreed to seek the consent of WP.29 and AC.3 to mandate a new informal group to solve the pending issues for incorporating the FlexPLI in Phase 2 of the

UN GTR No. 9 and in the draft Regulation on pedestrian safety at the same time. The World Forum agreed to set up another IWG, subject to the submission to WP.29 of the appropriate terms of references.

144. The IWG began its work on 3 November 2011 with a constitutional meeting in Bonn (Germany) to draft the terms of references, the rules of procedure, the time schedule and the work plan. There, the participants also agreed with the proposal of the co-sponsors on the IWG position of Chair, Vice-Chair and Secretariat as mentioned in para. 142 above.

145. At the 155th session of WP.29 and the thirty-third session of AC.3, Germany and Japan informed delegates about the outcome of the constitutional meeting, the management of the group and the ongoing activities of the IWG (document WP.29-155-35). WP.29 and AC.3 noted that the first meeting of the IWG was planned for 1 and 2 December 2011 to start the technical discussion and to finalize the draft terms of references as well as the work plan for submission to GRSP at its December 2011 session.

146. The first meeting of IWG was held on 1 and 2 December 2011 in Geneva (Switzerland). Technical discussions began and the draft document on the terms of reference, the rules of procedures, the time schedule and the work plan for submission to GRSP in December 2011 were concluded. The first progress report was submitted to GRSP in December 2011 and to WP.29 at its 156th session as well as to AC.3 at its thirty-fourth session in March 2012. At its 156th session, the World Forum, endorsed, in principle, the noted terms of references, pending the adoption of the report of the December 2011 session of GRSP. AC.3 also endorsed, in principle, the terms of reference of the IWG and requested the secretariat to distribute WP.29-156-11 with an official symbol for consideration at its June 2012 session.

147. The second meeting of the IWG took place in Osaka (Japan) on 28 and 29 March 2012. The discussion focused on the technical aspects including the accident and benefit analysis. High priority was given to the activities on the further development of the certification procedures. A task force was established to initiate a further work item on the bumper test area for the lower legform impact.

148. The second progress report was submitted to GRSP in May 2012 and to WP.29 for consideration at its 157th session and to AC.3 at its thirty-fifth session in June 2012. During these sessions, the first progress report (ECE/TRANS/WP.29/2012/58) and the terms of references including the rules of procedures, the time schedule and the work plan were formally adopted. The second progress report (WP.29-157-21) was distributed with an official symbol at the November 2012 sessions of WP.29 and AC.3.

149. The third meeting of the IWG was held on 29 and 30 May 2012 in Paris. During the meeting, the experts discussed main topics related to accident data on pedestrian injuries, the cost-benefit assessment and the set-up of certification corridors.

150. The fourth meeting of the IWG took place on 17 to 19 September 2012 in Washington, D.C. The group resumed discussions from the third meeting, while the main focus was given to finalizing the certification corridors and the cost-benefit assessment for introducing the FlexPLI. Priority was given to agree on an international round robin vehicle test programme with the FlexPLI.

151. The draft third progress report was submitted to WP.29 at its 158th session and to AC.3 at its thirty-sixth session. AC.3 requested the secretariat to distribute the draft third progress report (WP.29-158-28) with an official symbol for consideration at the next session and adopted the second progress report (ECE/TRANS/WP.29/2012/120).

152. The fifth meeting was held on 6 and 7 December 2012 in Bergisch Gladbach (Germany). Main discussions during this meeting were the review of the cost-benefit analysis, an exchange of information on the first results of the repeatability and reproducibility of the FlexPLI tests with vehicles, and a discussion on the threshold values for the injury criteria. Furthermore, the IWG agreed to seek the consent of GRSP and WP.29/AC.3 for extending the mandate (working schedule) to take all test results into account for the amendment of the UN GTR.

153. Delegates noted that GRSP had adopted the revised terms of reference of the IWG group as reproduced in Annex II to the GRSP report (ECE/TRANS/WP.29/GRSP/52) during the 159th session of WP.29 and at the thirty-seventh session of AC.3. The World Forum endorsed the extension of the mandate of the IWG until June 2014 (expected adoption at WP.29/AC.3) and, in principle, the revised terms of references, pending the adoption of the GRSP report of its December 2012 session at the 160th session of the World Forum in June 2013.

154. The third progress report (ECE/TRANS/WP.29/2013/36) was recalled at the 159th session of WP.29 and at the thirty-seventh session of AC.3 together with the amendments proposed by GRSP (WP.29-159-20) at the December 2012 session. AC.3 adopted ECE/TRANS/WP.29/2013/36, as amended by Annex III of the report of that session of WP.29 (ECE/TRANS/WP.29/1102).

155. The sixth meeting of the IWG took place in Washington, D.C. from 19 to 20 March 2013. The group agreed on the approach to review the FlexPLI drawing package to prepare the addendum of the Mutual Resolution No. 1 (M.R.1). The review of the controversial discussion on the cost-benefit studies was finalised and the results of the different regions and laboratories on the vehicle repeatability and reproducibility tests were discussed.

156. The draft fourth progress report of the IWG was presented at the fifty-third session of GRSP. It was added that the group had made good progress and that it was ready to submit an official proposal to the December session of GRSP with possible pending decisions on threshold values of injury criteria. GRSP agreed to resume consideration of this subject on the basis of a proposal submitted by the IWG.

157. At the 160th WP.29 session, the experts were informed by the representative of the United States of America that GRSP was expected to recommend that Amendment 2 (Phase 2) of the UN GTR on pedestrian safety, aimed at including the FlexPLI and the definition of the head form impact point into the UN GTR No. 9 test as well as into Regulation No. 127. He also announced the submission of another amendment to the UN GTR on pedestrian safety on an updated definition of the head form impact point.

158. At the same session, the representative of Japan, Vice-Chair of the IWG on Phase 2 of UN GTR No. 9, introduced the fourth progress report of the group together with a presentation. He confirmed the good progress of the IWG and that an official proposal for incorporating the flexible pedestrian legform impactor would be submitted to the December 2013 session of GRSP. AC.3 adopted the fourth progress report and requested the secretariat to distribute it with an official symbol at its November 2013 session.

159. The seventh meeting of the IWG (3 July 2013) was a telephone and online meeting. The group discussed specific issues, especially regarding the threshold values for the injury criteria, the definition of the rebound phase and the tolerances of FlexPLI output values during the free-flight phase. The latter two were agreed in principle while a decision on the threshold values is still pending. A further work item agreed would be an analysis on the necessity and possibility of introducing certification corridors for the femur bending moment.

160. The eighth meeting of the IWG was held on 9 and 10 September 2013. The meeting was dedicated mainly to discuss the open items like the injury criteria, the femur certification corridors and to review the preamble and the regulatory text of the UN GTR No. 9.

161. The ninth meeting of the IWG was held on 16 and 17 December 2013. During the meeting open items were discussed and resolved during the final review of the proposed amendment of the text for the UN GTR and Regulation No. 127. One pending issue on the performance limits for the injury criteria had to be discussed within GRSP involving all Contracting Parties.

162. The tenth meeting of the IWG was held on 24 November 2017. Main purpose of the meeting was to finalize all open issues with the text of the GTR since its adoption had been delayed for a longer time. The meeting was organised to discuss the amendments in detail and to agree on them.

3. Requirements

(a) Assessment of biofidelity

163. Japan Automobile Standard Internationalization Centre (JASIC) highlighted the improved biofidelity of the FlexPLI compared to the legform impactor currently used in UN GTR No. 9. The superior biofidelity was shown at component and assembly level using both the testing and the simulation tools. The improvements in the knee and tibia area were presented. A comparison study of the FlexPLI and post-mortem human subject (PMHS) test data was done for the performance limits. The FlexPLI was shown as more human-like with regard to the injury mechanism of the tibia.

164. The biofidelity study was performed with data from Japan and the USA. Some concerns were raised by the Alliance of Automobile Manufacturers on the validity of the method used by JASIC in comparing the finite element models with human body models. These concerns were not shared by the expert from Japan.

165. The expert from United Kingdom of Great Britain and Northern Ireland (UK) expressed that the FlexPLI could have limitations in assessing knee injuries. The expert from Japan explained that both, knee injuries and tibia fractures could be assessed. But during the development, higher priority was given to tibia fractures as the knee injuries are less represented compared to tibia fractures according to the accident data analyses.

166. The IWG received additional information on the superior performance of the FlexPLI compared to the current lower legform impactor.

167. The discussion on the limitations of the FlexPLI in assessing knee injuries was closed pending the submission of new information on this subject.

(b) Cost benefit analysis

168. At the start of the IWG, participants were asked to provide accident data. This request was also raised at the fiftieth session of GRSP by the Chair of the IWG. The expert of the USA informed the IWG that they were investigating if information on accidents with pedestrians could be supplied for discussion.

169. The expert from the National Highway Traffic Safety Administration (NHTSA) informed delegations about a research project in the USA to investigate the accident situation for pedestrians using the Pedestrian Crash Data Study (PCDS) and the German In-Depth Accident Study (GIDAS). The analyses only covered AIS 3-6 injuries and looked at disabling injuries according to the Functional Capacity Index (FCI) based on AIS.

170. According to both data sources, bumper-caused injuries represent up to 40 per cent of all pedestrian injuries. Notwithstanding, there are notable differences between the two sources on the number of injuries to the different body regions: the number of injuries to lower extremities are primarily caused by the bumper; and is in both cases close to 100 per cent (94 per cent for PCDS and 99 per cent for GIDAS). The presentation also showed the ranking of injured body regions for serious and disabling injuries, with the most frequent combination being the lower extremity to bumper impact.

171. The German Federal Highway Research Institute (BASt) submitted detailed information on the expected reduction of costs in Germany due to the introduction of the FlexPLI within the test procedures according to UN GTR No. 9. The study was based on both, national accident data as well as German in-depth accident data, using the injury shifting method. Here, the assumption was made that in case of a pedestrian being struck by a passenger car equipped with a frontend system designed for the protection of pedestrians, all corresponding AIS 1-3 injuries related to lower extremities could be shifted downwards by -1. In total, 498 accidents were vehicle to pedestrian accidents in the German in-depth study GIDAS. As a result, the study concluded that due to pedestrian friendly bumper designs, 25 per cent of all Maximum Abbreviated Injury Scale (MAIS) 3 injuries could be shifted to MAIS 2, and almost 8 per cent of all MAIS 2 injuries could be shifted to MAIS 1. Finally, the portion of MAIS 1 injuries would increase by approximately 2.5 per cent.

172. Applying this shifted injury distribution to the national database, the introduction of pedestrian friendly bumper designs was estimated at an annual decrease of 11 fatally injured pedestrians and an annual decrease of 506 severely injured pedestrians. In the same period of time, the number of slightly injured pedestrians would increase by 231. Finally, the maximum annual cost reduction in Germany due to vehicles designed with pedestrian

friendly bumpers was calculated at approximately € 63.5 million. According to an injury risk function developed by JASIC based on Nyquist and Kerrigan PMHS data and using the Weibull survival model, a 30 per cent tibia bone fracture risk when complying with the proposed FlexPLI tibia bending moment requirement of 340 Nm was calculated at 330 Nm bending moment of the human tibia. Thus, under consideration of a 70 per cent injury risk that is consistently assumed to be covered by the FlexPLI, the annual cost reduction due to the introduction of the FlexPLI was calculated by BASt at approximately € 44.5 million.

173. JASIC introduced detailed information on the possible benefit to tibia injuries that can be expected with the introduction of the FlexPLI. Based on accident data, it was presumed that tibia fractures mainly occur due to indirect loading (approximately 80 per cent). Only in a minor number of cases, the fracture of the tibia occurs due to direct loading of the bumper. It was also shown that the most significant improvement can be achieved by mitigation of leg fractures.

174. It was concluded that the FlexPLI can provide improved biofidelity for the tibia and the knee. Compared to the currently used legform impactor the cost savings due to mitigation of tibia fractures were estimated to be 100 million United States dollars for the USA and 50 million United States dollars for Japan based on calculation models using the annual medical costs for such types of injuries.

175. At the second meeting, the experts again reviewed the information from JASIC on the benefit of the FlexPLI, showing a significantly better biofidelity of the FlexPLI compared to the current legform impactor. It was concluded that the cost savings due to mitigation of tibia fractures were estimated to be around 77 million United States dollars for Japan based on calculation models using the annual economic cost for such types of injuries.

176. The expert from the Alliance of Automobile Manufacturers in the USA (Alliance) explained that the USA accident data used in the study might be processed in another way, as the current procedure is using the police-reported injury severity system KABCO (K - fatal, A – incapacitating, B – non-incapacitating, C – possible injury, O – no-injury) to classify injury severity might not be correct for pedestrian injuries. The expert from JASIC admitted that for some cases the injury severity classification based on the KABCO scale used for the study was not correct. A modified version of the study showed better results than the original document.

177. At the third and fourth meeting the pedestrian experts again reviewed JASIC information on the benefit of the FlexPLI. The Alliance had undertaken an investigation of the methodology that was presented by JASIC. One major concern of the Alliance was that the data used in the JASIC analysis does not correctly reflect the current accident situation in the United States due to the outdated data set and the assumptions for the injury levels taken as a basis for the benefit calculation.

178. During the fifth and the sixth meeting, the pedestrian experts further reviewed information from JASIC and the Federal Highway Research Institute of Germany (BASt) on calculating the benefits that would result from introducing the FlexPLI. The Alliance of Automobile Manufacturers in the USA repeated the concerns that the two approaches presented may not be valid for every market depending on the situation of accidents and the vehicle fleet.

179. The IWG finally agreed that this argument may be valid for some regions which would result in the need to undertake, within the individual countries or regions, a cost-benefit analysis using their national or regional data on accidents and the situation of the domestic vehicle fleet to verify the scope of the new provisions and the possible introduction of the FlexPLI in their territory.

(c) Technical specifications (drawings) and PADI (user manual)

180. Several items were raised on the user manual for the FlexPLI. An updated user manual incorporating the proposals was drafted including additional information for a visual inspection of the impactor.

181. Experts were informed that the drawings and specifications of the FlexPLI would be needed before the regulatory text can be approved by GRSP and adopted by WP.29 and AC.3. Humanetics confirmed that this is well known and such information would be submitted to the IWG.

182. The expert from OICA asked for more transparent documentation on the set-up of the flexible pedestrian legform impactor. The expert from Humanetics confirmed that information would be provided if the documentation for the FlexPLI could be made available for the informal group with a disclaimer against its use for commercial purposes.

183. The expert from the UK informed the participants about the ongoing activity at WP.29 to set up a repository that would form a kind of library for dummies and other test devices used in regulations. He informed GRSP that the experts from the UK and the USA were jointly preparing a mutual resolution (M.R.1.) of the 1958 and 1998 Agreements on the description and performance of test tools and devices necessary for assessing the compliance of wheeled vehicles, equipment and parts according to the technical prescriptions specified in Regulations and global technical regulations.

184. The IWG GTR9-PH2 was informed about a proposal of global technical regulation No. 7 (UN GTR No. 7) Phase II on the BioRID developed by the IWG working on this subject, where it was agreed that engineering drawings of dummies and dummy parts would be shared but not production drawings. The current proposal foresees that drawings would be made available during the discussion period only for information purposes and covered by a disclaimer that it may not be used for commercial purposes. The disclaimers would be withdrawn when dummies and dummy parts were agreed upon and engineering drawings would be made available.

185. It was then noted that the M.R.1 was adopted on 14 November 2012 by WP.29 (ECE/TRANS/WP.29/1101) and that contracting parties and manufacturers refer to this Mutual Resolution when establishing the suitability of their test tools and devices for assessing compliance with the prescriptions of Regulations or global technical regulations in the framework of the 1958 or 1998 Agreements respectively.

186. Humanetics provided a full drawing package for the FlexPLI in December 2012. The group discussed the plan to review the drawing package. It was agreed that a comparison of 100 per cent of the parts of one impactor would be done with the drawings. Additionally, the drawings would be checked for conformity with the requirements as defined by the IWG on Head Restraints Phase II, the IWG on Child Restraint Systems and the IWG GTR9-PH2. Only minor remarks for corrections resulted from the review of the drawing package.

187. IWG also reviewed the user manual for compliance with the defined requirements. Humanetics updated the drawings and the user manual with guidance from the IWG. A draft proposal for an addendum to the Mutual Resolution No. 1 (M.R.1) was prepared by the IWG.

(d) Evaluation of durability

188. The expert from OICA presented information on the long-term durability of the FlexPLI. Several items were mentioned, of which the durability of the bone core material led to extensive discussions. The bone core material suffers small cracks of the material during the testing. While several experts mentioned that the performance is still acceptable with these minor damages, information was given by the company Bertrandt that deviations in the performance may be seen during calibration of the legform impactor. Experts would further investigate this issue and present further information on the long-term performance at the second meeting of the IWG. Investigations showed no major issue.

189. The expert from the USA presented further information on the durability of the FlexPLI. Comparison tests of the earlier and the current versions of the FlexPLI found that the durability had improved for the current version of the impactor and, therefore, was not a major issue for the moment.

(e) Test procedure

190. The experts from BASt, JASIC and OICA presented proposals to define the rebound phase for the FlexPLI test. While JASIC and OICA were of the opinion that a definition cannot be currently introduced into UN GTR No. 9, BASt showed a procedure to define a biofidelic assessment interval (BAI). The IWG finally agreed to introduce an assessment interval (AI) as the current most appropriate method to objectively determine the valid maxima of the measurements.

191. The expert from OICA presented a proposal for the vehicle set-up in terms of riding height. The proposal to cover tolerances in built-up, adjustment and alignment of a test vehicle in actual testing recommends including the concept of the primary reference mark. The definitions would give clearer guidelines needed to perform the type approval or self-certification tests of vehicles.

192. The experts from BASt and OICA proposed to define the tolerance of FlexPLI output values during the free-flight phase for vehicle tests. Based on a BASt proposal, a definition for the free flight phase was introduced in the amendment.

(f) Certification tests

193. The IWG agreed to establish a task force, chaired by Japan, for reviewing and updating the certification corridors (TF-RUCC) to resolve issues with the current certification test procedures. Certification tests were performed with several legforms in a limited number of labs to check the performance of the flexible pedestrian legform impactors. The objective of the task force was to prepare a recommendation for the IWG on the certification procedures and the corridors to be used for the certification of the FlexPLI.

194. The results showed a good and repeatable performance of the three flexible pedestrian legform impactors with the final build level (three "master legs") tested. A round robin certification test series confirmed a stable performance of the legform impactors. The task force finalized the work and succeeded in proposing updated certification corridors based on proposals made by BASt for the dynamic tests and by Japan Automobile Research Institute (JARI) for the static tests for the certification of the flexible legform impactors on the assembly and component level.

195. The corridors were agreed by the IWG as final. It was also indicated that an evaluation of the stability of performance of the flexible legform impactors would be done during vehicle testing.

(g) Review of test results

196. The expert from OICA introduced results of impactor to vehicle tests. He added that the results were quite promising but for some peak values a deviation of up to 20 per cent was observed. IWG discussed if the impactors as well as the vehicles would really be comparable as the test results presented were generated during a period of several years (2009 - 2011), during which the impactors and the vehicles may have undergone some changes.

197. The Concept Tech GmbH presented information on the influence of friction in the test device used for inverse testing. Further information from the different laboratories investigating their own test apparatus was shown. Based on the presentations and the conclusions, the IWG agreed on the limit for the friction of test devices for inverse testing.

(h) Evaluation of reproducibility and repeatability

198. The IWG started an international round robin vehicle test programme in September 2012. The vehicle testing was finalised by March 2013. Results were presented by test houses from Europe, Republic of Korea and USA. Apart from minor issues, the results of the different test houses showed a stable performance of the legform impactors with a good repeatability. Problems in durability did not occur during vehicle testing. During the vehicle tests at BASt, the lower test results with the FlexPLI with the final build level (named "master legs" during the process of establishing certification corridors) compared with the test results with former prototype flexible legform impactors, but tested with the same cars, led to discussions about the threshold values for the impactor. However, OICA showed an example for test results with the FlexPLI against a test rig, where the output values were not lower than the results during the tests with the former flexible legform impactors. The IWG finally agreed to keep the limit values for the impactor unchanged.

(i) Performance / injury criteria and threshold values

199. JASIC introduced information on the performance and injury criteria for the FlexPLI (GTR9-1-05r1, GTR9-1-06r1). The validation of criteria for the tibia fracture and the medial collateral ligament (MCL) and anterior

cruciate ligament (ACL) failure was presented in detail and compared to the legform impactor currently used in UN GTR No. 9. The results are mainly based on data from different sources of specimen testing and from which a probability function for the injury risk was developed. Performance limits for the tibia bending moment, the ACL and the MCL proposed by the Technical Evaluation Group (TEG) on FlexPLI were presented to participants.

200. The expert from the USA raised some concerns regarding the injury thresholds that were chosen for the FlexPLI in relation to the EEVC legform impactor. With the ability of the flexible impactor it may be possible to achieve better protection with more stringent criteria. The USA do not see a necessity to just achieve a protection level that is comparable to the EEVC legform impactor. NHTSA will investigate this in more detail.

201. The IWG started discussion on the injury threshold values at its fifth meeting. The experts agreed on the injury criteria, but had an in-depth discussion on the threshold values for the different injury criteria and the injury probability that is chosen using risk curves. BASt proposed to lower the threshold values because of the FlexPLI with the final build level producing lower output values than the prototype legform impactors in inverse certification tests. This would also lead to difficulties in validating the original FlexPLI FE model against the FlexPLI prototype impactors that were used to establish the first dynamic certification corridors. BASt explained that it might be necessary to review all test results from former round robin test series prototype impactors. OICA was supportive of keeping the threshold values as proposed by the Technical Evaluation Group (TEG) on FlexPLI.

202. At the sixth meeting of the IWG, OICA presented further test data obtained using their FlexPLI with the final build level used for the round robin testing. These tests showed higher output values than those measured with the three FlexPLI with the final build level during vehicle tests.

203. In addition to the discussion on the injury threshold values, the IWG also begun discussing the underlying injury risk functions. NHTSA requested information from which the proposed threshold values were derived, because the injury probability needs to be estimated for their cost-benefit analysis. At the Technical Evaluation Group (TEG) on FlexPLI, two different approaches to derive threshold values were used, one proposed by BASt and the other proposed by JASIC. Upon request from NHTSA, BASt and JASIC provided information on the derivation of the injury risk function using their own approaches (GTR9-6-08r1, GTR9-6-26). Since BASt used a direct correlation between the knee bending angle of the EEVC legform impactor and the MCL elongation of the FlexPLI to derive the threshold value for MCL failure as well as the FlexPLI knee geometry to derive the threshold value for ACL/PCL failure, focus of the IWG discussion was given to the risk functions for tibia fracture.

204. From data on the peak human leg bending moment in dynamic 3-point lateral bending tests conducted by Nyquist et al., BASt used data for male subjects. Geometric data scaling was applied to the dataset using the standard length obtained from the German Industry Standard anthropometric database (DIN). As the used data was normally distributed according to the Shapiro Wilk Normality Test, the injury risk function for tibia fracture was derived from a normally distributed probability density function (GTR9-6-08r1).

205. JASIC chose to use both male and female data from the Nyquist study on the basis of past studies not showing significant difference in bone material property between males and females. In addition, JASIC also used more recently conducted leg 3-point bending test data from Kerrigan et al. The standard lengths taken from the anthropometric study by the University of Michigan Transportation Research Institute (UMTRI), which was also referred to when determining the legform dimensions, were used to geometrically scale the data. Since the peak moment data from the Nyquist study were attenuated by filtering, the survival model was applied to the dataset and the data from the Nyquist study were treated as right censored data, as opposed to the data from the Kerrigan study which were treated as uncensored (exact) data. Weibull distribution was assumed to allow asymmetric probability density distribution.

206. At the sixth meeting of the IWG, a comparison of both approaches carried out by BASt revealed that the calculated threshold values depend on various factors such as the underlying set of PMHS data, the scaling

method, the particular anthropometrical database for human data scaling, the injury risk to be covered, and the statistical procedure used for the development of the injury risk function (GTR9-6-08r1). At the same meeting, JASIC presented a complete description of their approach by referring to the SAE technical paper already presented at the 2012 SAE World Congress (GTR9-6-26).

207. At the seventh meeting of the IWG, NHTSA preferred to recommend one single approach. JASIC, therefore, investigated and BASt contributed to further clarifications of their approaches by providing additional technical information to NHTSA (GTR9-7-07), but an effort to come up with one common proposal was not successful. Therefore, JASIC investigated the effect of each factor (human data sources, standard lengths for geometric data scaling, statistical procedure, etc.) on the injury risk function, so that any interested Contracting Party could refer to the provided information and determine its preferred approach. The information was shared by JASIC at the eighth meeting of the IWG (GTR9-8-11).

208. In the investigation done by JASIC, it was found that the choice of the standard length used to scale the human data is one of the most significant contributors to the injury risk functions. The dimensions of the EEVC legform impactor and the FlexPLI were determined from the anthropometric measurements for fiftieth percentile male conducted by the University of Michigan Transportation Research Institute (UMTRI). NHTSA pointed out at the eighth meeting that for any test dummy the scaling of a risk curve (in theory) should be consistent with the actual anthropometry represented by the dummy. Accordingly, the legform injury risk scaling based on the fiftieth percentile UMTRI anthropometry would probably be the most accurate.

209. In previous work under the TEG, FlexPLI measurements were correlated with human injury risks with the aid of human body computer models and FlexPLI computer models. An injury transfer function was developed from the results of paired simulations of vehicle to pedestrian interactions. These simulations, which consisted of pedestrian leg or FlexPLI impacts into a series of simplified vehicle front-ends, provided a human vs. FlexPLI comparison under conditions that match actual UN GTR tests. BASt acknowledged the good correlation between the human FE model and the FlexPLI version GTR FE model of the final build level in terms of the tibia loadings; however, the knee correlation especially for the MCL elongation still shows potential for being further improved. The Alliance members indicated that the MCL response correlation between the two simulation models (FlexPLI & human body) would not be considered statistically significant ($R < 0.8$). Therefore, it is proposed that the threshold limits for the MCL based on this analysis should be used as a guide but should not be used as a pass fail criterion.

210. At the seventh meeting of the IWG, JASIC presented results that validated their FlexPLI FE model against the final build level of the actual FlexPLI (GTR9-7-08). The validation was based on the certification tests and corridors agreed to by the IWG. It was also shown that the injury threshold values derived using the transfer functions determined from the FlexPLI FE model were virtually the same as those proposed earlier by the TEG. BASt questioned how the original FlexPLI FE model could be validated against the FlexPLI prototype impactors – i.e. not the mass production model – that were used to establish the first dynamic certification corridors because of the master legs producing slightly different output values than the prototypes in inverse certification tests.

211. At the eighth meeting, the IWG agreed that the methodologies for establishing the injury risk curves should be stated for all Contracting Parties in the preamble of the UN GTR no. 9. In addition the injury threshold values proposed by the TEG should be incorporated into the document taking into account the text of the preamble, that for some Contracting Parties further studies may be necessary to evaluate the appropriateness of the thresholds for their domestic regions.

212. In order to meet the time line of Phase 2, the USA proposed at the ninth meeting of the IWG to include both the current injury threshold values in the UN GTR as well as the injury risk curves in the regulatory text of the UN GTR as Contracting Party options. The injury risk curves would be used by contracting parties selecting that option to determine the injury threshold values based on their domestic benefit assessment. The IWG deferred the discussion on this to GRSP to get all Contracting Parties involved in the discussion and the decision.

(j) Evaluation of vehicle countermeasures

213. During the fifth and sixth IWG meetings, information on the technical feasibility and possible vehicle countermeasures was provided by the experts from OICA, JASIC and NHTSA. OICA informed IWG that the feasibility may be a problem for some small volume products for which currently no detailed information on the performance with the FlexPLI was available.

214. Automakers from the USA explained that, for some heavier trucks and Sport Utility Vehicles (SUV), there would be a conflict between the customer requests for the US-market and the pedestrian requirements in the bumper area. The IWG agreed that, for some markets, it may be necessary to further consider the scope of the UN GTR and to review, for specific vehicles, the lead time for the transposition of UN GTR No. 9 into regional or national law.

(k) Other items

Finite element models

215. The European Association of Automotive Suppliers (CLEPA) requested information on the development of finite element models for the FlexPLI. It was decided that the IWG would not develop such models but would serve as a platform for a regular exchange of information on this subject. This task was started at the second meeting of the IWG.

216. The expert from Humanetics informed participants about the status of work on developing a finite element model for the FlexPLI. Currently a model is available for purchase. The further development of the model is currently stopped and would be restarted as soon as the status of the impactor is final.

4. Key elements of the amendment

217. The key elements introduced by this amendment to the UN GTR No. 9 are:

(a) the introduction of the flexible pedestrian legform impactor;

(b) the introduction of new dynamic certification corridors;

(c) the introduction of new static certification corridors;

(d) the process of using an assessment interval for identifying maximum measurements.

(e) and modification of the definition of the bumper test area.

5. Recommendations and limitations for introducing the flexible lower legform impactor

218. At the sixth meeting of the IWG, the United States of America noted that while they would be in a position to agree with the injury risk curves within the timeline of the amendment 2 of this UN GTR, they may not be in a position to agree to injury risk values without delaying the timeline. The expert from United States of America suggested that, given that cost benefits may vary depending on the fleets of different countries, the UN GTR should include only the injury risk curves, with contracting parties choosing appropriate injury assessment reference values (IARVs) when implementing the UN GTR in national legislation. In subsequent discussions to this meeting, the United States of America agreed to accept the proposed IARVs so Phase 2 of the GTR could move forward, and would propose changes to the IARVs through the normal United Nations process if changes to the IARVs are warranted by a benefits analysis conducted during the adoption process in their domestic regulations.

219. While the IWG rejected the suggestion of including only the injury risk curves, it is understood that the United States of America will conduct a full analysis of the impacts of the IARVs of the UN GTR. The United States of America will conduct fleet testing with the FlexPLI to evaluate the benefits. It would be also examined possible incremental improvements, such as the effect of lowering injury threshold values. These efforts could

result in future recommendations to adjust the injury risk values and other aspects of this UN GTR. The United States of America will report back to the United Nations any recommended adjustments to the IARVs once its cost benefit analysis has been completed.

220. As described in paragraphs 74 and 75, in some domestic regions introducing pedestrian safety requirements automobile manufacturers may find it challenging if offering a specific vehicle design. Therefore an adequate lead time during the transposition of the requirements of UN GTR No. 9 should be foreseen on the basis of cost benefit data. This is indispensable especially for contracting parties without any requirements on pedestrian safety for vehicles and planning the introduction of pedestrian safety requirements. Regarding the applicability of phase 2 of this UN GTR it should be noted that the requirements of the draft UN GTR are substantially more severe than any existing legislation at the time of adoption of the GTR. In addition, many countries do not yet have pedestrian safety requirements. It is therefore recommended that contracting parties implementing this UN GTR allow adequate lead time before full mandatory application, considering the necessary vehicle development time and product lifecycle.

221. Without obligation, it is recommended for contracting parties without requirements on pedestrian safety in their domestic law, to use the FlexPLI rather than the EEVC LFI during the transposition process due to its superior performance compared to the lower legform impactor of Phase 1 of the UN GTR No. 9.

222. The IWG noted that the simultaneous application of the EEVC legform impactor and the FlexPLI in various regulative and consumer rating requirements worldwide can lead to market distortions and an unnecessary burden on manufacturers. Therefore it is recommended that contracting parties implement this amendment for compliance at the earliest possible date as an option at the choice of the car manufacturer. However in those regions where there is existing legislation relating to legform testing with the EEVC legform impactor, vehicles fulfilling the requirements of Phase 1 of this legislation already provide protection of the lower leg. Where this is the case, a review of costs and benefits of changing to the use of the FlexPLI may not be justified if it were to require a general redesign of existing vehicle types. contracting parties should consider exempting vehicles from meeting FlexPLI requirements when these vehicles were designed and proven to comply with the requirements for the EEVC LFI.

6. Task Force Bumper Test Area (TF-BTA)

223. On request of the expert from the European Commission a discussion on the current bumper test area, mostly for the lower legform impact, took place. The necessity of improving and notably widening the test area on the bumper for the lower legform test was shown as the area of the bumper is quite restricted as a result of angled front fascia designs and protrusions or other features on the fascia of some vehicles that interact with the 60 degree planes that in the current test procedure define the test area. The decision was to discuss the whole subject in detail in a specific task force on the bumper test area (TF-BTA).

224. The IWG agreed to establish such a task force. The expert of the European Commission chaired the task force’s discussions.

225. The task force met eight times between September 2012 and November 2014 in face-to-face and web meetings. First results of the task force’s work showed that, for newer vehicles, the test areas for the lower legform impact were narrower than in the past. A contractor therefore was requested to further investigate possibilities to solve this issue, in cooperation with stakeholders. Based on the investigations of the contractor, members of the task force made different proposals on how to possibly modify the bumper test area.

226. The different proposals were discussed in the task force and finally led to proposed further amendments to this UN GTR regarding the determination of the bumper test area (document ECE/TRANS/WP.29/2014/30). These proposed amendments are subsequently considered for phase 2 of the GTR, together with some further slight modifications proposed by GRSP.

227. The definition of the bumper beam is based on the structural cross member, which is usually covered by the bumper fascia. Parts linked or connected to the structural cross member only belong to the cross member if the stiffness of such parts is not significantly lower compared to the stiffness of the cross member.

7. List of documents discussed in the IWG on UN GTR No. 9 – Phase 2

Doc. No.	*Rev.*	*Name*
GTR9-C-01	1	Agenda of the Constitutional Meeting of the Informal Group on GTR No. 9 - Phase 2 (IG GTR9-PH2)
GTR9-C-02	1	Minutes of the Constitutional Meeting of the Informal Group on Global Technical Regulation No. 9 – Phase 2 (IG GTR9-PH2)
GTR9-C-03		Informal document GRSP-49-38: Draft terms of reference for the informal group on pedestrian safety phase 2 (IG PS2)
GTR9-C-04	1	History of Development of the FlexPLI
GTR9-C-05		Review of the FlexPLI TEG Activities (2 parts)
GTR9-C-06		Comments on the Draft Terms of Reference for the Informal Group on Pedestrian Safety Phase 2 (28/10/2011)
GTR9-C-07	1	Final Operating Principles and Terms of Reference for the IG GTR9-PH2
GTR9-C-08		TEG document matrix
GTR9-1-01	1	Agenda for the 1st meeting of the Informal Group on Global Technical Regulation No. 9 - Phase 2 (IG GTR9-PH2)
GTR9-1-02	1	Minutes of the 1st meeting of the Informal Group on Global Technical Regulation No. 9 - Phase 2 (IG GTR9-PH2)
GTR9-1-03	1	Document TF-RUCC-K-03-Rev.1: Work plan of Task Force Review and Update of Certification Corridors (TF-RUCC)
GTR9-1-04	c	FlexPLI Version GTR Prototype SN-02 - Durability Assessment
GTR9-1-05	1	Technical Discussion – Biofidelity
GTR9-1-06	1	Technical Discussion – Injury Criteria
GTR9-1-07	1	Technical Discussion – Benefit
GTR9-1-08	1	FlexPLI GTR Status, 1 – 2 December 2011
GTR9-1-09		Informal document WP.29-155-35: Report to the November session of WP.29 on the activities of the IG GTR9-PH2
GTR9-1-10	c	Changes to Flex PLI GTR Since Prototype Build, Status Dec. 2010
GTR9-1-11		Scatter of pendulum test results, 09.11.2010
GTR9-1-12		Informal document GRSP-49-23: Update on Pedestrian Leg Testing
GTR9-2-01	1	Agenda for the 2nd meeting of the Informal Group on Global Technical Regulation No. 9 - Phase 2 (IG GTR9-PH2) - Final

GTR9-2-02	1	Minutes of the 2nd meeting of the Informal Group on Global Technical Regulation No. 9 – Phase 2 (IG GTR9-PH2) - Final
GTR9-2-03		Proposal for a Modification of the Bumper Test Area for Lower and Upper Legform to Bumper Tests
GTR9-2-04	1	Robustness of SN02 prototype test results - Revision 1
GTR9-2-05		Comparison of Filter Classes for FlexPLI
GTR9-2-06		Technical Specification and PADI
GTR9-2-07	1	Technical Discussion – Benefit (Update of document GTR9-1-07 Rev. 1)
GTR9-2-08		FlexPLI GTR meeting actions
GTR9-2-09		FlexPLI GTR – FE model v2.0
GTR9-2-10	2	FlexPLI Comparison - test experiences with different impactors (completed during the 3rd meeting)
GTR9-2-11		Informal document WP.29-156-11: First progress report of the informal group on Phase 2 of gtr No. 9
GTR9-2-12		Re-examination of Number of Pedestrians by Injury Severity
GTR9-2-13		FLEX PLI Update for Alliance of Automobile Manufacturers
GTR9-2-14		Updated Japan Progress Report: Review and Update Certification Test Corridors and Test Methods (added pendulum Test data)
GTR9-3-01	1	Agenda for the 3rd meeting of the Informal Group on Global Technical Regulation No. 9 - Phase 2 (IG GTR9-PH2) - Final
GTR9-3-02	1	Minutes of the 3rd meeting of the Informal Group on Global Technical Regulation No. 9 - Phase 2 (IG GTR9-PH2) – Final
GTR9-3-03		Informal document GRSP-51-15: Draft second progress report of the informal group on Phase 2 of UN GTR No. 9 (IG GTR9 - PH2)
GTR9-3-04		Flex PLI GTR User Manual Rev. C
GTR9-3-05		FlexPLI Prototype SN04 Robustness Test results
GTR9-3-06		Proposal for a future vehicle test matrix
GTR9-4-01	1	Agenda for the 4th meeting of the Informal Group on Global Technical Regulation No. 9 - Phase 2 (IG GTR9-PH2) - Final
GTR9-4-02	1	Minutes of the 4th meeting of the Informal Group on Global Technical Regulation No. 9 - Phase 2 (IG GTR9-PH2) - Final
GTR9-4-03	3	Status of activity list items
GTR9-4-04		1994 EEVC WG10 Report
GTR9-4-05		1996 EEVC WG10 report to the 15th ESV conference
GTR9-4-06		1998/2002 EEVC WG17 Report
GTR9-4-07		TF-RUCC Activity Report 6 Sept. 2012

GTR9-4-08		Guidelines To Conduct FlexPLI Round Robin Car Test Smoothly and Effectively
GTR9-4-09		FlexPLI Round Robin Test Results
GTR9-4-10		Informal document WP.29-157-16: Proposal for the establishment of Special Resolution No. 2 on description and performance of test tools and devices necessary for the assessment of compliance
GTR9-4-11		Investigation of the Influences of Friction within the Inverse Certification Test Setup
GTR9-4-12		Report addressing the Pedestrian Research performed by JASIC
GTR9-4-13		JP Research Summary: JASIC Flex Injury Estimate
GTR9-4-14		Comparison of FlexPLI Performance in Vehicle Tests with Prototype and Series Production Legforms
GTR9-4-15		Informal document WP.29-157-21: Second progress report of the informal group on Phase 2 of gtr No. 9 (IG GTR9 - PH2)
GTR9-4-16	1	Pedestrian Lower Extremity Injury Risk - Revision 1
GTR9-4-17		FlexPLI Round Robin Car Test Schedule
GTR9-4-18		FlexPLI vs. EEVC LFI Benefit Estimation
GTR9-4-19		Overview of NHTSA Pedestrian Activities
GTR9-4-20		Validation of Pedestrian Lower Limb Injury Assessment using Subsystem Impactors (IRCOBI conference, 12th – 14th Sept. 2012)
GTR9-4-21		OSRP Pedestrian Lower Leg Response Research test series
GTR9-4-22		Checklist for Vehicle Testing
GTR9-5-01	1	Agenda for the 4th meeting of the Informal Group on Global Technical Regulation No. 9 - Phase 2 (IG GTR9-PH2) - Final
GTR9-5-02	1	Minutes of the 5th meeting of the Informal Group on Global Technical Regulation No. 9 – Phase 2 (IG GTR9-PH2) - Final
GTR9-5-03		Pedestrian Injuries By Source: Serious and Disabling Injuries in US and European Cases (Mallory et al. Paper for 56th AAAM Annual Conference)
GTR9-5-04		Flex PLI GTR User Manual Rev. D, Oct. 2012
GTR9-5-05	2	FlexPLI - Round Robin Tests
GTR9-5-06		Informal document WP29-158-28: Draft 3rd progress report
GTR9-5-07	c2	Discussion on Feasibility of FlexPLI Countermeasures
GTR9-5-08		Proposal for Procedure to Process FlexPLI Measurements in Rebound Phase
GTR9-5-09		Applicability Information
GTR9-5-10		FlexPLI Durability Against Larger Vehicles
GTR9-5-11		FlexPLI Repeatability in Car Tests
GTR9-5-12		Experimental Validation of Human and FlexPLI FE Models

GTR9-5-13		FlexPLI vs. EEVC LFI Correlation
GTR9-5-14		Benefit and Cost; Additional Analysis based on GTR9-2-07r1
GTR9-5-15		Moving Ram Friction Effect
GTR9-5-16	1	Round Robin Test Result (E-Leg)
GTR9-5-17		FlexPLI Test Results (SN-03)
GTR9-5-18		Flex PLI Logbook for the IG GTR9-PH2 Round Robin Tests
GTR9-5-19		Estimation of Cost Reduction due to Introduction of FlexPLI within GTR9
GTR9-5-20		Verification of Draft FlexPLI prototype impactor limits and application to FlexPLI serial production level
GTR9-5-21		US Round Robin Test Status
GTR9-5-22		Information on vehicle data used in NHTSA's studies
GTR9-5-23	c	Initial comments of OICA representatives to the 5th IG GTR9-PH2 meeting in response to document GTR9-5-20
GTR9-5-24		Height tolerance for pedestrian protection
GTR9-5-25		Flex PLI Inverse Test Setup - Moving Ram Friction
GTR9-5-26		Investigation of the Influences of Friction within the Inverse Certification Test Setup of the FlexPLI - Lower Legform Impactor
GTR9-5-27		Clarification of Injury Threshold Determination Process Used by JAMA
GTR9-5-28		Operating Principles and Terms of Reference for the IG GTR9-PH2, updated version 5th meeting
GTR9-5-29		Draft gtr No 9 amendment, version 2012-12-06
GTR9-5-30		Discussion of the Rebound Issue, ACEA comments
GTR9-5-31	1	FlexPLI version GTR drawing package
GTR9-6-01	1	Agenda for the 6th meeting of the Informal Group on Global Technical Regulation No. 9 – Phase 2 (IG GTR9-PH2) - Final
GTR9-6-02	2	Minutes of the 6th meeting of the Informal Group on Global Technical Regulation No. 9 – Phase 2 (IG GTR9-PH2) - Final
GTR9-6-03		FlexPLI Testing: Propelling Accuracy
GTR9-6-04		Guidelines for the development of drawings for a test tool to be added as an Addendum to UN Mutual Resolution No. 1 (M.R.1) - (ECE/TRANS/WP.29/1101)
GTR9-6-05		Schedule to prepare an Addendum for FlexPLI for the M.R.1
GTR9-6-06		FlexPLI GTR User Manual Rev. E 2013
GTR9-6-07		Definition of FlexPLI Biofidelic Assessment Interval
GTR9-6-08	1	Derivation of FlexPLI thresholds
GTR9-6-09		FlexPLI Drawings

GTR9-6-10		FlexPLI Pre- & Post-Test Procedure
GTR9-6-11		Consideration of the Rebound Phase
GTR9-6-12		Validation of Flex-GTR model
GTR9-6-13		Proposal for a wording to consider tolerances of the normal ride height
GTR9-6-14	1	FlexPLI Round Robin Testing
GTR9-6-15	1	Summary JPR Report Evaluating the Methodology and Assumptions Made in Doc. GTR9-5-14 and GTR9-5-19
GTR9-6-16		JPR Report Evaluating the Methodology and Assumptions Made in Doc. GTR9-5-14 and GTR9-5-19
GTR9-6-17		Large Truck/SUV Challenges
GTR9-6-18		FlexPLI Round Robin Test Results
GTR9-6-19	1	FlexPLI Round Robin Test Results
GTR9-6-20		Discussion on Impactor Thresholds
GTR9-6-21		Flex-PLI Rebound Issue: Industry Proposal (Update)
GTR9-6-22		FlexPLI Drawing Review (Surface Level)
GTR9-6-23	2	FlexPLI Drawings Review
GTR9-6-24		Durability Study SN-03
GTR9-6-25		Comments on GTR9-6-15 (JP Research review of JASIC & BASt FlexPLI Injury Reduction Estimate)
GTR9-6-26		Development of Injury Probability Functions for the Flexible Pedestrian Legform Impactor
GTR9-6-27		Comments on Alliance and JP Research Documents (GTR9-6-15 and GTR9-6-16)
GTR9-6-28		Certification test results of the OEM legform used in document GTR9-6-20
GTR9-7-01	1	Agenda for the 7th meeting of the Informal Group on Global Technical Regulation No. 9 – Phase 2 (IG GTR9-PH2) - Final
GTR9-7-02	1	Minutes of the 7th meeting of the Informal Group on Global Technical Regulation No. 9 – Phase 2 (IG GTR9-PH2) - Final
GTR9-7-03		Draft running order of the provisional agenda
GTR9-7-04		Information on drawing package kindly provided by Humanetics
GTR9-7-05	c	Result of drawing review (surface level)
GTR9-7-06	c	Result of manual review
GTR9-7-07		Injury Probability Function for Tibia Fracture and MCL Failure
GTR9-7-08		Development of Flex-GTR Master Leg FE Model and Evaluation of Validity of Current Threshold Values
GTR9-7-09		Flex-GTR Master Leg Level Impactor Test Data - Pendulum Test

GTR9-7-10		FlexPLI Logbook - legform SN-01
GTR9-7-11		FlexPLI Logbook - legform SN-03
GTR9-7-12		FlexPLI Logbook - legform E-Leg
GTR9-7-13		FlexPLI Rebound Phase
GTR9-7-14		Detailed Review of Drawing Package and Itemized Check against Master Leg Impactor SN03
GTR9-7-15		BASt comments on GTR9-7-13: JASIC position on FlexPLI rebound phase
GTR9-7-16	1	Collation of FlexPLI Pendulum Certification Test Results
GTR9-7-17	1	Collation of FlexPLI Inverse Certification Test Results
GTR9-8-01	1	Agenda for the 8th meeting of the Informal Group on Global Technical Regulation No. 9 – Phase 2 (IG GTR9-PH2) - Final
GTR9-8-02	1	Minutes of the 8th meeting of the Informal Group on Global Technical Regulation No. 9 – Phase 2 (IG GTR9-PH2) - Final
GTR9-8-03	1	GTR9 – Draft Working Document of IG GTR9 PH2; Version 1, 04 Sept. 2013
GTR9-8-04		GTR9 – Draft Preamble of IG GTR9 PH2; Version 1, 04 Sept. 2014
GTR9-8-05		(not used)
GTR9-8-06		OICA comments on the draft text, based on document GRSP-53-29
GTR9-8-07		Proposed Title Block for Regulation Drawings
GTR9-8-08	1	Comments to JAMA presentation GTR9-7-06c and Proposed Changes, FlexPLI GTR Manual
GTR9-8-09		Comments to Cellbond Flex PLI Drawing Check Document GTR9-6-23
GTR9-8-10		Comments to JASIC's Comments Provided with Document GTR9-7-05c
GTR9-8-11		Comparison of Effect of Different Approaches on Injury Risk Functions
GTR9-8-12		Possible Influence of Temperature and Humidity on the FlexPLI Behavior
GTR9-8-13		FlexPLI Manual: FlexPLI Preparation before Car Testing
GTR9-8-14		Request for Transitional Provisions for FlexPLI Usage
GTR9-8-15		FlexPLI Biofidelic Assessment Interval (BAI): Open Issues
GTR9-8-16		Change to foam flesh used by EEVC lower & upper legforms
GTR9-8-17		FlexPLI Version GTR - Testing of Vehicles with Different Bumper Systems
GTR9-8-18		Femur Certification Corridors for the Inverse Test (Zero Cross Timing)
GTR9-8-19		Femur Certification Corridors for the Pendulum Test (Zero Cross Timing)
GTR9-9-01	1	Agenda for the 8th meeting of the Informal Group on Global Technical Regulation No. 9 – Phase 2 (IG GTR9-PH2) - Final

GTR9-9-02	1	Minutes of the 9th meeting of the Informal Group on Global Technical Regulation No. 9 – Phase 2 (IG GTR9-PH2) - Final
GTR9-9-03		FlexPLI weight tolerances, Reduction of proposed weight tolerances
GTR9-9-04	1	Lower Legform Test Area, Justification of the Need for a Relaxation Zone
GTR9-9-05		Proposal of 01 series of amendments to Regulation No. 127: Transitional provisions
GTR9-9-06		Proposed amendments of the three-point bending certification test figure
GTR9-9-07	3	FlexPLI GTR User Manual Rev. F 2013
GTR9-9-08	1	Updates to Flex PLI Manual Rev E to Rev F
GTR9-9-09		Max and Min Femur Certification Analysis FlexPLI
GTR9-9-10		FlexPLI Weight Tolerance Review
GTR9-9-11	4	Reviewed FlexPLI version GTR drawing package
GTR9-9-12		Details of Drawing Updates to Flex GTR9 Regulation Drawings
GTR9-9-13		Flesh Neoprene Corridor FlexPLI
GTR9-9-14		Dimensional Tolerance Review FlexPLI
GTR9-9-15		Confor Foam Change [for EEVC Impactors]
GTR9-9-16		Flex PLI Drawing Review
GTR9-9-17		Legform Tests, Results from Round 2, FlexPLI
GTR9-10-01	1	Agenda for the 10th meeting of the Informal Group on Global Technical Regulation No. 9 – Phase 2 (IG GTR9-PH2) – Final
GTR9-10-02		Minutes of the 10th meeting of the Informal Group on Global Technical Regulation No. 9 – Phase 2 – Draft
GTR9-10-03		Consolidated version of GTR9 incl. all amendments before 10th meeting (for reference only)
GTR9-10-04	c1	JASIC review of document DRAFT_GRSP-2014-15-Rev1e
GTR9-10-05		FlexPLI Drawing and User Manual Status
GTR9-10-06		Comments by TRL - Correction List
GTR9-10-07		Comments by TRL – Request for more details
DRAFT_GRSP/2014/15 Rev.1		Draft for amendment to document UNECE/WP.29/GRSP/2014/15
DRAFT_GRSP/2014/16 Rev.1		Draft for amendment to document UNECE/WP.29/GRSP/2014/16

C. Phase 3

228. Paragraphs 228 to 304 reflect the development of Phase 3 of UN GTR No. 9 and relate to the development of test provisions for vehicles equipped with Deployable Pedestrian Protection Systems (DPPS), including

prerequisites, without changing the headform impactors and their corresponding parameters for tests to the bonnet top.

1. Introduction and General Background

229. During the fifty-sixth session of GRSP (9–12 December 2014) the expert from the Republic of Korea proposed to develop test provisions for active devices to further improve vehicle safety performance. It was noted that guidelines already existed for testing active bonnets (INF GR/PS/141 Rev. 1), however these were considered to be insufficient, and consent was sought from WP.29 and AC.3 to extend the IWG mandate on the development of phase 2 to UN GTR No. 9.

230. The proposal from the Republic of Korea to develop an amendment to UN GTR No. 9 on the testing of provisions for deployable systems of the outer surface to ensure an adequate protection of pedestrians was endorsed by AC.3 at its forty-eighth session (17 November 2016) and the IWG mandate on the development of Phase 2 was extended until December 2017.

231. The development of the test provisions for deployable systems was initiated by a Task Force under the umbrella of the IWG on phase 2 (TF-DPPS). After four meetings of TF-DPPS, the IWG mandate expired. Subsequently, AC.3 endorsed at its fifty-second session (14 March 2018) the transformation of TF-DPPS into a new Informal Working Group (IWG-DPPS).

232. TF-DPPS held the following meetings:

(a) 27–28 February 2017: Paris, France;

(b) 28–29 March 2017: Paris France;

(c) 7 September 2017: virtual;

(d) 21–23 November 2017: Berlin, Germany.

233. IWG-DPPS held the following meetings:

(a) 18–20 April 2018: Frankfurt/Main, Germany;

(b) 5–7 September 2018: Brussels, Belgium;

(c) 10 December 2018: Geneva, Switzerland;

(d) 12–14 March 2019: Paris, France;

(e) 3–4 September 2019: London, United Kingdom;

(f) 28 November 2019: virtual;

(g) 4–5 March 2020: virtual;

(h) 15–17 September 2020: virtual;

(i) 18 November 2020: virtual;

(j) 20–21 January 2021: virtual;

(k) 9–10 March 2021: virtual;

(l) 27–28 April 2021: virtual;

(m) 29–30 June 2021: virtual;

(n) 14–15 September 2021: virtual;

(o) 16–17 November 2021: virtual;

(p) 9–10 February 2022: virtual;

(q) 5–6 April 2022: virtual;

(r) 2–3 June 2022: hybrid;

(s) 18–20 October 2022: Paris, France: hybrid;

(t) 8–9 November 2022: virtual;

(u) 15–16 November 2022: virtual;

(v) 31 January–2 February 2023: Brussels, Belgium: hybrid;

(w) 12-13 April 2023: virtual;

(x) 24 April 2023: virtual;

(y) 19 June 2023: virtual;

(z) 29, 31 August 2023: virtual;

(z1) 8, 9 November 2023: virtual.

234. The meetings were attended by representatives of: Austria, European Commission, France, Germany, Italy, Japan, Republic of Korea, the Netherlands, Spain, United Kingdom of Great Britain and Northern Ireland, United States of America, European Association of Automotive Suppliers (CLEPA) and OICA.

235. The meetings were chaired by the expert of the Republic of Korea, while the secretariat has been provided by the experts of OICA since February 2017 (TF-DPPS1).

236. ECE/TRANS/WP.29/GRSP/2023/6 was proposed at the seventy-third session of GRSP as a complete draft of Amendment 3 to the UN GTR No. 9, including Part 1 and Part 2 and supersedes the other drafts previously submitted.

2. Principle of Deployable Pedestrian Protection Systems

237. DPPS should be activated as intended for pedestrian protection when the pedestrian is hit by a vehicle. To achieve this goal, the IWG agreed that requirements were needed to ensure that:

(a) The collision with the pedestrian is detected; and

(b) The existing headform requirements in the GTR are met for a 35km/h head impact velocity as well as for vehicle speeds below the deployment threshold of the DPPS.

Only contact sensors are taken into consideration for detection on current DPPS. It was discussed that in a possible "DPPS phase 2", non-contact sensors could be explored and comprehended.

238. This Regulation aims to improve protection from injury caused by a collision between the vehicle front and the pedestrian. To assure that a DPPS operates properly and offers at least the same level of pedestrian protection as a conventional passive system, the IWG agreed that the system provisions listed in paragraph 237 are a minimum requirement. Additionally, the IWG discussed the need for two other system requirements:

(a) Higher speeds - Assurance that a DPPS system will deploy safely at pedestrian impact speeds above 40 km/h;

(b) Body loading - Assurance that pedestrian body loading of a DPPS will not compromise its effectiveness prior to head impact.

These needs may exist for DPPS systems in particular, as opposed to conventional passive systems. Members of the IWG expressed their concerns that at higher speeds, actuator limitations may prevent the timely deployment of a DPPS, while the negative effects of body loading may be exacerbated by a deployed system without sufficient support. Some IWG members found that a reasonable bonnet clearance at the location and prior to the head impact is needed to prevent a hard head contact due to a collapsing bonnet.

239. At this time, the IWG agreed that a regulatory need is not known with enough certainty to warrant the development of test procedures and requirements related to higher impact speeds and body loading. In other words, current DPPS systems that meet the requirements listed in paragraph 237 may also account for higher impact speeds and body loading. However, further research or the development of future DPPS may result in insights for which the effect of pedestrian body loading and protection at higher speeds may be required.

Additionally, future accidentology may reveal a prominent safety need exists in current DPPS systems due to body loading and impacts at higher speeds. In either case, the UN GTR will be reviewed and adapted if and where necessary.

240. At the request of the United States of America, the IWG decided that, based on a determination by each Contracting Party or regional economic integration organization, the fulfilment of all provisions may be required to be demonstrated using the dynamic test option of Annex 1.

241. Since the expressions "static test" and "dynamic test" are not uniformly used within different Regulations and their test procedures, the IWG DPPS discussed about how to define static and dynamic tests for the purpose of testing DPPS. In this context, the IWG agreed upon defining a "static test" as a headform impact on the DPPS in the deployed position. A "dynamic test", on the other hand, is defined as a synchronised headform impact on the DPPS during its deployment.

3. Detection test area, lateral offset leg versus head

242. As one of the fundamental prerequisites to account for the potential safety benefits of DPPS, the pedestrian needs to be detected during an accident prior to head impact on the vehicle. IWG discussed the required width of the area on the vehicle front where a pedestrian needs to be detected in order to purposefully initiate the system.

243. An earlier Task Force study was recalled by the expert from Germany in which it was shown that pedestrian impacts take place over the entire vehicle width (TF-BTA-6-07). In a later IWG meeting, Laboratoire d'Accidentologie et Biomécanique (LAB) presented an analysis of fatal French accidents contained within the Etudes Détaillées d'Accidents (EDA) database (IWG-DPPS-18-08). The LAB analysis revealed that for all cases in which a pedestrian was struck outside the longitudinal frame rails of the vehicle, accounting for approximately 15–20 per cent of the vehicle width, there were no subsequent head impacts to the bonnet (though about one-third of the cases did result in pelvis impacts to the bonnet). Thus, in principle and ideally, a detection of pedestrians in nowadays DPPS should be required accordingly.

244. Also, it was agreed that in many cases the pedestrian may tend to spin off at the outer widths of typically angled or V-shaped vehicle front-end surfaces, without a head-to-bonnet impact. This effect is even more present when using a leg impactor as pedestrian surrogate without attaching any mass of a pedestrian hip, torso, arms, neck, and head, consequently limiting the load on the sensing system and therefore not being representative for a pedestrian.

245. In the light of these observations, IWG investigated further definitions of a detection area.

246. The expert from Japan proposed that the detection area – which would differ from the leg test area – be the area in the lateral direction of the vehicle in which activation of the DPPS is ensured in a vehicle-to-pedestrian impact (Task Force Document DPPS-3-03). The reason given was that: only in this area a head impact test would be allowed with activated DPPS, while outside this area the DPPS was supposed to remain deactivated.

247. The expert from Germany suggested to use the bumper test area (BTA) as already defined for the lower extremity injury risk assessment based on the tests with the lower legform impactor. Since the FlexPLI was also chosen as verification impactor and the BTA is well elaborated and established, the expert from Germany reasoned that existing definitions could be applied.

248. The expert from European Automobile Manufacturers' Association (ACEA) also referred to current regulatory definitions and proposed to apply the lower leg test area defined in Amendment 2 to UN GTR No. 9 as required confirmation of sensing capabilities for DPPS homologation or self-certification (IWG-DPPS-1-08).

249. The expert from OICA suggested to define the outer boundaries of the detection area by the width between the corner reference points (CRPs) (CRPs, the intersections of the side reference lines and the bonnet leading edge reference line), projected to the upper bumper reference line (IWG-DPPS-4-05). It was noted by IWG participants that when a vehicle has multiple or continuous intersections between the Bonnet Leading Edge Reference Line

(BLERL) and the Side Reference Line (SRL), the most outboard point is used as the CRP. It was also noted that the distance between right and left CRPs can be narrowed easily by a minor, cosmetic redesign of the vehicle front end. Such a redesign would have no effect on the legform test zone but could lead to large differences in CRP locations and thus greatly affect the DPPS detection test area. Therefore, the IWG abstained from further discussions on the use of the CRP in defining the detection test area.

250. An independent expert proposed a required percentage of the vehicle width (around the longitudinal vertical centreplane as its centre) as detection test area, with a subtraction of no more than 12.5 per cent of the vehicle width but a maximum of 250mm at each side of the vehicle. The independent expert's proposal also stated that the detection test area should be no less than the BTA (IWG-DPPS-5-09). It was explained that with a percentage all vehicles would be equally treated, regardless their effective width: however, big cars should not be allowed to further reduce the detection area, beyond 250mm on each side.

251. The expert from Germany subsequently provided an update to the independent expert's proposal wherein the vehicle width was defined as the width at the cross-section of the front axle, without rear view mirrors or rear-view mirror substitute systems, so that the proposed detection test area was not linked to the width of the deployed area of the DPPS (IWG-DPPS-7-10). Examples of four current vehicle models were displayed to show how the detection test area based on the 12.5 per cent stipulation was greater than the BTA.

252. Japan investigated the outer most boundary of the detection area proposed by Germany and confirmed that it covers the headform test area for vehicles equipped with DPPS currently available on the market in Japan. Thus, Japan accepted the detection area proposed by Germany. However, it was suggested that in cases where the sensing width is narrower than the width of the detection area, the DPPS would be allowed to only be activated within the sensing width (IWG-DPPS-9-09). The rationale was that the lateral offset between the lower extremity impact and the pedestrian head impact after wrap around would be considered as small.

253. The group examined indications regarding the possible lateral offset between lower extremities and head in pedestrian accidents with passenger cars.

254. The expert from Japan presented Post-Mortem-Human-Subject (PMHS) tests, Human Body Models (HBM) finite element simulations and dummy tests where the lateral movement of the pedestrian's head until the head impact on the vehicle front was small. It was concluded that the impact locations of the head and the leg would not differ much (IWG-DPPS-10-04).

255. The expert from Germany examined some cases from the German In-Depth Accident Study (GIDAS) database for real world trajectories of pedestrians. The sample showed in several cases a significant lateral offset between the first leg impact and the subsequent pedestrian head impact. They concluded that laboratory test conditions with stationary test specimen do not always reflect real world impact conditions in an appropriate way (IWG-DPPS-10-09).

256. The expert from Japan proposed that the pedestrian accidents scenario assumed under the current UN GTR No. 9 is the case when the vehicle impacts the stationary pedestrian from the side, and the pedestrian accidents scenario assumed in the test for DPPS should be the same as the current UN GTR No. 9. Japan found that, for consistency reasons, a consideration of the pedestrian kinematics with significant lateral offset between the pedestrian's leg impact and the subsequent head impact would require such a consideration with modified impact angles also during component tests. However, Japan also showed that this would not be in the scope of the IWG and beyond the minimum requirements as specified in the GTR (IWG-DPPS-11-03).

257. The expert from Germany clarified the objective of the IWG, which was not limited to clarification of the current practice, but also to develop new and more detailed requirements, where needed, to ensure a correct activation and design for vulnerable road user protection. Since the detection of pedestrians is one of the indispensable prerequisites and DPPS needs to be correctly activated, real world conditions under consideration of pedestrian trajectories with a considerable offset between leg and head impact need to be taken into account to provide for at least the same level of protection as conventional systems without DPPS (IWG-DPPS-11-05).

258. The expert from Japan presented a literature review of real-world accident data and concluded that a walking pedestrian hit laterally by a vehicle would be a representative accident scenario (IWG-DPPS-12-07) which is reflected by the current UN GTR No. 9 test procedures. The expert from Germany found that also a large number of oblique impacts were included in the share of given lateral impacts which need to be taken into consideration with respect to the leg versus head offset. It was added that UN GTR No. 9 would not only cover lateral but also oblique impacts, since the outer skin of the vehicle front would be, in most cases, not parallel to the moving trajectory of the crossing pedestrian and thus not perpendicular to the velocity vector of the impactor during the impact. Regarding the pedestrian accidents scenario assumed for DPPS, other contracting parties supported Germany's proposal, but Japan did not accept it. However, because the detection area proposed by Germany covers the headform test area for vehicles equipped with DPPS currently available on the market in Japan, Japan accepted the detection area proposed by Germany regardless the difference of assumption for the pedestrian accidents' scenario for the tests.

259. The expert from German Association of the Automotive Industry (VDA) explained possible shortcomings of the BTA definition when applied to the DPPS detection test area. For the lower leg injury assessment, the BTA is defined by the greatest of the following areas: (a) the area limited by the corners of bumper, moving on either side 42mm inboard; (b) the outermost ends of the bumper beam, moving on either side 42mm inboard. The expert from VDA took exception to the use of the bumper beam in defining the detection test area for DPPS applications (IWG-DPPS-14-04). They presented conditions that exist on two production vehicles, in which structures are appended to the bumper beam, but only for certain markets, in order to fulfil corresponding crash test requirements. These structures have the effect of extending the BTA. Hence, if the structures were used to stipulate the DPPS detection test area, there would exist different detection test areas for different markets.

260. Further discussion on bumper beam structures ensued. The expert from OICA described the structures as "optional" and insufficient to serve as a pressure tube backstop. Additionally, they extend outboard into an area in which the fascia covering is curved (outboard to the corners of bumper as defined by the 30° gauge). These two factors preclude the ability to install a sensing tube that could generate enough signal to trigger a DPPS actuator as described in a previous VDA analysis (IWG-DPPS-12-08).

261. A working subgroup of the IWG analysed current examples of DPPS on the market to guide a decision on how to proceed with a suitable definition for the detection test area. This survey included twelve production vehicles with different sizes and body styles. For each vehicle, the following widths were noted: the car manufacturer-reported width of sensing, a possible detection test area determined by the 12.5 per cent/250mm stipulation, and a possible detection test area determined via the lower leg Bumper Test Area (BTA) criteria: the 30° gauge and the bumper beam (IWG-DPPS-18-07).

262. The survey revealed that the width of sensing can also extend outboard of the detection test area when defined by the relevant 30° corner gauge contact points and into an area where a glancing blow will occur. In the vehicle survey, the 12.5 per cent-based width of the detection test area was wider than the corner gauge-based "geometry" in most of the vehicles surveyed. This shows that – at least to a certain extent – it is feasible to overcome the "spin off/low signal" issue brought up in IWG-DPPS-12-08.

263. The vehicle survey showed one instance where the corner gauge-based detection test area was greater than the 12.5 per cent-based detection test area. In this case, the reported width of sensing was even greater. This shows that it is feasible to enforce the corner gauge-based detection test area when it is wider than the 12.5 per cent-based width.

264. The vehicle survey also showed that some of the vehicles had reported widths of sensing that would not have met the width requirement of the detection test area as determined by the 12.5 per cent stipulation or the corner gauge. This means that with phase 3 of UN GRT No. 9, new vehicles will have a greater width of sensing relative to many vehicles not fulfilling this requirement.

265. Based on the discussions, the bumper beam has been excluded from the stipulation for the DPPS detection test area. Furthermore, the exclusion is consistent with a performance-based standard. If it was included, it would partly act to prescribe the sensing tube technology and the form of the bumper beam itself. Originally, the bumper beam was considered because sensing technology that uses a pressure tube typically operates by using the beam as a hard surface to "back up" the tube. It was reasoned that if the beam is of a certain length, it is feasible to require the tube (and the sensing area) to be the same length: in three vehicles of the survey, the width of the bumper beam underlying the fascia exceeded the 75 per cent stipulation. However, this misleadingly assumes that the beam will always be made of a rigid, tubular structure and that pressure tube technology is used. In fact, the survey showed that accelerometers were used in four of the vehicles. A Regulation should not prescribe a particular technology or stand in the way of new technologies, such as different sensing technologies or bumper beams that take on different materials, shapes and functions.

266. The IWG finally agreed upon the minimum width of the detection test area being the vehicle width minus 12.5 per cent (but not more than 250mm) on each side but extending at least up to the points 42 mm inboard of each corner of bumper.

4. Test procedures for the sensing systems of DPPS and selection of the verification impactor

267. For verification of the functionality of the DPPS sensing system, component tests will be performed with the FlexPLI, representing the lower extremities of a 50th percentile male for injury assessment of knee and tibia injuries. The use of the FlexPLI as sensing impactor was agreed following extensive investigations.

268. Contact biofidelity was considered to be an indispensable property of such a sensing impactor. The IWG-DPPS found that, when verifying the ability of a contact sensor to detect a pedestrian, the relevant properties of an impactor are the total mass, mass distribution, moments of inertia, centre of gravity, impactor width, bending stiffness and the local stiffness / compression behaviour in impact direction were highly relevant properties of an impactor for the signals in use with contact sensors. While most properties of the FlexPLI were accepted to be very reliable due to its design specifications, two complementary studies were carried out to ensure its biofidelic and repeatable local stiffness.

269. The first study, carried out by Concept Tech, investigated time histories of different pedestrian surrogates and HBMs for identical load cases. It concluded the FlexPLI had, in principle, an appropriate contact biofidelity to work as a representative pedestrian surrogate for sensing issues (IWG-DPPS-3-03).

270. The second study, carried out by The Federal Highway Research Institute (BASt) and an independent expert in cooperation with ACEA members, focused on the intrusion during inverse tests at impact speeds typical for the lower deployment threshold of DPPS within the typical time interval for detection of pedestrians. Here, two different setups were used, covering the height dimensions as required by Research Council for Automobile Repairs and UN Regulation No. 42 (Front and rear protection devices) which need to be fulfilled by a high number of vehicles. It could be shown that the double integral of the filtered impactor acceleration signal, representing the intrusion, was within a small range with satisfactory coefficients of variation (IWG-DPPS-6-04, IWG-DPPS-7-09 and IWG-DPPS-9-11).

271. The IWG-DPPS concluded that the FlexPLI was currently the best available pedestrian surrogate which could be used as an impactor for the sensing verification of the system for the time being.

272. The IWG emphasized that, due to the complexity of testing the DPPS, the test provisions laid down represent a limited range of typical load cases. It is therefore seen as due care of the vehicle manufacturer that any DPPS would ensure the necessary protection (e.g. for a variation of speeds and pedestrian statures) in order to act as intended in the event of a collision with a pedestrian for a variety of pedestrian statures.

5. Determination of Head Impact Time and wrap around distance

273. The pedestrian Head Impact Time (HIT) is defined as the elapsed time subsequent to the time of first contact of the Pedestrian surrogate (neglecting forearms and hands) with the vehicle outer surface and the time of first contact of its head with the vehicle outer surface. IWG decided to use two kinds of HIT determination depending on the application (static or dynamic test) as a compromise of technical feasibility (i.e. for pedestrian airbags) and the consideration of worst-case scenarios:

(a) "HIT_for decision" (HIT_d) represents the worst case and is determined with the deployed DPPS. The HIT_d is compared to the total response time and used to decide whether a test on the DPPS is to be done on a deploying (dynamic test) or on a statically deployed system (static test).

(b) HIT_s (s for synchronisation) is determined with the undeployed DPPS, as for airbags a deployed position is difficult to define. HIT_s is used to synchronise the test trigger time of a dynamic test in the case of a deploying system.

274. The IWG discussed three methods of determining HIT:

(a) Use of HBM simulations;

(b) Use test dummies and physical testing;

(c) Use of a "generic" approach.

275. IWG ultimately agreed to propose a procedure using HBM simulations based upon a procedure on Euro New Car Assessment Programme No. TB024, as an initial DPPS amendment.

276. For DPPS to work as intended, it is necessary that the system in question is activated in due time.

277. The HIT_d of pedestrians of the relevant statures needs to be compared with the total response time (TRT) of the DPPS.

278. This comparison provides the basis for whether headform tests to the vehicle front are performed with the DPPS either statically in undeployed or in deployed position, or dynamically onto a deploying system.

279. The IWG DPPS understood that HBM simulations were the common method for determination of the HIT. In order to ensure comparability and applicability of HBM for that purpose, a qualification procedure for HBMs was developed within a subgroup of the IWG DPPS, led by the experts of Austria and OICA. All Generic Vehicle (GV) Models used for the qualification are made available as Addendum 5 of Mutual Resolution No. 1 (M.R.1) of the 1958 and the 1998 Agreements ECE/TRANS/WP.29/1101 by Austria.

280. Given its limitations, the IWG recognized the qualification procedure being applicable for the determination of HIT and Wrap Around Distance (WAD) only.
The simulation procedure described is limited to HBM qualification for the determination of HIT and related WAD and not suited to qualify for injury assessment in any pedestrian or other crashworthiness regulations.

281. A subgroup of the IWG on DPPS investigated the corridors required for the qualification of HBMs for HIT and WAD determination. It was unanimously agreed that realistic HIT and WAD strongly depend on biofidelic whole-body kinematics of the HBM. However, a required location of the Acetabulum (AC) at the time of head impact, as e.g. described by means of a corridor, is not seen as an appropriate kinematic criterion. It was therefore, for the time being, abstained from introducing AC corridors for the HBMs at HIT; however, the values will be recorded for monitoring purposes and evaluated during a possible next phase of GTR9 towards the introduction of significant, meaningful kinematic criteria for biofidelic and reliable HBM trajectories.

282. In order to create an independent baseline, reference simulations have been used to determine requirements and tolerances described in Addendum 5 of M.R.1 "Specifications for the Qualification of Human Body Models for Pedestrian HIT Determination for DPPS (DPPS HBMs)". The HBMs that were used for these reference simulations have been validated by comparing their simulation responses (HIT, kinematics) with PMHS tests. The

background concerning the validation of reference HBMs are available in Annex B of Addendum 5 of M.R.1 of the 1958 and the 1998 Agreements ECE/TRANS/WP.29/1101.

283. For the qualification of HBMs, GV Models are used. These GV Models are available on the ECE website and described in M.R.1 Addendum 5. The GV Models are only used to run simulations with the individual HBM on the specific computational environment and compare the results with reference corridors. This comparison is needed to ensure similar head kinematics as the reference HBMs. Therefore, the individual HBM qualifies as tool for HIT determination for the DPPS assessment. The GV Models were developed as very simplified impact structures which show consistent behaviour when applied in the different Finite Element (FE) software packages, are robust and approximate impact conditions within the range full vehicles, while not representing any specific car with all details. The GV Models are developed to simulate the kinematics of HBMs up to the time of head contact. The information on the development of the GV Models is documented in IWG-DPPS-25-05.

284. A plausibility check of the GV Models using a rigid impactor is described in Annex C of Addendum 5 of M.R.1 of the 1958 and the 1998 Agreements ECE/TRANS/WP.29/1101. In the GV Models, a hard stop is implemented as contact between the outer and inner layer to avoid instabilities of the foam and increase the robustness and comparability between codes of the GV Models. The foam material’s stiffness is exponentially increasing after ~80 per cent compression to additionally avoid negative volumes within the foam in case of high local deformation.

The intrusion with HBMs on the GV Models may be different from that with the rigid impactor because of the bigger HBM surface contacting the GV Models. Therefore, it is unknown, if bottoming out with HBM on the GV Model occurs frequently and has an influence on either HIT or head position result for reference HBMs or a new HBM.

285. The IWG discussed the different levels of abstraction when assessing a DPPS system. The real vehicle to pedestrian accident (first level) is replicated with a physical headform test against the vehicle front (second level). For the determination of the test conditions and the state of the DPPS during the headform test (fully deployed, deploying or undeployed), the HIT is determined during HBM simulations on the actual vehicle model (third level). In order to ensure the HBM working as intended, they are qualified during simulations against GV Models by comparing their HIT with those of validated reference HBM (fourth level). The GV Models used during this qualification process should approximate trajectories and HIT comparable to those in simulations against actual vehicle frontends. This verification is done by means of impactor tests vs the GV Models (fifth level). Since the aim is a comparison of simulation tools (HBM), the GV Model does not have to represent any actual vehicle model but just to produce robust and to some extent comparable (not identical) results, to ensure that a HBM is working as intended with regard to its trajectories.

286. However, the injury assessment abilities of HBMs are not validated. Therefore, and as of now, HBMs may not be used for injury assessment in any pedestrian or other crashworthiness regulation.

287. The simulation procedures with the qualified HBMs and the actual vehicle model for HIT determination are described in Annex 2 "HIT determination simulation".

288. Concerning the linear regression explanation, IWG has discussed different options to determine HIT versus WAD graphs for the decision of a static or a dynamic test mode and a system triggering for the dynamic testing. IWG considered two methods – the "linear regression method" and the "dot-to-dot method" – and recognized that each method has pros and cons. Both Japan and Korea preferred the "dot-to-dot method" because the method is mathematically more accurate, by using the simulated HIT values as agreed upon in Annex 2, and the linear regression method may have low correlation (IWG-DPPS-5bis-04 (Japan), IWG-DPPS-6-05 (Korea)). Other Contracting Parties and NGOs were in favour of the "linear regression method", because the method is practical to figure out the corresponding HIT values to WADs and may also be used to extrapolate HIT for WADs lower than six years old or higher than 95th percentile adults (IWG-DPPS-6-06 (OICA)). In this context, for reasons of

likelihood and practicability, it is assumed that the HIT always increases with increasing WAD. Finally, the IWG agreed on the "linear regression method" to figure out HIT values after extensive discussions on both methods.

289. Japan proposed to allow the use of physical test tools to predict HIT in addition to numerical tools. Performance requirements and test procedures for full scale pedestrian dummies have been specified in a published Society of Automotive Engineers (SAE) technical standard (SAE J 2782) and test results for an existing pedestrian dummy have been reported in SAE J 2868. Due to the availability issue of the SAE standards and the upcoming update of J 2868, Japan also proposed to seek a way to transpose the contents of the SAE standards into UN GTR No. 9. As a result of discussion, IWG agreed to consider developing further modifications to this amendment to incorporate the allowance of the use of physical test tools to predict HIT after the phase 1 of this activity is complete.

6. Protection at speeds below lower deployment velocity threshold

290. To protect the head of a pedestrian in the event of a collision, DPPS usually provide additional clearance between the bonnet and underlying hard structure. However, since DPPS are only activated at and above a lower deployment velocity threshold, head protection at head impact velocities equivalent to the vehicle speed below this threshold must be demonstrated in order to ensure the same level of protection as conventional passive systems. For that purpose, headform tests are performed at these impact velocities on the undeployed DPPS and their results compared to the biomechanical limits which also apply for the compliance tests.

291. Members of the IWG found wide variations of the ratio between head impact velocity and vehicle impact speed (between 0.68 and 1.5 for a car impact speed of 40km/h) in former studies. When taking into account the ratio for the legal requirements (head impact velocity of 35 km/h corresponding to a vehicle speed of 40km/h), the IWG finally decided to use a rounded ratio of 0.9 for the verification tests at lower deployment velocity threshold. Therefore, for the DPPS to demonstrate the fulfilment of this prerequisite, head impact tests are to be performed at an impact velocity which is 0.9 times the vehicle lower deployment velocity threshold.

7. Total Response Time measurement

292. Since it is critical that the DPPS is deployed before the pedestrian’s head contacts the vehicle for pedestrian protection in a pedestrian-to-vehicle collision, how to measure TRT (ST+DT) needs to be verified. The specific conditions for the verification test were reviewed. The impact speed and location were decided as equal to the conditions for defining HIT, and the FlexPLI, which is the same impactor for sensing system verification, was selected for the test.

293. There were concerns about how to conduct a headform test when the bonnet is still deploying towards its maximum height. Whereas static tests are able to save time but may differ from actual test results with moving DPPS, dynamic tests can be performed in actual conditions but may take a longer time. The IWG concluded that dynamic tests should be conducted for the case, subsequently, TRT should be measured from the time of first contact of the FlexPLI with the vehicle outer surface to the time that DPPS reaches its maximum deployment height first.

8. Headform test options

294. Depending on the degree of fulfilment of the prerequisites, the compliance tests with adult and child headform impactors are performed on the static DPPS in either the undeployed or the deployed state, or on the deploying DPPS.

295. The IWG discussed several approaches to define tolerances for the validation tests of the different height and time values specified by the manufacturer. Two main sources for tolerances were identified: a) the variation of parts due to production process (geometry, assembly, material properties, micro gas generator propellant), b) different methods to measure the Sensing Time during compliance testing. Experts of VDA explained that

Deployment Height, Deployed Position and Deployment Time strongly depend on the variation of parts and their assembly so that a great portion of the specified values are related to the specified tolerances during production processes. Therefore, a percentage of the specified values would appear to be most convenient as tolerances for Deployment Height, Deployed Position as well as Deployment Time. VDA added that the scatter of Sensing Time, on the other hand, strongly depends on the procedure of time measurement (electronical, visual, etc.) in the test lab. It was recommended to introduce tolerances on this measurement as absolute values. Therefore, a time corridor around the specified Sensing Time was introduced.

296. A protection at speeds below the lower deployment velocity threshold and the appropriate detection of a pedestrian are both indispensable requirements for vehicle approval or compliance.

297. Furthermore, only those headform tests qualify for being performed on the deployed DPPS, where during simulations with qualified HBM on the deployed DPPS, the HIT_d is proven to be greater than the TRT.

298. Where a pedestrian is not detected or any relevant HBM fails the qualification procedure, all tests are to be performed on the undeployed DPPS.

299. In the remaining cases, where the HIT_d is smaller than or equals the TRT, or when requested by the contracting party, dynamic tests are to be performed on the deploying DPPS. Synchronisation of the headform impactor and the DPPS during dynamic tests are to be derived from the generated regression line out of HIT_s as a function of WAD during simulations on the undeployed DPPS.

300. For the dynamic test option, the IWG investigated possible misinterpretations of the calculated and actual launching time of the headform impactor during dynamic tests. Since the fire delay between the initiation of the launch of the headform and the triggering of the DPPS actuators is, amongst other things, based on the HIT of the HBM on the undeployed DPPS (HIT_s), but the actual headform impact takes place on the deploying DPPS (see Figure 1 (a)), the actual launching duration will deviate from the calculated one, as depicted in Figure 1 (b).

This needs to be taken into account in the course of verification of ambient conditions for dynamic tests.

Figure 1 (a) **Transfer of real life accident situation with deploying DPPS to ambient conditions for impactor testing.**

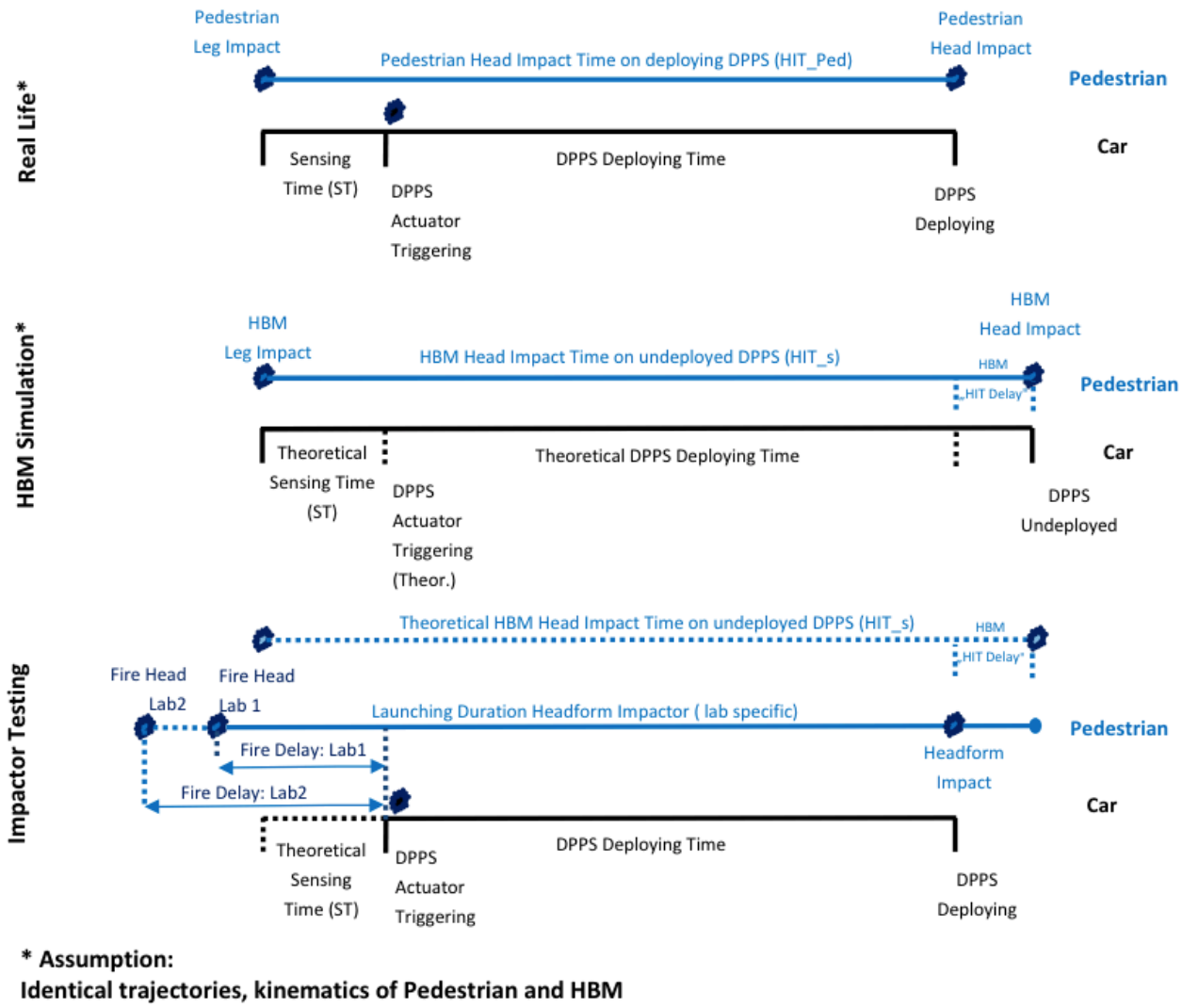
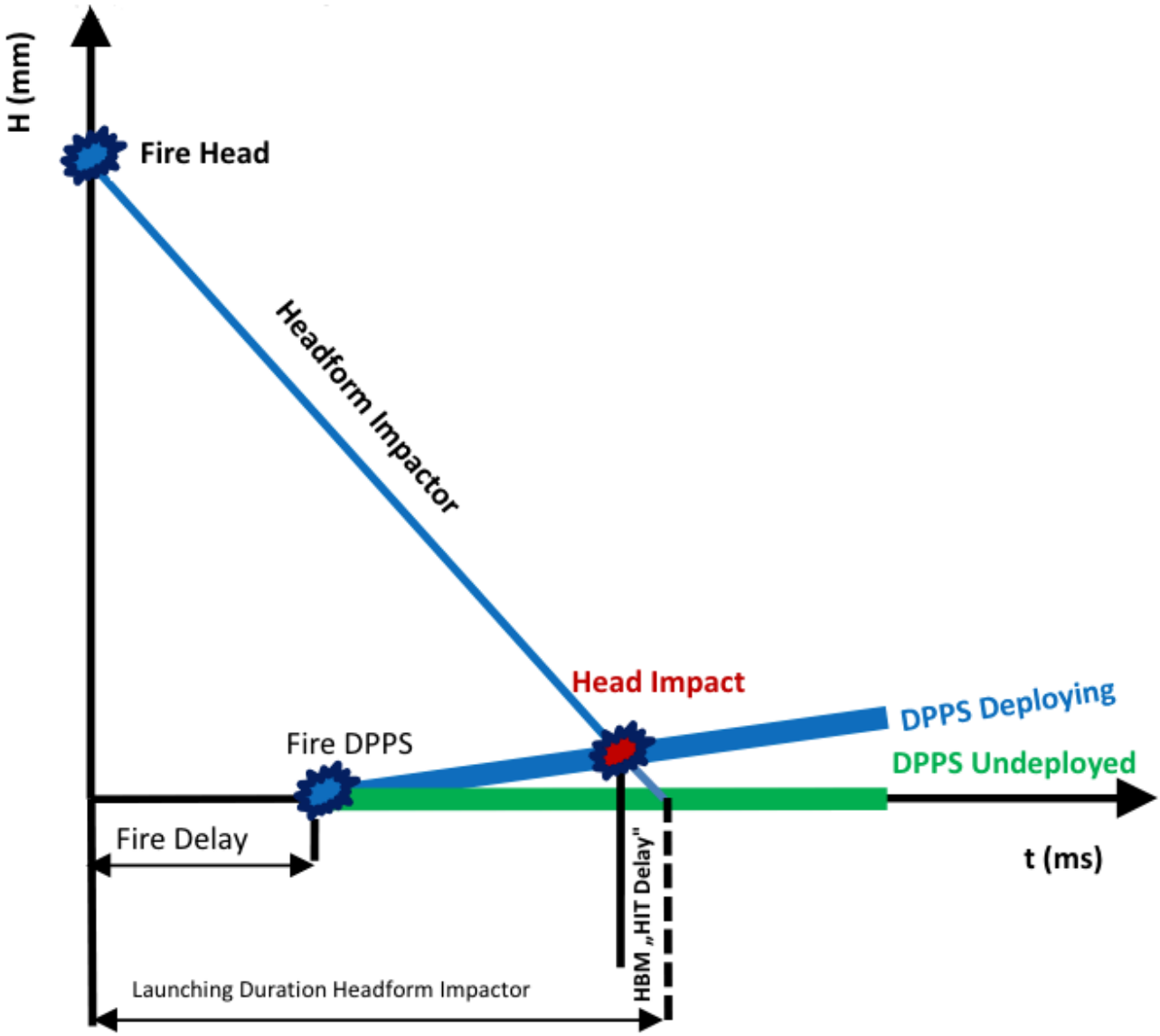


Figure 1 (b) **Effect of DPPS Deployment on Launching Duration Headform Impactor (Example).**



9. Head test area

301. Two approaches to define the head test area for vehicles fitted with DPPS were discussed, either in an undeployed position, or in a deployed position.

302. All IWG members agreed on defining the area in an undeployed position for dynamic testing of DPPS, including pedestrian airbag systems.

303. The expert from Korea proposed that the head test area should always be defined in an undeployed position for consistency. The Korean expert was concerned about an inconsistent test area depending on the test mode (static or dynamic), and also pointed out that there might be pragmatic issues to define the test area in the deployed position for a static headform test, especially, when the static test has to be partially conducted. On the other hand, manufacturers claimed that the test area defined in a deployed position makes more sense because the headform actually contacts a deployed bonnet in case of the static test.

304. After extensive discussions, the IWG decided to define the head test area in an undeployed position at all times.

10. List of documents discussed in the TF-DPPS and IWG-DPPS on UN GTR No. 9 Amendment 3

Doc. No.	*Rev.*	*Name*
DPPS-1-01		1st Meeting Agenda
DPPS-1-02	1	Minutes of the First Meeting
DPPS-1-03		Task Force Outline
DPPS-1-04		Test Procedure of Deployable Systems for Pedestrian Protection (Korea)
DPPS-1-05	1	OICA proposal based on GRSP-58-31 as revised during the meeting – 20170227
DPPS-1-06		Comments on Scope (Japan)
DPPS-1-06-Appendix		Comments on Scope – Appendix (Japan)
DPPS-1-07		Euro NCAP Pedestrian Testing Protocol v8.3 December 2016 (Germany)
DPPS-1-08		JNCAP Ped Active Device Test Protocol (unofficial) (Japan)
DPPS-1-09		Testing Deployable Bonnet Systems within Euro NCAP (Germany)
DPPS-1-10		Bonnet Deflection of Deployable Bonnet Systems (Germany)
DPPS-1-11		Industry Understanding on Rulemaking (OICA)
DPPS-1-12		Input bonnet deflection discussion (OICA)
DPPS-2-01	1	2nd meeting agenda
DPPS-2-02	1	Minutes of the 2nd meeting
DPPS-2-02-Annex		Annex to the minutes of the 2nd meeting: Attendance list
DPPS-2-03	3	Requirements Overview DPPS (Korea)
DPPS-2-04		Prerequisites for Deployable Bonnet Systems in Deployed State (Germany)
DPPS-2-05	1	Comments on OICA proposal (Japan)
DPPS-2-06		Comments on document TF-DPPS/1/05 Rev. 20170227 (Japan)
DPPS-2-07		Comments BASt on OICA Input Presentation (Germany)
DPPS-2-08		OICA comment for static and dynamic test (OICA)
DPPS-2-09		Marking of Deployable Bonnets: Differences of Bonnet Marking Positions and Challenges in Performance Assessment (OICA)
DPPS-2-10		Explanation JNCAP details for Items for DPPS Amendment (Japan)
DPPS-2-11		Development Head Test Procedure (Germany)
DPPS-2-12		Text for validation of simulation methods (OICA)
DPPS-2-13		Comments on document TF-DPPS/2/04 (OICA)

DPPS-2-14		Development of a Head Impact Test Procedure for Pedestrian Protection (Germany)
DPPS-2-15		Validity of a Headform to be used for a Specific Impact Test Speed Condition (Japan)
DPPS-2-16		Comments on document TF-DPPS/2/13 (Germany)
DPPS-3-01	Corr 1	3rd meeting agenda
DPPS-3-02	Corr 1	Minutes of the 3rd meeting
DPPS-3-03		Definition of sensing area (Japan)
DPPS-3-04		Proposal for Definition of Head Impact Test Area (Japan0
DPPS-4-01	1	4th meeting agenda
DPPS-4-02		Minutes of the 4th meeting
DPPS-4-03		Scope and Limitations of the PDI-2 (OICA)
DPPS-4-04		Static and Dynamic Testing of Deployable Systems (OICA)
DPPS-4-05		Marking + Deployed Position (OICA)
DPPS-4-06		JASIC proposals for document TF-DPPS/1/05-Rev.1 (Japan)
DPPS-4-07		Validity of Applying the Current Headform at Low Impact Speed (Japan)
DPPS-4-08		Dynamic Headform Test (Synchronization) (Korea)
DPPS-4-09		Discussion Issues for DPPS Testing (Korea)
DPPS-4-10		Alternative Determination of Head Impact Time (BGS)
IWG-DPPS-1-01		1st IWG-DPPS meeting agenda
IWG-DPPS-1-02	1	1st IWG-DPPS meeting notes
IWG-DPPS-1-03	2	IWG-DPPS Terms of Reference IWG-DPPS-1-03bis ECE-TRANS-WP29-2018-162e ToR official
IWG-DPPS-1-04	Corr 1	Presentation of the Euro NCAP CoHerent Project (Tu Graz)
IWG-DPPS-1-05		Comments: Deploy Height vs. Fully Deployed (OICA)
IWG-DPPS-1-06		Comments: Dynamic Testing (OICA)
IWG-DPPS-1-07		Comments: Pedestrian Sensing Impactor (OICA)
IWG-DPPS-1-08		ACEA Input: Definition of Sensing Width (ACEA)

IWG-DPPS-1-09		Summary of Compliance Test Procedure for Pedestrian Protection (Korea)
IWG-DPPS-1-11		Head Impact Time of Human Body Models (BASt)
IWG-DPPS-2-01	1	2nd IWG-DPPS meeting agenda
IWG-DPPS-2-02	1	2nd IWG-DPPS Meeting notes
IWG-DPPS-2-03		Summary Report Meeting 14 June 2018 (Sub-group Prerequisites)
IWG-DPPS-2-04	2	Contracting Parties' positions on DPPS amendments
IWG-DPPS-2-05	2	Proposal: Decision on Deployed Testing of DPPS (OICA)
IWG-DPPS-2-06	1	Proposal for a Rev. 4 of Document TF-DPPS/2/03
IWG-DPPS-2-07		Summary of SAE Standard for Full-Scale Pedestrian Dummy (Japan)
IWG-DPPS-2-08		Quick check of proposed logic to not activate DPPS outside of sensing width (OICA)
IWG-DPPS-2-09		2nd IWG-DPPS Attendance list
IWG-DPPS-2-10		JLR Presentation on synchronisation comparison (OICA)
IWG-DPPS-3-01		3rd IWG-DPPS Agenda
IWG-DPPS-3-02		4th IWG-DPPS Minutes
IWG-DPPS-3-03		Leg impactors and HBM simulation comparison for detection (CONCEPT)
IWG-DPPS-3-04		Study of Application of Upper Leg form to sensing test (Japan)
IWG-DPPS-3-05		Principle of a test procedure for Human Body Model numerical simulation (OICA)
IWG-DPPS-3-06		Attendance list
IWG-DPPS-4-01	1	4th IWG-DPPS Agenda
IWG-DPPS-4-02	1	Draft minutes +BAST comments

IWG-DPPS-4-03		Task 27 (deploy bonnet) (OICA)
IWG-DPPS-4-04		System Information Requirement (Korea)
IWG-DPPS-4-05		Sensing width proposal (OICA)
IWG-DPPS-4-06		Upper Leg Form Sensing Update (JASIC)
IWG-DPPS-4-07		Ped Dummy Test Procedure proposal (OICA)
IWG-DPPS-4-08		Marking of bonnet deployed/undeployed (OICA)
IWG-DPPS-4-09		Attendance list
IWG-DPPS-5-01		Draft agenda
IWG-DPPS-5-02		Draft minutes
IWG-DPPS-5-03		Marking (Korea)
IWG-DPPS-5-04		1st tentative draft
IWG-DPPS-5-05		Intended height test condition (JASIC)
IWG-DPPS-5-06		Sensing Impactors comparison (JASIC)
IWG-DPPS-5-07		Basis for General Wording Proposal (Germany)
IWG-DPPS-5-08		GRSP & WP29 report
IWG-DPPS-5-09		Detection Area Width (Germany)
IWG-DPPS-5bis-01		Draft agenda
IWG-DPPS-5bis-02		Draft skype minutes
IWG-DPPS-5bis-03		IDIADA -HIT calculation feedback (Spain)
IWG-DPPS-5bis-04		HIT calculation feedback (JASIC)

IWG-DPPS-5bis-05		ULF study 2013 (Altran)
IWG-DPPS-5bis-06		ULF study 2013-conclusion for detection (Altran)
IWG-DPPS-6-01	2	6th DPPS draft agenda
IWG-DPPS-6-02		6th IWG-DPPS Minutes
IWG-DPPS-6-03		V2 of draft text proposal (IDIADA)
IWG-DPPS-6-04		Sensing Impactor for DPPS (Germany)
IWG-DPPS-6-05		HIT-WAD calculation (Korea)
IWG-DPPS-6-06	2	HIT-WAD _Timing (OICA)
IWG-DPPS-6-07		Positioning of Ped HBM-v0 (OICA)
IWG-DPPS-6-08		Draft text proposal- updated (JASIC)
IWG-DPPS-6-09		Marking-up sketches (OICA)
IWG-DPPS-7-01		draft agenda
IWG-DPPS-7-02		IWG-DPPS 7-draft minutes
IWG-DPPS-7-03		Dynamic Static Test (Korea)
IWG-DPPS-7-04		Test Area (Korea)
IWG-DPPS-7-05		Draft-Annex (Korea)
IWG-DPPS-7-06		THUMS Overview (Toyota)
IWG-DPPS-7-07		Positioning of Ped HBM (OICA)
IWG-DPPS-7-08		GHBMC_M50-PS_Mo (GHBMC)
IWG-DPPS-7-08add		GHBMC addendum-publications (GHBMC)

IWG-DPPS-7=09		Flex-PLI as Sensing Impactor for UN-R127 - Contact Fidelity (Germany)
IWG-DPPS-7-10		Detection Area Width (Germany)
IWG-DPPS-7-11		Generic-Vehicle-Models (TU Graz)
IWG-DPPS-8-01		IWG - draft agenda
IWG-DPPS-8-02		Draft minutes
IWG-DPPS-8-03		Draft text Annex 2 organisation (Korea)
IWG-DPPS-8-04		FlexPLI Biofidelity for Detection - intermediate report (BASt/BGS)
IWG-DPPS-9-01		Draft Agenda
IWG-DPPS-9-02	2	Official minutes
IWG-DPPS-9-03		GTR9 Preamble for FlexPLI as detection impactor (BASt/BGS)
IWG-DPPS-9-04		Proposal GTR9 DPPS Sensor Detection (BASt/BGS)
IWG-DPPS-9-05		UNR127 amendment justification for FlexPLI as detection impactor (BASt/BGS)
IWG-DPPS-9-06		GRSP report
IWG-DPPS-9-07		Dynamic Static Test comparison (Korea)
IWG-DPPS-9-08		Comments on Korea Proposal for Draft Amendment (Japan)
IWG-DPPS-9-09		Proposal for Condition of Activation of DPPS (Japan)
IWG-DPPS-9-10	1	OICA – HIT simulation (OICA)
IWG-DPPS-9-11		Sensing FlexPLI Impactor Final Evaluation (BASt – BGS)
IWG-DPPS-9-12		Comment on IWG-DPPS-9-09 (BASt)
IWG-DPPS-10-01		Draft Agenda

IWG-DPPS-10-02		Draft Minutes
IWG-DPPS-10-03		IDIADA -Explanation_pressure_data-requirement (Spain)
IWG-DPPS-10-04		Leg_Head_Impact_Location_JASIC (Japan)
IWG-DPPS-10-05		HIT-HBM -TB024 simplification for Regulation (OICA)
IWG-DPPS-10-06		Ped-HBM-Certification for HIT Draft (OICA)
IWG-DPPS-10-07		Decision list
IWG-DPPS-10-08		UN webpages proposals to store Generic Vehicle Models (UN Secretariat)
IWG-DPPS-10-09		Detection Area- Lateral offset of head-Accident Data (BASt)
IWG-DPPS-11-01		Draft Agenda
IWG-DPPS-11-02		Draft Minutes
IWG-DPPS-11-03		Pedestrian Kinematic Assumptions GTR9 (Japan)
IWG-DPPS-11-04		Suggestion on Introduction of HIT Numerical Simulation (Japan)
IWG-DPPS-11-05		Clarification of IWG-DPPS scope (BASt)
IWG-DPPS-11-06		Comment for Detection Area (ACEA)
IWG-DPPS-11-07		Decision list
IWG-DPPS-12-01		Draft agenda
IWG-DPPS-12-02	1	Minutes
IWG-DPPS-12-03		Sensitivity Analysis Pressure Data Requirement (IDIADA)
IWG-DPPS-12-04		Sensitivity Analysis Pressure Data Requirement-GTR9 (IDIADA)
IWG-DPPS-12-05		Pressure Data Requirement-UNR127 (IDIADA)

IWG-DPPS-12-06		Comments on Priority of HIT Prediction Method (Japan)
IWG-DPPS-12-07		Pedestrian Kinematic Assumptions GTR9 (Japan)
IWG-DPPS-12-08		Detection Area II (VDA_SMMT-CCFA)
IWG-DPPS-12-09	1	Clarification for HIT Regression (VDA-SMMT-CCFA)
IWG-DPPS-13-01		Draft agenda
IWG-DPPS-13-02	1	Minutes
IWG-DPPS-13-03	1	Updated Decision_List
IWG-DPPS-13-04		Consolidated draft proposal of DPPS GTR9 amendment -210906
IWG-DPPS-13-05		revised 5-04 Pressure Data Requirement (IDIADA)
IWG-DPPS-13-05	1	Rev1 wording for CP options
IWG-DPPS-13-06		System Specifications Proposal (IDIADA)
IWG-DPPS-13-07		Participant list
IWG-DPPS-14-01	1	Official agenda
IWG-DPPS-14-02		Draft minutes
IWG-DPPS-14-03		Comment on HIT calculation-tool and HIT-WAD diagram (Japan)
IWG-DPPS-14-04		Sensing-width (VDA-CCFA)
IWG-DPPS-14-05		DPPS-HBM qualification procedure status (IWG subgroup)
IWG-DPPS-15-01	1	Draft agenda
IWG-DPPS-15-02		Draft & official minutes
IWG-DPPS-15-03		HBM qualification for GTR draft (TF-HBM subgroup)

IWG-DPPS-15-04	3	Draft1 & 2 technical requirements, then working ECE-TRANS-WP.29-GRSP-2022-02e tech requirements
IWG-DPPS-15-05	1	Preamble
IWG-DPPS-16-01	1	Draft & official agenda
IWG-DPPS-16-02	1	Draft & official minutes
IWG-DPPS-16-03		Draft Preamble
IWG-DPPS-16-04		Modified ECE-TRANS-WP.29-GRSP-2022-02e tech requirements
IWG-DPPS-16-05		Annex2_pedestrian_Human_Body_Model_qualification
IWG-DPPS-16-06		Annex3_HIT_determination_simulation
IWG-DPPS-16-07		Proposal HIT vs TRT requirement- OZ
IWG-DPPS-16-08		HIT vs TRT Explanation- OZ
IWG-DPPS-16-09		HBM-Simulations_Flow-Chart_AB
IWG-DPPS-16-10		Condition_for_Static_Test_in_Overshoot_Duration_JAMA
IWG-DPPS-17-01	1	Draft agenda
IWG-DPPS-17-02		Draft minutes
IWG-DPPS-17-03		GRSP-71-26e - DPPS status report
IWG-DPPS-17-04		Decision list
IWG-DPPS-17-05		Status_Nr_Simulation subgroup
IWG-DPPS-17-06		Annex2_Pedestrian_Human_Body_Model_Qualification
IWG-DPPS-17-07		Annex3_HIT_Determination_Simulation
IWG-DPPS-17-08		Draft preamble

IWG-DPPS-17-09		modified ECE-TRANS-WP.29-GRSP-2022-02e tech requirements
IWG-DPPS-17-10		Overall Flowchart DPPS_ALIGNED with Annex 23
IWG-DPPS-18-01	1	Draft & official agenda
IWG-DPPS-18-02	1	Draft & official minutes
IWG-DPPS-18-03		Annex2_Pedestrian_Human_Body_Model_Qualification
IWG-DPPS-18-04		Annex3_HIT_Determination_Simulation
IWG-DPPS-18-05		modified ECE-TRANS-WP.29-GRSP-2022-02e tech requirements
IWG-DPPS-18-06		Deployment test procedure doubts
IWG-DPPS-18-07		Sensing width-anonymised - Industry
IWG-DPPS-18-08		LAB_Pedestrian_DPPS_area detection width.pptx
IWG-DPPS-18-09		Action list
IWG-DPPS-18-10		Draft Wording Preamble GTR9 wrt Detection Area.docx
IWG-DPPS-18-11		Decision list
IWG-DPPS-18-12		Status Annexes 2 3 Subgroup
IWG-DPPS-18-13		Vehicle width additional fender - definition of RVW
IWG-DPPS-19-01	1	Draft & official agenda
IWG-DPPS-19-02	1	Draft & official minutes
IWG-DPPS-19-03		IDIADA wording subgroup results
IWG-DPPS-19-04		Marking undeployed
IWG-DPPS-19-05		Technical requirements 9Nov

IWG-DPPS-19-06		Preamble (17-08 merged with 18-10)
IWG-DPPS-19-07		Action list
IWG-DPPS-19-08		Annex3_HIT_Determination_Simulation
IWG-DPPS-19-09		Annex2_Pedestrian_Human_Body_Model_Qualification
IWG-DPPS-20-01	1	Draft & official agenda
IWG-DPPS-20-02	1	Draft & official minutes
IWG-DPPS-20-03		Action list
IWG-DPPS-20-04		Preamble 16Nov22
IWG-DPPS-20-05		Technical requirements 16Nov22
IWG-DPPS-20-06		(OZ) Proposal Overshoot Phase
IWG-DPPS-20-07		(BH-IDIADA)_Dynamic Testing Sync
IWG-DPPS-21		ECE-TRANS-WP.29-GRSP-2023-xx final GTR0-03 proposal IWG-DPPS.docx
IWG-DPPS-21		ECE-TRANS-WP29-GRSP-2023-yy MR1 Amend-4 from IWG-DPPS.docx
IWG-DPPS-21-01	1	Draft & rev1 agenda
IWG-DPPS-21-02	1	Draft & official minutes
IWG-DPPS-21-03		Action list
IWG-DPPS-21-04		Preamble GRSP-72-08
IWG-DPPS-21-05		Technical requirements GRSP-72-09
IWG-DPPS-21-06		Status report GRSP-72-11
IWG-DPPS-21-07	7	consolidated draft.rev1-7- updated 24Jan-15 Feb.2023
IWG-DPPS-21-07		consolidated final - updated 15 Feb.2023

IWG-DPPS-21-08		Dummy_comparison_with_HBM_rev1 Japan-draft
IWG-DPPS-21-09		Dummy_proposal_for_text_of_preamble Japan
IWG-DPPS-21-10	7	Proposal Overshoot Phase (BAST)
IWG-DPPS-21-11	1	Decision_List & Decision_List with text check
IWG-DPPS-21-12	1	smallest_HBM (BAST)
IWG-DPPS-21-13	1	Comments to Document Comparison bet. HBMs (Annex 2) and Pedestrian Dummy (BAST)
IWG-DPPS-21-14	1	only_Annex2-3 update (CK)
IWG-DPPS-22	1	ECE-TRANS-WP.29-GRSP-2023-06e.pdf GTR9-03 Proposal
IWG-DPPS-22	1	ECE-TRANS-WP.29-GRSP-2023-07e.pdf MR1 Addendum 5 proposal
IWG-DPPS-22-01	1	Draft agenda
IWG-DPPS-22-02	1	Draft & official minutes
IWG-DPPS-22-03	1	Head impact time verification (VDA)
IWG-DPPS-22-04	1	consolidated Doc with all changes GTR and UNR- 13Apr23
IWG-DPPS-22-05	1	AC in preamble proposal
IWG-DPPS-22-06	1	Action list updated
IWG-DPPS-23-01	1	Draft agenda
IWG-DPPS-23-02	1	Draft minutes
IWG-DPPS-23-03	1	Test Rig Synchronization for DPPS
IWG-DPPS-23-04	1	Overall Flowchart DPPS_23
IWG-DPPS-23-05	1	consolidated doc with all changes GTR and UNR 24Apr23
IWG-DPPS-23-06	1	modified_ECE-TRANS-WP29-1101-Amend-5

IWG-DPPS-23-07	1	OZ small group on tolerances Proposal on 28Apr2023
IWG-DPPS-23-08	1	GRSP-73-10 MR1-Amend4 Addendum5 GVM consolidated 28Apr, 11May
IWG-DPPS-23-09	1	GRSP-73-11 MR1-Amend5 Addendum6 HBM informal 28Apr
IWG-DPPS-23-10	1	GRSP-73-12 GTR9-03 informal doc
IWG-DPPS-23-11	1	GRSP-73-13 IWG-DPPS final status report
IWG-DPPS-24-01	1	draft agenda
IWG-DPPS-24-02	1	Draft minutes
IWG-DPPS-24-03	1	MR1_related_to_IWG-DPSS -CK
IWG-DPPS-24-04	1	2023_06_28_GV-Models_DPSS
IWG-DPPS-24-05	1	2023.06.16 - Structure_IM
IWG-DPPS-25-01	1	Draft_agenda
IWG-DPPS-25-02	1	Draft & official minutes
IWG-DPPS-25-03	5	GTR9-03 DPPS amendment draft
IWG-DPPS-25-04	4	MR1_related_to_IWG-DPSS draft
IWG-DPPS-25-05		Documentation Generic Vehicle Models
IWG-DPPS-25-06 IWG-DPPS-25-07		OICA draft preamble on GV Models-f OICA background on preamble on GV Models
IWG-DPPS-26-01	2	Draft_agenda
IWG-DPPS-26-02	1	Draft minutes
IWG-DPPS-26-03	1	GTR9-03 DPPS amendment draft – last modifications with comments

IWG-DPPS-26-04	1	GTR9-03 DPPS amendment draft – consolidated last modifications-clean
IWG-DPPS-26-05	1	GTR9-03 DPPS final status report – updated for Dec23 GRSP
IWG-DPPS-26-06		MR1 amend 4 draft – consolidated last modifications

II. Text of the Regulation

1. PURPOSE

1.1. The purpose of this global technical regulation is to bring about an improvement in the construction of certain parts of the front of vehicles which have been identified as causing injury when in collision with a pedestrian or other vulnerable road user.

1.2. The vehicles to be tested under the regulation are representative of the majority of vehicles in circulation in the urban environment, where there is a greater potential for collision with pedestrians and other vulnerable road users, and include passenger cars, vans and light trucks.

2. APPLICATION / SCOPE

2.1. This global technical regulation (gtr) shall apply to the frontal surfaces of power driven vehicles of category 1-1 with a gross vehicle mass exceeding 500 kg; and of category 1-2 with a gross vehicle mass exceeding 500 kg but not exceeding 4,500 kg; and of category 2 with a gross vehicle mass exceeding 500 kg but not exceeding 4,500 kg. [2] However, power driven vehicles of category 1-2 and category 2, where the distance, measured longitudinally on a horizontal plane, between the transverse centre line of the front axle and the R-point of the driver's seat is less than 1,100 mm, are exempt from the requirements of this regulation. Contracting Parties can exempt category 1-1 vehicles where the distance, measured longitudinally on a horizontal plane, between the transverse centre line of the front axle and the R-point of the driver's seat is less than 1,100 mm and having the components of the frontal structure that are interchangeable with the above-mentioned category 1-2 and category 2 vehicles.

All definitions of Special Resolution No. 1 shall apply as necessary.

[2] A contracting party may restrict application of the requirements in its domestic legislation if it decides that such restriction is appropriate.

3. Definitions

When performing measurements as described in this Part, the vehicle shall be positioned in its normal ride attitude.

In case of the vehicle equipped with a deployable pedestrian protection system (DPPS)as defined in paragraph 3.16., the measurements shall be taken with the system undeployed.

If the vehicle is fitted with a badge, mascot or other structure, which would bend back or retract under an applied load of maximum 100 N, then this load shall be applied before and/or while these measurements are taken.

Any vehicle component which could change shape or position, other than suspension components or active devices to protect pedestrians, shall be set to their stowed position.

For the purposes of this regulation:

3.1. "Adult headform test area" is an area on the outer surfaces of the front structure. The area is bounded, in the front, by a wrap around distance (WAD) of 1,700 mm and, at the rear, by the rear reference line for adult

headform and, at each side, by the side reference line.

3.2. "A-pillar" means the foremost and outermost roof support extending from the chassis to the roof of the vehicle.

3.3. The "Assessment Interval" (AI) of the flexible lower legform impactor is defined and limited by the time of first contact of the flexible lower legform impactor with the vehicle and the timing of the last zero crossing of all femur and tibia segments after their first local maximum subsequent to any marginal value of 15 Nm, within their particular common zero crossing phases. The AI is identical for all bone segments and knee ligaments. In case of any bone segment not having a zero crossing during the common zero crossing phases, the time history curves for all bone segments are shifted downwards until all bending moments are crossing zero. The downwards shift is to be applied for the determination of the AI only.

3.4. "Bonnet leading edge" means the edge of the front upper outer structure of the vehicle, including the bonnet and wings, the upper and side members of the headlight surrounds and any other attachments. The reference line identifying the position of the bonnet leading edge is defined by its height above the ground reference plane and by the horizontal distance separating it from the bumper (bumper lead).

3.5. "Bonnet leading edge height" means, at any point on the bonnet leading edge, the vertical distance between the ground reference plane and the bonnet leading edge reference line at that point.

3.6. "Bonnet leading edge reference line" means the geometric trace of the points of contact between a straight edge 1,000 mm long and the front surface of the bonnet, when the straight edge, held parallel to the vertical longitudinal plane of the car and inclined rearwards by 50° from the vertical and with the lower end 600 mm above the ground, is traversed across and in contact with the bonnet leading edge (see Figure 1).

For vehicles having the bonnet top surface inclined at 50°, so that the straight edge makes a continuous contact or multiple contacts rather than a point contact, determine the reference line with the straight edge inclined rearwards at an angle of 40° from the vertical.

For vehicles of such shape that the bottom end of the straight edge makes first contact with the vehicle then that contact is taken to be the bonnet leading edge reference line, at that lateral position.

For vehicles of such shape that the top end of the straight edge makes first contact with the vehicle then the geometric trace of 1,000 mm wrap around distance, will be used as bonnet leading edge reference line at that lateral position.

The top edge of the bumper shall also be regarded as the bonnet leading edge if it is contacted by the straight edge during this procedure.

3.7. "Bonnet rear reference line" means the geometric trace of the most rearward points of contact between a 165 mm diameter sphere and the front structure of the vehicle when the sphere is traversed across the front structure of the vehicle while maintaining contact with the windscreen (see Figure 2). The wiper blades and arms are removed during this process.

Where the bonnet rear reference line and the side reference line do not intersect, the bonnet rear reference line should be extended and/or modified using a semi-circular template, of radius 100 mm. The template should be made of a thin flexible sheet material that easily bends to a single curvature in any direction. The template should, preferably, resist double or complex curvature where this could result in wrinkling. The recommended material is a foam backed thin plastic sheet to allow the template to "grip" the surface of the vehicle. The template should be marked up with four points "A" through "D", as shown in Figure 3, while the template is on a flat surface.

The template should be placed on the vehicle with Corners "A" and "B" coincident with the side reference line. Ensuring these two corners remain coincident with the side reference line, the template should be slid progressively rearwards until the arc of the template makes first contact with the bonnet rear reference line. Throughout the process, the template should be curved to follow, as closely as possible, the outer contour of the vehicle's bonnet top, without wrinkling or folding of the template. If the contact between the template and bonnet

rear reference line is tangential and the point of tangency lies outside the arc scribed by points "C" and "D", then the bonnet rear reference line is extended and/or modified to follow the circumferential arc of the template to meet the bonnet side reference line, as shown in Figure 4.

If the template cannot make simultaneous contact with the bonnet side reference line at points "A" and "B" and tangentially with the bonnet rear reference line, or the point at which the bonnet rear reference line and template touch lies within the arc scribed by points "C" and "D", then additional templates should be used where the radii are increased progressively in increments of 20 mm, until all the above criteria are met.

3.8. "Bonnet top" is the area which is bounded by (a), (b) and (c) as follows:

(a) the bonnet leading edge reference line;

(b) the bonnet rear reference line;

(c) the side reference lines.

3.9. "Bumper" means the front, lower, outer structure of a vehicle. It includes all structures that are intended to give protection to a vehicle when involved in a low speed frontal collision and also any attachments to this structure. The reference height and lateral limits of the bumper are identified by the corners and the bumper reference lines.

3.10. "*Bumper beam*" means the structural cross member, rearward of the bumper fascia if present, protecting the front of the vehicle. The beam does not include foam, cover support or any pedestrian protection devices.

3.11. "Bumper lead" means for any longitudinal section of a vehicle, the horizontal distance in the vehicle longitudinal plane between the upper bumper reference line and the bonnet leading edge reference line

3.12. "*Bumper test area*" means either the front vehicle fascia between the left and right corner of bumper as defined in paragraph 3.14., minus the areas covered by the distance of 42 mm inboard of each corner of bumper, as measured horizontally and perpendicular to the longitudinal median plane of the vehicle, or between the outermost ends of the bumper beam as defined in paragraph 3.9. (see Figure 5D), minus the areas covered by the distance of 42 mm inboard of each end of the bumper beam, as measured horizontally and perpendicular to the longitudinal median plane of the vehicle, whichever area is wider.

3.13. "*Centre of the knee joint*" of the lower legform impactor is defined as the point about which the knee effectively bends.

3.14. "Child headform test area" is an area on the outer surfaces of the front structure. The area is bounded, in the front, by the front reference line for child headform, and, at the rear, by the WAD1700 line, and by the side reference lines.

3.15. "*Corner of bumper*" means the transversal position of the vehicle's point of contact with a corner gauge as defined in Figure 5B.

For determination of the corner of bumper, the front surface of the corner gauge is moved parallel to a vertical plane with an angle of 60° to the vertical longitudinal centre plane of the vehicle (see Figures 5A and 5C) at any height of the centre point of the corner gauge between:

(a) Equal to and above the point found on the vertical line intersecting the Lower Bumper Reference Line at the assessment position in transversal direction or at 75 mm above the ground reference plane, whichever is higher.

(b) Equal to and below the point found on the vertical line intersecting the Upper Bumper Reference Line at the assessment position in transversal direction or at 1,003 mm above the ground reference plane, whichever is lower.

For determination of the corner of bumper, the gauge is moved to contact the outer contour/front fascia of the vehicle touching at the vertical centre line of the gauge. The horizontal centre line of the gauge is kept parallel to the ground plane.

The corners of bumper on both sides are subsequently defined as the outermost points of contact of the gauge with the outer contour/front fascia of the vehicle as determined in accordance with this procedure. Any points of contact on the top and the bottom edges of the gauge are not taken into account. The external devices for indirect vision and the tyres shall not be considered.

3.16. "Deployable Pedestrian Protection System (DPPS)" means a technical system, which is activated for head protection of a pedestrian in the event of a collision of the vehicle with a pedestrian. It comprises a deployment module, together with other related components required for its function, such as bonnet, sensors, or wiring, etc.

3.17. "Deployment module" means a unit, comprising components, such as airbags, springs, or pyrotechnic actuators, etc., that are used to change the vehicle outer surface from a position of normal use in the vehicle to a deployed position.

3.17.1. "Initiation of the deployment module" means, at the option of the manufacturer, either the moment when visible movement of the actuator is initially detected, or the moment when the triggering signal is sent from the electronic control unit to the deployment module.

3.18. "Deployment Time (DT)" means the duration from the initiation of the deployment module(s) until the DPPS reaches its maximum deployment height for the first time. Measurement shall be done on the outer surface of the DPPS, in the area above the lifting device(s).

3.18.1. "Deployed position" means the position of the vehicle outer surface equipped with a DPPS that can be maintained by the system after its activation, as shown in Figure 1-1 of Annex 1.

3.18.2. "Undeployed position" means the position of the vehicle outer surface equipped with a DPPS when the DPPS is not activated.

3.19. "Detection test area" is the area designated to detect a pedestrian in order to activate the deployable system. The width of the detection test area shall be the relevant vehicle width, minus a distance from each side of 12.5 per cent of the relevant vehicle width, but not more than 250mm from each side. The detection test area must not be smaller than the area inboard of the corners of bumper (CoB) minus a distance of 42mm on each side, as measured horizontally and perpendicular to the longitudinal median plane of the vehicle. (see Figure 11).

3.20. "*Femur*" of the lower legform impactor is defined as all components or parts of components (including flesh, skin covering, instrumentation and brackets, etc. attached to the impactor for the purpose of launching it) above the knee.

3.21. "Front reference line for child headform" means the geometric trace as described on the vehicle front structure using a WAD1000 line. In the case of vehicles where the wrap around distance to the bonnet leading edge reference line, is more than 1,000 mm at any point, then the bonnet leading edge reference line will be used as the front reference line for child headform at that point.

3.22. "Front structure" means all outer structures of the vehicle except the windscreen, the windscreen header, the A-pillars and structures rearward of these. It therefore includes, but is not limited to, the bumper, the bonnet, wings, scuttle, wiper spindles and lower windscreen frame.

3.23. "Ground reference plane" means a horizontal plane, either real or imaginary, that passes through the lowest points of contact for all tyres of a vehicle while the vehicle is in its normal ride attitude. If the vehicle is resting on the ground, then the ground level and the ground reference plane are one and the same. If the vehicle is raised off the ground such as to allow extra clearance below the bumper, then the ground reference plane is above ground level.

3.24. "The pedestrian Head Impact Time (HIT)" is defined as the elapsed time subsequent to the time of first contact of the pedestrian surrogate (neglecting forearms and hands) with the vehicle outer surface and the time of first contact of its head with the vehicle outer surface.

There are two kinds of HIT :

3.24.1. HIT_d : to decide whether the physical head test on the deployable system can be done dynamically or statically. The HIT_d is determined with the deployed DPPS.

3.24.2. HIT_s : to synchronise the test rig for dynamic testing. The HIT_s is determined with the undeployed DPPS.

3.25. "Head Injury Criterion (HIC)" means the calculated result of accelerometer time histories using the following formula:

$$HIC = \left[\frac{1}{t_2 - t_1} \int_{t_1}^{t_2} a \, dt \right]^{2.5} (t_2 - t_1)$$

Where:

a	is the resultant acceleration measured in units of gravity "g" (1 g = 9.81 m/s²);
t_1 and t_2	are the two time instants (expressed in seconds) during the impact, defining an interval between the beginning and the end of the recording period for which the value of HIC is a maximum (t2 - t1 ≤ 15 ms)

3.26. "Impact point" means the point on the vehicle where initial contact by the test impactor occurs. The proximity of this point to the target point is dependent upon both the angle of travel by the test impactor and the contour of the vehicle surface (see point B in Figure 6).

3.27. "Lower bumper height" means the vertical distance between the ground reference plane and the lower bumper reference line, with the vehicle positioned in its normal ride attitude.

3.28. "Lower bumper reference line" means the lower limit to significant points of pedestrian contact with the bumper. It is defined as the geometric trace of the lowermost points of contact between a straight edge 700 mm long and the bumper, when the straight edge, held parallel to the vertical longitudinal plane of the car and inclined forwards by 25° from the vertical, is traversed across the front of the car, while maintaining contact with the ground and with the surface of the bumper (see Figure 7).

3.29. "Normal ride attitude" means the vehicle positioned on a flat horizontal surface with its mass in running order (as defined in Annex 3, paragraph 3 of Special Resolution No. 1), with the tyres inflated to manufacturer recommended pressures, the front wheels in the straight-ahead position and with a passenger mass (as defined in Annex 3, paragraph 6.2. of Special Resolution No. 1) placed on the front passenger seat. The front seats are placed at the nominal mid-track position. The suspension shall be set in normal running condition as specified by the manufacturer for a speed of 40 km/h.

3.30. "Outer surface" means those components of the vehicle, which may be contacted by the pedestrian in case of an accident. The outer surface may include the bumper, the bonnet, the fenders, but also external airbags or other components.

3.32. "Primary reference marks" means holes, surfaces, marks and identification signs on the vehicle body. The type of reference mark used and the vertical (Z) position of each mark relative to the ground shall be specified by the vehicle manufacturer according to the running conditions specified in paragraph 3.22. These marks shall be selected such as to be able to easily check the vehicle front and rear ride heights and vehicle attitude.

The primary reference marks shall be within ± 25 mm of the design position in the vertical (Z) axis. All tests are conducted with either the vehicle or all further measurements adjusted to simulate the vehicle being in the design position. This position shall be considered to be the normal ride attitude.

3.33. "Rear reference line for adult headform" means a geometric trace as described on the front structure of the vehicle using a WAD2100 line.

3.34. "Relevant Vehicle Width (RVW)" is the maximum width of the vehicle without devices for indirect vision, measured on or in front of a vertical transverse plane passing through the front axle of the vehicle.

3.35. "Sensing Time (ST)" means the duration from the time of the first contact of the flexible lower legform impactor (FlexPLI) with the vehicle outer surface to the initiation of the deployment module.

3.36. "Sensors" are pedestrian contact sensors that detect a pedestrian contact with the front of the vehicle. These sensors include, but are not limited to, accelerometers, fibre optic sensors, pressure sensors, etc.

3.37. "Side reference line" means the geometric trace of the highest points of contact between a straight edge 700 mm long and the sides of the vehicle, when the straight edge, held parallel to the transverse vertical plane of the vehicle and inclined inwards by 45°, is traversed down, and maintains contact with the sides of the front structure (see Figure 8).

3.38. "Target point" means the intersection of the projection of the headform longitudinal axis with the front surface of the vehicle (see point A in Figure 6.).

3.39. "Tibia" of the lower legform impactor is defined as all components or parts of components (including flesh, skin covering, instrumentation and brackets, pulleys, etc. attached to the impactor for the purpose of launching it) below the level of the centre of the knee. Note that the tibia as defined includes allowances for the mass, etc., of the foot.

3.40. "Testing of the DPPS":

The headform impact tests on the DPPS can be performed in three ways: statically, dynamically or combined.

3.40.1. "Static testing" means the launch of the headform on a DPPS being in the deployed position.

3.40.2. "Dynamic testing" means the synchronised launch of the headform onto the deploying DPPS at the appropriate HIT_s.

3.40.3. "Combined testing" means a mixed set of tests on a DPPS in which a given test is run either statically or dynamically.

3.41. "Testing time" for static time constraint means the timeframe after the DPPS reaches its deployed position in which the headform test to the DPPS is to be performed (see Figure 1-2 of Annex 1).

3.42. "Total Response Time (TRT)" means the duration from the time of first contact of the FlexPLI with the vehicle outer surface to the time the DPPS reaches its maximum deployment height for the first time. It is the sum of the ST and DT.

3.43. "Upper bumper reference line" means the upper limit to significant points of pedestrian contact with the bumper. For vehicles with an identifiable bumper structure it is defined as the geometric trace of the uppermost points of contact between a straight edge and the bumper, when the straight edge, held parallel to the vertical longitudinal plane of the car and inclined rearwards by 20° to the vertical, is traversed across the front of the car, while maintaining contact with the surface of the bumper (see Figure 9).

For vehicles with no identifiable bumper structure it is defined as the geometric trace of the uppermost points of contact between a straight edge 700 mm long and the bumper area, when the straight edge, held parallel to the vertical longitudinal plane of the car and inclined rearwards by 20° from the vertical is traversed across the front of the car, while maintaining contact with the ground and with the surface of the bumper area (see Figure 9).

3.44. "Wrap Around Distance (WAD)" means the geometric trace described on the outer surface of the vehicle front structure by one end of a flexible tape, when it is held in a vertical longitudinal plane of the vehicle and traversed across the front structure. The tape is held taut throughout the operation with one end held at the same

level as the ground reference plane, vertically below the front face of the bumper and the other end held in contact with the front structure (see Figure 10). The vehicle is positioned in the normal ride attitude.

This procedure shall be followed, using alternative tapes of appropriate lengths, to describe wrap around distances of 1,000 mm (WAD1000), of 1,700 mm (WAD1700) and of 2,100 mm (WAD2100).

3.45. "Windscreen" means the frontal glazing of the vehicle situated between the Apillars.

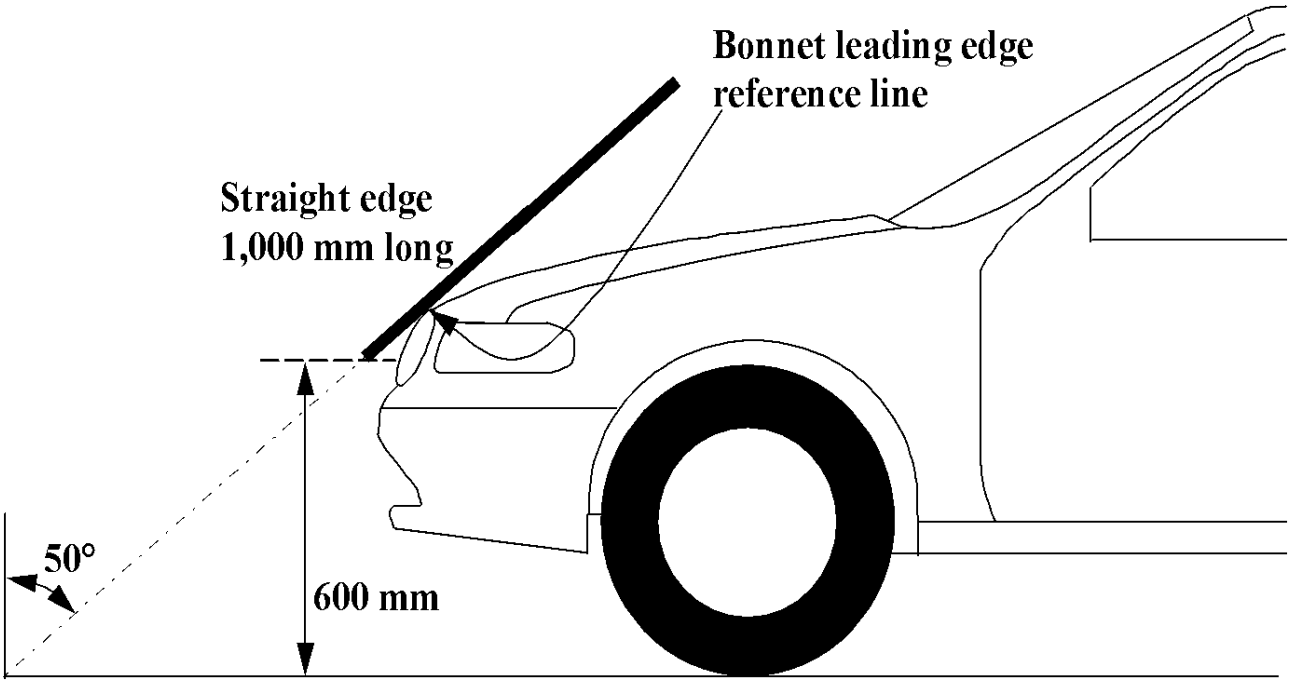


Figure 1. Bonnet leading edge reference line (see paragraph 3.6.)

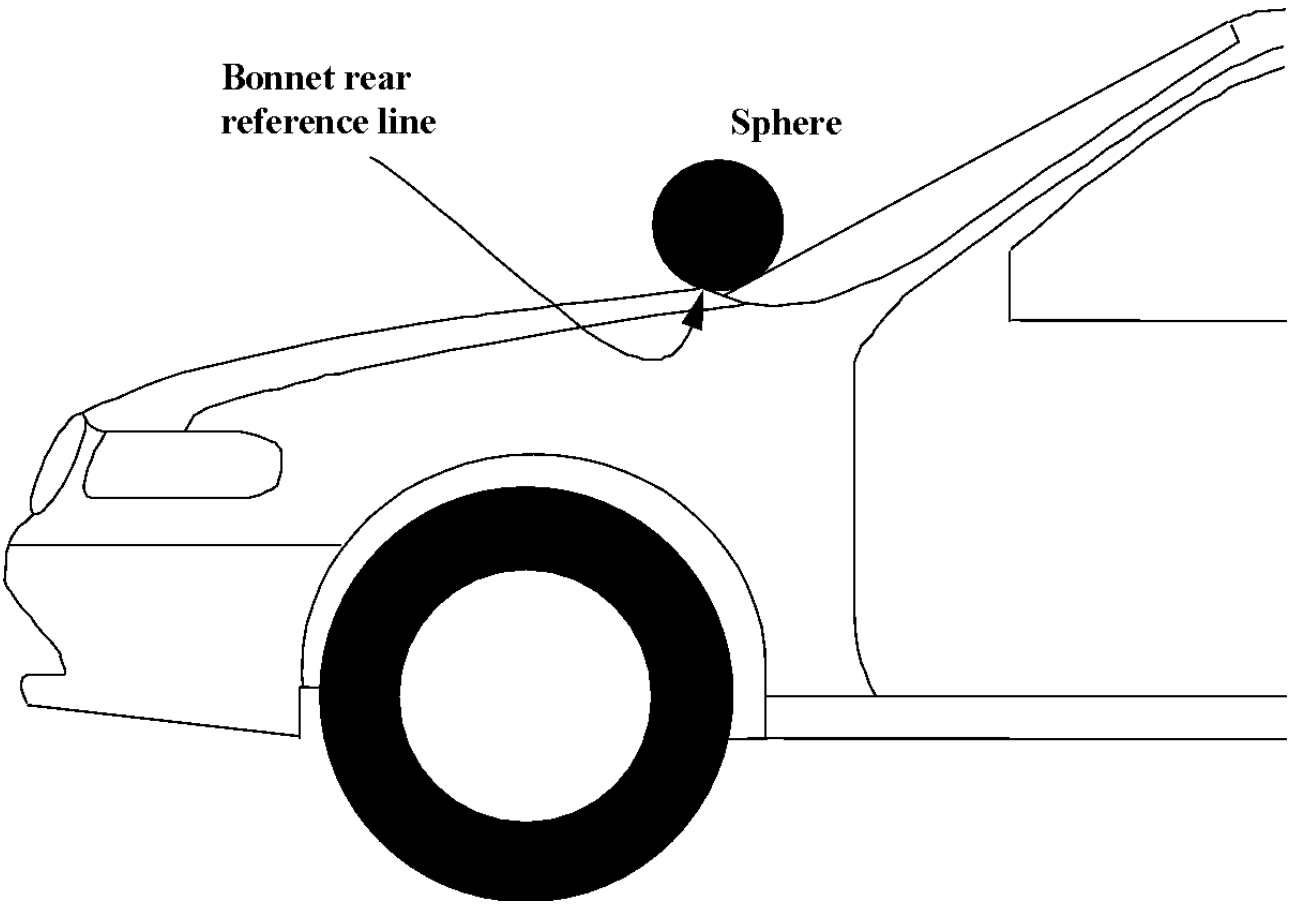


Figure 2. Bonnet rear reference line. (see paragraph 3.7.)

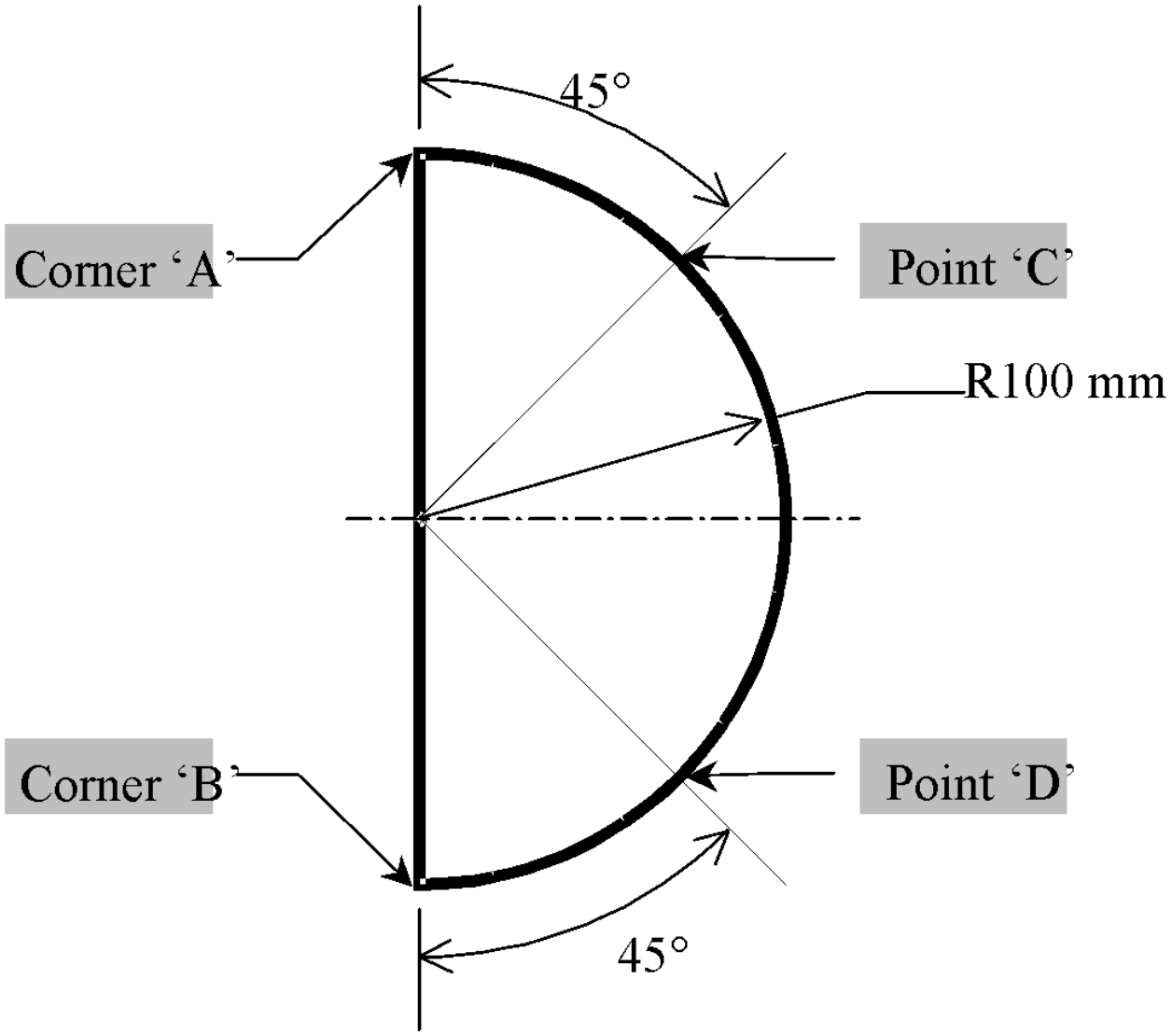


Figure 3: Template (see paragraph 3.7.)

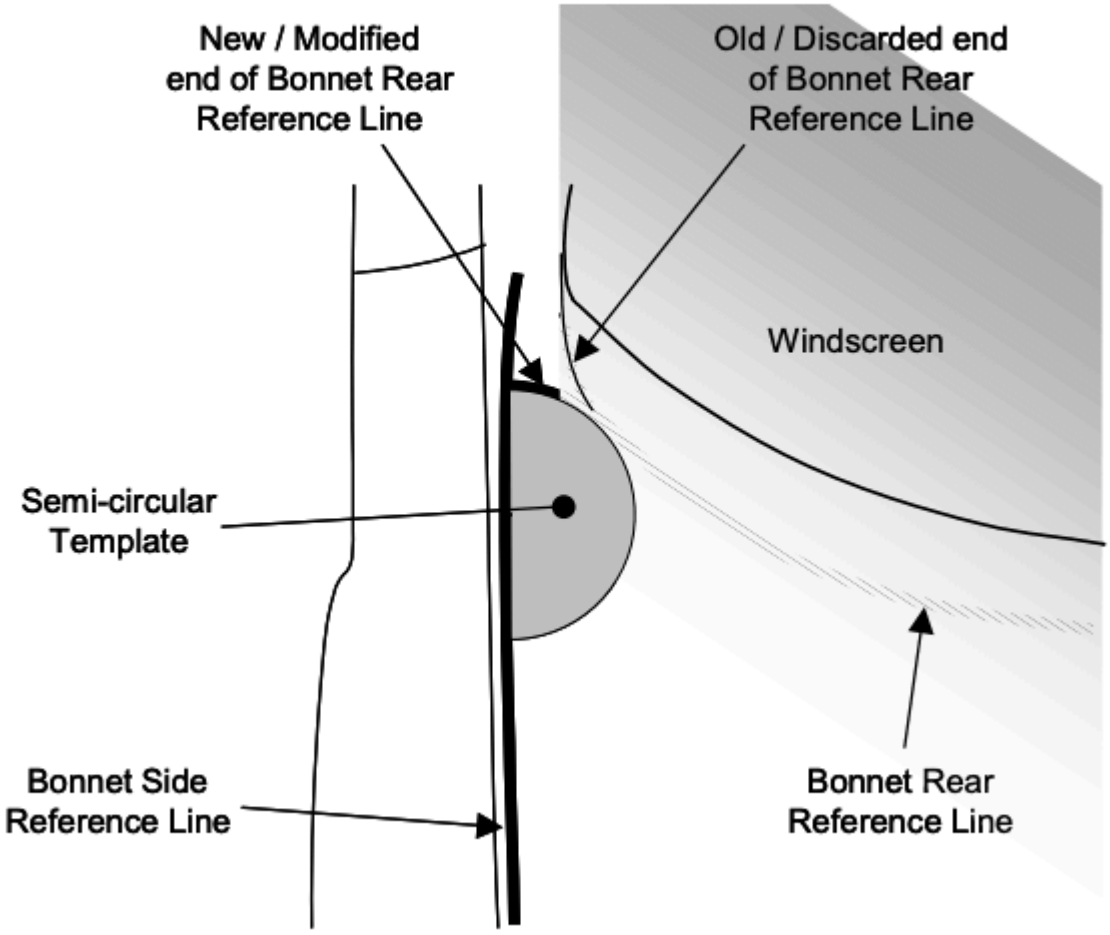


Figure 4. Marking of intersection between bonnet rear and side reference lines (see paragraph 3.7.)

Figure 5A. **Corner of bumper example (see paragraph 3.15., note that the corner gauge is to be moved in vertical and horizontal directions to enable contact with the outer contour /front fascia of the vehicle)**

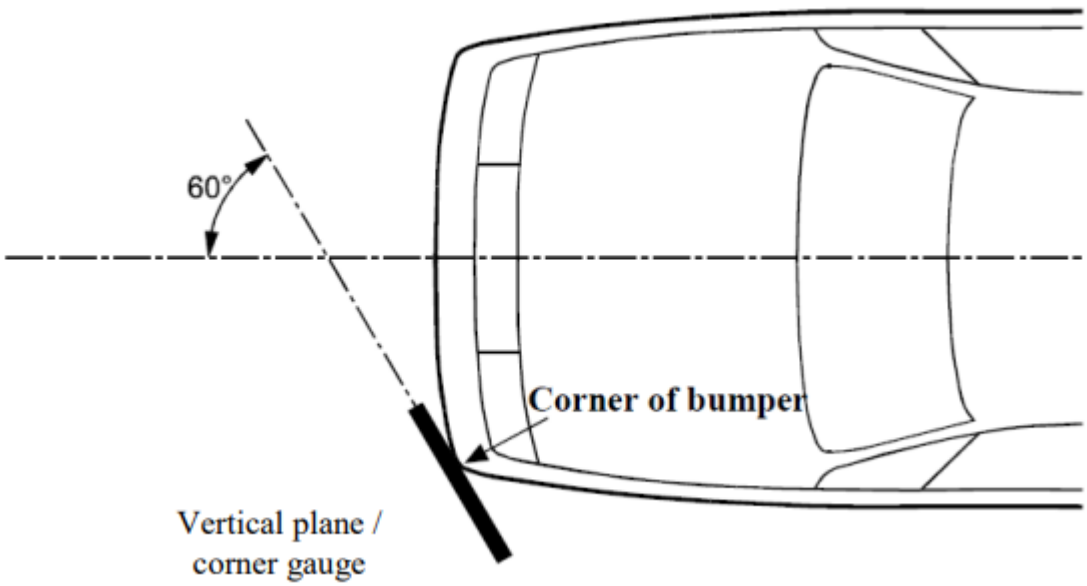
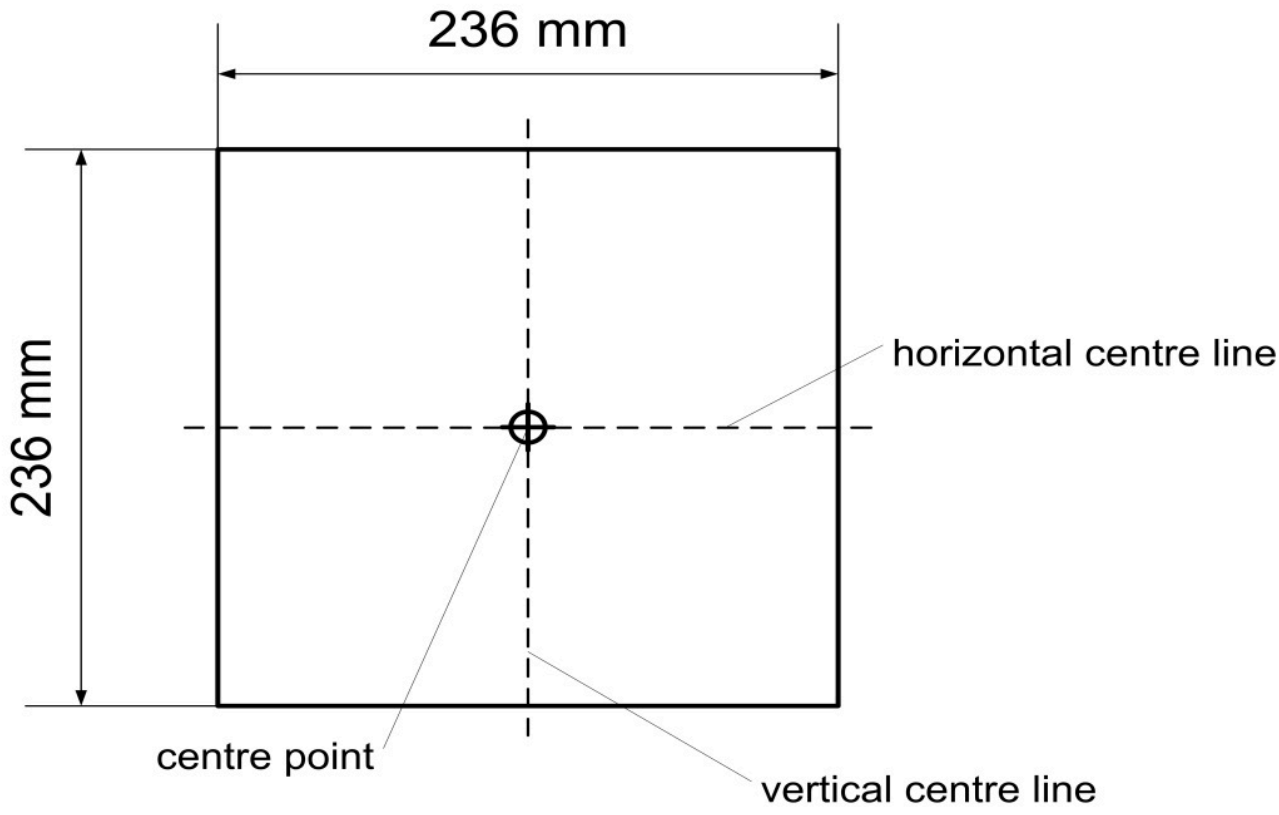


Figure 5B **Corner gauge**



The front surface of the corner gauge is flat.

The centre point is the intersection of the vertical and horizontal centre lines on the front surface.

Figure 5C **Determination of the corner of bumper with the corner gauge (shown in random location)**

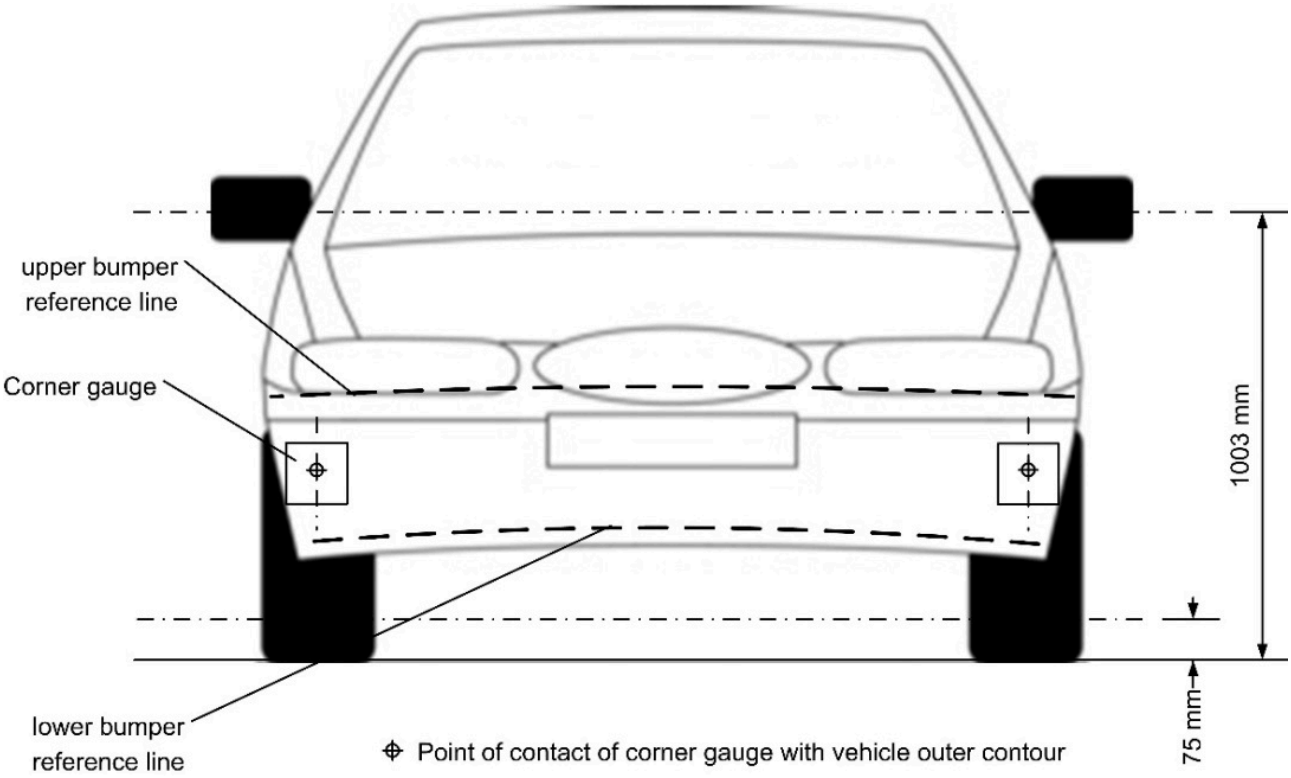


Figure 5D **Determination of bumper test area (note that the corner gauges are to be moved in vertical and horizontal directions to enable contact with the outer contour/front fascia of the vehicle)**

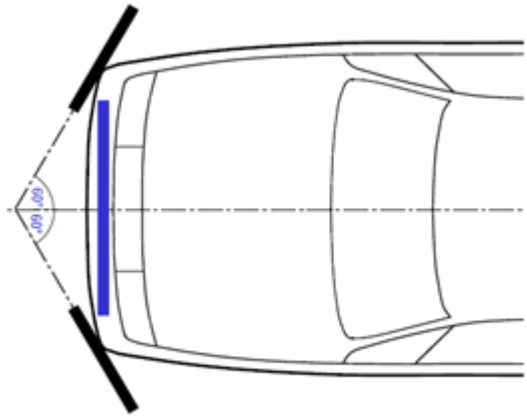


Figure 6. **Impact and target point (see paragraphs 3.26. and 3.38.)**

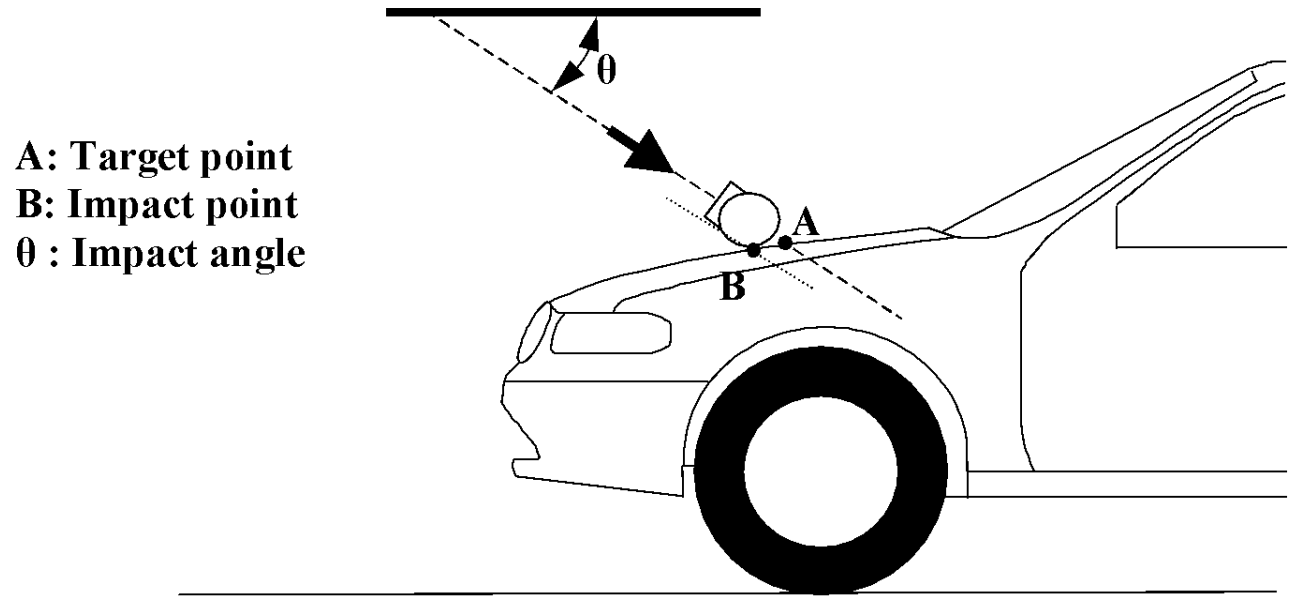


Figure 7. **Lower bumper reference line, LBRL (see paragraph 3.28.)**

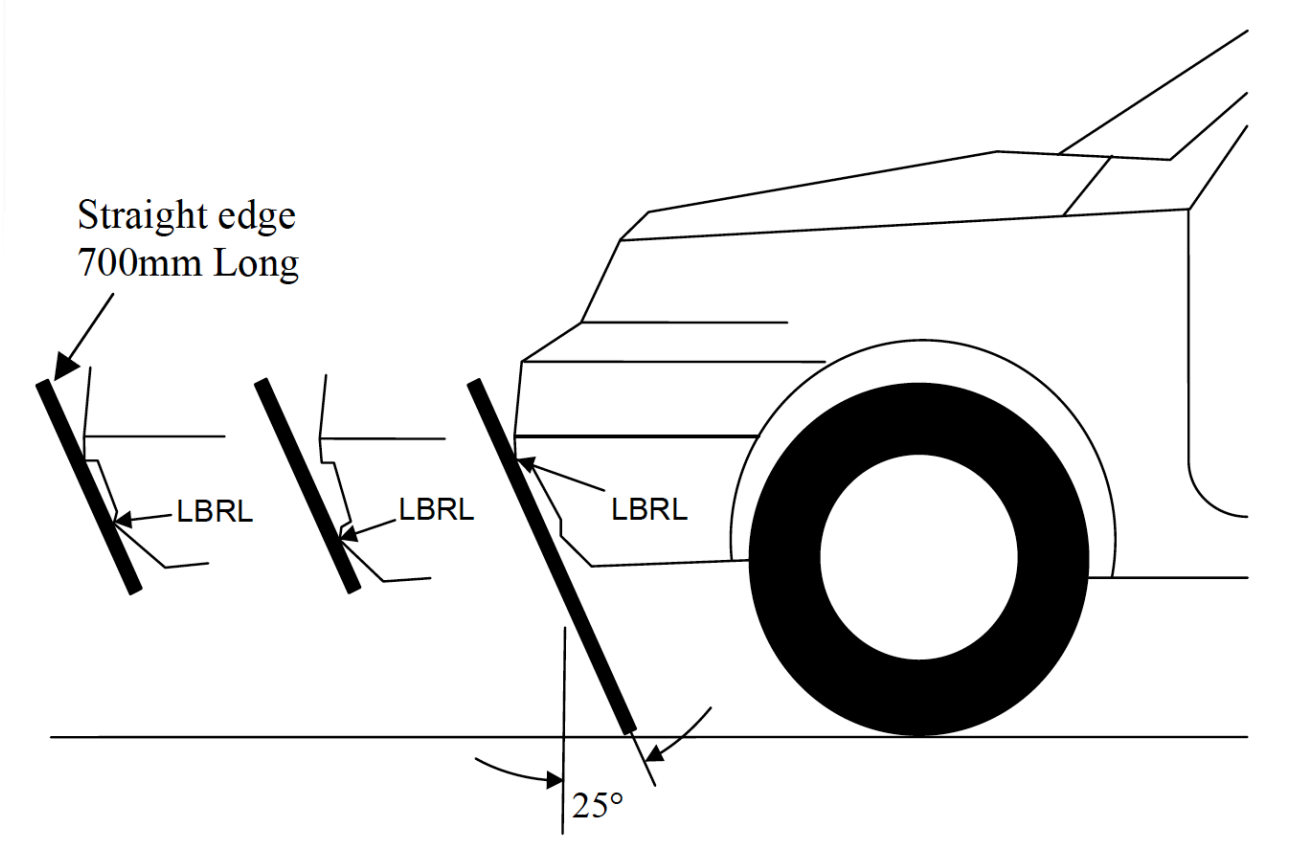


Figure 8. **Side reference line (see paragraph 3.37.)**

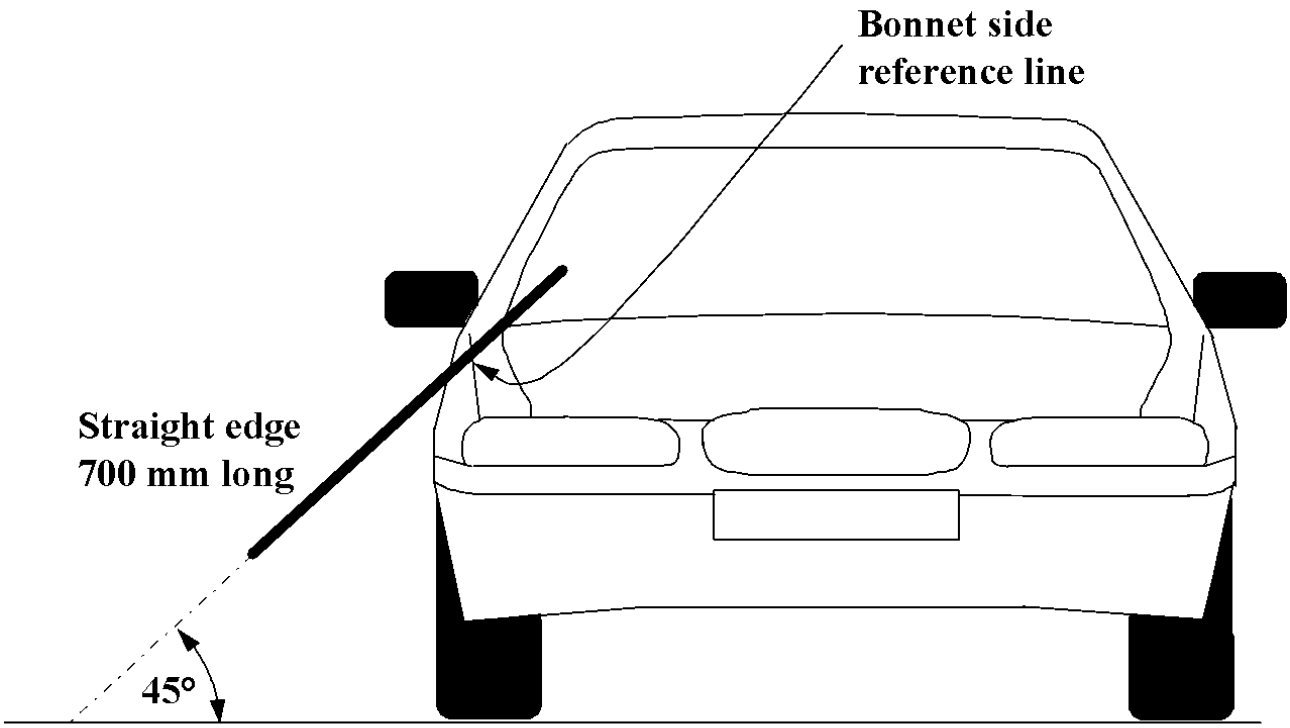


Figure 9. **Upper bumper reference line, UBRL (see paragraph 3.43.)**

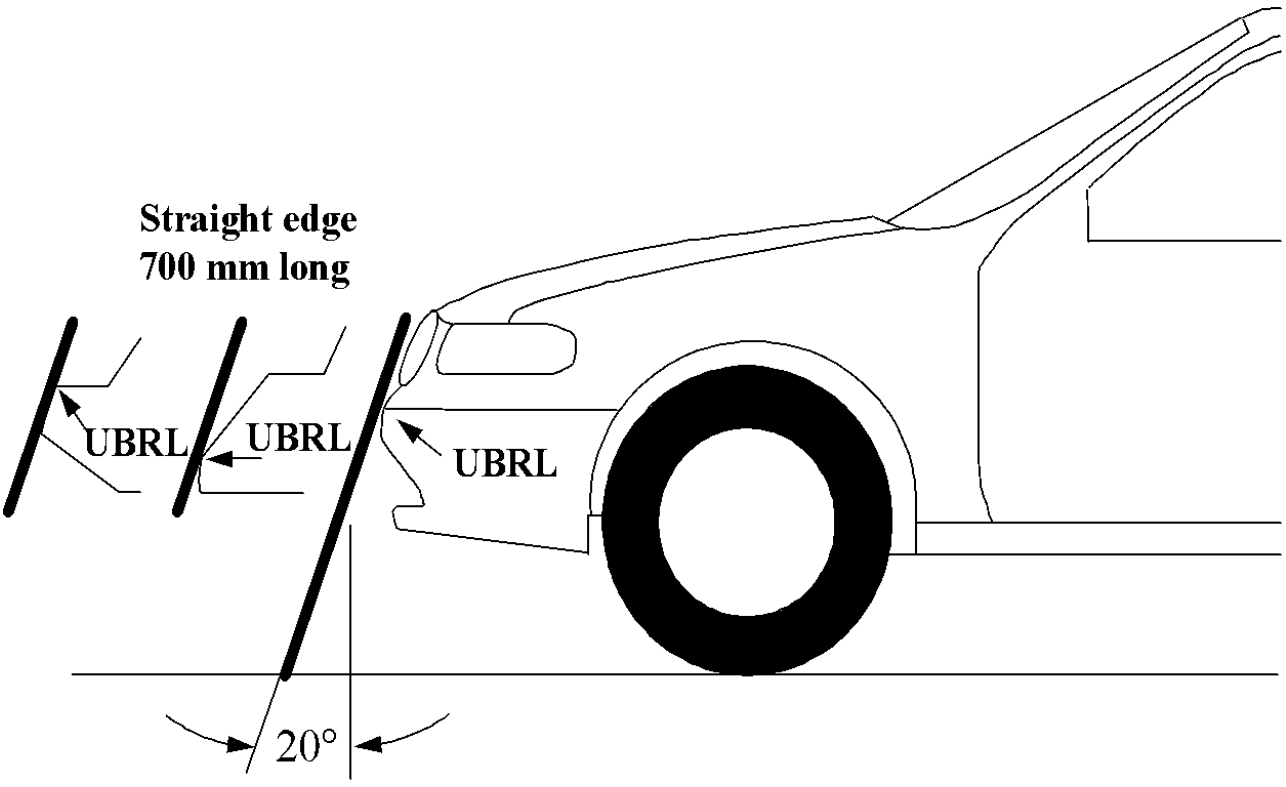


Figure 10 **Wrap around distance measurement (see paragraph 3.44.)**

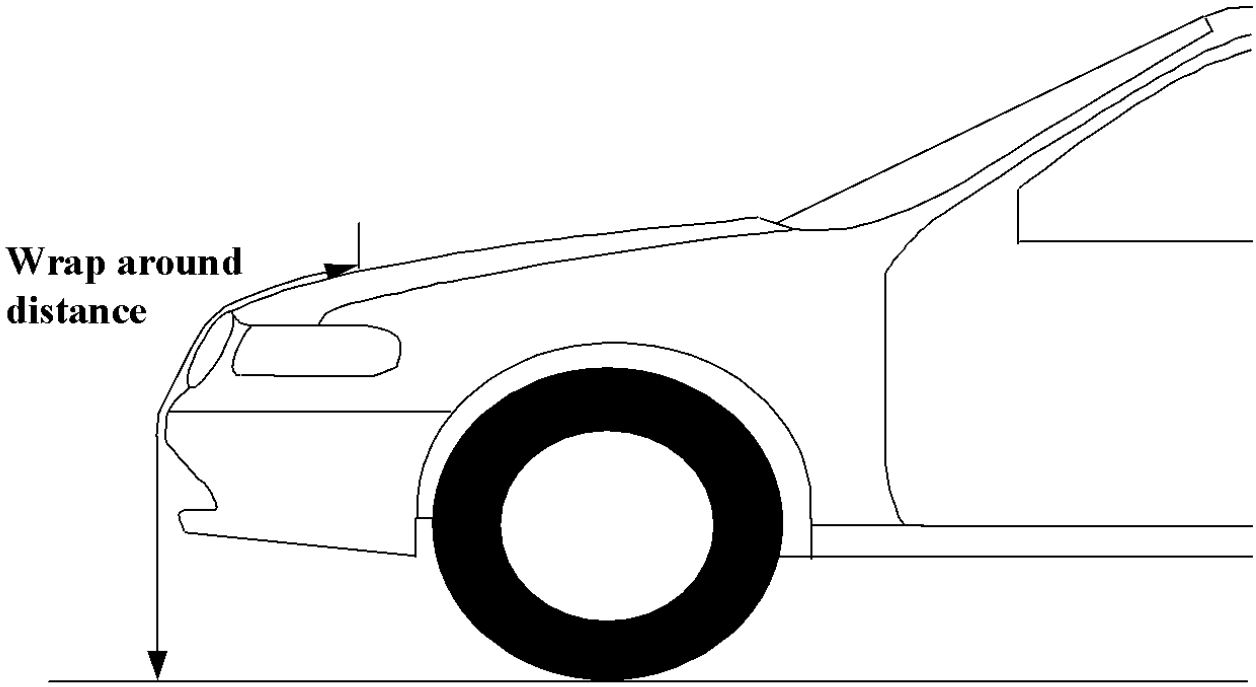
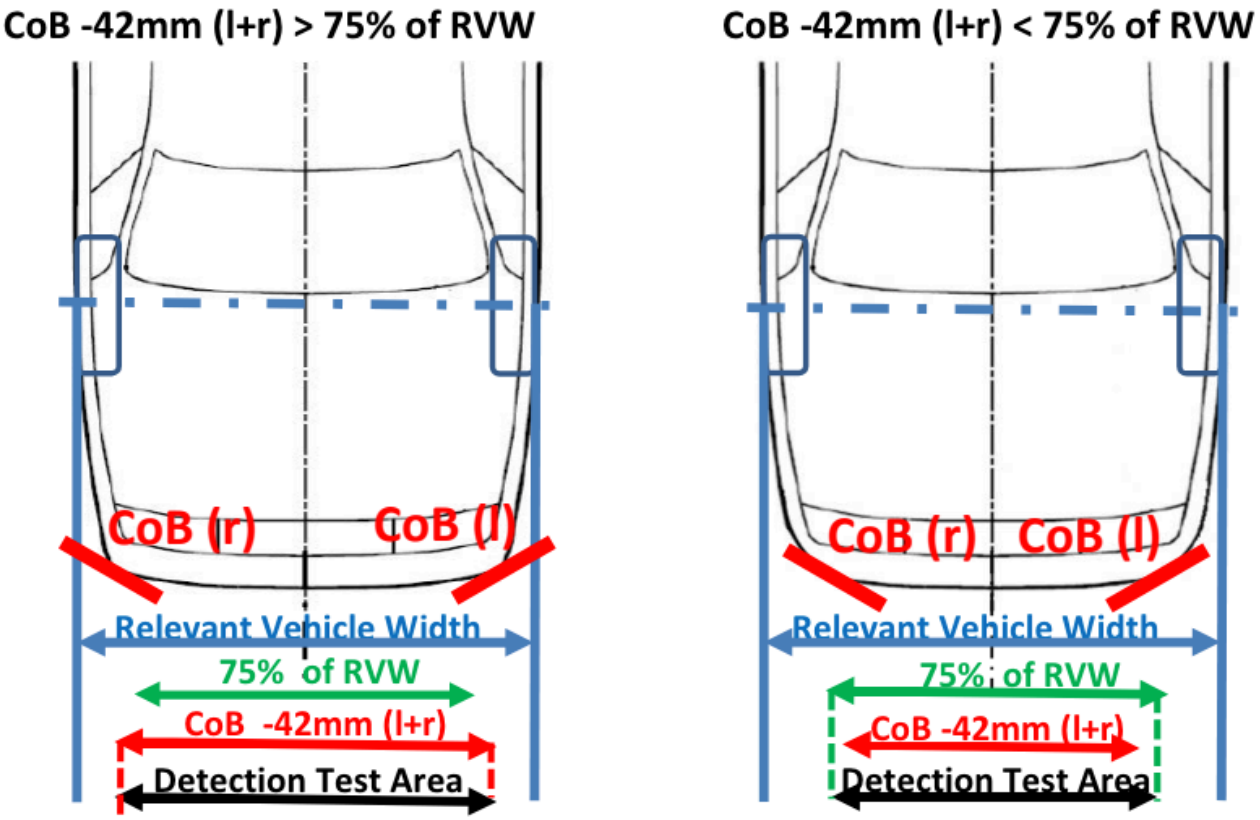


Figure 11 **Determination of the Detection Test Area (see paragraph 3.19): examples**



4. GENERAL REQUIREMENTS

This global technical regulation specifies the following tests to verify compliance of vehicles.

4.1. Legform test to bumper:

For vehicles with a lower bumper height of less than 425 mm the requirements of paragraph 4.1.1. shall be applied.

For vehicles with a lower bumper height which is greater than, or equal to 425 mm and less than 500 mm the requirements of either paragraph 4.1.1. or 4.1.2., at the choice of the manufacturer, shall be applied.

For vehicles with a lower bumper height of greater than, or equal to, 500 mm the requirements of paragraph 4.1.2. shall be applied.

4.1.1. Lower legform to bumper:

To verify compliance with the performance requirements as specified in paragraph 5.1.1., both the test impactor specified in paragraph 6.3.1.1. and the test procedures specified in paragraph 7.1.1. shall be used.

4.1.2. Upper legform to bumper:

To verify compliance with the performance requirements as specified in paragraph 5.1.2., both the test impactor specified in paragraph 6.3.1.2. and the test procedures specified in paragraph 7.1.2. shall be used.

4.2. Child headform impact:

To verify compliance with the performance requirements as specified in paragraph 5.2.1., both the test impactor specified in paragraph 6.3.2.1. and the test procedures specified in paragraphs 7.2. and 7.3. shall be used.

4.3. Adult headform impact:

To verify compliance with the performance requirements as specified in paragraph 5.2.2., both the test impactor specified in paragraph 6.3.2.2. and the test procedures specified in paragraphs 7.2. and 7.4. shall be used.

5. PERFORMANCE REQUIREMENTS

5.1. Legform to bumper:

5.1.1. When tested in accordance with paragraph 7.1.1. (flexible lower legform to bumper), the absolute value of the maximum dynamic medial collateral ligament elongation at the knee shall not exceed 22 mm, and the maximum dynamic anterior cruciate ligament and posterior cruciate ligament elongation shall not exceed 13 mm. The absolute value of dynamic bending moments at the tibia shall not exceed 340 Nm. In addition, the manufacturer may nominate bumper test widths up to a maximum of 264 mm in total where the absolute value of the tibia bending moment shall not exceed 380 Nm. A Contracting Party may restrict application of the relaxation zone requirement in its domestic legislation if it decides that such restriction is appropriate.

5.1.2. When tested in accordance with paragraph 7.1.2. (upper legform to bumper), the instantaneous sum of the impact forces with respect to time shall not exceed 7.5 kN and the bending moment on the test impactor shall not exceed 510 Nm.

5.2. Headform tests

When tested as a DPPS, the test conditions and requirements in Annex 1 shall apply.

5.2.1. Child headform to the front structure:

When tested in accordance with paragraphs 7.2., 7.3. and, if applicable, Annex 1, the HIC shall comply with paragraph 5.2.3.

5.2.2. Adult headform to the front structure:

When tested in accordance with paragraph 7.2., 7.4. and, if applicable, Annex 1, the HIC shall comply with paragraph 5.2.3.

5.2.3. The HIC recorded shall not exceed 1,000 over a minimum of one half of the child headform test area and 1,000 over two thirds of the combined child and adult headform test areas. The HIC for the remaining areas shall

not exceed 1,700 for both headforms.

In case there is only a child headform test area, the HIC recorded shall not exceed 1,000 over two thirds of the test area. For the remaining area the HIC shall not exceed 1,700.

5.2.4. Splitting of headform test zone

5.2.4.1. The manufacturer shall identify the zones of the bonnet top where the HIC must not exceed 1,000 (HIC1000 Zone) or 1,700 (HIC1700 Zone) (see Figure 11).

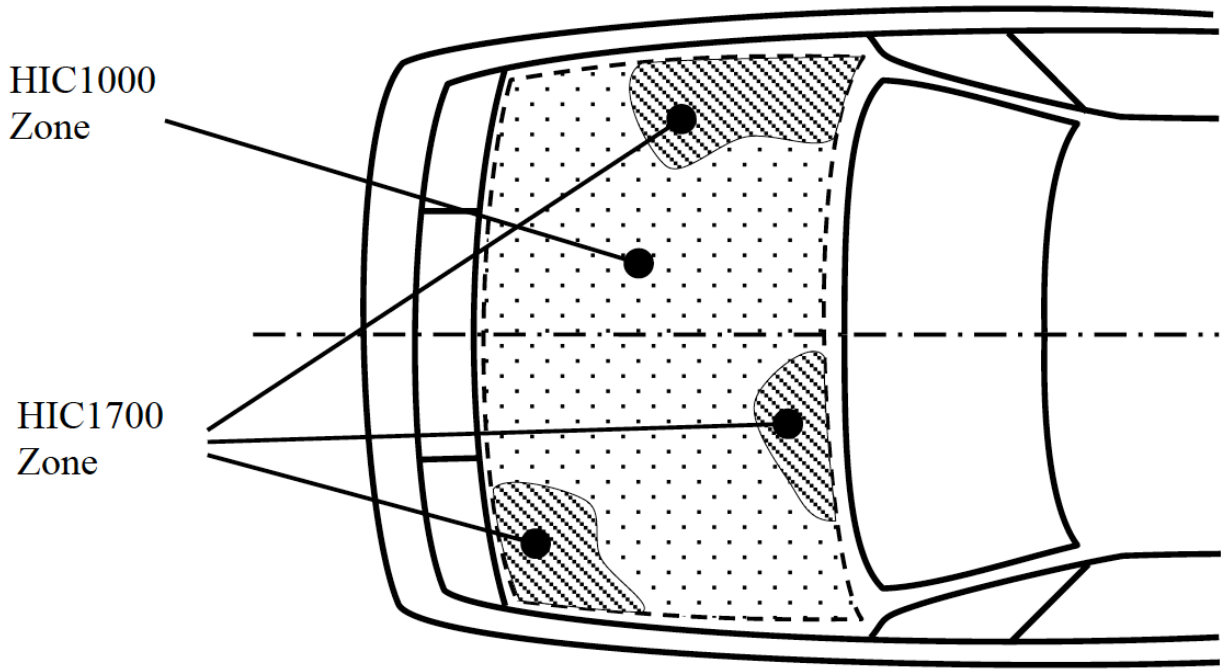


Figure 12: Example of marking of HIC1000 zone and HIC1700 zone

5.2.4.2. Marking of the "bonnet top" impact area as well as "HIC1000 Zone" and "HIC1700 Zone" will be based on a drawing supplied by the manufacturer, when viewed from a horizontal plane above the vehicle that is parallel to the vehicle horizontal zero plane. A sufficient number of x and y co-ordinates shall be supplied by the manufacturer to mark up the areas on the actual vehicle while considering the vehicle outer contour in the z direction.

5.2.4.3. The areas of “HIC1000 Zone” and “HIC1700 Zone” may consist of several parts, with the number of these parts not being limited. The determination of the impacted zone is done by the first contact point of the head form with the “bonnet top."

5.2.4.4. The calculation of the surface of the impact area as well as the surface areas of "HIC1000 Zone" and "HIC1700 Zone" shall be done on the basis of a projected bonnet when viewed from a horizontal plane parallel to the horizontal zero plane above the vehicle, on the basis of the drawing data supplied by the manufacturer.

6. TEST SPECIFICATIONS

6.1. General test conditions

6.1.1. Temperature and humidity

At the time of testing, the test facility and the vehicle or sub-system shall have a relative humidity of 40 percent ± 30 percent and stabilized temperature of 20 ± 4 ºC.

6.1.2. Impact test site

The test site shall consist of a flat, smooth and hard surface with a slope not exceeding 1 percent.

6.2. Preparation of the vehicle

6.2.1. Either a complete vehicle, or a cut-body, adjusted to the following conditions shall be used for the test.

6.2.1.1. The vehicle shall be in its normal ride attitude, and shall be either securely mounted on raised supports or at rest on a flat horizontal surface with the parking brake applied.

6.2.1.2. The cut-body shall include, in the test, all parts of the vehicle front structure, all under-bonnet components and all components behind the windscreen that may be involved in a frontal impact with a vulnerable road user, to demonstrate the performance and interactions of all the contributory vehicle components. The cut-body shall be securely mounted in the normal vehicle ride attitude.

6.2.2. All devices designed to protect vulnerable road users when impacted by the vehicle shall be correctly activated before and/or be active during the relevant test. It shall be the responsibility of the manufacturer to show that any devices will act as intended in a pedestrian impact.

6.2.3. For vehicle components which could change shape or position, other than active devices to protect pedestrians, and which have more than one fixed shape or position shall require the vehicle to comply with the components in each fixed shape or position.

6.2.4. When tested as a DPPS, the vehicle shall be adjusted as specified in the test procedure defined in Annex 1.

6.3. Test impactor specifications

6.3.1. Legform impactors:

6.3.1.1. Flexible lower legform impactor:

The flexible lower legform impactor shall consist of the flesh and skin, flexible long bone segments (representing femur bone and tibia bone), and the knee joint as shown in Figure 12.[2]

The assembled impactor shall have a total mass of 13.2 ± 0.4 kg. The dimensions of the fully assembled impactor shall be as defined in Figure 12, measured at the vertical centre line.

Brackets, pulleys, protectors, connection parts, etc. attached to the impactor for the purposes of launching and/or protection may extend beyond the dimensions and tolerances shown in Figures 12 and 13.

[2] Technical specifications, including detailed drawings and procedures for assembly/disassembly of the flexible lower legform impactor are specified in Addendum 3 of Mutual Resolution No. 1.

6.3.1.1.1. The cross-sectional shape of the femur main body segments, the tibia main body segments and their impact faces shall be as defined in Figure 13 (a).

6.3.1.1.2. The cross-sectional shape of the knee joint and its impact face shall be as defined in Figure 13 (b).

6.3.1.1.3. The masses of the femur and the tibia without the flesh and skin, including the connection parts to the knee joint, shall be 2.46 ± 0.12 kg and 2.64 ± 0.13 kg respectively. The mass of the knee joint without the flesh and skin shall be 4.28 ± 0.21 kg. The assembled mass of the femur, the knee joint and the tibia without the flesh and skin shall be 9.38 ± 0.3 kg. The screws that attach femur and tibia to the knee are part of the knee assembly.

The centres of gravity of the femur and tibia without the flesh and skin, including the connection parts to the knee joint, shall be as defined in Figure 12. The centre of gravity of the knee joint shall be as defined in Figure 12.

The moment of inertia of the femur and the tibia without the flesh and skin, including the connection parts inserted to the knee joint, about the X-axis through the respective centre of gravity shall be 0.0339 ± 0.0016 kgm² and 0.0486 ± 0.0023 kgm² respectively. The moment of inertia of the knee joint about the X-axis through the respective centre of gravity shall be 0.0180 ± 0.0009 kgm².

6.3.1.1.4. For each test, the impactor (femur, knee joint and tibia without flesh and skin) shall be covered by the flesh and skin composed of synthetic rubber sheets (R1, R2) and foamed neoprene sheets (N1F, N2F, N1T, N2T,

N3) as shown in Figure 14. The size of the sheets shall be within the requirements described in Figure 14. The sheets are required to have compression characteristics as shown in Figure 15. The compression characteristics shall be checked using material from the same batch as the sheets used for the impactor flesh and skin.

6.3.1.1.5. All impactor components shall be stored for a sufficient period of time in a controlled storage area with a stabilized temperature of 20 ± 4°C prior to impactor removal for testing. After removal from the storage, the impactor shall not be subjected to conditions other than those in the test area as defined in paragraph 6.1.1.

6.3.1.1.6. Lower legform instrumentation

6.3.1.1.6.1. Four transducers shall be installed in the tibia to measure bending moments at locations within the tibia.

Three transducers shall be installed in the femur to measure bending moments applied to the femur. The sensing locations of each of the transducers are as defined in Figure 16.

6.3.1.1.6.2. Three transducers shall be installed in the knee joint to measure elongations of the Medial Collateral Ligament (MCL), Anterior Cruciate Ligament (ACL), and Posterior Cruciate Ligament (PCL). The measurement locations of each transducer are shown in Figure 16. The measurement locations shall be within ± 4 mm along the X-axis from the knee joint centre.

6.3.1.1.6.3. The instrumentation response value Channel Frequency Class (CFC), as defined in ISO 6487:2002, shall be 180 for all transducers. The Channel Amplitude Class (CAC) response values, as defined in ISO 6487:2002, shall be 30 mm for the knee ligament elongations and 400 Nm for the tibia and femur bending moments. This does not require that the impactor itself is able to physically elongate or bend until these values.

6.3.1.1.6.4. The determination of all flexible lower legform impactor peak tibia bending moments and ligament elongations shall be limited to the assessment interval (AI) as defined in paragraph 3.24.

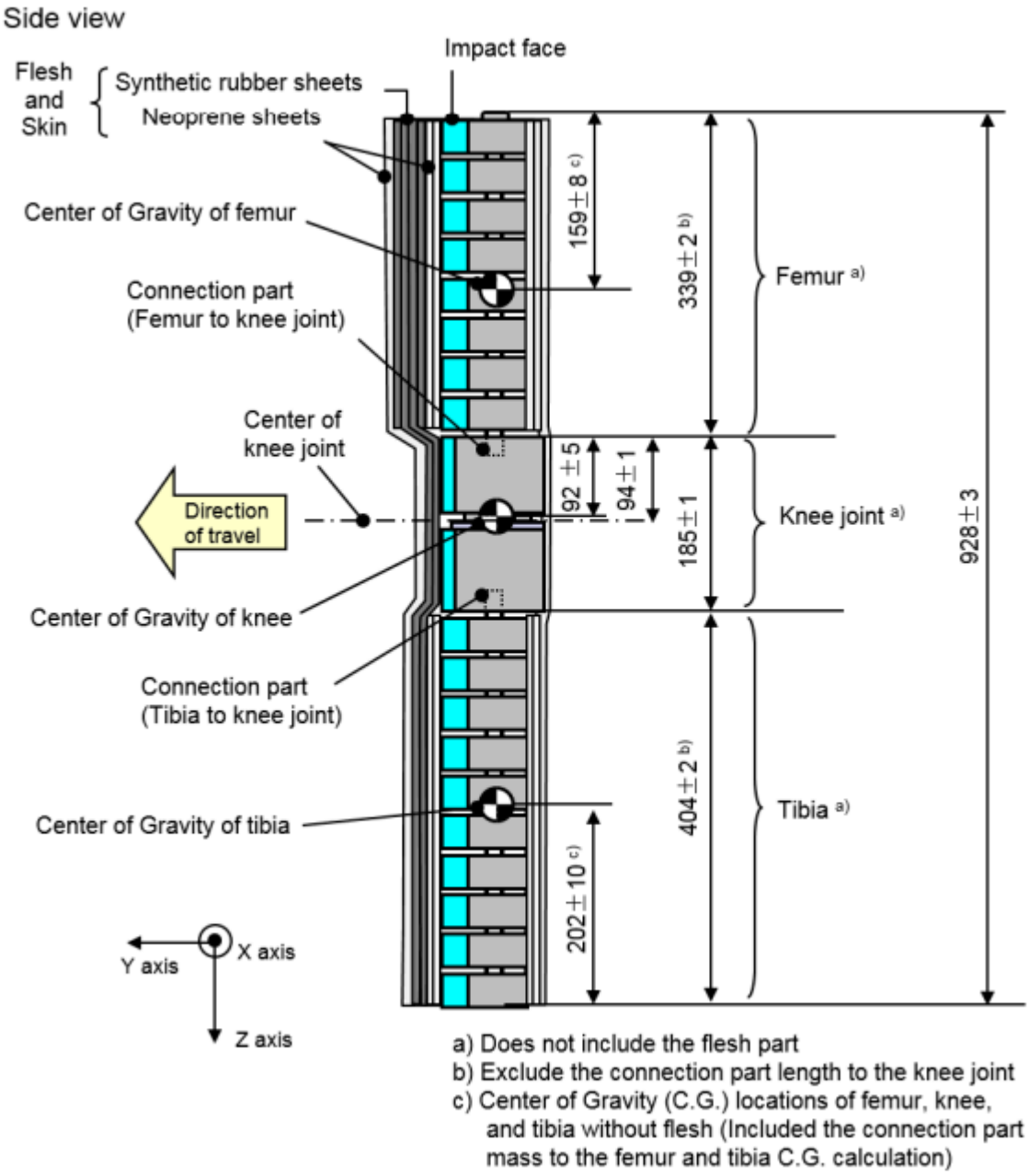
6.3.1.1.7. Flexible lower legform impactor certification

6.3.1.1.7.1. The flexible lower legform impactor shall meet the performance requirements specified in paragraph 8.

6.3.1.1.7.2. The impactor shall be certified using two certification tests as follows: First, the certification shall be conducted according to the inverse certification (IC) test procedure prescribed in paragraph 8.1.3 before starting a vehicle test series. Second, after a maximum of 10 vehicle tests, certification should be conducted according to the pendulum certification (PC) test procedure prescribed in paragraph 8.1.2. Ongoing certification testing then shall constitute the sequence IC – PC – PC – IC – PC – PC – etc. with a maximum of 10 tests between each certification.

In addition, the impactor shall be certified according to the procedures prescribed in paragraph 8.1. at least once a year.

Figure 13 **Flexible lower legform impactor: Dimensions and centre of gravity locations of femur, knee joint and tibia (side view)**



All dimensions in millimetres (mm)

Figure 14 **Flexible lower legform impactor: Schematic plan views of femur, tibia and knee dimensions (top view, main body segments)**

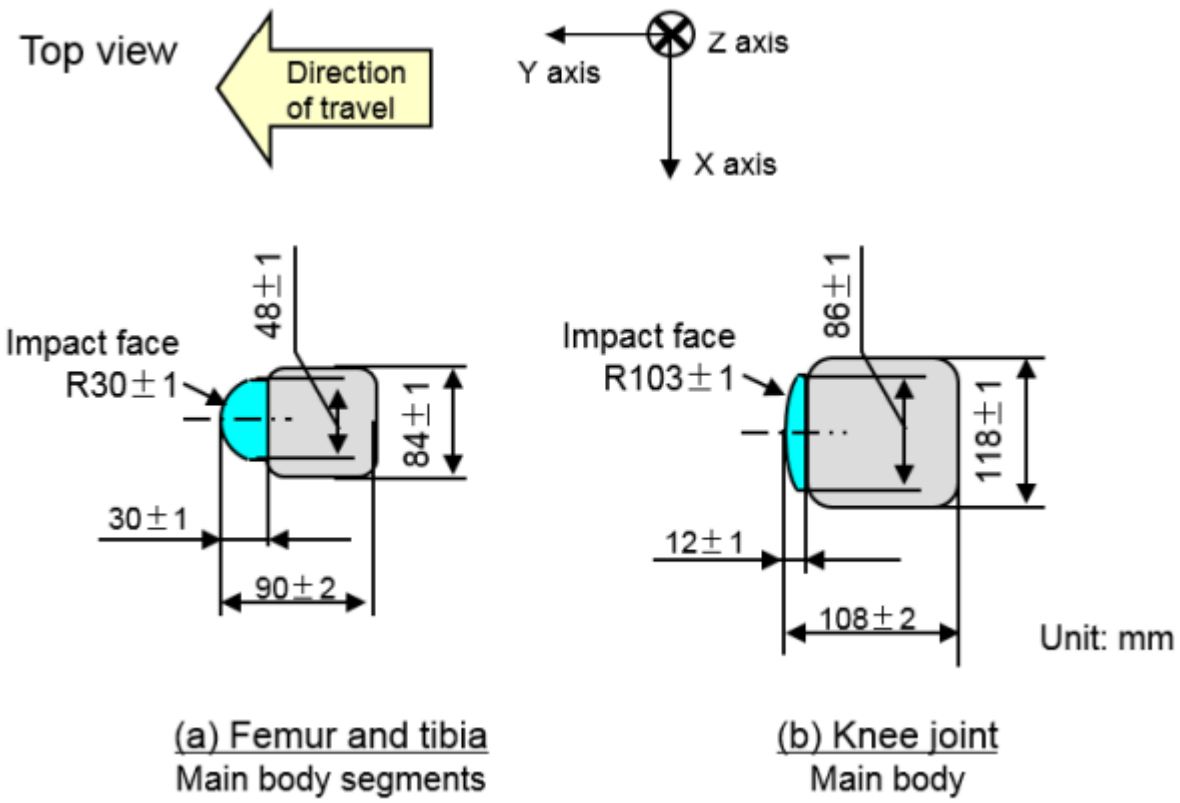


Figure 15 **Flexible lower legform impactor: Flesh and skin dimensions**

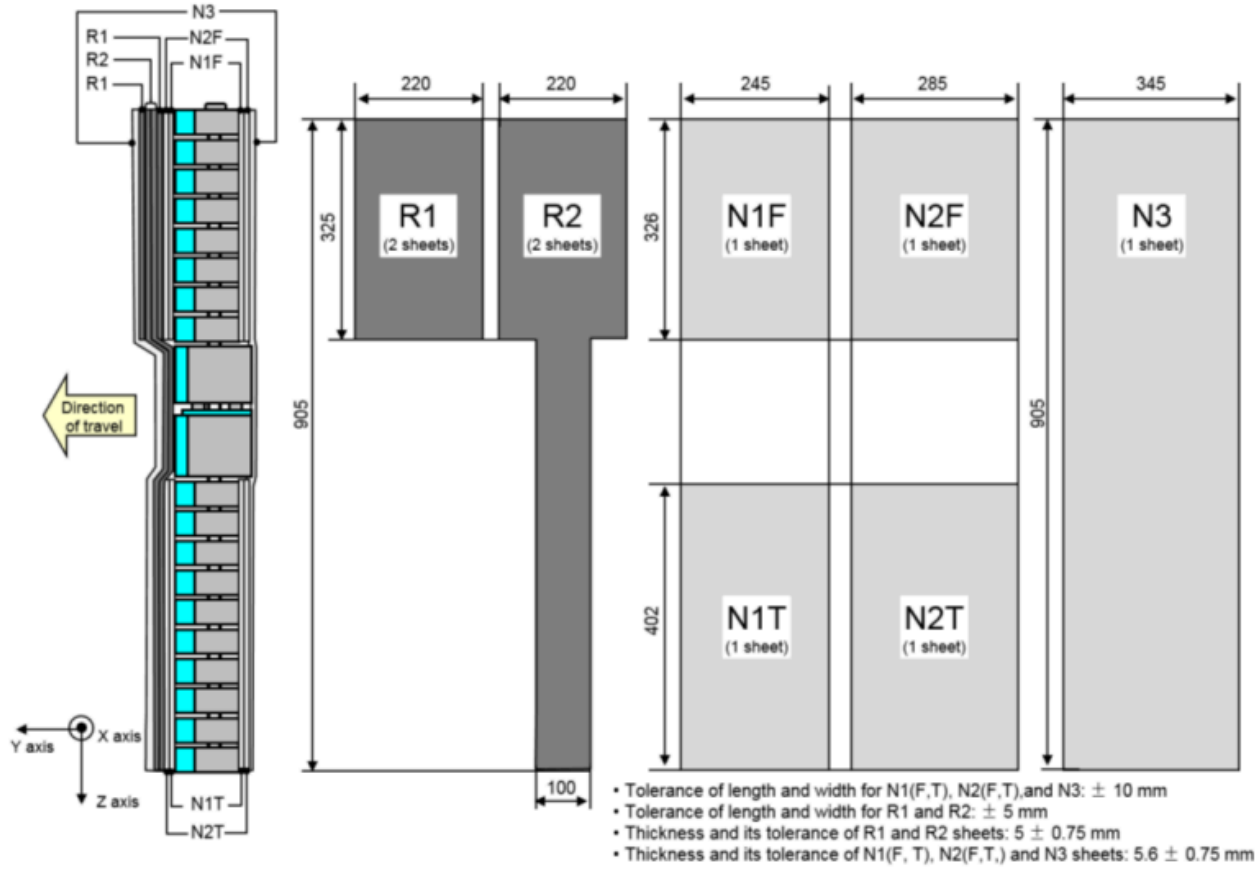
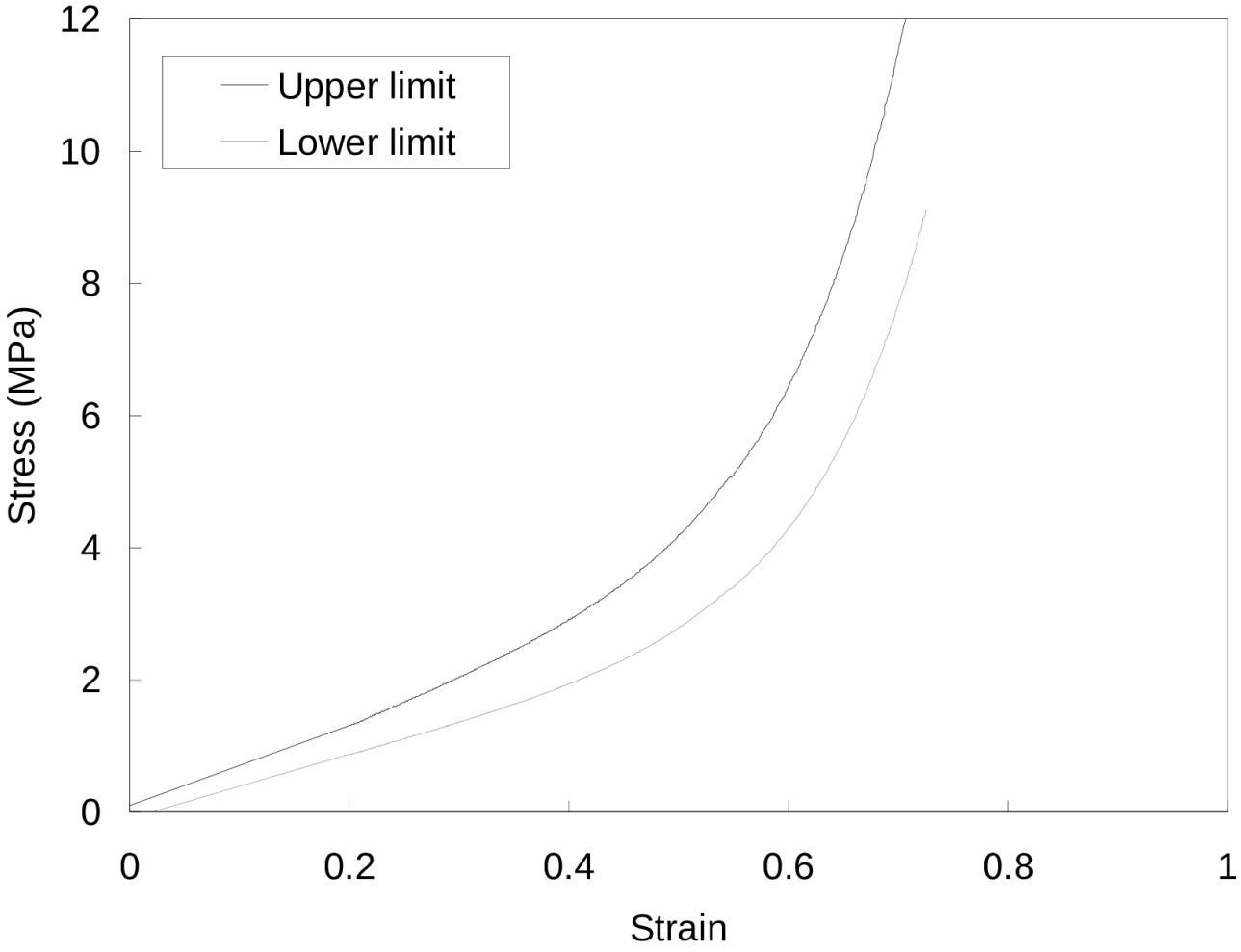


Figure 16 **Flexible lower legform impactor: Flesh and skin compression characteristics**

(a) Synthetic rubber sheets



(b) Foamed neoprene sheets

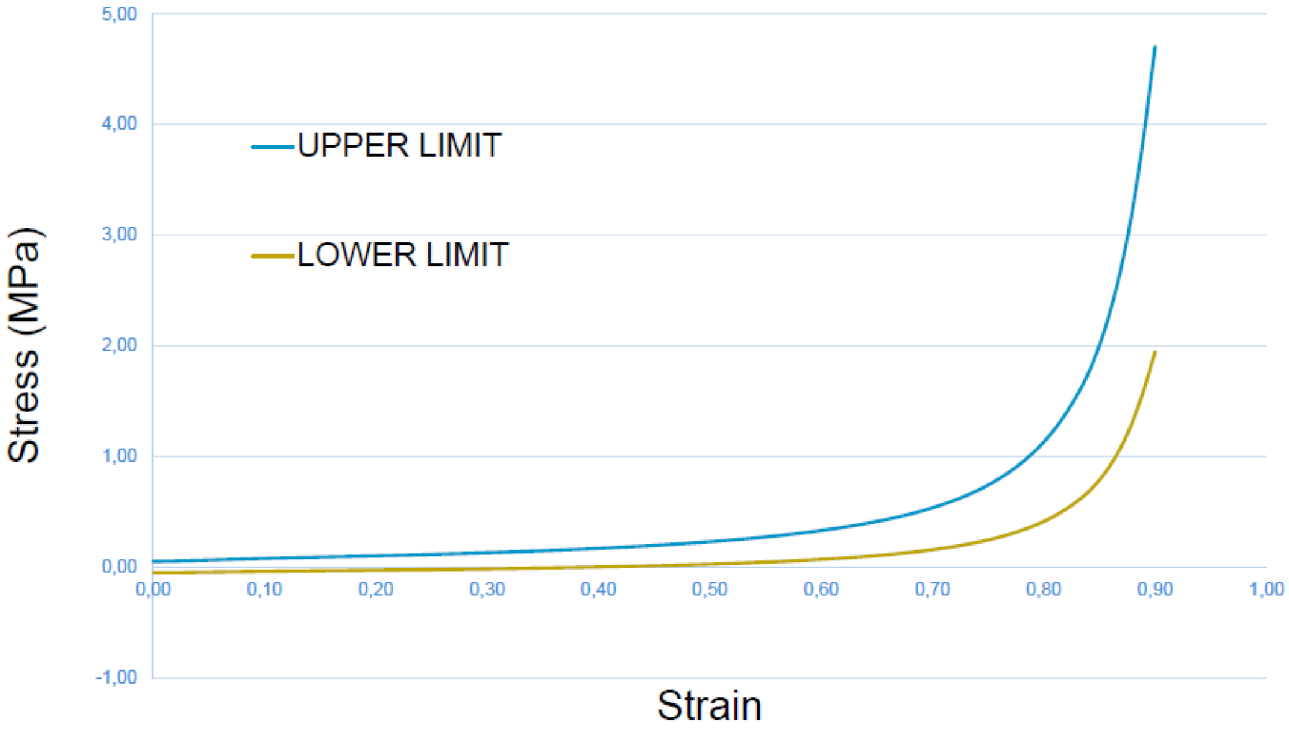
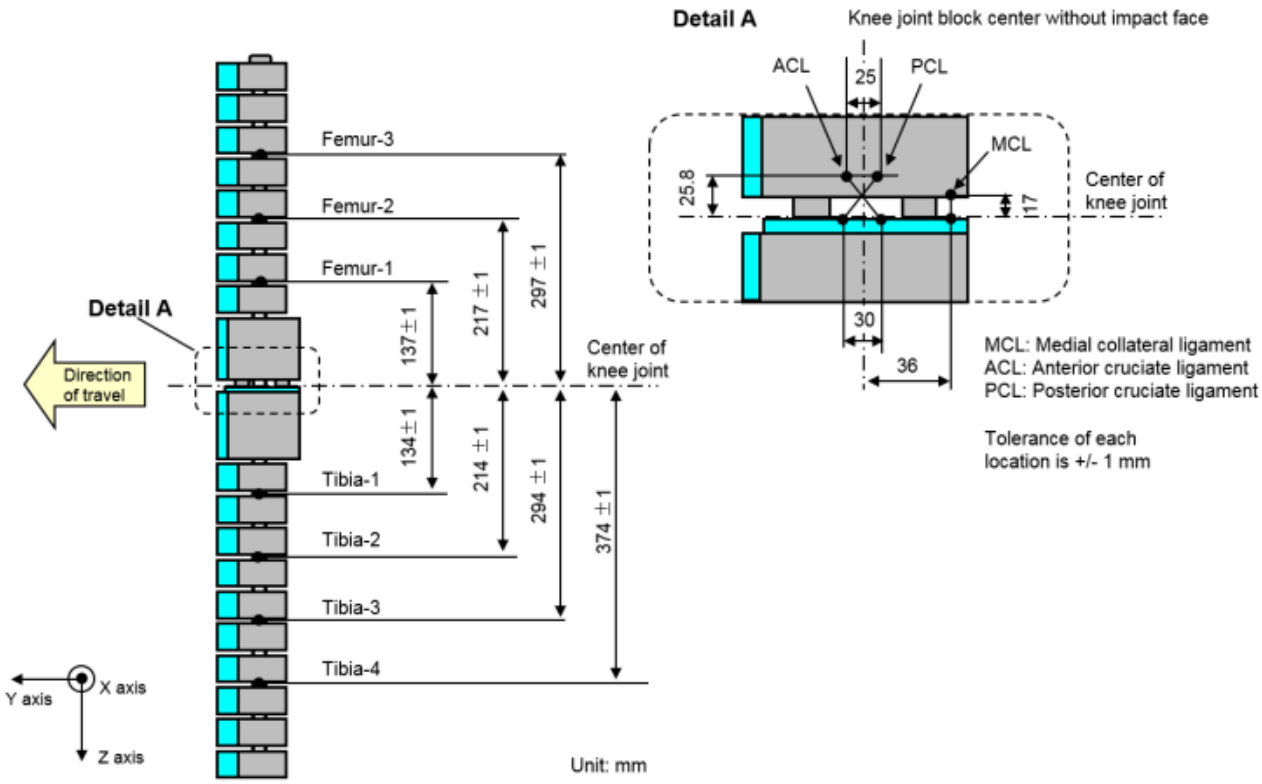


Figure 17 **Flexible lower legform impactor: Location of the transducers**



6.3.1.2. Upper legform impactor:

The upper legform impactor shall be rigid, foam covered at the impact side, and 350 ± 5 mm long (see Figure 17).

6.3.1.2.1. The total mass of the upper legform impactor including those propulsion and guidance components which are effectively part of the impactor during the impact shall be 9.5 kg ± 0.1 kg.

6.3.1.2.2. The total mass of the front member and other components in front of the load transducer assemblies, together with those parts of the load transducer assemblies in front of the active elements, but excluding the foam and skin, shall be 1.95 ± 0.05 kg.

6.3.1.2.3. The upper legform impactor for the bumper test shall be mounted to the propulsion system by a torque limiting joint and be insensitive to off-axis loading. The impactor shall move only in the specified direction of impact when in contact with the vehicle and shall be prevented from motion in other directions including rotation about any axis.

6.3.1.2.4. The torque limiting joint shall be set so that the longitudinal axis of the front member is vertical at the time of impact with a tolerance of ± 2°, with the joint friction torque set to a minimum of 650 Nm but no more than 850 Nm.

6.3.1.2.5. The centre of gravity of those parts of the impactor which are effectively forward of the torque limiting joint, including any weights fitted, shall lie on the longitudinal centre line of the impactor, with a tolerance of ± 10 mm.

6.3.1.2.6. The length between the load transducer centre lines shall be 310 ± 1 mm and the front member diameter shall be 50 ± 1 mm.

6.3.1.2.7. For each test the foam flesh shall be two new sheets of 25 mm thick foam type CF45 or equivalent, which shall be cut from the sheet of material used for the dynamic certification test. The skin shall be a 1.5 mm thick fibre reinforced rubber sheet. The mass of the foam and the rubber skin together shall be 0.6 ± 0.1 kg (this excludes any reinforcement, mountings, etc. which are used to attach the rear edges of the rubber skin to the rear member). The foam and rubber skin shall be folded back towards the rear, with the rubber skin attached via

spacers to the rear member so that the sides of the rubber skin are held parallel. The foam shall be of such a size and shape that an adequate gap is maintained between the foam and components behind the front member, to avoid significant load paths between the foam and these components.

6.3.1.2.8. The test impactor or at least the foam flesh shall be stored during a period of at least four hours in a controlled storage area with a stabilized humidity of 35 percent ± 15 percent and a stabilized temperature of 20 ± 4 °C prior to impactor removal for test. After removal from the storage the impactor shall not be subjected to conditions other than those pertaining in the test area.

6.3.1.2.9. Upper legform instrumentation

6.3.1.2.9.1. The front member shall be strain gauged to measure bending moments in three positions, as shown in Figure 17, each using a separate channel. The strain gauges are located on the impactor on the back of the front member. The two outer strain gauges are located 50 ± 1 mm from the impactor's symmetrical axis. The middle strain gauge is located on the symmetrical axis with a ± 1 mm tolerance.

6.3.1.2.9.2. Two load transducers shall be fitted to measure individually the forces applied at either end of the upper legform impactor, plus strain gauges measuring bending moments at the centre of the upper legform impactor and at positions 50 mm either side of the centre line (see Figure 17).

6.3.1.2.9.3. The instrumentation response value CFC, as defined in ISO 6487:2002, shall be 180 for all transducers. The CAC response values, as defined in ISO 6487:2002, shall be 10 kN for the force transducers and 1,000 Nm for the bending moment measurements.

6.3.1.2.10. Upper legform certification

6.3.1.2.10.1. The upper legform impactor shall meet the performance requirements specified in paragraph 8.

6.3.1.2.10.2. The certified impactor may be used for a maximum of 20 impacts before re-certification (this limit does not apply to propulsion or guidance components). The impactor shall also be re-certified if more than one year has elapsed since the previous certification or if any impactor transducer output, in any impact, has exceeded the specified CAC.

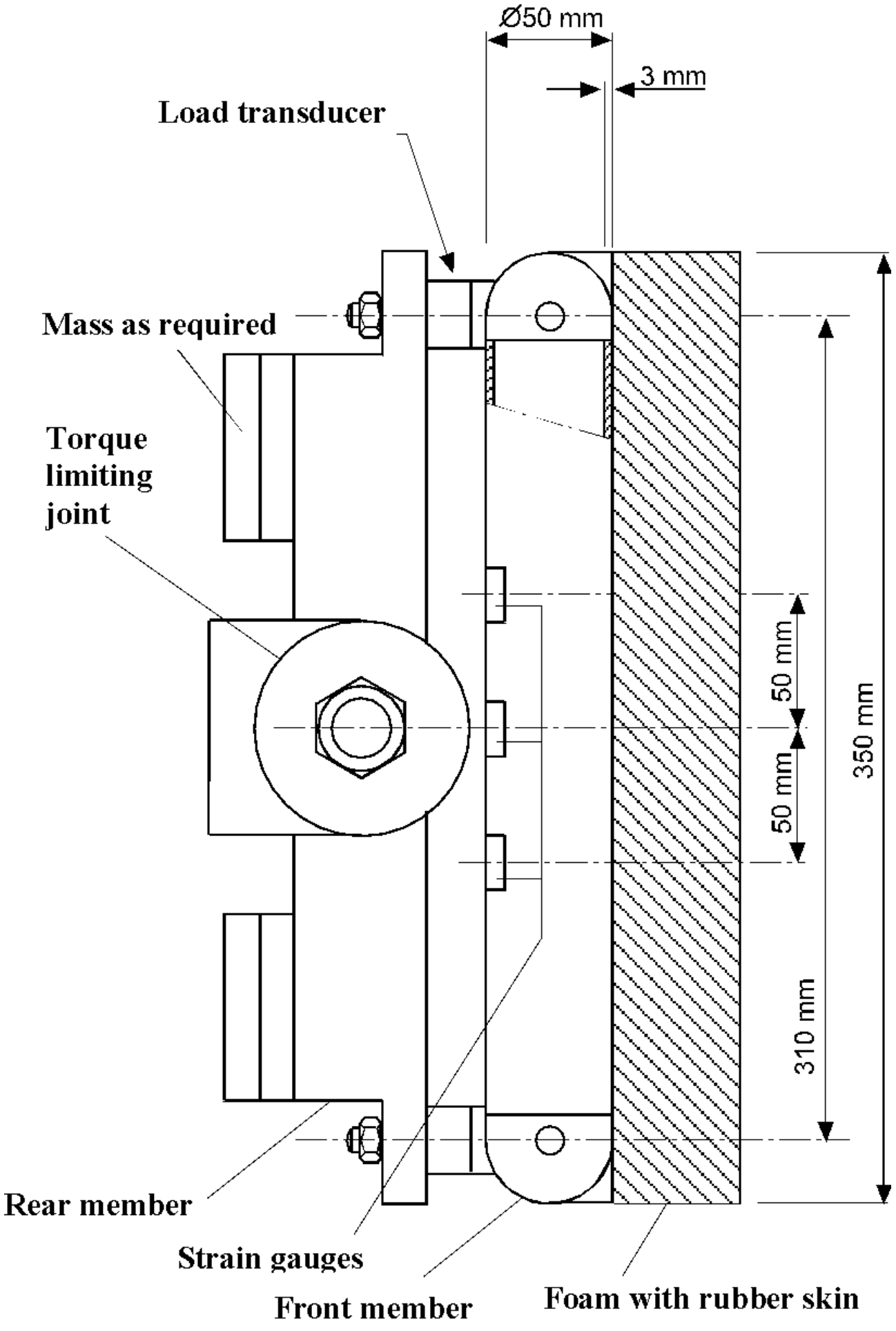

Ø50 mm
3 mm
Load transducer
Mass as required
Torque limiting joint
50 mm
50 mm
350 mm
310 mm
Rear member
Strain gauges
Front member
Foam with rubber skin

Figure 18: Upper legform impactor (see paragraph 6.3.1.2.)

6.3.2. Child and adult headform impactors

6.3.2.1. Child headform impactor (see Figure 18)

The child headform impactor shall be made of aluminium, be of homogenous construction and be of spherical shape. The overall diameter shall be 165 ± 1 mm. The mass shall be 3.5 ± 0.07 kg. The moment of inertia about an axis through the centre of gravity and perpendicular to the direction of impact shall be within the range of 0.008 to 0.012 kgm^2. The centre of gravity of the headform impactor including instrumentation shall be located in the geometric centre of the sphere with a tolerance of ± 2 mm.

The sphere shall be covered with a 14 ± 0.5 mm thick synthetic skin, which shall cover at least half of the sphere.

6.3.2.1.1. Child headform instrumentation

A recess in the sphere shall allow for mounting one triaxial or three uniaxial accelerometers within ± 10 mm seismic mass location tolerance from the centre of the sphere for the measurement axis, and ± 1 mm seismic mass location tolerance from the centre of the sphere for the perpendicular direction to the measurement axis.

If three uniaxial accelerometers are used, one of the accelerometers shall have its sensitive axis perpendicular to the mounting face A (see Figure 18) and its seismic mass shall be positioned within a cylindrical tolerance field of 1 mm radius and 20 mm length. The centre line of the tolerance field shall run perpendicular to the mounting face and its mid-point shall coincide with the centre of the sphere of the headform impactor.

The remaining accelerometers shall have their sensitive axes perpendicular to each other and parallel to the mounting face A and their seismic mass shall be positioned within a spherical tolerance field of 10 mm radius. The centre of the tolerance field shall coincide with the centre of the sphere of the headform impactor.

The instrumentation response value CFC, as defined in ISO 6487: 2002, shall be 1,000. The CAC response value, as defined in ISO 6487:2002, shall be 500g for the acceleration.

6.3.2.1.2. First natural frequency

The first natural frequency of the headform impactor shall be over 5,000 Hz.

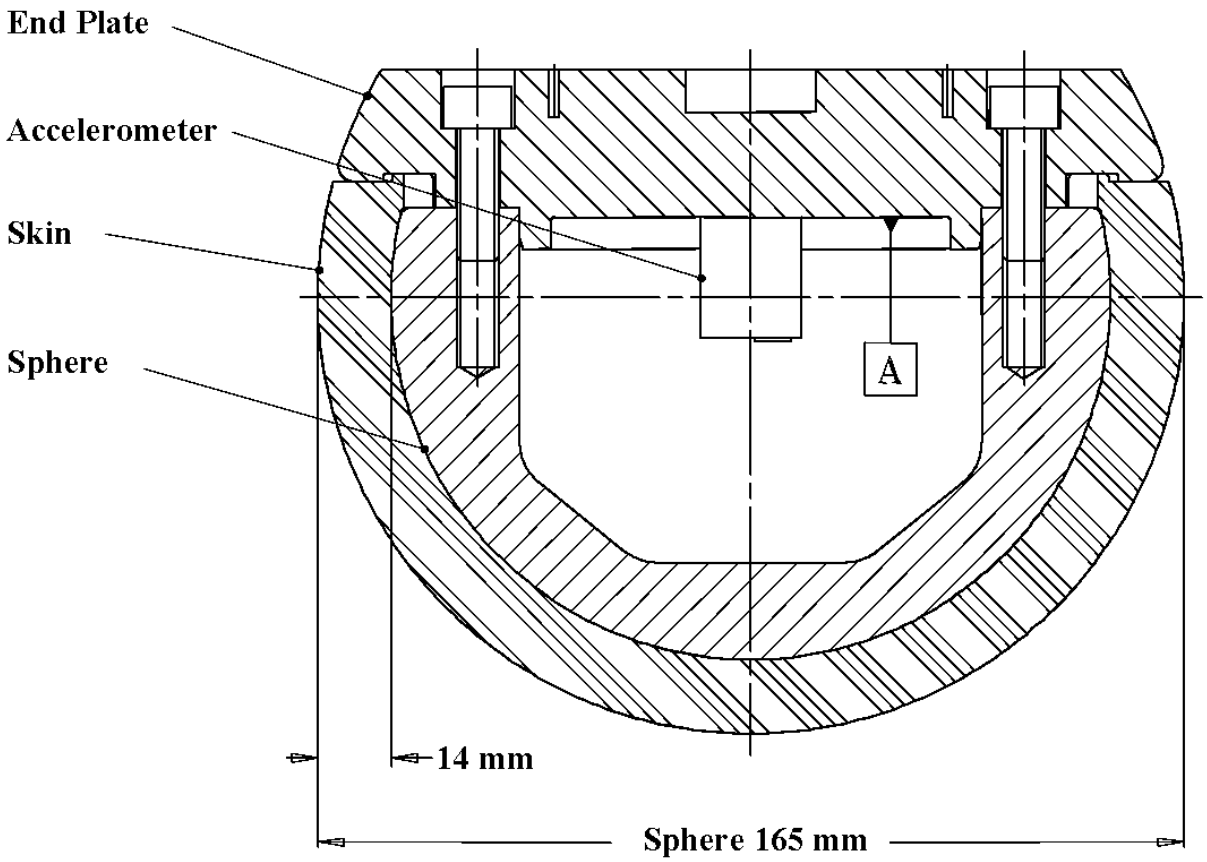

End Plate
Accelerometer
Skin
Sphere
A
14 mm
Sphere 165 mm

Figure 19: Child headform impactor (see paragraph 6.3.2.1.)

6.3.2.2. Adult headform impactor (see Figure 19)

The adult headform impactor shall be made of aluminium, be of homogenous construction and be of spherical shape. The overall diameter is 165 ± 1 mm as shown in Figure 19. The mass shall be 4.5 ± 0.1 kg. The moment of inertia about an axis through the centre of gravity and perpendicular to the direction of impact shall be within the range of 0.010 to 0.013 kgm^2. The centre of gravity of the headform impactor including instrumentation shall be located in the geometric centre of the sphere with a tolerance of ± 5 mm.

The sphere shall be covered with a 14 ± 0.5 mm thick synthetic skin, which shall cover at least half of the sphere.

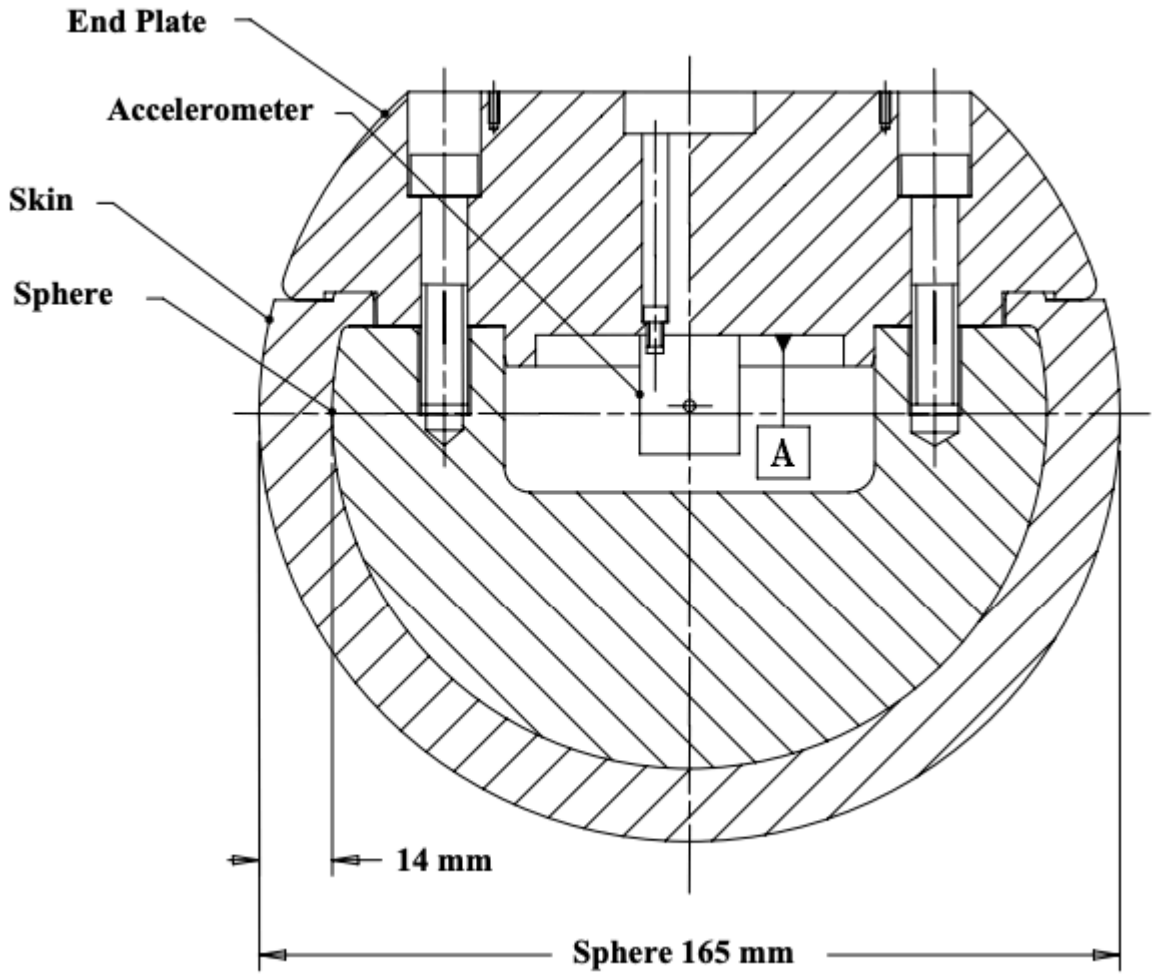

End Plate
Accelerometer
Skin
Sphere
A
14 mm
Sphere 165 mm

Figure 20: Adult headform impactor (see paragraph 6.3.2.2)

6.3.2.2.1. Adult headform instrumentation

A recess in the sphere shall allow for mounting one triaxial or three uniaxial accelerometers within ± 10 mm seismic mass location tolerance from the centre of the sphere for the measurement axis, and ± 1 mm seismic mass location tolerance from the centre of the sphere for the perpendicular direction to the measurement axis.

If three uniaxial accelerometers are used, one of the accelerometers shall have its sensitive axis perpendicular to the mounting face A (see Figure 19) and its seismic mass shall be positioned within a cylindrical tolerance field of 1 mm radius and 20 mm length. The centre line of the tolerance field shall run perpendicular to the mounting face and its mid-point shall coincide with the centre of the sphere of the headform impactor.

The remaining accelerometers shall have their sensitive axes perpendicular to each other and parallel to the mounting face A and their seismic mass shall be positioned within a spherical tolerance field of 10 mm radius. The centre of the tolerance field shall coincide with the centre of the sphere of the headform impactor.

The instrumentation response value CFC, as defined in ISO 6487: 2002, shall be 1,000. The CAC response value, as defined in ISO 6487: 2002, shall be 500g for the acceleration.

6.3.2.2.2. First natural frequency

The first natural frequency of the headform impactor shall be over 5,000 Hz.

6.3.2.3. Rear face of the headform impactors

A rear flat face shall be provided on the outer surface of the headform impactors which is perpendicular to the direction of travel, and typically perpendicular to the axis of one of the accelerometers as well as being a flat plate capable of providing for access to the accelerometers and an attachment point for the propulsion system.

6.3.2.4. Certification of the headform impactors

The headform impactors shall meet the performance requirements specified in paragraph 8. The certified impactors may be used for a maximum of 20 impacts before re-certification. The impactors shall be re-certified if more than one year has elapsed since the previous certification or if the transducer output, in any impact, has exceeded the specified CAC.

7. TEST PROCEDURES

7.1. Legform to bumper test procedures

7.1.1. Flexible lower legform impactor to bumper test procedure:

Each test shall be completed within two hours of when the impactor to be used is removed from the controlled storage area.

7.1.1.1. The selected target points shall be in the bumper test area.

7.1.1.2. The direction of the impact velocity vector shall be in the horizontal plane and parallel to the longitudinal vertical plane of the vehicle. The tolerance for the direction of the velocity vector in the horizontal plane and in the longitudinal plane shall be ± 2° at the time of first contact. The axis of the impactor shall be perpendicular to the horizontal plane, with a roll and pitch angle tolerance of ± 2° in the lateral and longitudinal plane. The horizontal, longitudinal and lateral planes are orthogonal to each other (see Figure 20).

7.1.1.3. The bottom of the impactor (without parts needed for the purposes of launching and/or protection) shall be 75 mm above the ground reference plane at the time of the first contact with the bumper (see Figure 21), with a tolerance of ± 10 mm. When setting the height of the propulsion system, an allowance shall be made for the influence of gravity during the period of free flight of the impactor.

7.1.1.3.1. The lower legform impactor for the bumper tests shall be in 'free flight' at the moment of impact. The impactor shall be released to free flight at such a distance from the vehicle that the test results are not influenced by contact of the impactor with the propulsion system during the rebound of the impactor.

The impactor may be propelled by any means that can be shown to meet the requirements of the test.

7.1.1.3.2. At the time of first contact the impactor shall have the intended orientation about its vertical axis, for the correct operation of its knee joint, with a yaw angle tolerance of ± 5° (see Figure 20).

7.1.1.3.3. At the time of first contact the centre line of the impactor shall be within a ± 10 mm tolerance to the selected impact location.

7.1.1.3.4. During contact between the impactor and the vehicle, the impactor shall not contact the ground or any object which is not part of the vehicle.

7.1.1.4. The impact velocity of the impactor when striking the bumper shall be 11.1 ± 0.2 m/s. The effect of gravity shall be taken into account when the impact velocity is obtained from measurements taken before the time of first contact.

Figure 21 **Tolerances of angles for the flexible lower legform impactor at the time of the first impact…..**

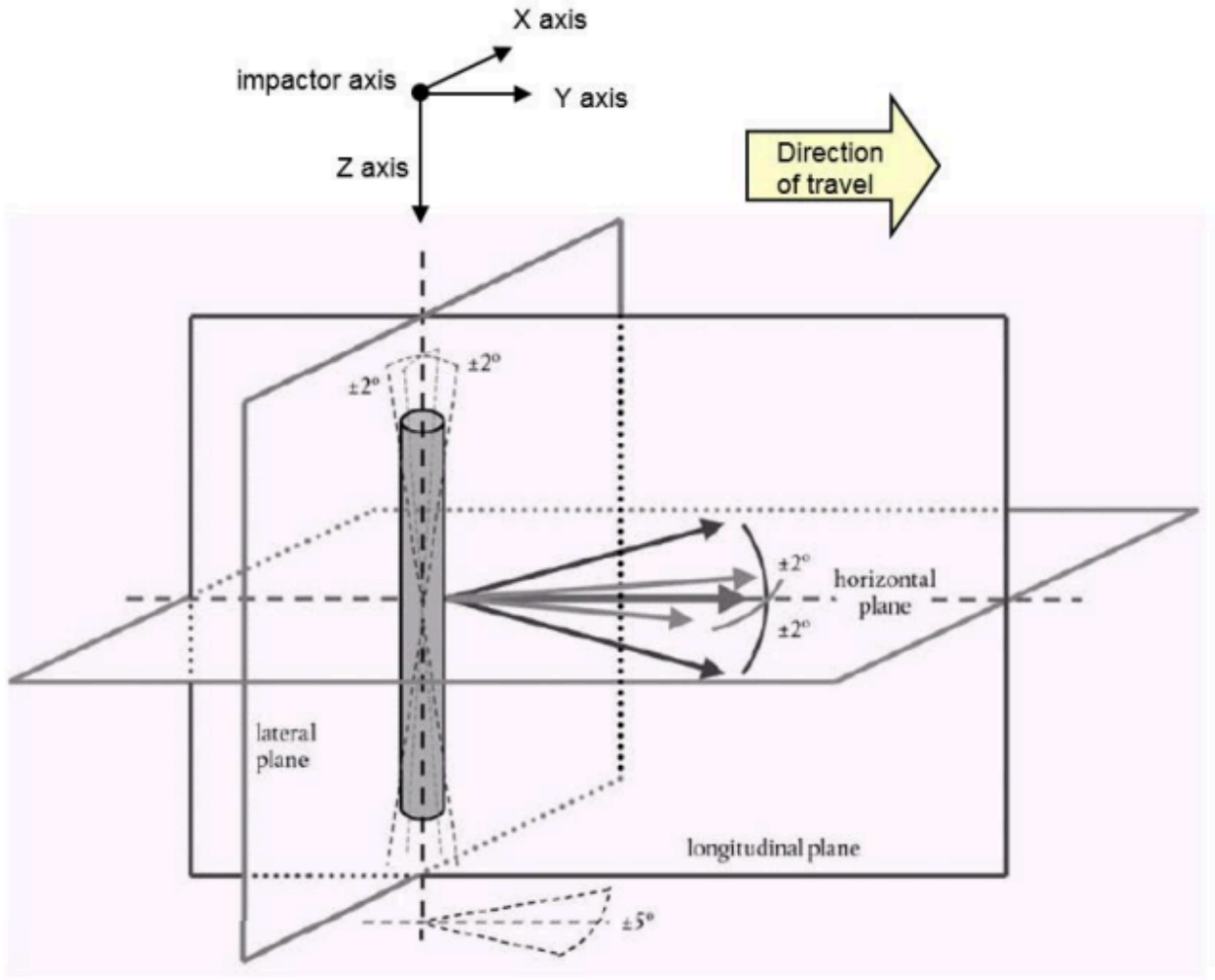
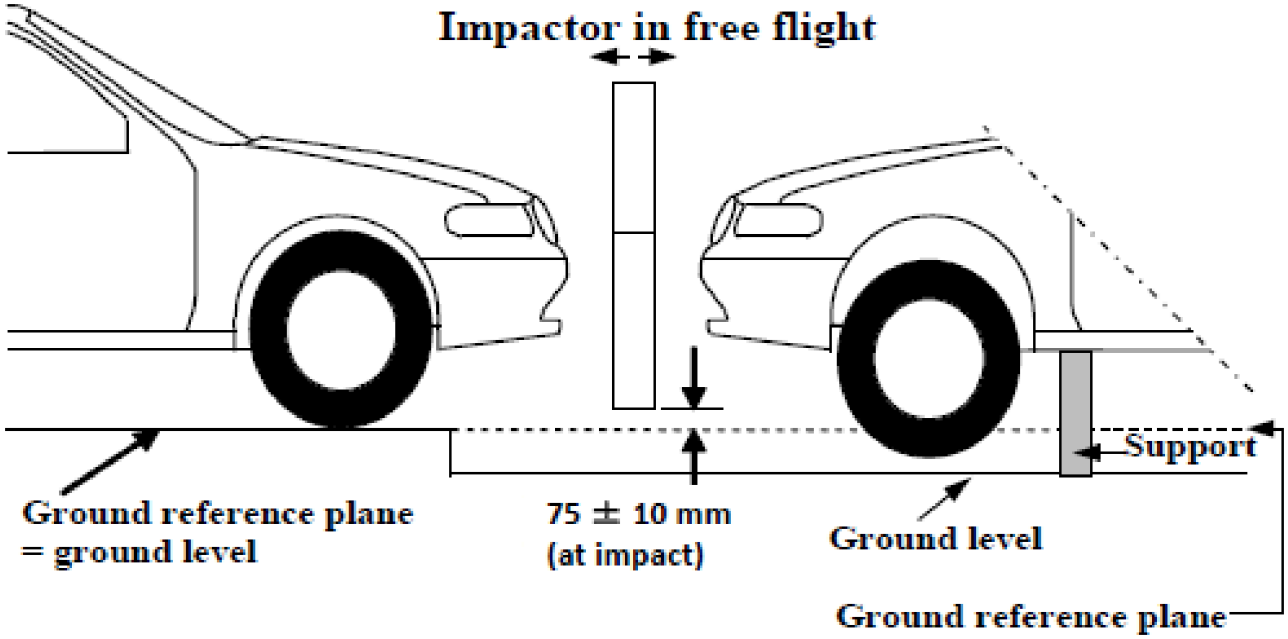


Figure 22 **Flexible lower legform impactor to bumper tests for complete vehicle in normal ride attitude (left) and for cut-body mounted on supports (right) (see paragraph 7.1.1.3.)**



7.1.1.5 The tibia bending moments shall not exceed ± 15 Nm within an evaluation interval of 30 ms immediately prior to impact.

7.1.1.6 The offset compensation shall be done with the flexible lower legform impactor in resting position prior to the test / acceleration phase.

7.1.2. Upper legform to bumper test procedure:

Each test shall be completed within two hours of when the impactor to be used is removed from the controlled storage area.

7.1.2.1. The selected target points shall be in the bumper test area as defined in paragraph 3.10.

7.1.2.2. The direction of impact shall be parallel to the longitudinal axis of the vehicle, with the axis of the upper legform vertical at the time of first contact. The tolerance to this direction is ± 2°.

At the time of first contact the impactor centre line shall be vertically midway between the upper bumper reference line and the lower bumper reference line with a ± 10 mm tolerance and the impactor vertical centre line shall be positioned laterally with the selected impact location with a tolerance of ± 10 mm.

7.1.2.3. The impact velocity of the upper legform impactor when striking the bumper shall be 11.1 ± 0.2 m/s.

7.2. Headform test procedures

7.2.1. Propulsion of the headform impactors

The headform impactors shall be in "free flight" at the moment of impact, at the required impact velocity (as specified in paragraphs 7.3.4. and 7.4.4.) and the required direction of impact (as specified in paragraphs 7.3.5. and 7.4.5.).

The impactors shall be released to "free flight" at such a distance from the vehicle that the test results are not influenced by contact of the impactor with the propulsion system during rebound of the impactor.

7.2.2. Measurement of impact velocity

The velocity of the headform impactor shall be measured at some point during the free flight before impact, in accordance with the method specified in ISO 3784:1976. The accuracy of velocity measurement shall be ± 0.01 m/sec. The measured velocity shall be adjusted considering all factors which may affect the impactor between the point of measurement and the point of impact, in order to determine the velocity of the impactor at the time of impact. The angle of the velocity vector at the time of impact shall be calculated or measured.

7.2.3. Recording

The acceleration time histories shall be recorded, and HIC shall be calculated. The first point of contact on the front structure of the vehicle shall be recorded. Recording of test results shall be in accordance with ISO 6487:2002.

7.3. Child headform test procedure

This test procedure is applicable with respect to the requirements of paragraphs 5.2.1. and 5.2.3.

7.3.1. Tests shall be made to the front structure within the boundaries as defined in paragraph 3.13. For tests on the rear area of the bonnet top, the headform impactor shall not contact the windscreen or A-pillar before impacting the bonnet top.

7.3.2. No impact point shall be located so that the impactor will impact the test area with a glancing blow resulting in a more severe second impact outside the test area.

Selected impact points on the bonnet for the child headform impactor shall be, at the time of first contact:

(1) a minimum of 82.5 mm inside the defined side reference lines, and;

(2) forward of the WAD1700 line, or,

a minimum of 82.5 mm forwards of the bonnet rear reference line,

- whichever is most forward at the point of measurement, and;

(3) be rearward of the WAD1000 line, or,

a minimum of 82.5 mm rearwards of the bonnet leading edge reference line,

- whichever is most rearward at the point of measurement.

These minimum distances are to be set with a flexible tape held tautly along the outer surface of the vehicle.

7.3.3. The point of first contact of the headform impactor shall be within a ± 10 mm tolerance to the selected impact point.

7.3.4. The headform velocity at the time of impact shall be 9.7 ± 0.2 m/s.

7.3.5. The direction of impact shall be in the longitudinal vertical plane of the vehicle to be tested at an angle of 50 ± 2° to the horizontal. The direction of impact of tests to the front structure shall be downward and rearward.

7.4. Adult headform test procedure:

This test procedure is applicable with respect to the requirements of paragraphs 5.2.2. and 5.2.3.

7.4.1. Tests shall be made to the front structure within the boundaries as defined in paragraph 3.1. For tests at the rear of the bonnet top, the headform impactor shall not contact the windscreen or A-pillar before impacting the bonnet top.

7.4.2. No impact point shall be located so that the impactor will impact the test area with a glancing blow resulting in a more severe second impact outside the test area.

Selected impact points on the bonnet for the adult headform impactor shall be, at the time of first contact:

(1) a minimum of 82.5 mm inside the defined side reference lines, and;

(2) forward of the WAD2100 line, or,

a minimum of 82.5 mm forward of the bonnet rear reference line,

whichever is most forward at the point of measurement, and;

(3) rearward of the WAD1700 line.

These minimum distances are to be set with a flexible tape held tautly along the outer surface of the vehicle.

7.4.3. The point of first contact of the headform impactor shall be within a ± 10 mm tolerance to the selected impact point.

7.4.4. The headform velocity at the time of impact shall be 9.7 ± 0.2 m/s.

7.4.5. The direction of impact shall be in the longitudinal vertical plane of the paragraph of the vehicle to be tested at an angle of 65° ± 2° to the horizontal. The direction of impact of tests to the front structure shall be downward and rearward.

8. CERTIFICATION OF IMPACTORS

The impactors that are used in the tests described in this gtr are required to comply with the following performance requirements.

The requirements for the lower legform impactor are specified in paragraph 8.1., the upper legform impactor requirements are specified in paragraph 8.2. and the adult and child headform impactors requirements are specified in paragraph 8.3.

8.1. Flexible lower legform impactor certification

8.1.1. Static certification tests

8.1.1.1. The femur and the tibia of the flexible lower legform impactor shall meet the requirements specified in paragraph 8.1.1.2. when tested according to paragraph 8.1.1.4. The knee joint of the lower legform impactor shall

meet the requirements specified in paragraph 8.1.1.3. when tested according to paragraph 8.1.1.5. The stabilized temperature of the impactor during the certification tests shall be 20° ± 2°C.

The CAC response values, as defined in ISO 6487:2002, shall be 30 mm for the knee ligament elongations and 4 kN for the applied external load. For these tests, low-pass filtering at an appropriate frequency is permitted to remove higher frequency noise without significantly affecting the measurement of the response of the impactor.

8.1.1.2. When the femur and the tibia of the impactor, without flesh and skin, are loaded in bending in accordance with paragraph 8.1.1.4., the applied moment and the generated deflection at the centre of the femur and the tibia (M_c and D_c) shall be within the corridors shown in Figure 22.

8.1.1.3. When the knee joint of the impactor, without flesh and skin, is loaded in bending in accordance with paragraph 8.1.1.5., the MCL, ACL, and PCL elongations and applied bending moment or the force at the centre of the knee joint (M_c or F_c) shall be within the corridors shown in Figure 23.

8.1.1.4. The edges of the femur and tibia, without flesh and skin, not bending parts shall be mounted to the support rig firmly as shown in Figure 24 and Figure 25. The Y-axis of the impactor shall be parallel to the loading axis within 180 ± 2° tolerance. To obtain repeatable loading, low friction Polytetrafluoroethylene (PTFE) plastic pads are used under each support (see Figures 24 and 25).

The centre of the loading force shall be applied at the centre of the femur and the tibia within ± 2mm tolerance along the Z-axis. The force shall be increased so as to maintain a deflection rate between 10 and 100 mm/minute until the bending moment at the centre part (M_c) of the femur or tibia reaches 380 Nm.

8.1.1.5. The ends of the knee joint, without flesh and skin, shall be mounted to the support rig firmly as shown in Figure 26. The Y-axis of the impactor shall be parallel to the loading axis within ± 2° tolerance. To obtain repeatable loading, low friction Polytetrafluoroethylene (PTFE) plastic pads are used under each support (see Figure 26). To avoid impactor damage, a foamed neoprene sheet shall be set between the loading ram and the impactor face of the knee joint, which is described in Figure 26, shall be removed. The foamed neoprene sheet used in this test shall have compression characteristics as shown in Figure 15.

The centre of the loading force shall be applied at the knee joint centre within ± 2°mm tolerance along the Z-axis (see Figure 12). The external load shall be increased so as to maintain a deflection rate between 10 and 100 mm/minute until the bending moment at the centre part of the knee joint (M_c) reaches 400 Nm.

8.1.2. Dynamic certification tests (pendulum test)

8.1.2.1. The assembled flexible lower legform impactor shall meet the requirements according to paragraph 8.1.2.3. when tested as specified in paragraph 8.1.2.4.

8.1.2.2. Certification

8.1.2.2.1. The test facility used for the certification test shall have a stabilized temperature of 20 ± 2 °C during the test.

8.1.2.2.2. The temperature of the certification area shall be measured at the time of certification and recorded in a certification report.

8.1.2.3. Requirements

8.1.2.3.1. When the flexible lower legform impactor is used for a test according to paragraph 8.1.2.4., the absolute value of the maximum bending moment of the tibia at

(a) Tibia-1 shall be 235 Nm ≤ 272 Nm,

(b) Tibia-2 shall be 187 Nm ≤ 219 Nm,

(c) Tibia-3 shall be 139 Nm ≤ 166 Nm,

(d) Tibia-4 shall be 90 Nm ≤ 111 Nm.

The absolute value of the maximum elongation of MCL shall be

(a) MCL shall be 20.5 ≤ 24.0 mm,

(b) ACL shall be 8.0 ≤ 10.5 mm,

(c) PCL shall be 3.5 ≤ 5.0 mm.

For all these values for the maximum bending moment and the maximum elongation, the readings used shall be from the initial impact timing to 200 ms after the impact timing.

8.1.2.3.2. The instrumentation response value CFC, as defined in ISO 6487:2002, shall be 180 for all transducers. The CAC response values, as defined in ISO 6487:2002, shall be 30 mm for the knee ligament elongations and 400 Nm for the tibia bending moments. This does not require that the impactor itself is able to physically elongate or bend until these values.

8.1.2.4. Test procedure

8.1.2.4.1. The flexible lower legform impactor, including the flesh and skin, shall be suspended from the dynamic certification test rig 15 ± 1 ° upward from the horizontal as shown in Figure 27. The impactor shall be released from the suspended position and fall freely against the pin joint of the test rig as shown in Figure 27.

8.1.2.4.2. The knee joint centre of the impactor shall be 30 ± 1 mm below the bottom line of the stopper bar, and the tibia impact face without the flesh and skin shall be located 13 ± 2 mm from the front upper edge of the stopper bar when the impactor is hanging freely as shown in Figure 27.

8.1.3. Dynamic certification tests (inverse test)

8.1.3.1. The assembled flexible lower legform impactor shall meet the requirements according to paragraph 8.1.3.3. when tested as specified in paragraph 8.1.3.4.

8.1.3.2. Certification

8.1.3.2.1. The test facility used for the certification test shall have a stabilized temperature of 20 ± 2 °C during the test.

8.1.3.2.3. The temperature of the certification area shall be measured at the time of certification and recorded in a certification report.

8.1.3.3. Requirements

8.1.3.3.1. When the flexible lower legform impactor is used for the test according to paragraph 8.1.3.4., the absolute value of the maximum bending moment of the tibia at

(a) Tibia-1 shall be 230 Nm ≤ 272 Nm,

(b) Tibia-2 shall be 210 Nm ≤ 252 Nm,

(c) Tibia-3 shall be 166 Nm ≤ 192 Nm,

(d) Tibia-4 shall be 93 Nm ≤ 108 Nm.

The absolute value of the maximum elongations of

(a) MCL shall be 17.0 ≤ 21.0 mm,

(b) ACL shall be 8.0 ≤ 10 mm,

(c) PCL shall be 4.0 ≤ 6.0 mm.

For all these values for the maximum bending moment and the maximum elongation, the readings used shall be from the initial impact timing to 50 ms after the impact timing.

8.1.3.3.2. The instrumentation response value CFC, as defined in ISO 6487:2002, shall be 180 for all transducers. The CAC response values, as defined in ISO 6487:2002, shall be 30 mm for the knee ligament elongations and 400 Nm for the tibia bending moments.

8.1.3.4. Test procedure

8.1.3.4.1. The assembled flexible lower legform impactor (with the flesh and skin) shall be hung vertically and freely suspended from a test rig as shown in Figure 28. It is then impacted by the upper edge of a linearly guided aluminium honeycomb impactor, covered by a thin paper cloth with a maximum thickness of 1 mm, at an impact speed of 11.1 ± 0.2 m/s. The legform shall achieve a free flight condition within 10 ms after the time of first contact of the honeycomb impactor.

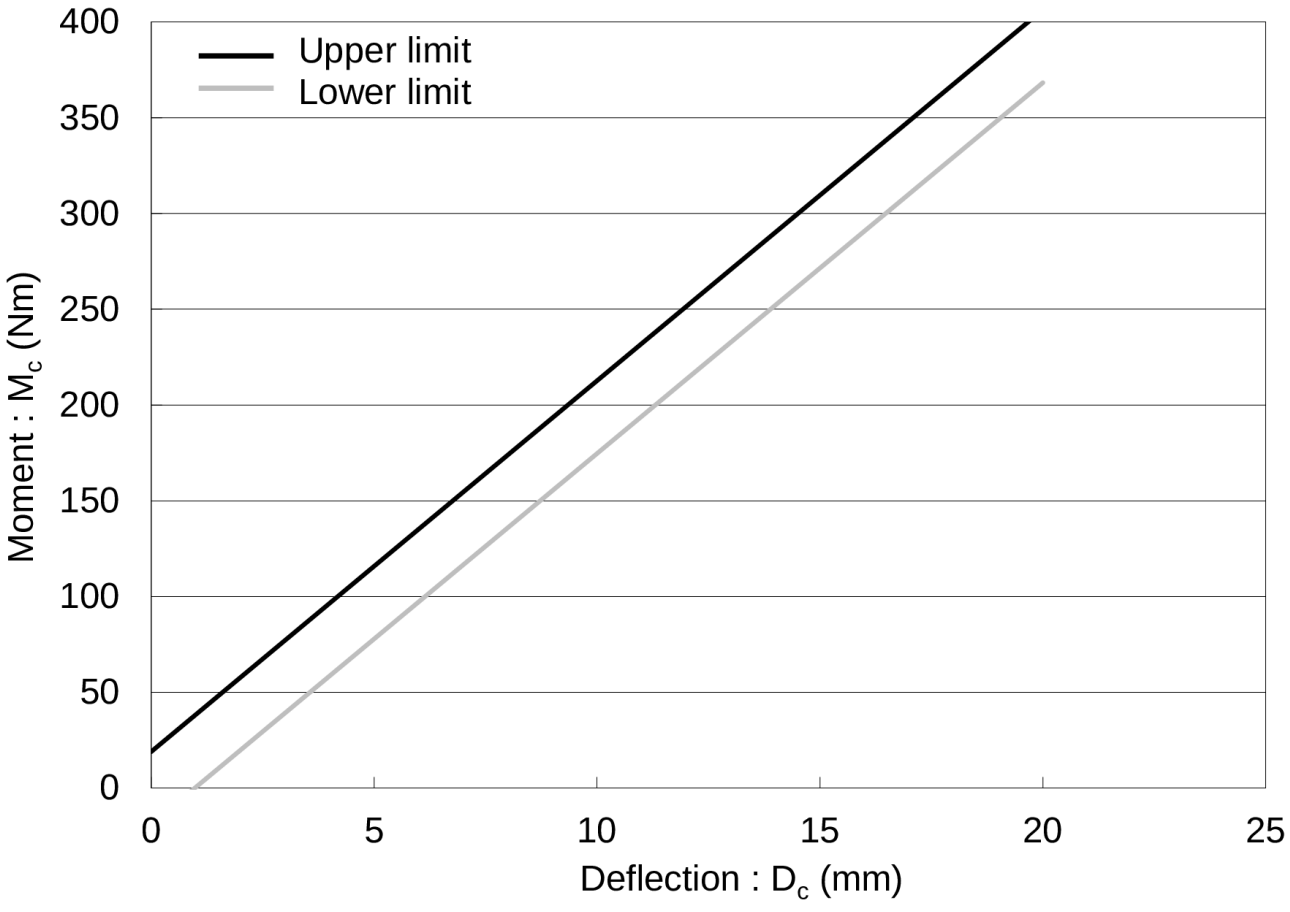
8.1.3.4.2. The honeycomb of 5052 alloy, which is attached in front of the moving ram, shall be 200 ± 5 mm wide, 160 ± 5 mm high and 60 ± 2 mm deep and shall have a crush strength of 517.1 kPa ± 10 per cent (75 pound per square inch (psi) ± 10 per cent). The honeycomb should have cell sizes of either 4.76 mm (3/16 inch) or 6.35 mm (1/4 inch) and a density of 32.0 kg/m³ (2.0 pound per cubic foot (pcf)) for the 4.76 mm (3/16 inch) cell size or a density of 36.8 kg/m³ (2.3 pound per cubic foot (pcf)) for the 6.35 mm (1/4 inch) cell size.

8.1.3.4.3. The upper edge of the honeycomb face shall be in line with the rigid plate of the linearly guided impactor. At the time of first contact, the upper edge of the honeycomb shall be in line with the knee joint centre line within a vertical tolerance of ± 2 mm. The honeycomb shall not be deformed before the impact test.

8.1.3.4.4. At the time of the first contact, the flexible lower legform impactor pitch angle (rotation around the Y-axis) and, therefore, the pitch angle of the velocity vector of the honeycomb impactor shall be within a tolerance of ± 2° in relation to the lateral vertical plane. The flexible lower legform impactor roll angle (rotation around the X-axis) and, therefore, the roll angle of the honeycomb impactor shall be within a tolerance of ± 2° in relation to the longitudinal vertical plane. The flexible lower legform impactor yaw angle (rotation around the Z-axis) and, therefore, the yaw angle of the velocity vector of the honeycomb impactor shall be within a tolerance of ±2°.

Figure 23 **Flexible lower legform impactor: Requirement corridors of the femur and the tibia, without flesh and skin, in the static certification test (see paragraph 8.1.1.2.)**

(a) Femur bending moment corridor



(b) Tibia bending moment corridor

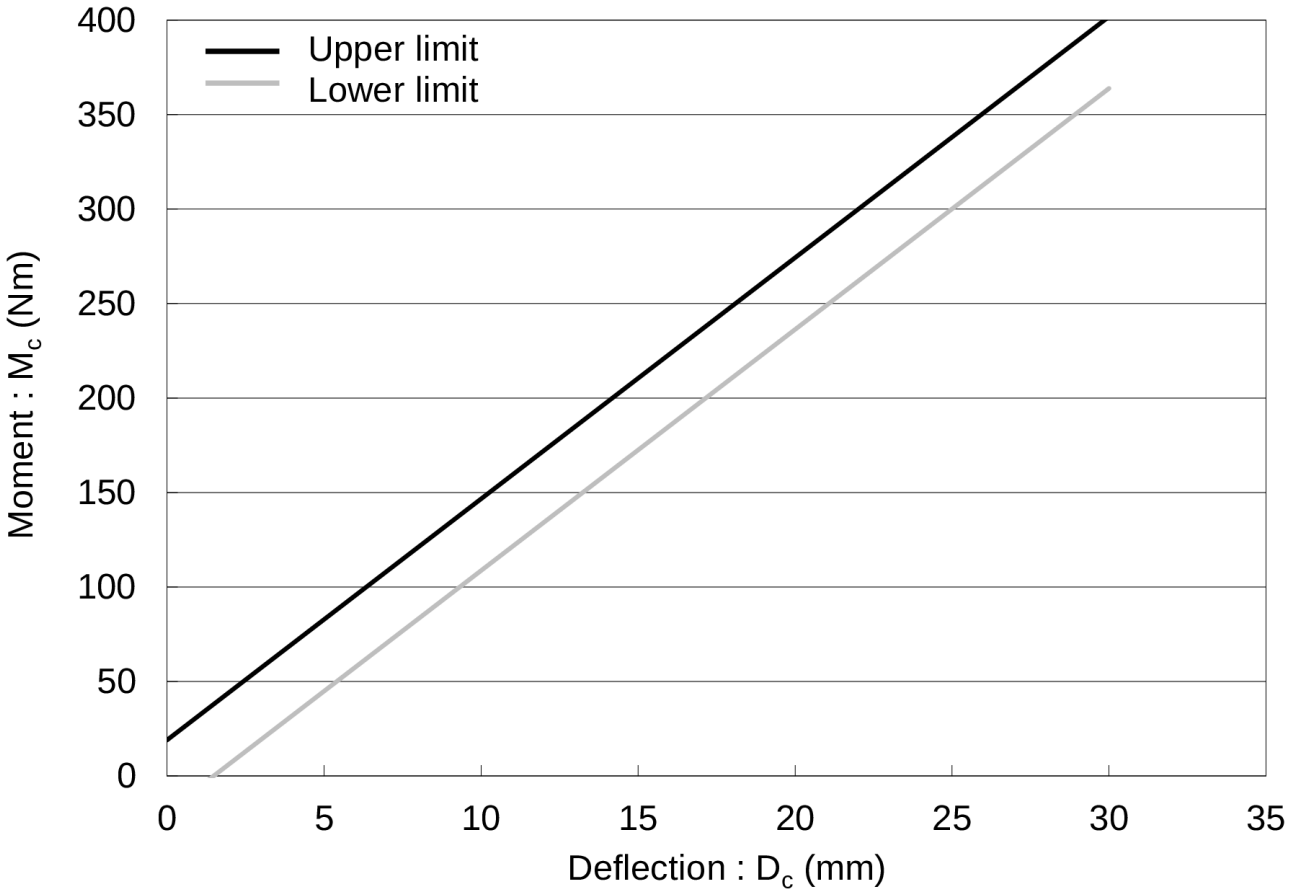
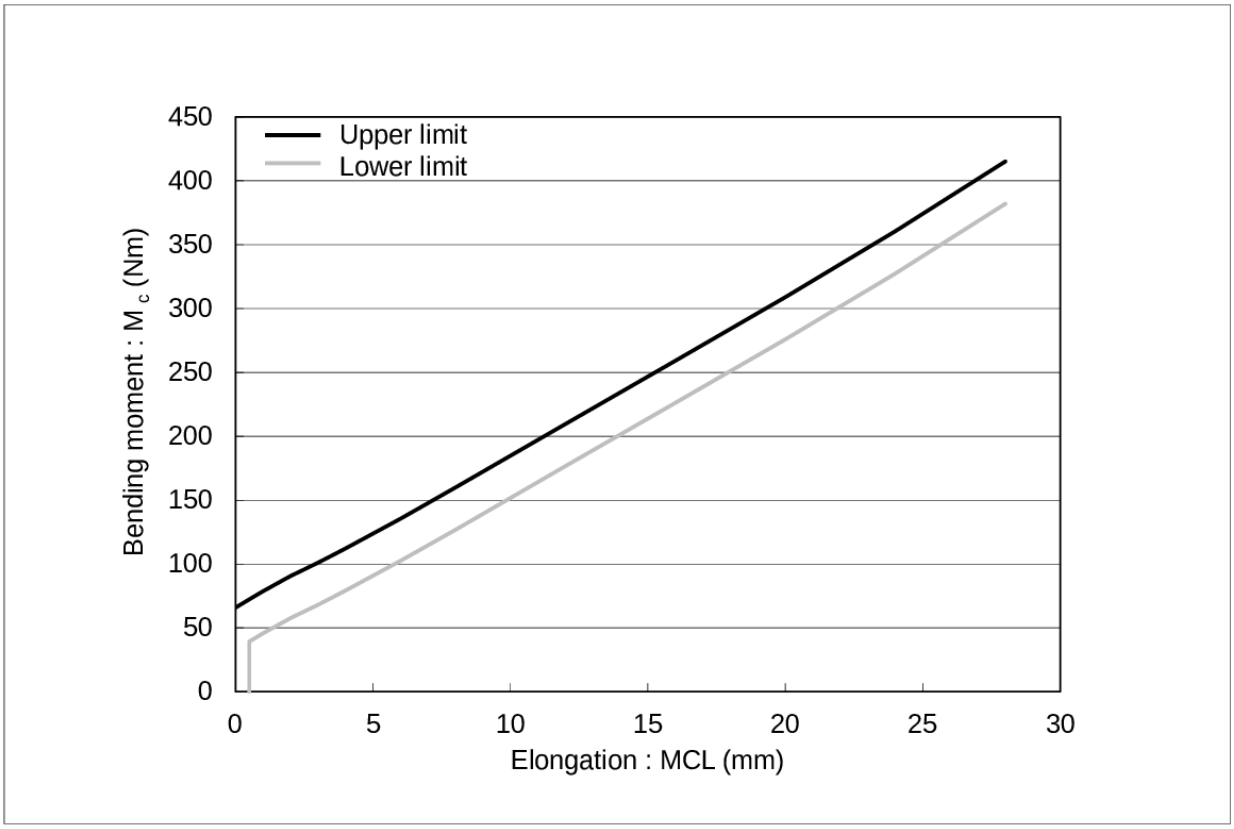
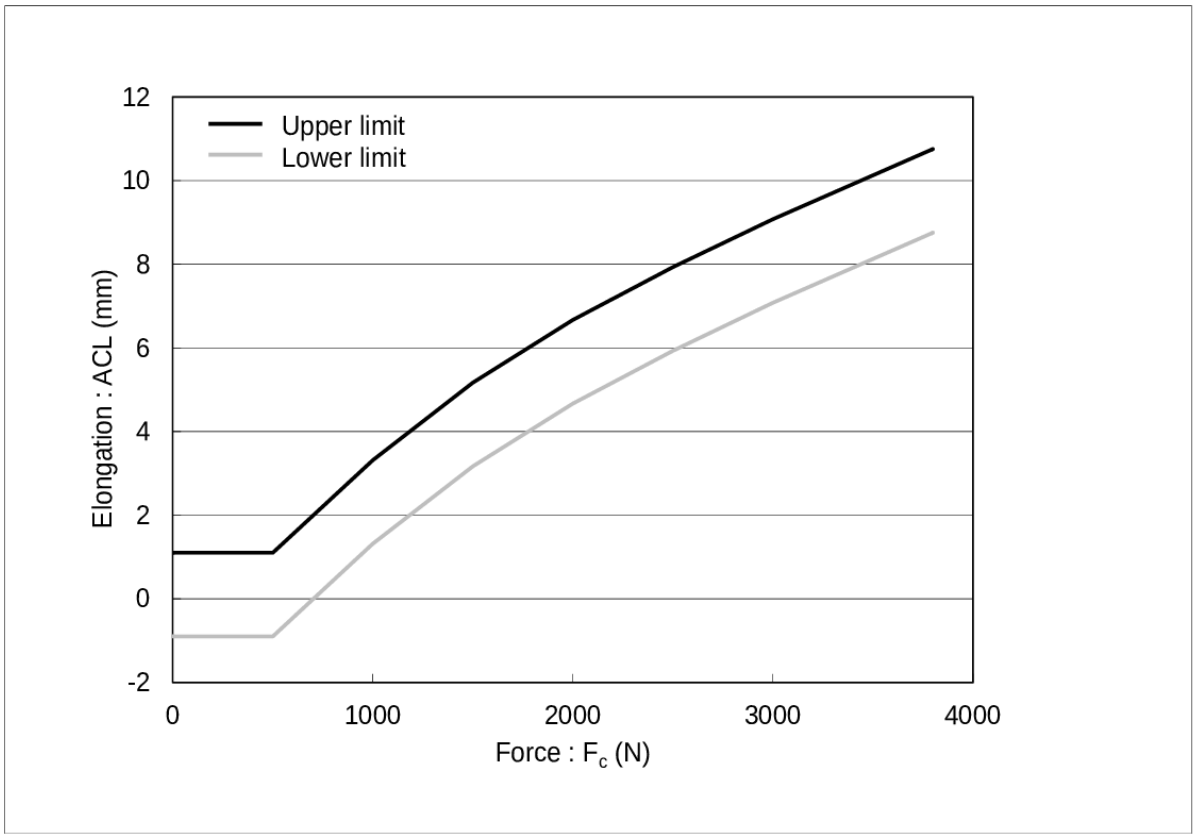


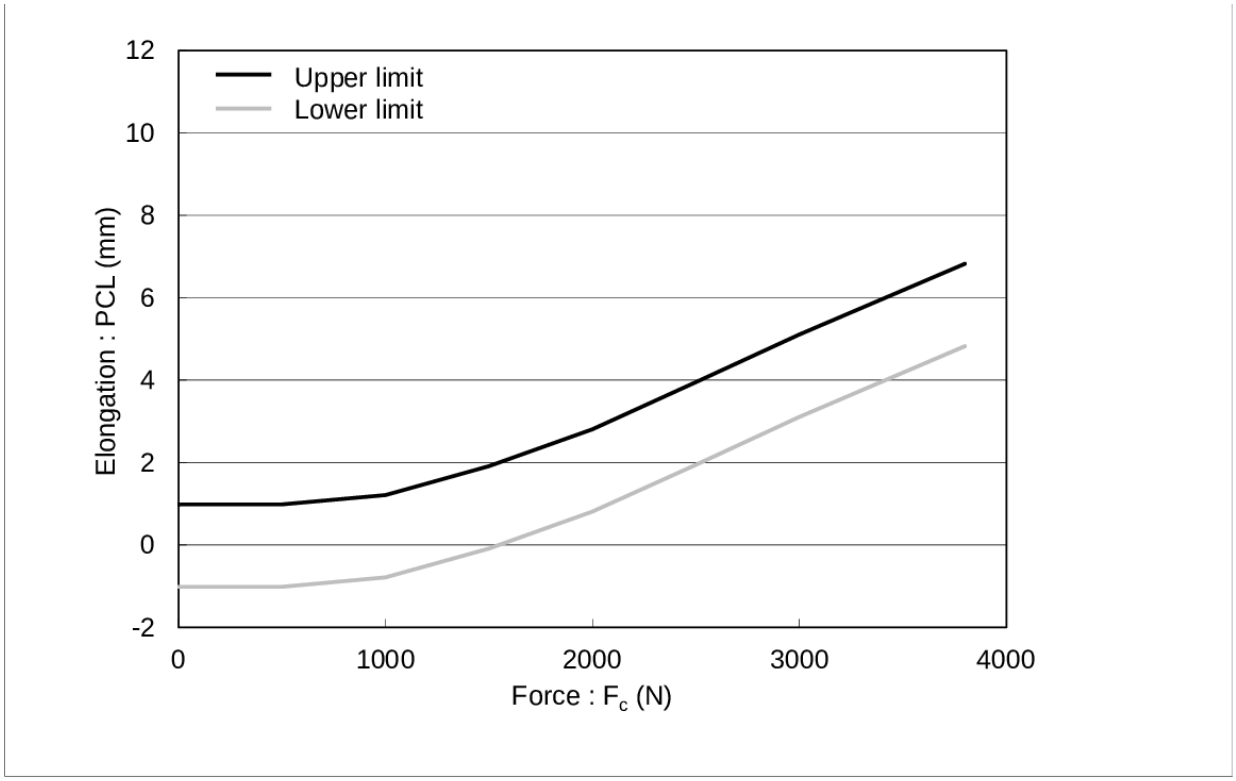
Figure 24 **Flexible lower legform impactor: Requirement corridors for the knee joint, without flesh and skin, in the static certification test (see paragraph 8.1.1.3.)**



(a) for MCL



(b) for ACL



(c) for PCL

Figure 25 **Flexible lower legform impactor: Test set-up for the femur in the static certification test (see paragraph 8.1.1.4.)**

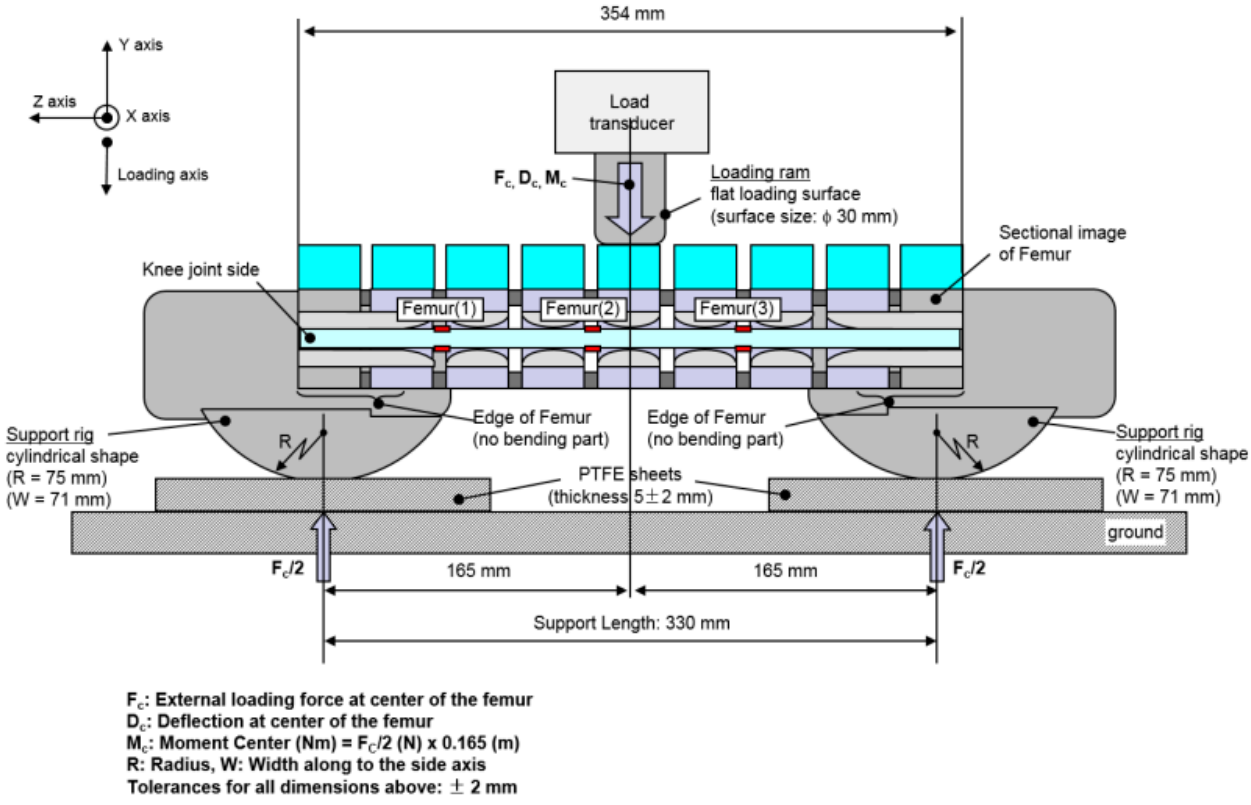


Figure 26 **Flexible lower legform impactor: Test set-up for the tibia in the static certification test (see paragraph 8.1.1.4.)**

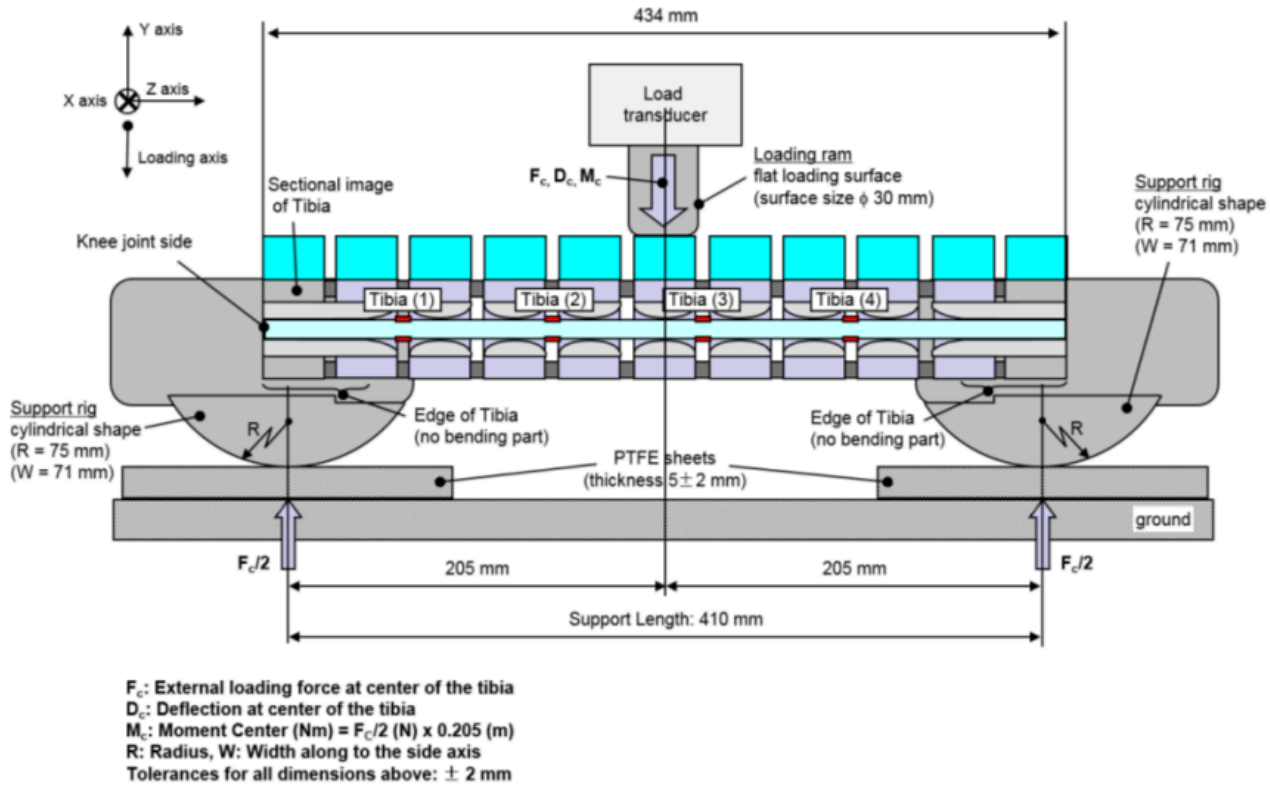


Figure 27 **Flexible lower legform impactor: Test set-up for the knee joint in the static certification test (see paragraph 8.1.1.5.)**

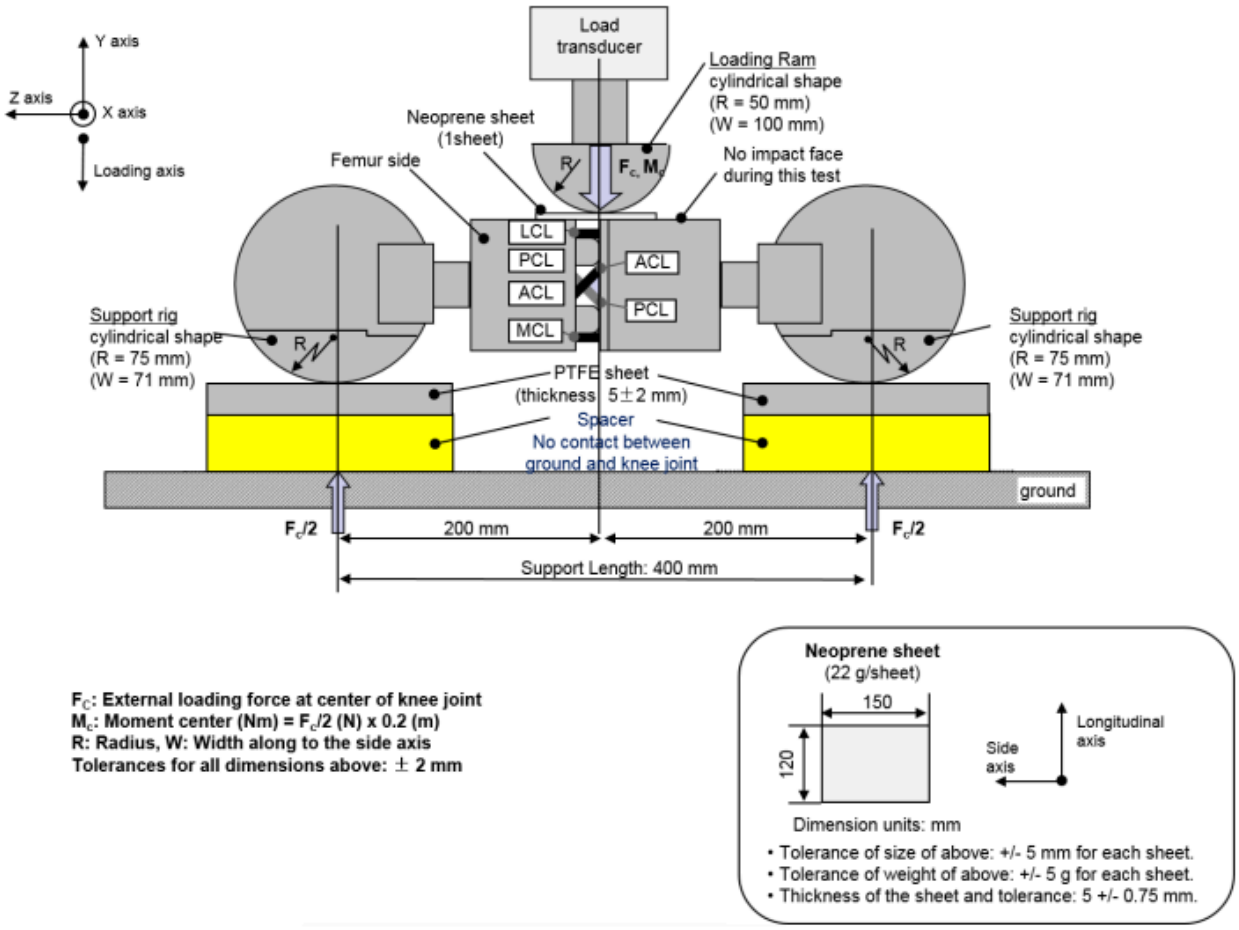


Figure 28 **Flexible lower legform impactor: Test set-up for the dynamic lower legform impactor certification test (pendulum test, see paragraph 8.1.2.4.)**

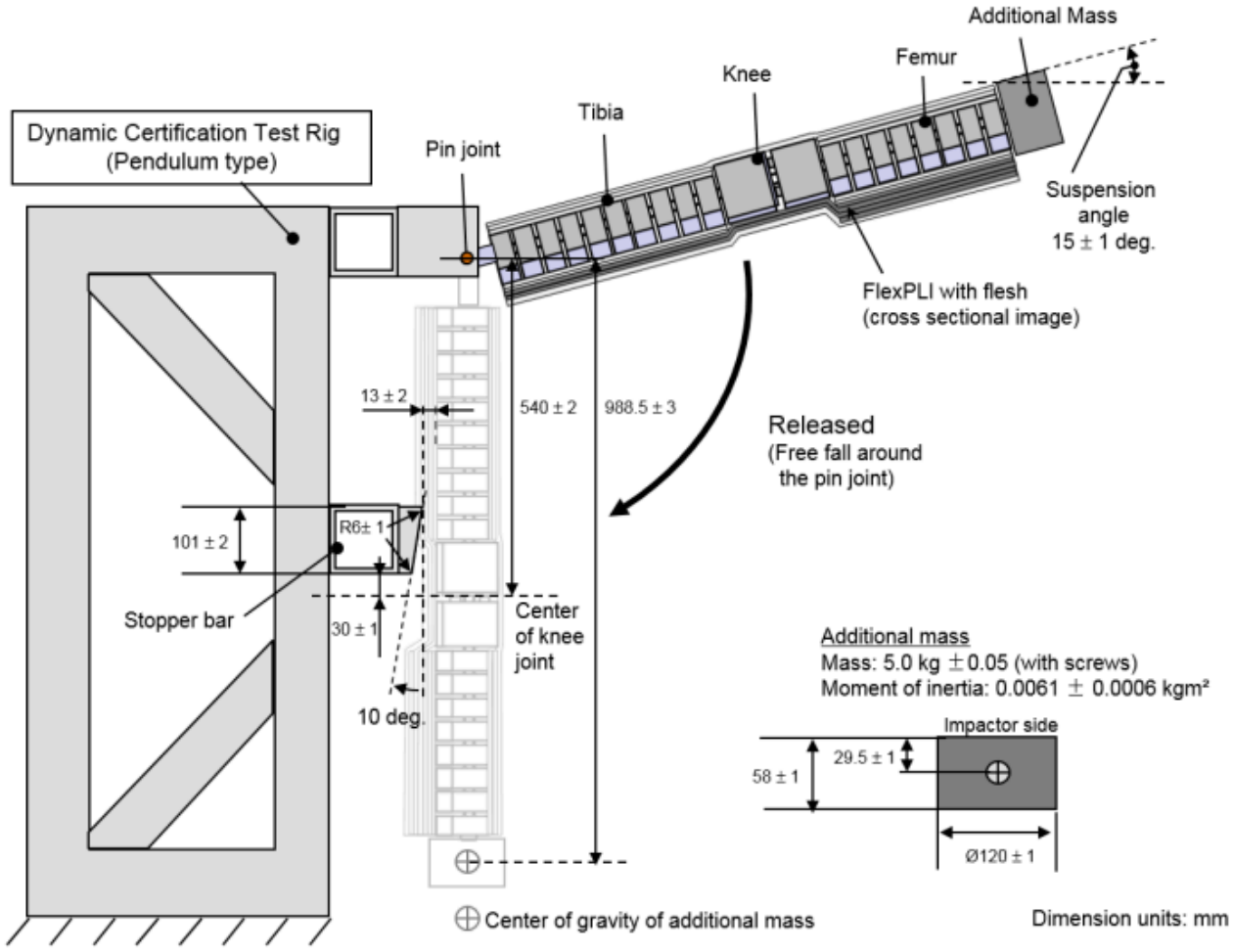
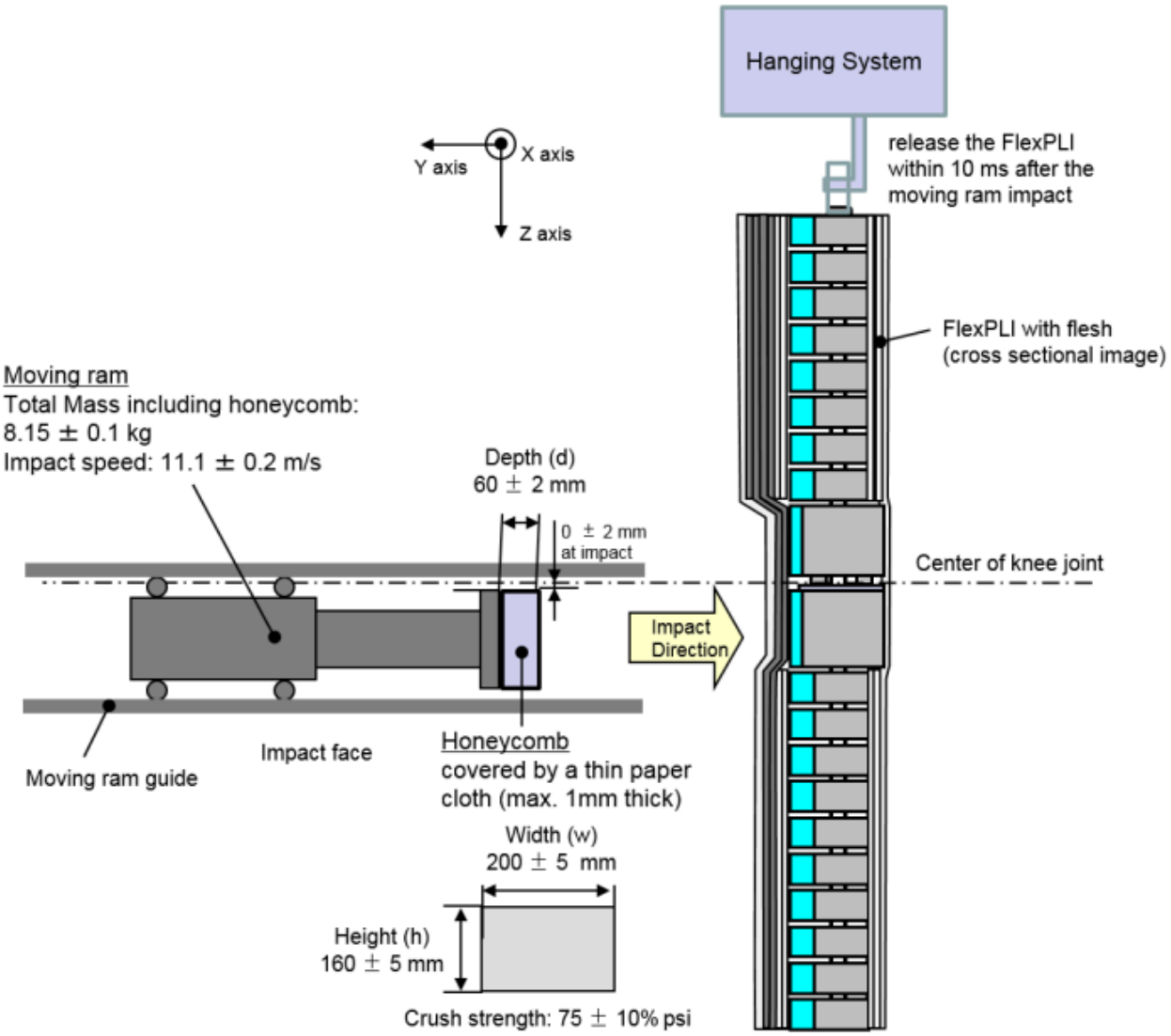


Figure 29 **Flexible lower legform impactor: Test set-up for the dynamic lower legform impactor certification test (inverse test, see paragraph 8.1.3.4.)**



8.2. Upper legform impactor certification

8.2.1. The upper legform impactor shall meet the requirements specified in paragraph 8.2.3. when tested as specified in paragraph 8.2.4.

8.2.2. Calibration

8.2.2.1. The foam flesh for the test impactor shall be stored for a period of at least four hours in a controlled storage area with a stabilized humidity of 35 ± 10 percent and a stabilized temperature of 20° ± 2°C prior to impactor removal for calibration. The test impactor itself shall have a temperature of 20° ± 2°C at the time of impact. The temperature tolerances for the test impactor shall apply at a relative humidity of 40 ± 30 percent after a soak period of at least four hours prior to their application in a test.

8.2.2.2. The test facility used for the calibration test shall have a stabilized humidity of 40 ± 30 percent and a stabilized temperature of 20° ± 4°C during calibration.

8.2.2.3. Each calibration shall be completed within two hours of when the impactor to be calibrated is removed from the controlled storage area.

8.2.2.4. The relative humidity and temperature of the calibration area shall be measured at the time of calibration and recorded in the calibration report.

8.2.3. Requirements

8.2.3.1. When the impactor is propelled into a stationary cylindrical pendulum the peak force measured in each load transducer shall be not less than 1.20 kN and not more than 1.55 kN and the difference between the peak forces measured in the top and bottom load transducers shall not be more than 0.10 kN. Also, the peak bending moment measured by the strain gauges shall not be less than 190 Nm and not more than 250 Nm on the centre position and not less than 160 Nm and not more than 220 Nm for the outer positions. The difference between the upper and lower peak bending moments shall not be more than 20 Nm.

For all these values, the readings used shall be from the initial impact with the pendulum and not from the arresting phase. Any system used to arrest the impactor or pendulum shall be so arranged that the arresting phase does not overlap in time with the initial impact. The arresting system shall not cause the transducer outputs to exceed the specified CAC.

8.2.3.2. The instrumentation response value CFC, as defined in ISO 6487:2002, shall be 180 for all transducers. The CAC response values, as defined in ISO 6487:2002, shall be 10 kN for the force transducers and 1000 Nm for the bending moment measurements.

8.2.4. Test procedure

8.2.4.1. The impactor shall be mounted to the propulsion and guidance system, by a torque limiting joint. The torque limiting joint shall be set so that the longitudinal axis of the front member is perpendicular to the axis of the guidance system, with a tolerance of ± 2°, with the joint friction torque set to a minimum of 650 Nm but no more than 850 Nm. The guidance system shall be fitted with low friction guides that allow the impactor to move only in the specified direction of impact, when in contact with the pendulum.

8.2.4.2. The impactor mass shall be adjusted to give a mass of 12 ± 0.1 kg, this mass includes those propulsion and guidance components which are effectively part of the impactor during impact.

8.2.4.3. The centre of gravity of those parts of the impactor which are effectively forward of the torque limiting joint, including the extra masses fitted, shall lie on the longitudinal centreline of the impactor, with a tolerance of ± 10 mm.

8.2.4.4. The impactor shall be certified with previously unused foam.

8.2.4.5. The impactor foam shall not be excessively handled or deformed before, during or after fitting.

8.2.4.6. The impactor with the front member vertical shall be propelled horizontally at a velocity of 7.1 ± 0.1 m/s into the stationary pendulum as shown in Figure 29.

8.2.4.7. The pendulum tube shall have a mass of 3 ± 0.03 kg, a wall thickness of 3 ± 0.15 mm and an outside diameter of 150 mm $^{+1\ mm}/_{-4\ mm}$. Total pendulum tube length shall be 275 ± 25 mm. The pendulum tube shall be made from cold finished seamless steel (metal surface plating is permissible for protection from corrosion), with an outer surface finish of better than 2.0 micrometer. It shall be suspended on two wire ropes of 1.5 ± 0.2 mm diameter and of 2.0 m minimum length. The surface of the pendulum shall be clean and dry. The pendulum tube shall be positioned so that the longitudinal axis of the cylinder is perpendicular to the front member (i.e. level), with a tolerance of ± 2°, and to the direction of impactor motion, with a tolerance of ± 2°, and with the centre of the pendulum tube aligned with the centre of the impactor front member, with tolerances of ± 5 mm laterally and ± 5 mm vertically.

8.3. Child and adult headform impactors certification

8.3.1. Drop test

8.3.1.1. Performance criteria

The headform impactors shall meet the requirements specified in paragraph 8.3.2. when tested as specified in paragraph 8.3.3.

8.3.2. Requirements

8.3.2.1. When the headform impactors are dropped from a height of 376 ± 1 mm in accordance with paragraph 8.3.3. the peak resultant acceleration measured by one triaxial (or three uniaxial) accelerometer (accelerometers) in the headform impactor shall be:

(a) for the child headform impactor not less than 245g and not more than 300g;

(b) for the adult headform impactor not less than 225g and not more than 275g.

The acceleration time curve shall be uni-modal.

8.3.2.2. The instrumentation response values CFC and CAC for each accelerometer shall be 1,000 Hz and 500g respectively as defined in ISO 6487:2002.

8.3.2.3. Temperature conditions

The headform impactors shall have a temperature of 20 ± 2°C at the time of impact. The temperature tolerances shall apply at a relative humidity of 40 ± 30 percent after a soak period of at least four hours prior to their application in a test.

8.3.2.4. After complying with the certification test, each headform impactor can be used for a maximum of 20 impact tests.

8.3.3. Test procedure

8.4.3.1. The headform impactor shall be suspended from a drop rig as shown in Figure 30.

8.3.3.2. The headform impactor shall be dropped from the specified height by means that ensure instant release onto a rigidly supported flat horizontal steel plate, over 50 mm thick and over 300 x 300 mm square which has a clean dry surface and a surface finish of between 0.2 and 2.0 micrometers.

8.3.3.3. The headform impactor shall be dropped with the rear face of the impactor at the test angle specified in paragraph 7.3.5. for the child headform impactor and in paragraph 7.4.5. for the adult headform impactor with respect to the vertical as shown in Figure 30. The suspension of the headform impactor shall be such that it does not rotate during the fall.

8.3.3.4. The drop test shall be performed three times, with the headform impactor rotated 120° around its symmetrical axis after each test.

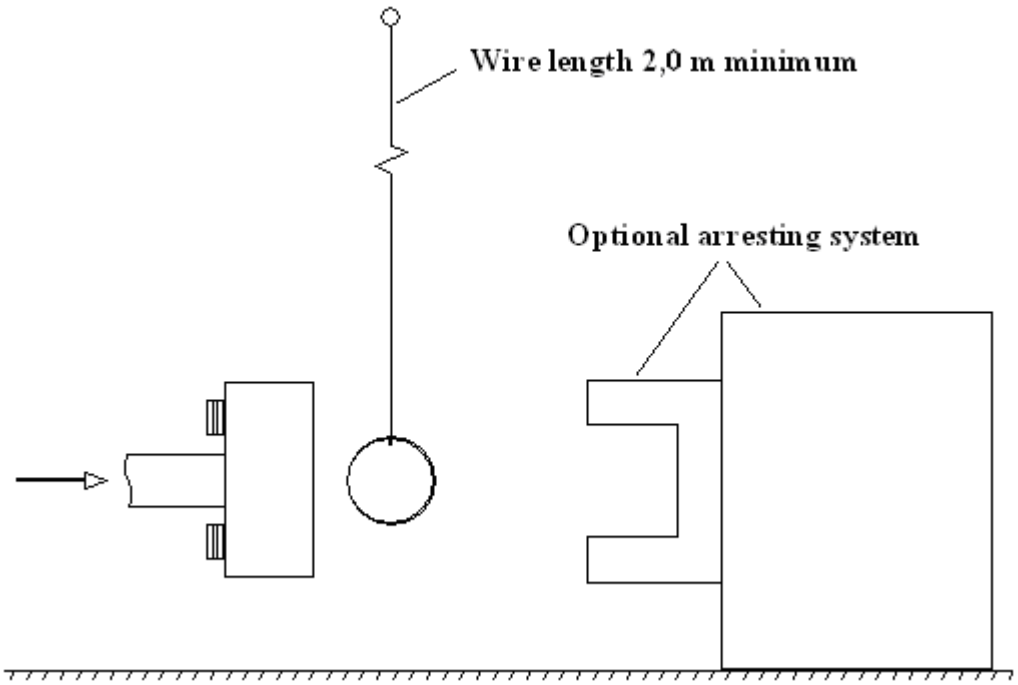


Figure 30. Test set-up for dynamic upper legform impactor certification test (see paragraph 8.2.4.6)

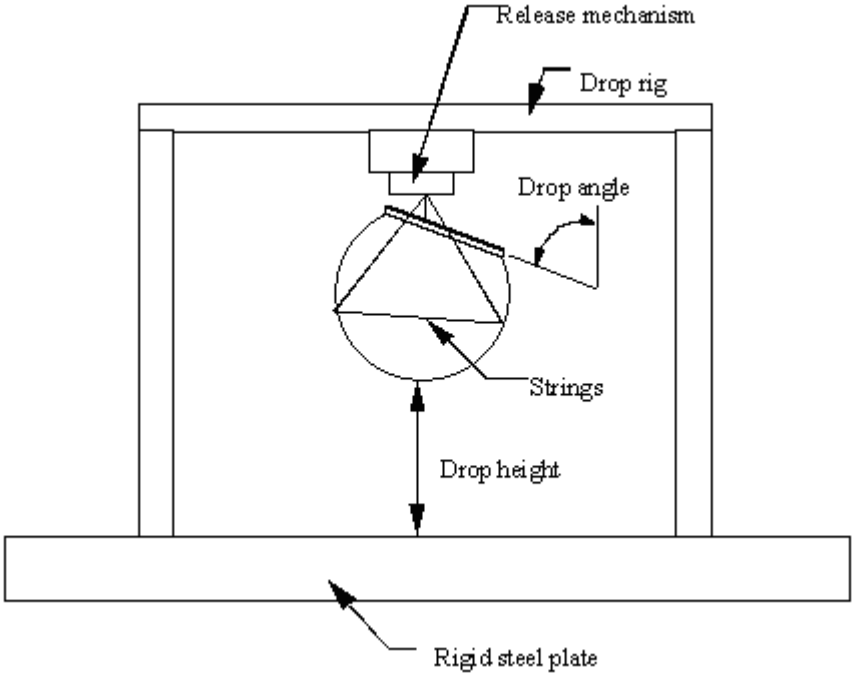


Figure 31. Test set-up for dynamic headform impactor biofidelity test (see paragraph 8.3.3.1.)

Annex 1 Test Procedure for the Deployable Pedestrian Protection Systems

1. Prerequisites

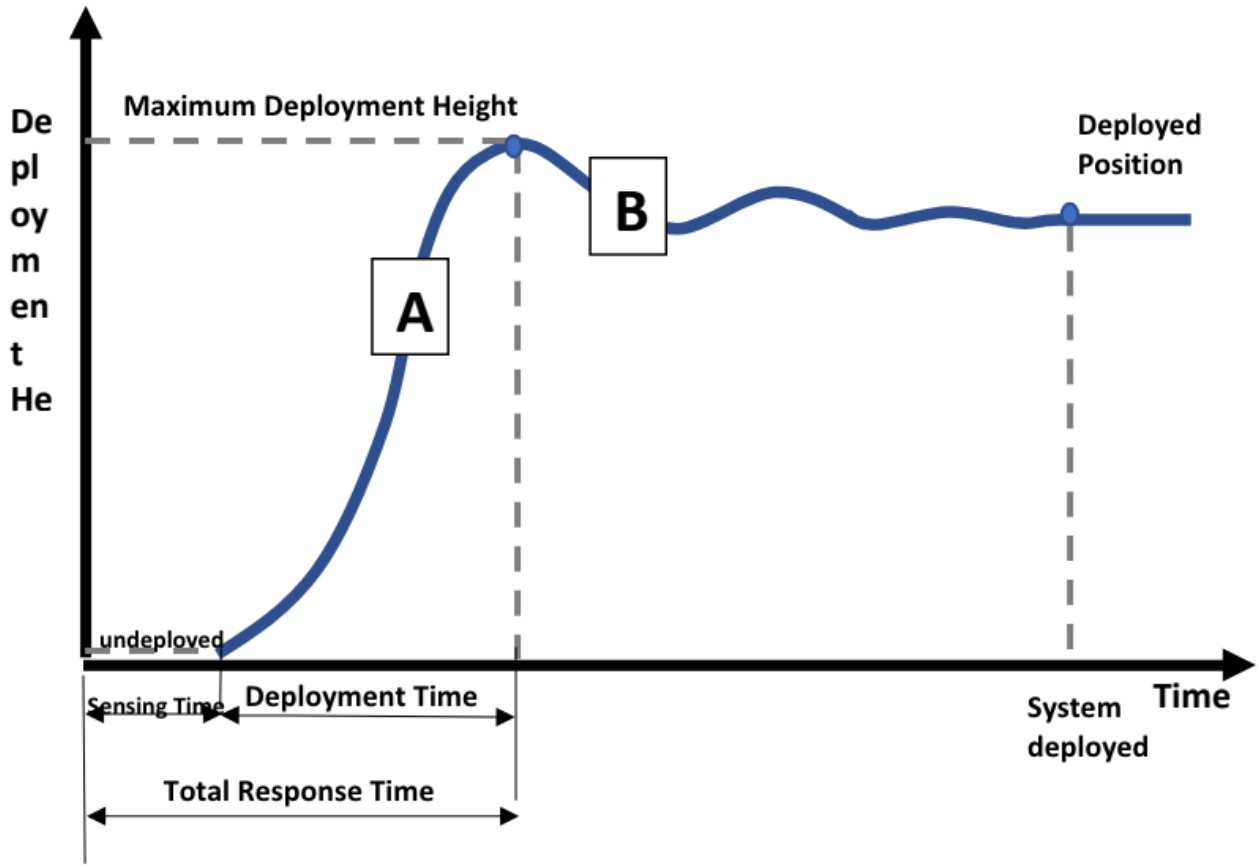
If all of the following prerequisites are met, the vehicle shall be tested with the DPPS activated as intended (as described below) for the entire headform test area.

Based on a determination by each Contracting Party, a Contracting Party may either allow static tests, dynamic tests, and a combination thereof, or stipulate dynamic tests only.

For DPPS to be assessed statically, dynamically or combined, it will be necessary for the vehicle manufacturer to identify detailed information highlighted in this Annex before any testing begins. The vehicle manufacturer shall identify all necessary information regarding detection of pedestrians and the deployment of the system. Based on the evidence identified, activation of the system in the headform test will be determined.

The principle of testing the DPPS is as follows:

Figure 1-1 **Deployment Time History Curve**



(a) In case of HIT_d < TRT, the headform test shall be performed dynamically;

(b) In case of HIT_d ≥ TRT, the headform test may be performed either statically at a height no more than the deployed position or dynamically.

1.1. If any of the prerequisites from paragraphs. 1.2 to 1.7. are not met, the vehicle shall be tested with the DPPS in the undeployed position.

1.2. System specification:

As a Contracting Party option, a technical description of the DPPS components shall be identified by the manufacturer. This shall be accompanied by the following information:

1.2.1. For the sensing system:

(a) Sensor type (e.g. pressure, optical, acceleration, etc.);

(b) Sensor locations;

(c) Operation process (including the lower deployment threshold speed of the DPPS).

1.2.2. Deployment information:

(a) Technology of the DPPS (airbag, active bonnet, etc.);

(b) Mechanism explanation;

(c) Component description (lifting system (e.g. actuator), hinge, latch, etc.);

(d) Deployed position information (not required for dynamic testing);

(e) TRT (ST and DT separately) information (for dynamic testing, ST only);

(f) Evolution of system stability (e.g. pressure or force versus time diagram) (static testing, only).

1.3. The marking of the headform test areas of the DPPS shall always be done in undeployed position, for static, dynamic or combined testing.

1.4. Headform test for protection below the lower deployment threshold speed of the DPPS.

1.4.1. The vehicle outer surface shall remain in undeployed position.

1.4.2. The test procedures specified in paragraphs 7.2. to 7.4. of this Regulation shall apply with the impact speed specified at 0.9 times the lower deployment threshold speed. The allocation of the HIC1700 and HIC1000 Zones may differ from those at nominal velocity (9.7 m/s) headform impact tests according to paragraph 5.2.4 of this Regulation.

1.4.3. The HIC shall comply with paragraph 5.2.3. of the Regulation.

1.5. HBM shall be qualified according to Addendum 5 of Mutual Resolution No. 1 (M.R.1). HIT information shall be documented according to Annex 2 of this Regulation.[1]

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[1] Will be updated with DPPS Phase 2 (HIT determination by a generic approach option).

1.6. Verification of the prerequisites for deployed static tests: Deployed Position, Maximum Deployment Height, ST and DT as illustrated in the deployment time history curve (see Figure 1-1 of this Annex).

The values specified by the manufacturer shall be verified by using appropriate tracking means, such as high-speed videos, accelerometer, or laser at the reference points as indicated by the manufacturer (on the lifting devices). The tolerance for the ST is -5ms/+3ms on the specified value, whereas the other tolerances are ±20 percent on the specified values respectively. If a measured value is within the defined tolerances, the values specified by the manufacturer shall be used. Otherwise, the measured values shall be used.

1.7. Sensing System Verification

1.7.1. The vehicle manufacturer shall specify the lowest speed of activation (lower deployment velocity threshold) of the DPPS.

1.7.1.1. For the system deployment verification, sensor activation tests with the FlexPLI, as specified in paragraph 6.3.1.1. of this Regulation, shall be performed within the detection test area at the DPPS lower deployment velocity threshold.

1.7.2. A test with the FlexPLI shall be performed at nominal velocity (11.1m/s) at vehicle centreline (Y0).

1.7.3. Where a test is performed within the tolerances as specified in paragraph 3 of this Annex, but below the lower deployment velocity threshold or outside the detection test area and the system does not deploy, the test must be repeated.

1.7.4. If the system is not activated during any of the verification tests, all headform tests shall be conducted in undeployed position according to paragraphs 7.2. to 7.4. of this Regulation.

1.7.5. For tests with stationary vehicle: the vehicle shall be set to the normal running condition as specified by the manufacturer for a vehicle speed corresponding to the particular use case.

2. Verification of the Total Response Time and /or Sensing Time at Nominal Velocity

2.1. TRT shall be confirmed by using the FlexPLI at the vehicle speed at 11.1 m/s and at the vehicle centreline (Y0).

2.2. ST is measured either independently, or during a TRT measurement test, at the vehicle speed as specified in this Regulation and at the vehicle centreline (Y0).

2.2.1. If the measured ST is within a tolerance of -5ms/+3ms, the value specified by the manufacturer shall be used. Otherwise, the measured value shall be used for the test. For dynamic testing, only ST shall be verified.

2.2.2. For tests with stationary vehicle: the vehicle shall be set to the normal running condition as specified by the manufacturer.

3. Tolerances

For verification tests of paragraphs 1.7 and 2 of Annex 1 with the FlexPLI, the following tolerances shall apply:

3.1. For tests with a moving vehicle impacting the stationary impactor:
Target speed: ±0.6 m/s: impact accuracy: ±50 mm.

3.2. For tests with a propulsion system propelling the impactor against the stationary vehicle:

Target speed, impact accuracy, angle tolerances are those of the performance tests, as in paragraph 7.1. of the Regulation.

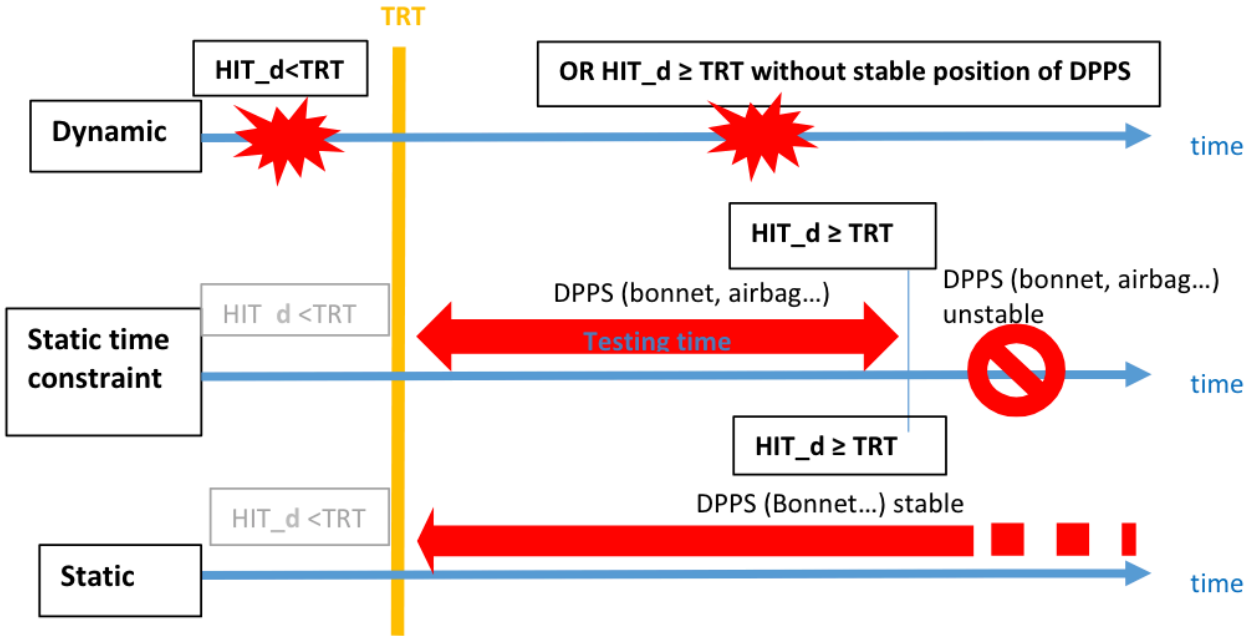
4. Headform Test Procedure at Nominal Velocity (9.7m/s)

The selection of impact points and the allocation of the HIC1700 and HIC1000 zones shall always be based on and related to the test area with undeployed DPPS.

4.1. Static test option:

If the vehicle manufacturer opts for the static test procedure, the following conditions shall be fulfilled. If so, the headform tests on the headform test area shall be performed statically.

Figure 1-2 **Timeline for dynamic, static time constraint and static testing representing real life conditions**



If any of the following conditions are not met, the headform tests on the headform test area shall be performed dynamically.

4.1.1. Where the vehicle manufacturer has demonstrated by numerical simulations on the deployed DPPS, that HIT_d ≥ TRT for the smallest selected pedestrian stature, as defined in Annex 2, then all tests may be performed statically.

4.1.2. The vehicle outer surface shall represent the deployed position (see Figure 1-1, B section) within the specified tolerances, while the resisting force is considered:

4.1.2.1. Static time constraint condition, linked to the resisting force:

When there is a constraint on time for the stability of the system and HIT_d ≥TRT, the launching time of the headform impactor shall ensure that the system remains stable (tolerance ±10 per cent of corresponding resisting force), as identified by the manufacturer (prerequisite in paragraph 1.2. of Annex 1).

Based on the evolution of system stability (see Figure 1-2), a decision can be made on how to perform the test. During the static tests it shall be ensured that the resisting force of the DPPS is equivalent to the actual situation at HIT.

4.1.2.2. Appropriate means (e.g. actuator surrogates) may be used to ensure that the corresponding resisting force of the DPPS is reached.

4.1.3. The test procedures specified in paragraphs 7.2. to 7.4. of this Regulation shall apply.

4.1.4. Test accuracy at impact location

4.1.4.1. Prior to conducting the static tests at 9.7 m/s, one headform test at the discretion of the test laboratory may be conducted on the undeployed DPPS to confirm that impact velocity and impact location are within tolerances.

4.1.4.2. If the tolerances for impact speed and location are met during the test on the undeployed DPPS, there is no requirement to prove that these tolerances are still met during the static tests, provided that test inputs remain the same.

4.1.4.3. Alternative methods to demonstrate the test accuracy may also be accepted.

4.2. Dynamic test option

4.2.1. The dynamic verification of a DPPS is based on a headform test performed on the DPPS, where the headform launch device and DPPS deployment are synchronised to achieve the correct HIT_s.

The following steps are conducted:

4.2.1.1. Test accuracy at impact location

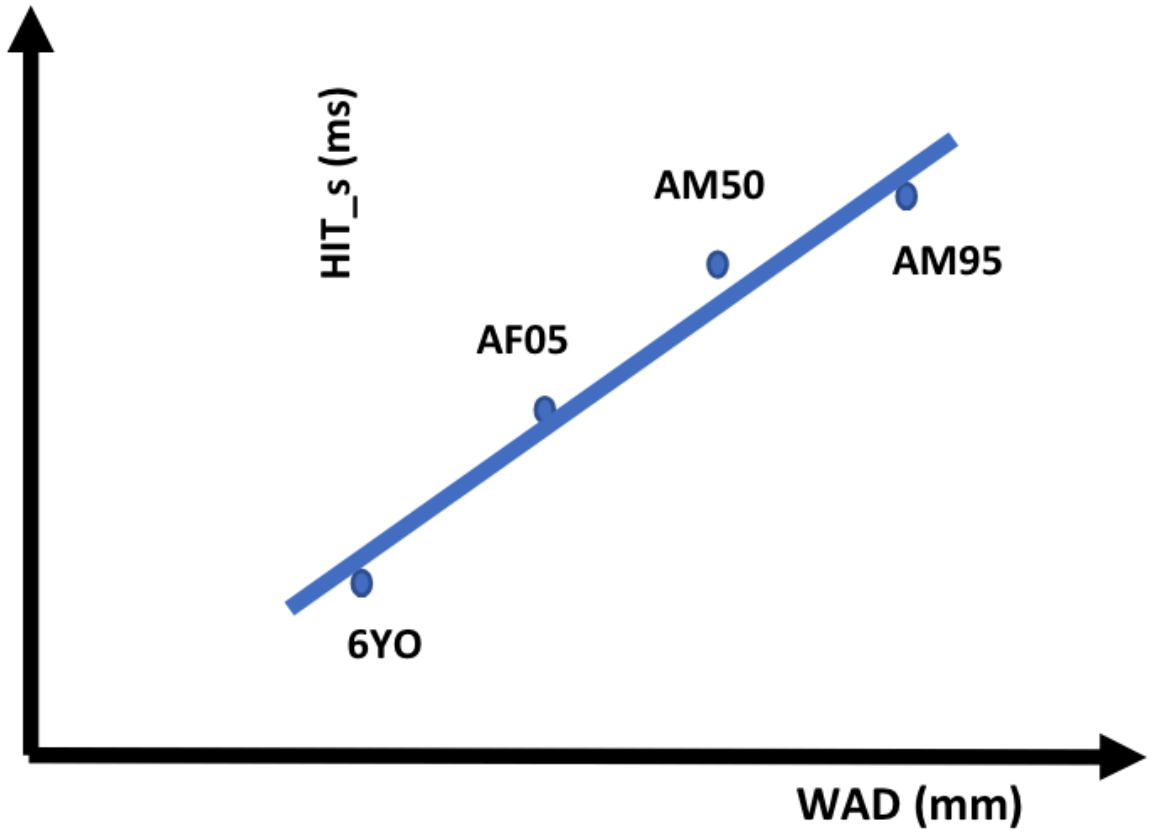
Prior to conducting the dynamic tests at 9.7m/s, one headform test at the discretion of the test laboratory shall be conducted on the undeployed DPPS to confirm that impact velocity and impact location are within tolerances.

If the tolerances for impact speed and location are met during the undeployed test, there is no requirement to meet these tolerances during dynamic tests, provided test inputs remain the same.

4.2.1.2. To enable dynamic testing to be conducted, HIT_s and ST are required inputs, which shall be established by the following:

(a) HIT_s (see Fig. 1-3 hereafter, obtained from Annex 2, Figure 2-3).

Figure 1-3 **Head Impact Time(s for synchronisation) versus Wrap Around Distance(HIT_s vs WAD)**



(b) ST is determined from the manufacturer prerequisite or sensor verification test, carried out at the vehicle centreline (Y0).

4.2.1.3. Fire Delay

The test facility shall ensure that the head impact occurs at the correct time relative to the deployment of the DPPS, taking into account the HIT_s for the corresponding WAD of the head impact point from Figure 1-3 and ST, as shown in the example in Figure 1-4 (a) below. "Fire Delay" is the elapsing time between the initiation of the headform launch and the initiation of the DPPS deployment module. It is determined according to the equation: Fire Delay = Launching Duration Headform Impactor – (HIT_s – ST). The "Launching Duration Headform Impactor" is rig-specific and is the time period between launching of the headform impactor and the theoretical time of head impact on the undeployed DPPS. Due to the DPPS deployment during testing, the actual launching duration of the headform impactor is expected to differ from the calculated launching duration headform impactor (time difference: see example in Figure 1-4 (b)).

Figure 1-4 (a) **Synchronization of Test Rig and DPPS Deployment (Example).**

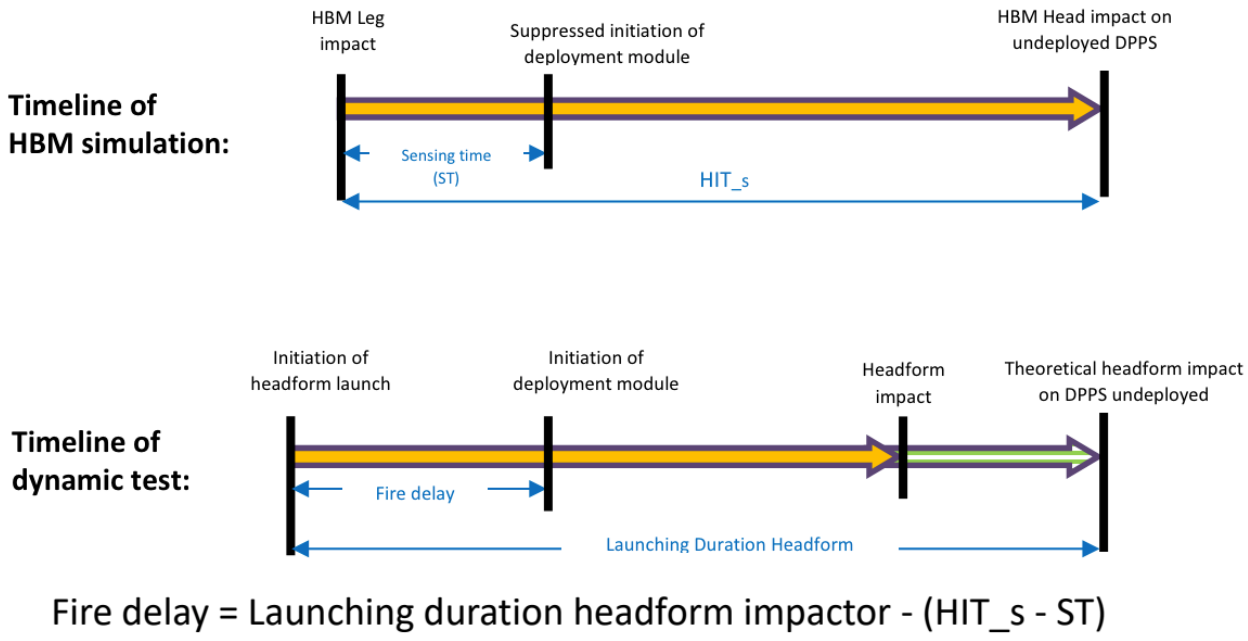
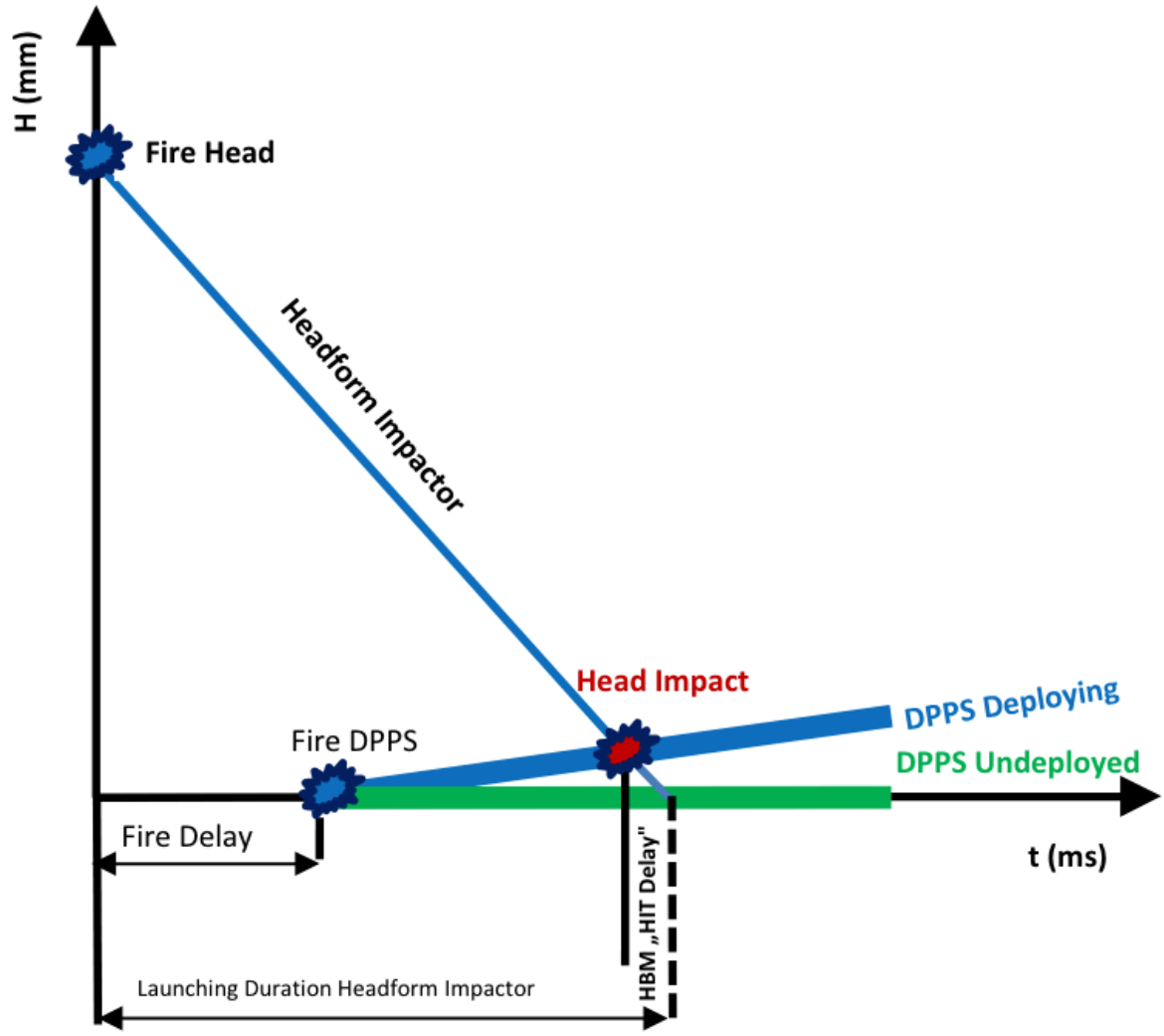


Figure 1-4 (b) **Effect of DPPS Deployment on Launching Duration Headform Impactor (Example).**

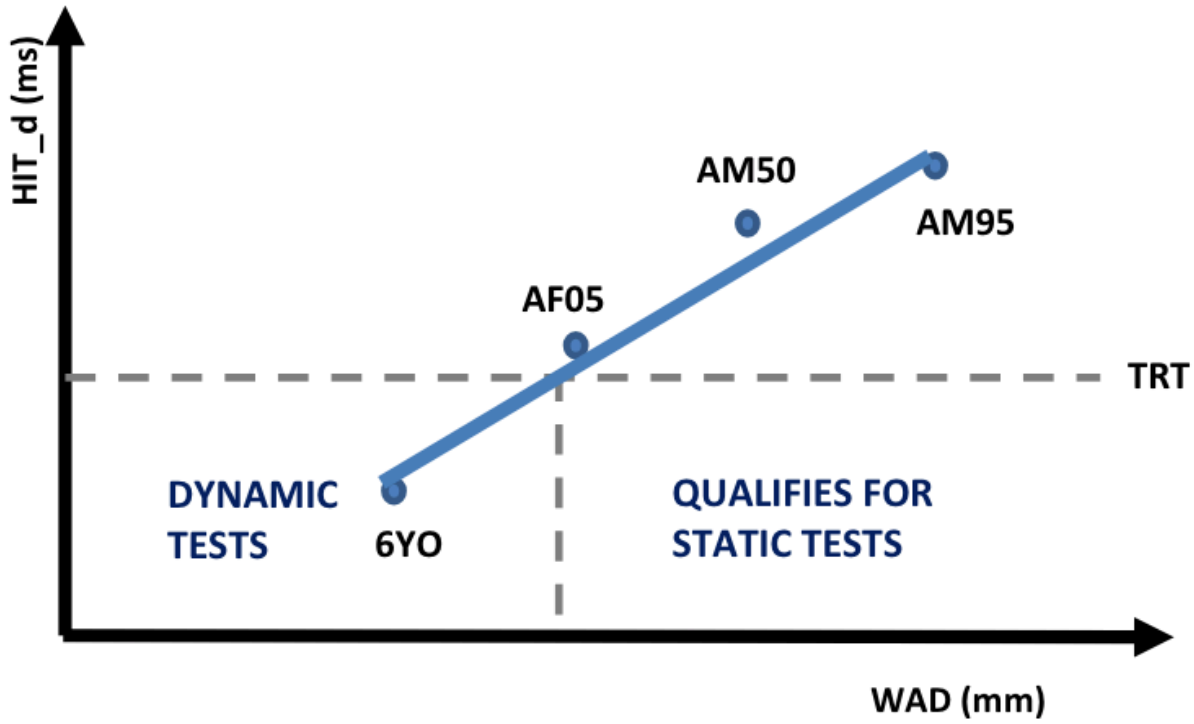


4.3. "Combined" test option:

Combined static and dynamic tests may apply, at manufacturer’s choice.

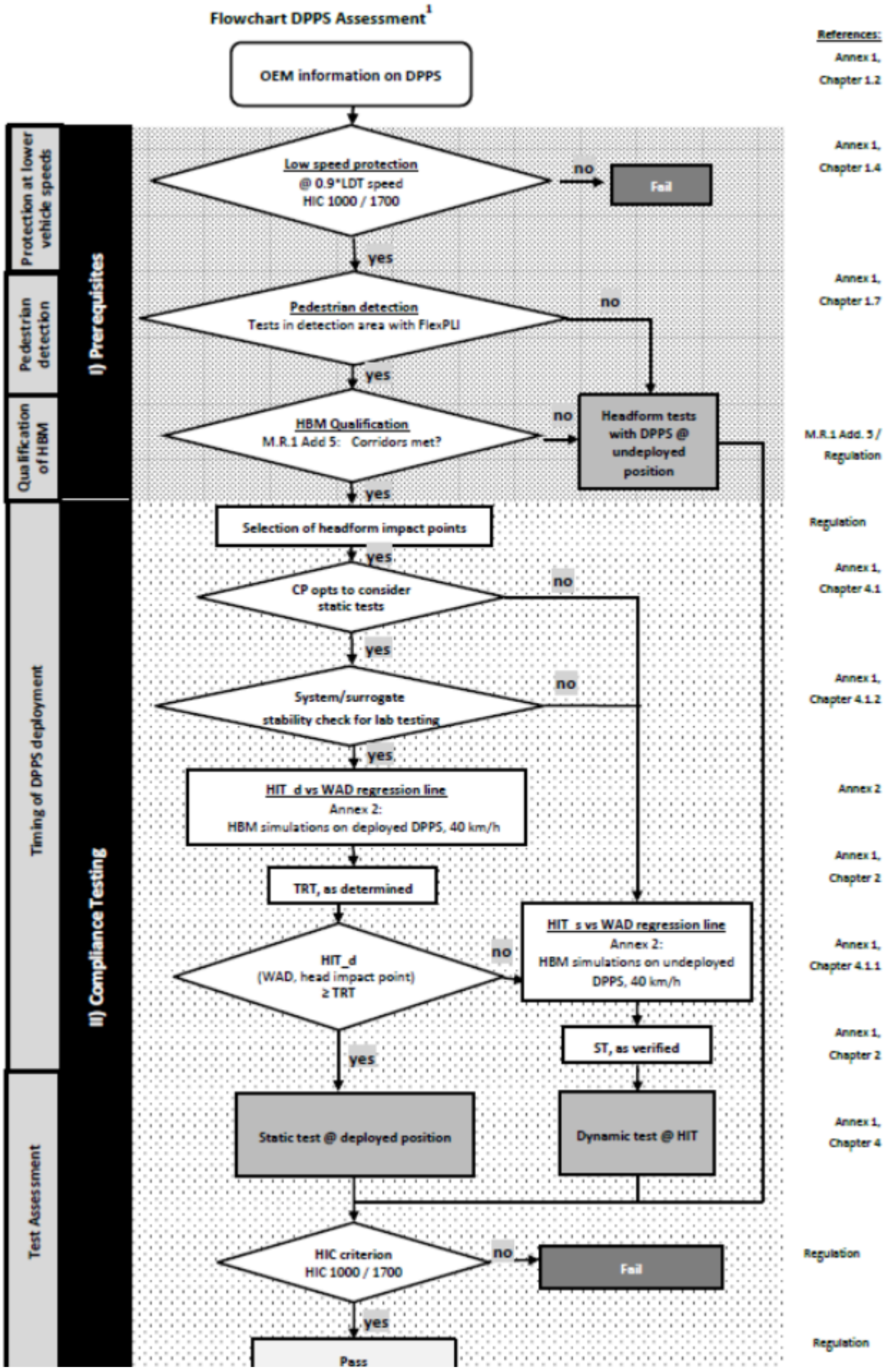
If the headform test area consists of sections where the HIT_d at the corresponding impact point is less than TRT (HIT_d <TRT) , as in the A section in Figure 1-1, and sections where the HIT_d at the corresponding impact point is greater than or equal to TRT (HIT_d ≥TRT, as in the B section in Figure 1-1, then all test points forward of the corresponding WAD (HIT_d < TRT) shall be tested dynamically. The remaining section of the headform test area may be tested statically (see Figure 1-5 below, obtained from Annex 2, Figure 2-2).

Figure 1-5 **Scheme of HIT_d vs WAD for Combined Testing**



Annex 1 - Appendix

Flowchart DPPS Assessment Guideline

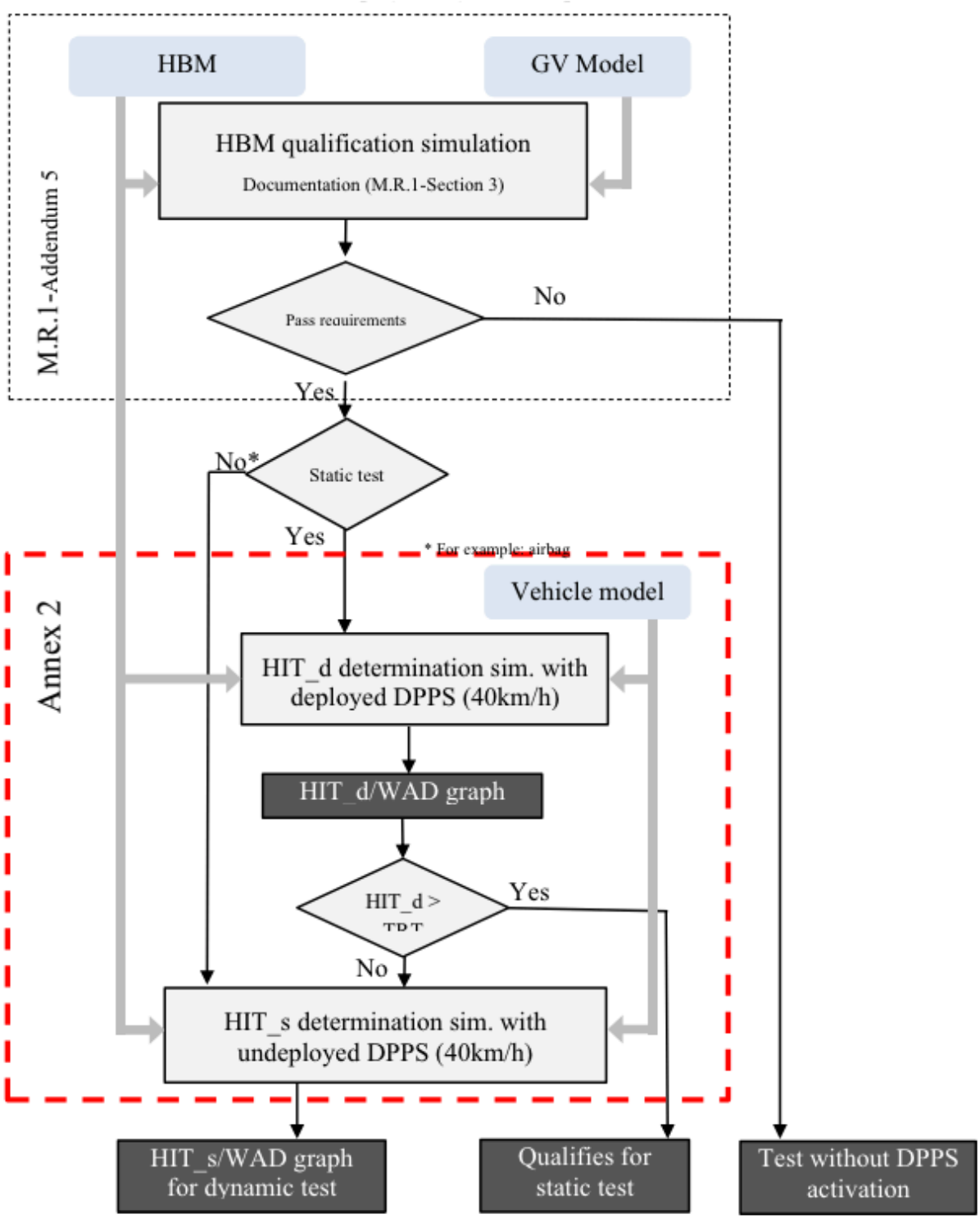

Flowchart DPPS Assessment[1]
References:
OEM information on DPPS
I) Prerequisites
Protection at lower vehicle speeds
Low speed protection
@ 0.9*LDT speed
HIC 1000 / 1700
no
Fail
Annex 1, Chapter 1.4
yes
Pedestrian detection
Pedestrian detection
Tests in detection area with FlexPLI
no
Annex 1, Chapter 1.7
yes
Qualification of HBM
HBM Qualification
M.R.1 Add 5: Corridors met?
no
Headform tests with DPPS @ undeployed position
M.R.1 Add. 5 / Regulation
yes
II) Compliance Testing
Timing of DPPS deployment
Selection of headform impact points
Regulation
yes
CP opts to consider static tests
no
Annex 1, Chapter 4.1
yes
System/surrogate stability check for lab testing
no
Annex 1, Chapter 4.1.2
yes
HIT_d vs WAD regression line
Annex 2:
HBM simulations on deployed DPPS, 40 km/h
Annex 2
TRT, as determined
Annex 1, Chapter 2
HIT_d
(WAD, head impact point)
≥ TRT
no
HIT_s vs WAD regression line
Annex 2:
HBM simulations on undeployed DPPS, 40 km/h
Annex 1, Chapter 4.1.1
yes
ST, as verified
Annex 1, Chapter 2
Test Assessment
Static test @ deployed position
Dynamic test @ HIT
Annex 1, Chapter 4
HIC criterion
HIC 1000 / 1700
no
Fail
Regulation
yes
Pass
Regulation

1: Will be updated with DPPS Phase 2 (Generic approach)

Annex 2 Head Impact Time (HIT)- Determination Simulation

1. Introduction

An HIT-Determination simulation is a computer simulation for determination of HIT over WAD in the DPPS vehicle model for deriving the test conditions for the assessment of deployable systems as specified in the Annex 1.



1.2. Definitions

For the purposes of this annex:

1.2.1. "Generic Vehicle (GV) Models" are generic replications of car fronts representing three vehicle categories: Family Cars (FCR), Roadsters (RDS), Sports Utility Vehicles (SUV). (The shape of the generic Multi-Purpose-Vehicle (MPVs) was found to lay in between the generic FCR and generic SUV and is therefore covered already.) The vehicle models provide representative shapes for the selected vehicle categories as well as median structural response upon pedestrian impact in terms of force- deflection characteristics and are modelled to be robust and transferable to all considered explicit Finite Element (FE) codes.

1.2.2. "Human Body Model" (HBM) is understood as a virtual geometric and mechanical representation of the human body, which takes the human anatomy into consideration. The procedure described in this Annex refers to HBMs used for the simulation of pedestrian impacts. Pedestrian models which are required for GTR No. 9 shall be selected from the following statures, a six-year-old (6YO), 5th percentile female (AF05), 50th percentile male (AM50) and 95th percentile male (AM95).

1.2.3. "HBM qualification simulation": A computer simulation (HBM vs. GV Model) providing evidence that the specific HBM simulation is comparable with reference simulations and shows consistent results – in particular referring to HIT and WAD. The reference simulations are based on models which have been validated by comparing their simulation response with PMHS tests. Another purpose is to make sure that models give comparable results with varying hardware or software environments when applied for a specific purpose.

1.3. General Requirements

1.3.1. It shall be ensured that the HBMs used in this Annex comply with all requirements within Addendum 5 of Mutual Resolution No.1 (M.R.1). The qualification results shall be documented as specified in Addendum 5 of M.R.1.

1.3.2. Only those HBM statures selected according to paragraph 2.2. of this Annex shall be qualified.

1.3.3. The pedestrian HBM that is qualified is the very same model as used for HIT determination simulations. This applies to:

(a) Version of the HBM;

(b) Node-Position of every single node of the HBM;

(c) Identical material cards (including fracture mode), contact cards, control cards and constraints;

If available:

(d) identical initial element stresses/strains;

(e) identical initial contact penetrations/contact forces.

1.3.4. Furthermore, all simulations (qualification and HIT determination) shall be performed with consistent settings. This applies to:

(a) Solver-Version and type (e.g. processing type, precision, parallelisation);

(b) The time-step used for simulations;

(c) Time-step settings (relating to initial and dynamic mass scaling);

(d) Contact settings (between HBM and Vehicle);

(e) Control settings which are affecting the pedestrian model.

2. Procedure

2.1. Impact Simulations

Pedestrian models shall be selected from the following statures, a six-year-old (6YO), 5th percentile female (AF05), 50th percentile male (AM50) and 95th percentile male (AM95). The pedestrian position and stance to be used in the model is defined in Addendum 5 of M.R.1. The pedestrian model has to be positioned, such that the head centre of gravity (CoG) is aligned with the vehicle centreline.

The vehicle model has to be positioned in the setup such that the vehicle ground level is aligned with the ground level used in the qualification simulations.

As described in Addendum 5 of M.R.1, the HBM shall be exposed to a vertical acceleration field constituting the gravitational loading.

A local vehicle coordinate system has to be initially aligned with the global coordinate system defined in Addendum 5 of M.R.1 and shall be connected to the vehicle model CoG.

The initial speed of the vehicle model has to be prescribed and is 40 km/h for all simulations. The y and z motion of the car has to be constrained and the motion in x-direction must not be constrained.

2.2. Selection of HBMs

The selected HBMs (needed to draw the WAD/HIT-line in the evaluation) are those HBMs where the head hits the DPPS properly, which is when:

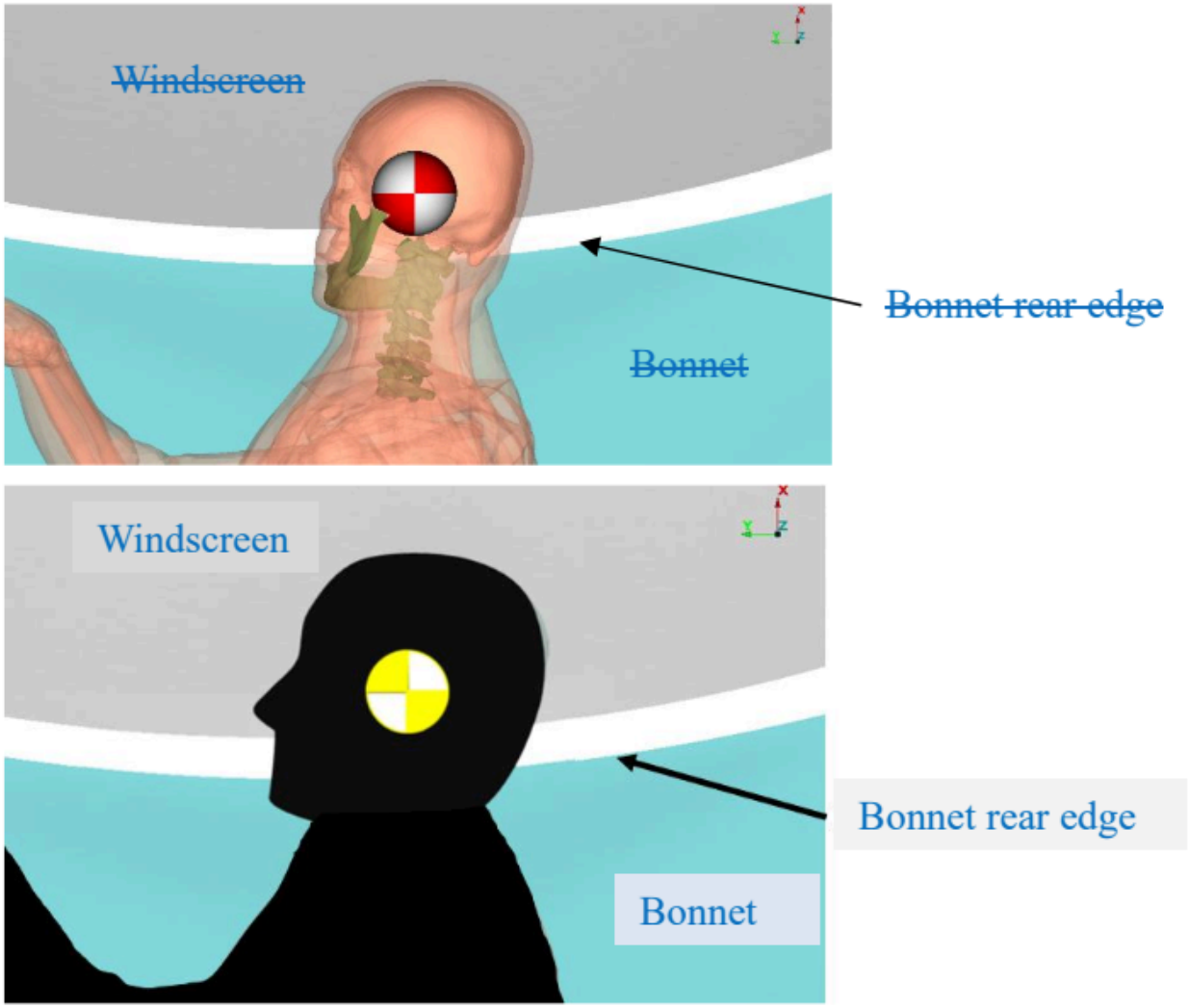
• there is a contact between the head and DPPS.

• at time of this contact the x-coordinate of the CoG of the head is between the smallest and largest x-coordinate of the DPPS at y=0.

Simulations with the next tallest HBM shall also be performed, but only to prove that this HBM does not hit the DPPS properly.

If only one HBM hits the DPPS properly, the next tallest HBM shall also belong to the selected HBMs.

Figure 2-1 **Example (where the CoG of the head lies behind the DPPS at time of contact. This HBM does not hit the DPPS properly (only contact with chin of HBM))**



2.3. Output Requirements

It shall be confirmed that the following outputs have been generated from each simulation, time history curves of:

(a) x and z coordinate of HC and AC in the global coordinate system;

(b) x displacement of vehicle CoG in the global coordinate system;

(c) Resultant acceleration of HC;

(d) Contact forces (between vehicle and HBM without upper extremities, vehicle and HBM head and total contact force);

(e) Total hourglass and internal energies of the total setup;

(f) Mass increase.

All shall be plotted every 0.1ms or less.

Furthermore, animations of the simulations shall be generated with an output interval of 1ms.

2.4. Quality Checks

The following Quality Checks shall be performed:

(a) Contact force (between HBM and vehicle) is zero at simulation start;

(b) Total energy remains constant within a 15 per cent tolerance;

(c) Hourglass energy ≤ 10 per cent of the total energy;

(d) Artificial mass increase is less than 3 per cent.

2.5. Calculation of Head Impact Time

Time of first contact is defined as the first time where the contact force is not zero anymore.

The Head Impact Time (HIT) is defined as the elapsed time subsequent to the time of first contact of the HBM (neglecting forearms and hands) with the vehicle outer surface and the time of first contact of its head with the vehicle outer surface.

If this method is for any reason not applicable, an appropriate alternative method may be applied and shall be reported.

2.6. Determination of WAD corresponding to HIT

For the determination of the WAD, a point on the surface of the vehicle is necessary. This point is defined as follows (all coordinates relative to the local vehicle coordinate system):

At time of first head contact with the DPPS the point

(X_{head}, 0, Z_{head})

where:

X_{head} is the x-coordinate; and

Z_{head} is the z-coordinate of the CoG of the head

will be projected orthogonally onto the surface of the undeployed vehicle. (If there are multiple projection points take the one with the highest x value.)

Compute the WAD for this point rounded the nearest full millimetre.

3. Documentation

3.1. General

The following information shall be documented:

(a) Date of report;

(b) Name of car manufacturer;

(c) Type and release version of software (FE-software package name, revision and version);

(d) Name and version of the HBM;

(e) Specification of car.

Images showing the front view and side view of the pedestrian, at t0 and at the time of head impact shall be added to the report.

3.2. Consistency with qualification simulations

For all simulations Table 2-1 shall be filled in.

Table 2-1 **Check of consistency between qualification and HIT determination simulations**

Checklist for simulation settings	Consistent between qualification and HIT determination simulations:
Identical HBM	Y/N
Solver Version	Y/N
Timestep	Y/N
All other control settings	Y/N

3.3. Quality Checks

For all simulations Table 2-2 shall be filled in.

Table 2-2 **Quality Checks**

Verification evaluation criteria	Allowed	Observed	Pass:
Coefficient of friction between Vehicle and HBM	0.3		Y/N
Head centre of gravity is positioned at vehicle centreline	Y=0 mm		Y/N
Contact force between HBM and vehicle at simulation start	0		Y/N
Change in total energy throughout simulation	≤15%		Y/N
Amount of hourglass energy relative to total energy	≤10%		Y/N
Artificial mass increase relative to total mass of the setup	≤3%		Y/N

3.4. Results of HIT determination simulations

For those HBMs that are selected according to paragraph 2.2. of this Annex, the computed HIT-Values and corresponding WADs have to be filled into the following Tables 2-3 and 2-4.

If HIT_d ≥ TRT for all HBMs, simulations on the undeployed DPPS are not required.

Table 2-3 **HIT_d Simulations on DPPS in Deployed Mode**

HBM	WAD (mm)	HIT_d (ms)
6YO		
AF05		
AM50		
AM95		

Table 2-4 **HIT_s Simulations on DPPS in Undeployed Mode**

HBM	WAD (mm)	HIT_s (ms)
6YO		
AF05		
AM50		
AM95		

For each simulation, the following diagrams shall be documented:

(a) ACx and HCx as a function of time;

(b) ACz and HCz as a function of time;

(c) HCz as a function of HCx and ACz as a function of ACx;

(d) Total Contact Force between HBM and vehicle as a function of time;

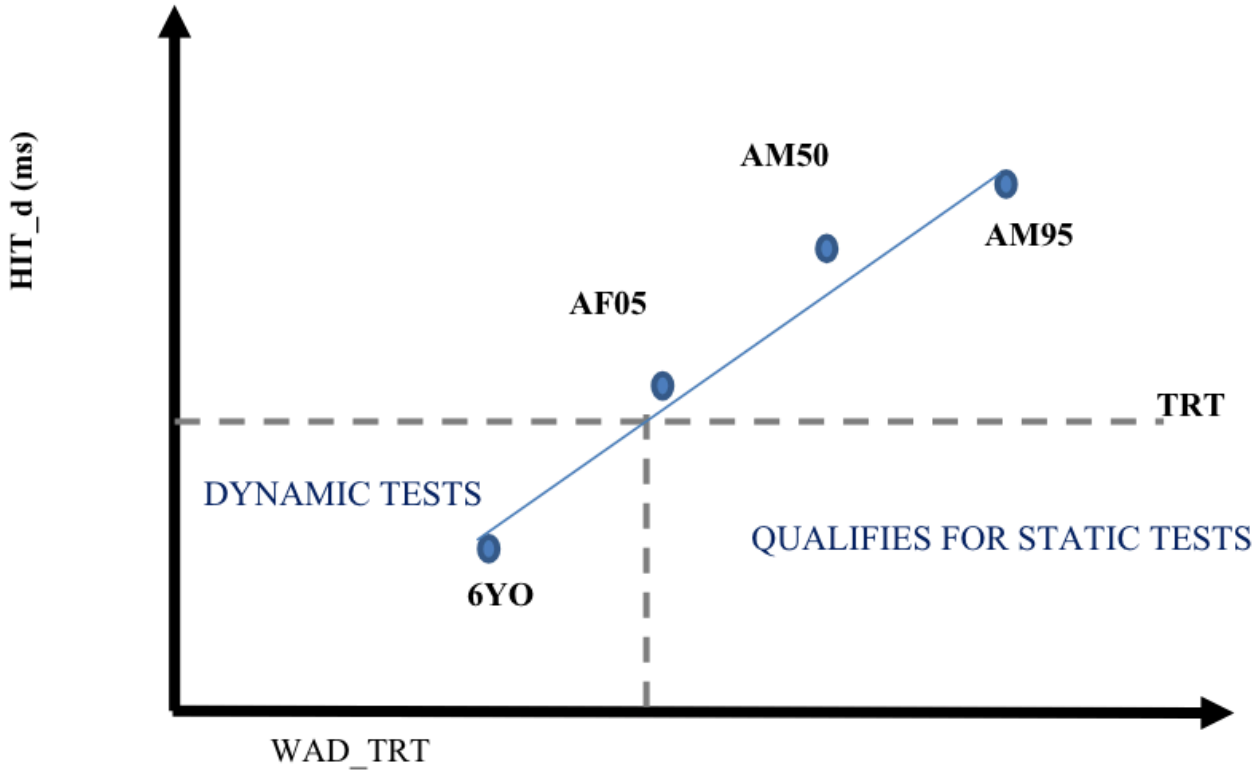
(e) Total, kinetic, internal and hourglass energy as a function of time.

4. Evaluation

4.1. HIT_d Simulations with Deployed DPPS

Based on the results of Table 2-3, a graph shall be plotted using a linear regression line for comparison with TRT in the diagram as shown in Figure 2-2.

Figure 2-2 **Wrap Around Distance versus Head Impact Time_for decision (WAD vs HIT_d)**



4.2. HIT_s Simulations with Undeployed DPPS

Based on the results of Table 2-4, a graph shall be plotted using a linear regression line as shown in Figure 2-3. The lines have to be extrapolated in both directions.

Figure 2-3 **Wrap Around Distance versus Head Impact Time_s (s for synchronisation) (WAD vs HIT_s)**

